

FROM THE AUTHOR OF
A IS FOR ARSENIC



IS FOR VENOM

AGATHA CHRISTIE'S
CHEMICALS OF DEATH

KATHRYN HARKUP

BLOOMSBURY

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V IS FOR VENOM

Agatha Christie's Chemicals of Death

Kathryn Harkup

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The Queen of Crime

Dame Agatha Mary Clarissa Christie (1890–1976) is the acknowledged ‘Queen of Crime’. She has also been called the ‘Mistress of Mystery’ and the ‘Duchess of Death’. She described herself as ‘a rather good writer of detective stories’, but she was far more than this. Agatha Miller (her maiden name) worked as a nurse and qualified as a pharmaceutical dispenser. Her singing voice was good enough that she could have made a career in concert halls, if she hadn’t suffered from debilitating stage fright. Agatha Mallowan, her married name after her second marriage, is honourably mentioned in the annals of archaeology for the work she did cleaning, photographing and cataloguing finds from her husband’s digs in Iraq and Syria. But most people know of Agatha Christie (her married name from her first marriage) from her prolific writing: 78 crime novels, 150-odd short stories, six ‘straight’ novels, four non-fiction books and 20 plays.

There are many statistics relating to Christie’s popularity and publishing prowess. In terms of influence, it has been vast. She was the inspiration behind Penguin paperback books, *The Murder of Roger Ackroyd* was the first ever audio book, and her detective Hercule Poirot is the only fictional character to have been given a front-page obituary in *The New York Times*. From a sales point of view, the numbers are so large and difficult to keep track of, thanks to the sheer number of editions and formats her words have appeared in around the world, that they become almost meaningless. One widely repeated statistic is that Agatha Christie is the third-most widely published author of all time, outsold only by the Bible and Shakespeare. It would be difficult if not impossible to verify this statistic, but there are plenty of others that make it believable. In 1959 UNESCO reported that, with her books available for sale in 102 countries, Christie was the

bestselling author in the English language. A survey of passengers on United Airlines in the 1970s found that 58.6 per cent carried hand luggage with them and 47.2 per cent carried an Agatha Christie book. By 1990, Christie had sold an estimated two billion books and had been translated into 104 languages.

If Christie's popularity in print was not enough, she is also the most successful female playwright, bar none. She is the only woman to have had three plays running concurrently in the West End.* A survey of productions staged in Britain's repertory theatres in 1983 revealed that out of 28 plays by women, Christie had written 22 of them.† Her play *The Mousetrap* opened in the West End in 1952 and is *still* running. Only a global pandemic could temporarily interrupt its 73-year run.‡ The play has set and broken so many records it is almost in a category of its own.

Then there are the big- and small-screen adaptations of Christie's works. There have been at least 11 film versions of *And Then There Were None*, made for English, Russian, Hindi and Tamil-speaking cinema-goers. The 1974 film of her novel *Murder on the Orient Express* broke box-office records. Her work has also been heard on radio, enjoyed in graphic novels and played as board games. Praise, prizes and awards have been heaped on Christie. It is almost 50 years since her death and the love for her writing shows no signs of abating.

Mayhem Parva

Why are Christie's stories so enduringly popular? There have been many theories put forward. In 1946, George Orwell mused on the fascination for accounts of real-life murders reported in newspapers throughout the nineteenth and early twentieth centuries. In his essay, 'The Decline of the English Murder', he outlined the key elements for the perfect murder that would have most appeal to these true crime fans. The murderer should be a professional man – a dentist, or solicitor – and he should 'go astray through cherishing a guilty passion for his secretary, or the wife of a rival professional man'. Having struggled with his conscience, he would decide on murder, and one that was fiendishly well planned out but for one small, overlooked detail that would lead to his detection. 'The means should, of

course, be poison.’ Many of Christie’s stories meet Orwell’s criteria, but then so do those written by other authors.

Several people have drawn attention to the settings where Christie’s murders take place. In his essay ‘The Simple Art of Murder’, Raymond Chandler contrasted the grittier fictional world of gangs, bullets and beautiful dames that he became famous for, with the murders among the English upper classes full of ingenious tricks and plodding policemen. Chandler’s fictional crimes were usually committed in the seedy hotel rooms and illegal gambling dens of America’s big cities. The corpses written about by Christie were more likely to be found in the manicured grounds or on the expensive rugs of English country houses. In his 1979 book *Snobbery with Violence*, Colin Watson coined the name ‘Mayhem Parva’* for the genteel, more rural style of many classic British crime novels.

All crime fiction has to have an element of detachment from the real world; otherwise, as Chandler pointed out, it would only be written and read by psychopaths. The world Christie created is perhaps a little more detached from horrible reality than others. But, again, many other writers have done the same, blotting their charming chocolate-box sceneries with large numbers of corpses. The bodies in these settings are usually relatively free of the blood and gore that would be expected in real life. The corpse is of interest, not as a comment on society or as an example of human beings’ worst instincts, but because it presents a puzzle to be solved. It is this puzzle element that most people cite as the source of Christie’s appeal to readers.

A Mass Observation report published during the Second World War found Christie a consistently popular choice for reading during an air raid. As one woman explained, ‘I like to have to concentrate. The working it all out – you know – it soothes your nerves.’ Christie’s deceptively simple prose hides a wealth of information that has to be carefully unpacked and properly interpreted if you want to work out ‘whodunit’ before Miss Marple or Hercule Poirot gather the suspects together for the denouement. The straightforward presentation of facts often tricks us into overcomplicating things. Christie’s ability to hide the solution in plain sight is second to none.

The simple sentence structure and plain English Christie used in all her stories also extends to the science of the poisons she was so fond of using. All the scientific information and forensic detail necessary for solving the mystery is presented fairly. Forensic reports, details of the compounds involved and symptoms they might produce are all summarised and presented in layman's terms. This was certainly necessary as the most avid reader of murder mysteries, even doctors and toxicologists, are unlikely to have come across the number and range of deadly substances Christie used to kill her characters. The details of these killer chemicals is not just accurate, but often woven into the plots in such a brilliant way that they almost become characters in their own right.

The use of poisons, so often and in such variety, is where Christie truly stands out. Such expert and deft use of toxic substances is one of her great strengths as a murder mystery writer. Where did Christie's detailed toxicological knowledge come from? From direct experience with poisons and a life-long interest in the subject, though not in the criminal sense.

The poisoner's apprentice

During the First World War Christie, like many women, volunteered as a nurse at her local hospital in Torquay. She enjoyed the work but the hours were very unsociable and her family worried about her walking back alone from the hospital late at night. When a new dispensary opened at the hospital it was suggested that she might prefer to work there as she would be working regular office hours.

Christie needed further training and to pass several exams before she would be qualified as an apothecaries' assistant, or dispenser. This was no easy task. The exams were difficult, as you would expect when they were testing someone for a job that involves very real risks. The drugs available in the early twentieth century were very different to what you might be prescribed today. Pharmacy shelves were stacked with bottles of arsenic, strychnine, coniine (from hemlock) and many other compounds that would be considered poisons rather than medicines today.

As medicine has moved on, many of the compounds Christie began her dispensing career with have disappeared from the pharmacopoeia because

their dangers outweigh any benefits they may offer and better alternatives have been discovered. However, some medicines are still with us from this time. Morphine, digitalis (in a refined form), atropine and barbiturates are still given to patients today, though not as freely as in the past, and with much better monitoring of their effects. Nevertheless, 'Poison is in everything, and no thing is without poison. The dosage makes it either a poison or a remedy', as the physician and founder of toxicology, Paracelsus (1493–1541), pointed out. Christie understood the difference between the therapeutic and lethal dose as well as appreciating the dangers of not doing her job properly.

When Christie trained to be a dispenser, drugs did not come ready formulated, pre-packaged or even pre-weighed. Everything was measured out, mixed together and processed into a cream, pill or tonic by hand and according to the doctor's prescription. Poisons and dangerous drugs were carefully weighed out and checked by colleagues before being dispensed. Other innocuous ingredients, such as colouring or flavouring, could be added according to personal taste. Get things wrong and the customer might complain their usual medicine didn't taste right or worse, the pills could fall apart in the bottle. But, as long as the drug was there in the right dose, this was not usually a serious problem.

The prescriptions themselves were not standard either. Every doctor had their own particular preferences for treatments and doses of drugs. Then there was the potential problem of mixing the wrong drugs together in the same pill or tonic, causing them to react together to form something new and possibly dangerous. Some drugs can also influence how others behave inside the body if they are swallowed together.

Adding to the complications, in the early twentieth century pharmacists in the UK were transitioning from imperial to metric measurements. Quantities of drugs had been traditionally measured in grains, rather than the modern system of grams (one grain is equivalent to 64.79891mg).^{*} Christie quoted quantities of poisons in grains throughout her writing career as she was not a fan of the metric system. 'The great danger of the metric system', she said, 'is that if you go wrong you go ten times wrong.'

This was something she appreciated from first-hand experience. As well as being tutored in practical and theoretical aspects of chemistry and

pharmacy by her colleagues at the dispensary, Agatha also received private tuition from a pharmacist in Torquay, the disconcerting Mr P. One day Mr P was showing her the correct way to make suppositories. He melted cocoa butter, added the drug and then demonstrated the precise moment to turn the suppositories out of the moulds. He then showed Agatha the correct way to box up and label the drugs correctly. Mr P wrote 'one in one hundred' on the box of suppositories but Christie was convinced the pharmacist had made a mistake and added a dose of *one in ten*, or ten times the prescribed dose and potentially dangerous. She surreptitiously checked his calculations and confirmed her suspicions. Unable to confront the pharmacist with his mistake, and frightened of the consequences of dispensing the dangerous medicine, she pretended to trip and knocked the suppositories to the floor where she trod on them firmly. After she had apologised profusely and cleared up the mess, a fresh batch was made, but this time at the correct dose.

It was not Mr P's inattention to detail that unnerved the young Agatha the most. One day he pulled a brown lump from his pocket and asked her what she thought it might be. Christie was perplexed but Mr P told her it was a lump of curare, a poison used by hunters on the tips of their arrows (see Chapter A, page 24). Curare, he explained, is completely safe to eat but deadly if introduced directly into the bloodstream. He said he carried it around with him because, 'it makes me feel powerful.'^{*}

*

All that studying and practice paid off in the end. Christie passed her exams in 1917 and went to work in the hospital dispensary.[†] She soon settled into the work and became familiar with the doctors' preferred prescriptions. She slipped into an easy routine that left her with time on her hands. Time to wonder and think.

Christie and her sister Madge were both fans of detective fiction. After reading *The Mystery of the Yellow Room* by Gaston Leroux (originally published in 1907), Christie had said she would like to try and write a detective novel herself. Her sister, a more successful writer than Agatha at that time, thought that it would be very difficult and bet Agatha that she

would not be able to do it. It was not a formal bet with terms and amounts but an idea had been seeded in her mind and started to sprout while she was sitting in the dispensary surrounded by deadly poisons.

Christie drew on her recently acquired knowledge and experience to write her first detective novel, *The Mysterious Affair at Styles*. She used not just one but a combination of three poisons (strychnine, potassium bromide and an opiate) to kill the unfortunate Mrs Inglethorpe. Her thorough understanding of drug interactions meant medicinal doses of two of the compounds mixed together could have a deadly effect. Her explanation for how these two drugs worked together to kill Mrs Inglethorpe is lifted almost word for word from *The Art of Dispensing*, a book she likely studied for her exams. The third drug delays the onset of symptoms, throwing everyone's carefully worked-out alibis into doubt.* Among the list of suspects in *Styles* are dispensers, doctors and pharmacists. This is also the novel where she introduced a retired Belgian police officer as her detective and, in the tradition of the detective stories she so enjoyed, she gave him a faithful sidekick. However, she had to wait a few years and try a number of publishing houses before the novel was accepted.

After publication in 1920, *The Mysterious Affair at Styles* received Christie's most cherished compliment in a review by the *Pharmaceutical Journal*: 'This novel has the rare merit of being correctly written'. The reviewer was certain the author must have had pharmaceutical training or called in an expert.

A criminal career

The year 1920 marked not just the start of Christie's long and successful writing career, but also the beginning of what would later be described as 'the Golden Age of detective fiction'. This is when detective stories became an established and very popular genre. To illustrate the point, in 1914 the *American Book Review Digest* reviewed fewer than a dozen crime stories. The number had grown to 97 by 1925 and an incredible 217 in 1939. As the genre developed so did expectations of what made a good detective story. In 1928 these criteria were formalised into a set of 10 rules by Ronald Knox,

priest, theologian, BBC broadcaster and writer of detective stories featuring his detective Miles Bredon. 'Knox's Ten Commandments' stated that:

1. *The criminal must be someone mentioned in the early part of the story, but must not be anyone whose thoughts the reader has been allowed to follow.*
2. *All supernatural agencies are ruled out as a matter of course.*
3. *Not more than one secret room or passage is allowable.*
4. *No hitherto undiscovered poisons may be used, nor any appliance which will need long scientific explanation at the end.*
5. *No Chinamen must figure in the story.*
6. *No accident must ever help the detective, nor must he ever have an unaccountable intuition that proves to be right.*
7. *The detective must not himself commit the crime.*
8. *The detective must not light on any clues that are not instantly disclosed to the reader.*
9. *The stupid friend of the detective, the Watson, must not conceal any thoughts that pass through his mind; his intelligence must be slightly, but very slightly, below that of the average reader.*
10. *Twin brothers, and doubles generally, must not appear unless we have been duly prepared for them.*

Two years after the writing of these rules Knox, along with Christie and a few other well-known authors, formed The Detection Club, a dining club for writers of detective fiction. Knox's rules were adopted by the club as a kind of code of ethics to be observed in their professional work. Members also had to swear an oath as part of an elaborate initiation ceremony.

Do you promise that your detectives shall well and truly detect the crimes presented to them using those wits which it may please you to bestow upon them and not placing reliance on nor making use of Divine Revelation, Feminine Intuition, Mumbo Jumbo, Jiggery-Pokery, Coincidence, or Act of God?

Agatha Christie was part of the detective writing establishment but she didn't always stick to their rules, though she mostly kept her oath. Few

people today would consider her to be a controversial writer. To modern readers her stories may seem conventional, even clichéd or stereotypical, but that is probably because so many people have copied, parodied and mimicked Christie's work. In fact, Christie was very unconventional. In particular, her 1926 novel, *The Murder of Roger Ackroyd*, caused severe consternation at the time of publication. Readers and newspaper articles accused Christie of cheating but several fellow detective fiction authors came to her defence. Today the novel is regarded as one of the best detective stories of all time.

The Murder of Roger Ackroyd was not an isolated incident. The rules Christie didn't break outright, she strained to their very limit. The only rule she adhered to faithfully was one not included in the list or oath: to play fair with the reader.* Christie was herself very proud of the fact that she never 'cheated'. Anyone surprised by her endings may be convinced something has been kept from them, but if they go back and look, the clues will always be there.

When people talk of the 'Golden Age' of detective fiction today, Agatha Christie is usually the first name to be mentioned, alongside Dorothy L. Sayers, Margery Allingham and Ngaio Marsh, as producers of the finest examples of writing from this era. Christie's ability to defy convention, challenge expectations and still retain a massive and loyal fanbase came from her thorough understanding of the genre.

Christie used her knowledge of all the well-known tropes of detective stories and twisted them to her advantage, as we shall see in the coming chapters. Her stories are not just packed with carefully constructed problems to please her readers; but there are also nods to other fictional detective stories and she repeatedly name-dropped real-life murderers. As well as her broad knowledge of detective fiction, Christie also read about real crime extensively. Occasionally she used real-life cases as direct inspiration for her plots. These added extras never detract from the main story but give another layer of interest for those in the know.

Some real-life cases

Fans of true crime will have no particular advantage over other readers of Christie's novels as she rarely used the exact same circumstances for the basis of her main plot.* Usually these real crimes were the starting point for a thought experiment. For example, what was the impact on those accused of killing Charles Bravo?

In 1875, Charles Bravo was taken ill after eating dinner with his wife, Florence, and her live-in companion Jane Cox. He was treated by Dr James Gully, his wife's former lover, but died three days later. When Bravo's death was judged to be homicide by poison, specifically antimony poisoning, the three people who had been with him in his final days became the chief suspects, but nothing could be proved. No one was ever convicted of the crime but suspicion hung around Florence Bravo, Jane Cox and Dr Gully for the rest of their lives. To quote Agatha Christie, 'Someone was guilty – and got away with it. But the others were innocent – and didn't get away with anything.'

Christie's 1958 novel, *Ordeal by Innocence*, explores what happens when the wrong person is convicted of murdering the mother of a large family but is later found to be innocent. A bereaved family has all their certainty and security taken away from them and they are left wondering who among them really did it.

In another example, the case of 'The Dandelion Poisoner' provided some of the details for the murder of Mrs Horton in Christie's 1939 novel *Murder is Easy*. Mrs Horton is described as a domineering wife who was hospitalised for a while but was discharged to complete her recovery at home. Though she appeared to be doing well, Mrs Horton suffered a sudden relapse and died.

The real-life example happened in 1919 when Mrs Armstrong was hospitalised suffering from neuritis. She made a recovery and was discharged, only to suffer a relapse and die. Her death was attributed to natural causes, but when her husband was accused of attempting to poison someone else the circumstances around her death were looked into more closely. Her body was exhumed and found to be full of arsenic. A large quantity of arsenic was also found in Armstrong's house, though he claimed he had bought it to kill dandelions in his lawn. He was not believed.

There are many other examples of classic cases in Christie's work for any true crime fans to look for, and not all of them relate to poisonings. Nevertheless, poison would find its way back into Christie's work again and again.

Agatha and poisons

Poison may be a recurring theme in Christie's stories, but it never becomes predictable or boring. She interspersed her chemical killings with many different methods of murder, and when she did go back to poison she changed the compounds she used and the method of administration. The foundations of this vast repertoire of toxic substances came from her dispensing days in the First World War, but she added to her knowledge throughout her life.

During the Second World War Christie volunteered again as a dispenser at the University College Hospital in London, despite the obvious dangers: 'Hospital still standing, though flattened buildings all around,' she wrote. It was over 20 years since she had passed her apothecaries' assistants' exams and medicine had changed considerably in the intervening period. The process of renewing her training would have added to her stock of potential poisons. More medicines were also now pre-prepared in standard formulations leaving her with ample thinking time during her shifts. She worked two full days a week, three half days and Saturday mornings. She also filled in when other workers were unable to get to the hospital. Despite the extra shifts Christie still had time on her hands. Her husband, Max Mallowan, had been posted to North Africa and many of her friends and relations had scattered to the relative safety of the countryside. To fill the empty hours and to distract herself from the air raids, Christie used her free time to think out new stories and invent fiendishly deceptive plots. She completed 12 novels during the war years.

After the war she divided her time between the UK and wherever her husband's archaeological work took him, mostly Iraq and Syria. She travelled, went to the theatre and enjoyed life. She also found time to complete a new book every year. Having left her dispensing work and direct involvement with medicines behind, Christie built up a considerable

medico-legal library. She also corresponded with experts to check her facts. The breadth and depth of her knowledge can be seen throughout her writing but is particularly evident in the lesser-known poisons she employed. That is what this book hopes to showcase: the accurate and inventive use of the more unusual means of chemical killing that makes Agatha Christie a true ‘Queen of Crime’.

Notes

- * *The Mousetrap*, *Witness for the Prosecution* and *Spider’s Web*.
- † Careful readers will notice a discrepancy with the number of Christie’s plays mentioned in the first paragraph. Christie wrote 20 plays that have been published, though not all have been performed. On top of this, she wrote several plays that have not been published or performed. Then there are a handful of Christie’s stories that were adapted for the stage by other playwrights and are still performed, but are now usually credited with joint authorship.
- ‡ As of 2025. *The Mousetrap* shows no sign of closing any time soon.
- * Parva from the Latin for ‘little’. There are several places in England given this name, such as Appleby Parva in Leicestershire, to distinguish it from a larger settlement nearby, in this case Appleby Magna.
- * This book will give equivalent measurements in grams (g) or milligrams (mg, thousandths of grams) and ounces (oz) for American readers. One grain is equivalent to 0.0023oz.
- * Nearly 50 years later Christie resurrected Mr P as the sinister pharmacist in her novel *The Pale Horse*.
- † She failed first time round, but only the practical part of the exams. She passed on the second try.
- * If you are interested in the scientific detail of Christie’s poison plot in *The Mysterious Affair at Styles*, my previous book, *A is for Arsenic*, discusses it at great length.
- * To play fair with anyone reading this book, I will not be giving away whodunit in Christie’s mysteries. If there is simply no way round it, I promise to give lots of warning to look away if you don’t want to read spoilers.
- * One possible exception is her novel *The Mirror Crack’d from Side to Side* (see Chapter C, page 69).



‘The Case of the Caretaker’ (Miss Marple’s Final Cases)

Poisons have always had a certain appeal to both the murder mystery writer and reader. With poisons, the murderer doesn’t always have to be on hand at the point of death, which opens up the field of suspects. Clues can be guessed from the symptoms displayed by the victim before they die. If and when the identity of the lethal substance is confirmed at autopsy, there is then speculation as to who would have knowledge of such poisons and the opportunity to get hold of them. The more obscure the poison, the more

difficult it is to identify and the more speculating can be done to keep the reader guessing.

Writers undoubtedly search for something new and unusual to intrigue their audiences. Perhaps this is why little-known compounds used as arrow poisons in far-flung parts of the world that leave no trace at autopsy became something of a cliché in murder mystery stories. Many such substances, both real and invented, have been employed to get rid of characters without leaving any obvious clues that might make the mystery too easy to solve.

While other murder mystery writers searched the globe and their imagination for rare killer compounds, Christie delved into the deepest recesses of the pharmacopoeia for her obscure deadly substances. She was certainly not above using arrow poisons in her stories, but she was also aware that her readers might have come to expect such things. Part of the appeal of Christie's writing is that she doesn't often do what you would expect her to do, and her use of arrow poisons is no exception.

The arrow poison story

Arrow poison is a broad term. For thousands of years people all over the world have tipped their weapons with toxic compounds. A smear of carefully chosen substances on a blade or arrow point can mean even a scratch from these weapons can prove fatal. Warriors and hunters have used all sorts of chemical advantages provided by plants and animals, depending on local availability and what the weapon is to be used for. If the goal is simply to kill your enemy, a fast-acting toxin is the order of the day. But, if you want to eat what you kill afterwards, you have to be careful your choice of poison doesn't kill you as well.

In some parts of South America, curare was the substance of choice to smear on hunters' weapons. It was also a popular choice with murder mystery writers around the turn of the century. Christie was well aware of the toxic extract of the *Strychnos toxifera* plant. She was introduced to it by her mentor at a Torquay pharmacy who was helping her to train for her apothecaries' assistants' exams (see page 14). As the pharmacist explained, curare was favoured by South American hunters because it was deadly

when introduced directly into the bloodstream, but it was not absorbed into the body from the gut, meaning it was completely safe to eat.

A curare-tipped arrow fired into an animal will cause it to collapse and become immobile very quickly. From the bloodstream, curare is quickly distributed around the body where it interacts with nerves controlling movement. The result is paralysis. The hunters, therefore, don't have to go very far to collect their prey, though the animal may not actually be dead at that point.* The paralysis that brings these creatures to a stop also affects the muscles that control breathing, which is what ultimately kills them.

Curare, and arrow poisons generally, are referenced in several Christie stories. When a victim suddenly falls dead on the floor for no apparent reason, she often has witnesses blaming 'the arrow poisons of the South American Indians'. These references are always very tongue in cheek and are quickly dismissed as red herrings, sometimes even before the real killer compound is revealed. She appears to be poking gentle fun at the arrow poison trope and she never used curare to kill any of her many victims. It was too well known and too obvious. But her pharmaceutical training introduced her to several other arrow poisons, some of which she was perfectly happy to use.

Because of their pronounced effects on the body, many arrow poisons have been investigated for their medical potential, and a few have been found to be very useful. Extracts of curare, for example, slightly chemically modified and appropriately used, are beneficial in surgery, particularly abdominal surgery, because the immobility produced means muscles can be stilled during an operation, causing less damage.

A very different substance, aconitine, a highly toxic alkaloid found in species of *Aconitum* plants, has been used in the past to tip arrows and poison wolves in Europe. It was also used by Agatha Christie to kill a character in her novel *4.50 from Paddington*.† This compound was briefly in vogue as a topical application for treating painful joints because its interaction with nerves caused a numbing or tingling sensation. When it became obvious that aconitine was far too toxic to be trusted even in tiny doses, and with minimal medical benefits, it was abandoned. Christie pointed out that aconitine had gone out of medical fashion when she wrote

about it in 1957, though she makes no mention of its ancient use as an arrow poison.*

Africa also had a handy supply of toxic substances for adding to arrow points. The Greek philosopher and ‘father of botany’ Theophrastus (c.371–c.287 BC) wrote about a toxic substance Ethiopians smeared on their arrows. This is believed to be the same poison used by Somalis, Rwandans and many other people living in Africa that came from a few species of plants in the Apocynaceae family. It was said to be so powerful it could kill an elephant.† The deadly addition to their weapons enabled more effective hunting of large game and also helped resist colonisers and enslavers.‡ Agatha Christie used the key component of this arrow poison, strophanthin, purified and prepared as a heart medicine, to kill three wives in three different stories between 1937 and 1958.

If the name strophanthin is unfamiliar to you, you are not alone. Few of Christie’s fans would have heard of it when her strophanthin stories were published. It was an unusual choice of heart medicine, and not even particularly well known in medical circles. The narrow gap between a therapeutic dose and a lethal dose of strophanthin was a distinct disadvantage in medicine but a boon for a murder mystery story. Such tiny amounts were needed to kill that toxicologists would have struggled to find evidence of it in a murder victim.* Strophanthin also has an exotic origin and a fascinating history behind it. It would seem to be the perfect poison for a crime writer.

The first of Christie’s three victims is Valentine Chantry, who dies after drinking a strophanthin-laced pink gin in the Hercule Poirot short story ‘Triangle at Rhodes’. Mrs Laxton is the second victim, this time injected with the same stuff to bring about her death in Miss Marple’s ‘The Case of the Caretaker’. Exactly as you would expect for a Christie story, working out who is responsible for these murders is the mystery readers are challenged to solve before her detectives reveal all. Christie’s third strophanthin death is where she subverts several expectations.

The title of Christie’s 1958 play, *Verdict*, may sound like a courtroom drama along the lines of her highly successful play *Witness for the Prosecution*, which had been turned into an Oscar-nominated film the previous year. Instead, it is a domestic drama played out in a book-filled

living room. Yes, there is a murder, but there is no mystery as to whodunit because Anya Hendryk is poisoned by Helen Rollander as the audience watches.

Anya, married to the brilliant Professor Hendryk, suffers from disseminated sclerosis (now more usually called multiple sclerosis), a degenerative disease that she may live with for years but will ultimately kill her. There is no cure and the only medical care available to her is palliative. Part of this ongoing treatment is a strong heart medicine, strophanthin. Helen, a young, attractive student, has fallen in love with Professor Hendryk. Thinking that the professor would be better off with her instead of his wife due to her physical impairment, Helen gives Anya an overdose of her heart medicine.

In all three stories, Christie emphasised the medical use of strophanthin and *only* its medical use. Untraceable arrow poisons were so overused in fiction that she may have thought no one would have believed stories of famous explorers and poisons known only to remote African tribes. But in the case of strophanthin, it was all true.

Strophanthin

From the fifteenth century onwards Europeans developed an increasing interest in the African continent, which had the side effect of them becoming increasingly exposed to local arrow poisons. Some ships carrying European invaders couldn't even make landing under the hail of deadly arrows from the shore. However, they appeared to learn very little from these experiences. In 1505, the Portuguese learned that the wood of the *Acokanthera schimperi* tree was used to make an arrow poison. It was the first new bit of information about African arrow poisons since Theophrastus's contribution, and little else would be added for another 350 years.*

In 1858, Dr David Livingstone (1813–1873), the famous Scottish explorer, missionary and physician, embarked on his expedition to the Zambezi to seek out strange new worlds and to boldly collect specimens of anything interesting he found there. Livingstone was accompanied by Dr John Kirk (1832–1922), another Scottish physician and a naturalist, who

had been given the task of finding plants that could be commercially exploited in some way.

It was during this six-year exploration of Africa's interior that the two Scots learned about kombé, a poison so powerful that a single arrow tipped with the preparation could bring down a buffalo. Hunters harvested the seeds from the long, thin fruits, or follicles, produced by *strophanthus* plants, a woody climber or liana that grows up through trees. The seeds were pounded into a paste in a mortar and then the milky juice from a euphorbia plant was added so the toxic mixture would stick to the arrows. A buffalo hit by these poisoned arrows would be tracked until it fell behind the herd. The hunters then had to wait patiently for the animal to collapse before they could approach it safely. The tissue immediately around the arrow wound would be cut away and discarded, but the rest of the animal could be eaten.

Kirk collected samples of the poison along with some arrows and stored them in a bag where he also kept his toothbrush. Brushing his teeth one morning he noticed a bitter taste. His pulse, which had been elevated due to a fever, decreased significantly. Clearly, whatever was in the poison had a powerful effect on the body and specifically the heart.

Kirk sent samples of kombé, as well as parts of the plant it was derived from, to the Royal Botanical Institute at Kew Gardens in London, where the plant was identified as *Strophanthus hispidus*. Thomas Richard Fraser (1841–1920), professor of clinical medicine at the University of Edinburgh, also took an interest in Kirk's samples. In 1870, Fraser managed to isolate a pure compound from the seeds of the plant *Strophanthus kombé*.^{*} He classified the compound as a glycoside, noted its pronounced effect on the heart and named it strophanthin.

There were many other scientists investigating the properties of arrow poisons and their plant origins around this time. In 1888, Léon-Albert Arnaud (1853–1915), a self-taught chemist working in Paris, investigated two different African arrow poisons. The first was used by the Somali, who made their arrow poison by boiling the bark and roots of the *Ouabaio* tree (*Acokanthera schimperi*, see above), mixing the extract with acacia juice and concentrating the result into a black tar-like substance they called *ouabe*. Arnaud isolated the active component of *ouabe* and named it

ouabain. He noted the effects this substance had on the heart and that it was similar to the strophanthin compound isolated by Fraser.

In a separate investigation, Arnaud isolated the active compound from a different arrow poison used by the Pahouins of northern Gabon that they called *inée* or *onaye*. This arrow poison was derived from the liana *Strophanthus gratus*, the same genus of plant Fraser had investigated. This compound, however, was different to Fraser's but identical to the ouabain Arnaud had isolated from the Somali tree.

The result was two toxic substances found in several different plant species. One compound, from *Strophanthus kombé*, became known as k-strophanthin to distinguish it from g-strophanthin that was found in *S. gratus*, *S. hispidus* and a few other strophanthus species as well as *Acokanthera schimperi*. Both these substances belong to the group of chemicals called glycosides and, because of their powerful effect on the heart, they are known as cardiac glycosides.



Cardiac glycosides are produced by a number of plants and a few animals including the oleander tree (*Nerium oleander*), the woodland plant lily of the valley (*Convallaria majalis*), and the cane toad (*Rhinella marina*). Around 200 different cardiac glycosides have been identified, but only a handful have any medical significance and the most well known of these are the digitalis glycosides, digitoxin and digoxin.

All these different glycoside molecules are made up of two parts: a sugar section, called the glycone, and a non-sugar component, called the aglycone (or genin) (see Fig. A.1). The different sugars attached to the aglycone affect how soluble these molecules are and therefore how easily they are absorbed into the body. The aglycone sections all have the same basic steroid structure at their core with only slight differences around the periphery. These small differences in the aglycone structures result in slightly different effects on the body.

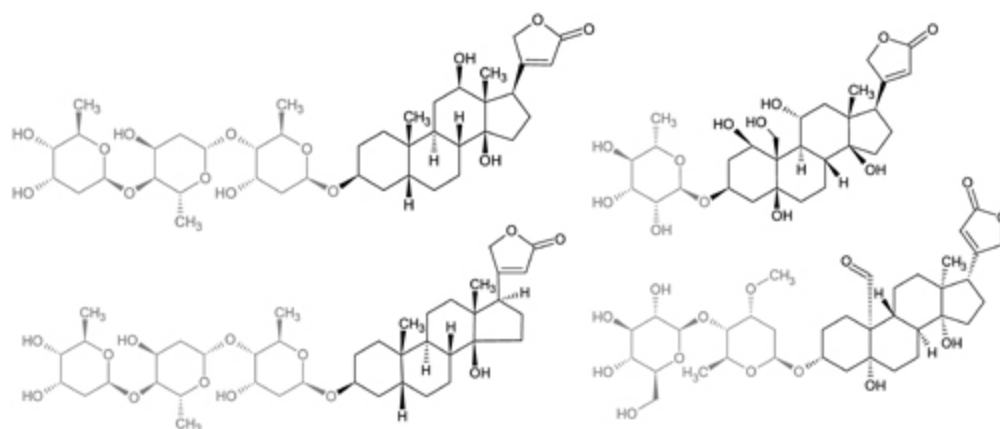


Fig. A.1. Comparison of four different cardiac glycosides with their glycones shaded grey and aglycones in black. From top left clockwise: digoxin, g-strophanthin, k-strophanthin (ouabain) and digitoxin.

All these aglycones interact with Na^+/K^+ -ATPase enzymes. These enzymes are responsible for moving sodium and potassium ions (Na^+ and K^+) in and out of cells – three Na^+ ions are pushed out and two K^+ ions are pumped into the cell in each cycle. Every cell of our body has these sodium-potassium pumps, but they are built slightly differently depending on where the cell is and what it does. In the gut, the sodium-potassium pumps help the absorption of glucose into the body. In nerves, the pumps set up the positive and negative regions that allow these cells to send their electrical signals (discussed in more detail in Chapter F, page 90).

Cardiac glycosides can interact with all of these different sodium-potassium pumps to a lesser or greater extent. Interactions with gut Na^+/K^+ -ATPase enzymes, for example, can cause diarrhoea, nausea and vomiting. The sodium-potassium pumps in nerve cells in the eye can be affected, resulting in a patient seeing everything with a yellow tint or haloes of colour appearing around points of light.* But these tend to be side effects because cardiac glycosides' strongest interactions are (unsurprisingly, given their name) with the Na^+/K^+ -ATPase enzymes in the heart.



The heart has four chambers. Muscle in the walls of the heart contracts, squeezing these chambers to force the blood inside them out. The muscle

then relaxes, and the chambers expand and refill with more blood so the process can repeat itself again and again and again. To move the blood efficiently and in the right direction, the muscle contractions have to be coordinated and the chambers have to be squeezed in the right order. Regulating these contractions is a network of nerves running through the entire organ. As mentioned before, Na^+/K^+ -ATPase enzymes are involved in generating nerve signals, but they are also critical for making the heart muscle contract.

Heart muscle is made up of bands of specialised cells called cardiac myocytes. Inside these large cells there are bundles of thick and thin fibres called myofibrils that slide over each other. This sliding makes the myofibrils shorter and the cell contracts. Lots of cells making little contractions together causes the muscle to make one large contraction that creates the heartbeat.

To make the fibres slide over each other requires calcium ions (Ca^{2+}). The flow of Ca^{2+} ions between these fibres is triggered by the presence of Na^+ ions, which is controlled by Na^+/K^+ -ATPase enzymes. When the sodium-potassium pumps push Na^+ ions out of the cell, the Ca^{2+} ions follow. With fewer Ca^{2+} ions between the thick and thin fibres, they slide back to their original position, the myofibrils lengthen and the muscle relaxes.

Experiments conducted by Fraser, the man who first isolated strophanthin, showed that just a tiny amount of the compound in the body slowed and strengthened the contractions of the heart. This was because strophanthin hinders the work of the sodium-potassium pump, the Na^+ ions stay inside the myocytes longer and, therefore, so do the Ca^{2+} ions. The lingering Ca^{2+} ions cause the myofibrils to stay in their shortened position longer and the muscle contractions become slower and more intense. Fraser didn't know the exact mechanism of strophanthin's action – the Na^+/K^+ -ATPase enzyme wasn't discovered until 1957 – but he was confident enough in its effects to recommend it for the treatment of certain heart conditions.

From 1886, 'Tincture of Strophanthus' was on sale in Europe. It was prescribed to patients with weak and rapid pulses and was recommended for the treatment of heart murmurs. In the initial enthusiasm for this new

treatment it was also sold as a remedy for nervous asthma, typhoid fever and pneumonia, though it would have had no beneficial effects on these conditions. With some syrup added to sweeten the bitter flavour, it was even given to children.

None of this would have come as a surprise to African healers who had been administering strophanthin remedies for centuries. They would soak the roots of strophanthus plants then ferment them to produce a bitter-tasting fluid. Small sips of this fluid were administered over days or weeks to treat wounds, constipation, food poisoning and sexual diseases as well as heart diseases. Europeans only caught on to the medical benefits much, much later. Strophanthin wasn't introduced into the British pharmacopoeia until 1898. But, despite eventually receiving the official stamp of approval among British and other Western medical institutions, strophanthin never became a popular prescription, for a number of reasons.



Strophanthus plants grew plentifully over vast stretches of the African continent, but not all species produced the desired strophanthin compounds. Those that did had the highest concentration of the compounds in their seeds, which was convenient for shipping large quantities of the raw material because the rest of the plant could be discarded. However, separated from the rest of the plant, strophanthus seeds are difficult to identify and the wrong seeds could easily find their way, deliberately or accidentally, into consignments headed for Europe. Chemical tests were designed to identify seeds with high concentrations of strophanthin but they were not reliable. The quality of strophanthus tinctures suffered as a result.

To make life easier, attempts were made to farm the commercially valuable plants, but without success. All the expertise in growing, harvesting and using strophanthus seeds was with the Africans, but Europeans did not trust them. Possession of poisoned arrows, even just the poison, had been outlawed under British rule. Though Africans were free to collect strophanthus seeds from the wild, encouraging them in the cultivation of these plants was considered too dangerous. The very people

who could have helped establish a trade in strophanthus were deliberately excluded.

By 1922, the mass production of strophanthus seeds had been largely abandoned. Yields remained low and the huge anticipated demand for strophanthin had never really materialised. Many doctors had been reluctant to prescribe medicines whose purity could not be relied upon. Even when quality improved, there were other problems with strophanthin as a drug.

Strophanthin is poorly absorbed into the body when it is ingested. This makes it a good choice as an arrow poison as you can kill your prey effectively but still eat the meat afterwards. Patients swallowing strophanthin tinctures were also unlikely to take an accidental overdose of the drug. Strophanthin was therefore considered safe, but it was also difficult to get the dose right.

In Germany, this difficulty with administration was overcome by injecting tiny quantities of strophanthin directly into a vein. Missing the vein meant the drug would be delivered into muscle, which was painful. Giving drugs intravenously in a hospital setting is fairly standard today, but a patient would have to be trained to give these injections if they were to continue strophanthin therapy after being discharged. Nevertheless, intravenous strophanthin therapy continued in Germany for much of the twentieth century, but it never really caught on elsewhere. Formulations were developed that increased the absorption of strophanthin from the gastrointestinal tract but the drug remained a niche product. The main thing preventing strophanthin's medical success was direct competition with another drug, digitalis.

Digitalis is a mixture of compounds extracted from foxglove plants that contains two important cardiac glycosides, digoxin and digitoxin. These glycosides had several advantages over strophanthin: they were easy to get hold of and already had a long history of use before Fraser started advocating for strophanthin. Foxgloves are native to Europe, where they grow abundantly and easily. They had been used in folklore remedies for hundreds of years before they were introduced into mainstream medical practice in 1785 by the British physician William Withering (1741–1799). The all-important cardiac glycosides within digitalis are also readily absorbed from the gut, meaning the drug is effective when swallowed. But this ease of gastrointestinal absorption was also one of the disadvantages of

digitalis – it was much easier to swallow an overdose. Digitalis glycosides are also less potent than strophanthin so larger doses, though still frighteningly small amounts, are needed to produce the same beneficial effects on the heart. Digitalis glycosides also produce more side effects.

As reliability in strophanthin supply improved along with better formulations, improvements were also being made in monitoring patients on digitalis drugs, making them much safer to take. Over the course of the twentieth century more and safer drugs were also becoming available to treat the same heart conditions.* Digoxin and digitoxin are still in use today, but by 2012 strophanthin had disappeared from the medical market completely.†

From Agatha Christie's point of view, digitalis would appear to be the better drug for a murder mystery story. It is easily swallowed and difficult to detect in the small quantities needed to kill. It also produces side effects that could be useful as clues for the detective and reader. Several of her characters did meet their end with foxgloves, digitalis and digitoxin, the most memorable probably being the formidable Mrs Boynton in *Appointment with Death* (1938). By choosing strophanthin for her stories, Christie appears to be making things more difficult for herself. By the time she was writing these stories, the drug was admittedly more readily available but not widely used. And when it came to killing her fictional victims, she also had to contend with the problems of administration.

How strophanthin kills

Mrs Laxton's death in 'The Case of the Caretaker' is the most accurate of Christie's three strophanthin deaths because her lethal dose is injected. If the murderer takes care to inject a vein, the drug would have travelled swiftly to the heart, where it would have interacted with the Na^+/K^+ -ATPase enzymes. With the sodium-potassium pumps effectively disabled, Na^+ ions could not be pumped out of the heart's muscle cells efficiently and the Ca^{2+} ions would stay where they were, keeping the muscle fibres shortened. The heart's contractions would have become permanent and blood would cease to be pumped around the body.

Christie's other two strophanthin victims die from swallowing the drug. In all the therapeutic investigations and applications of strophanthin, there were no recorded deaths from oral use of this drug. That is not to say it would be impossible to die by swallowing strophanthin, but a very large dose would be needed, at least in comparison to the amount that could be injected with the same result. Just 2mg (0.00007oz) of injected strophanthin have proved fatal, but patients have swallowed 30mg (0.001oz) with no ill effects. The lethal oral dose must, therefore, be at least 30mg and probably much larger.

Large doses would make the bitter taste of strophanthin more noticeable to the victim. For Valentine Chantry in 'Triangle at Rhodes' the flavour would have been disguised somewhat by the bitters in her pink gin, but she still notices and comments on the strange taste. Anya Hendryk in *Verdict* knows to expect the bitter flavour as she has to swallow strophanthin daily as a medicine, but she still comments on how strong her medicine tastes when she is given her fatal dose. This is not surprising as she is given the whole bottle of strophanthin when her medicinal dose is just four drops. How much she actually swallows depends on the size of the bottle, how diluted it was and how much was left, which we are not told.

Valentine Chantry's fatal dose is also a mystery. We know it was added to her pink gin from a 'packet', which implies a solid and potentially pure form of strophanthin. A few hundred milligrams of pure strophanthin is a physically small amount and would easily dissolve in her drink.*

The other problem with giving strophanthin orally was that it takes longer to be absorbed into the body and make its effects on the heart felt. In the 1950s, patients in Germany with suspected heart failure would be given two tablets containing 3mg (0.0001oz) of strophanthin in each. They would be told to let the tablets dissolve in their mouth, and patients with heart failure would feel a relief of their symptoms five to ten minutes later.

Valentine Chantry collapses before she can complete her comment on the flavour of her drink. She is dead five minutes later. Anya Hendrik takes a little longer to fall unconscious, giving her time to tell the audience she is struggling for breath and for her hand to move towards her heart. Even with their much larger doses, both Anya and Valentine die a little too quickly to

be scientifically accurate, but audiences probably appreciate not having to watch or read a slow, drawn-out death.

Mrs Laxton, Anya and Valentine, like the majority of Christie's murder victims, are dead before medical help can arrive, which is unfortunate for the victims but convenient when you need a murder to solve. Could timely medical intervention have ruined Christie's plots?

Is there an antidote?

The first step in any medical emergency is to treat life-threatening symptoms as quickly as possible. In the case of an overdose of cardiac glycosides this would involve ensuring oxygen levels in the body are maintained, which may mean artificial respiration is needed.

After the patient is out of immediate danger, the next step when an overdose of something has been swallowed is to remove as much of the substance as possible before any more can be absorbed into the body proper. This means inducing vomiting or washing out the stomach contents.* The victim is then likely to be given a slurry of activated charcoal to drink. Activated charcoal has lots of tiny holes in its structure and grinding it into a fine powder increases the surface area and the number of openings to these holes. The holes trap poison molecules inside and keep them there until the charcoal can be passed out of the body naturally.†

Treatment can then be given for the poison already absorbed into the body. In the case of poisons that have been injected, this will be the entire dose. The interaction of strophanthin, and other cardiac glycosides, with Na^+/K^+ -ATPase often means there is an excess of K^+ ions outside of the heart cells, because the pump is not working to move them back in. Drugs are available to remove this excess of K^+ ions. Other drugs, such as atropine (see Chapter S), can stimulate the heart to beat faster and lidocaine can be administered to maintain a normal rhythm. With treatment, anyone who survives 24 hours is likely to recover.

Not all of these drugs were available when Christie was overdosing her characters with strophanthin, but atropine certainly was and could have helped all three poisoned wives. Lidocaine was available when Anya's overdose was given in *Verdict* in 1958. There should also have been enough

time for medical help to arrive and treat both Anya and Valentine, who had swallowed their poison, because of the slow absorption and delayed effects. Little, however, could have been done to save Mrs Laxton.

What could have inspired Christie to use such an obscure and potentially awkward poison? It is unlikely to have come from a real murder case. Outside of colonial-era Africa, strophanthin deaths are exceptionally rare. There is one possible strophanthin murder from 1948, but the fact that strophanthin might have been involved was not revealed until many decades later, long after Christie was writing her strophanthin stories. It is interesting because it shows that Christie had chosen to write about a very dangerous, dreaded and mysterious poison.

A real-life case

Around 7 a.m. on 1 December 1948, the body of a man was found lying on the sand at Somerton beach, a seaside suburb of Adelaide in Australia. Medical opinion was that the man had died around 2 a.m. His head and shoulders rested against the sea wall and his legs were stretched out in front of him. A partially smoked cigarette had been trapped between the man's cheek and lapel as his head lolled to one side, but neither the cheek nor the lapel was burnt. Closer examination showed the deceased was between 40 and 50 years old, 180cm tall (5ft 11in) and in good physical condition. His light brown, sandy hair was greying slightly at the temples and had been brushed back from the forehead. His nails and toenails were neat and clean, suggesting the man looked after himself. His clothes were of fairly good quality but they contained nothing more than a few cigarettes in a pack, a box of matches, a pack of chewing gum, two combs, a train ticket from Adelaide to Henley, unused, and a bus ticket from Adelaide to Somerton. There was no money, wallet or identification. Any clothing labels that might have helped identify the deceased had been removed.

A second examination of the man's clothes found a scrap of paper in the trouser pocket that had been missed the first time. Two words were printed on it: 'Tamám Shud'. The piece of paper had been torn from the last page of a book, the second edition of *The Rubaiyat of Omar Khyyam*, a collection of

poems by the eleventh-century Persian poet-astronomer Omar Khyyam and translated by Edward Fitzgerald.* Tamám shud is Persian for 'the end'.

A detailed autopsy found no signs of violence on the body or external markings of any note. The condition of the internal organs suggested the immediate cause of death was heart failure, but the heart itself appeared to be healthy. The pathologist's thoughts therefore turned to poisons and so samples of the stomach contents, blood and urine were sent for analysis. Tests were performed for all the usual chemical suspects. Nothing was found.

Witnesses were questioned to see if they could reveal anything about the man's death. One man thought he saw the victim on the beach and that he raised his arm around 7.30 p.m. Half an hour later a couple sat on the steps a short distance from a man lying on the beach with his legs stretched out. But, from their position on the steps above the man, they could only see him from the waist down. He didn't seem to be moving despite the large number of mosquitoes around. It would appear he was alive at 7.30 p.m., unconscious at 8 p.m. and, according to medical opinion, he had died around 2 a.m. This gave a minimum 6.5-hour window for any poison to have taken effect.

Many different toxic substances were considered, but all either acted too slowly, too quickly, or would have left some tell-tale sign in the body that was not found. In his summing-up, the coroner, Thomas Erskin Cleland, judged the unknown man had died an unnatural death, but whether by his own hand or someone else's there was not sufficient evidence to decide. The most probable cause of death was due to poison, most likely a glycoside. These compounds could stop the heart pumping, as appeared to have happened to the deceased, and they could do this in such small quantities it would be difficult for an analyst to detect them.

Two types of glycoside were discussed in the inquest but they were only referred to as 'type 1' and 'type 2'. The names were written on a piece of paper for the benefit of the coroner but not revealed publicly because they were both very easy for people to obtain. No one wanted to help out any would-be poisoners who were looking for substances that were both lethal and difficult to trace. Only in the 1980s were the two glycosides identified as digitalis and strophanthin.

Glycoside poisoning was still not a perfect explanation. Individual responses to a poison are certainly possible, but normally glycosides would be expected to cause vomiting, and there was no vomit on or near the corpse. Also, given the poor solubility of strophanthin, a large dose would be needed to kill the man and the residue of this dose might be expected to be found in the stomach contents. They weren't. No injection marks were found on the body, but one could have been missed. On balance, digitalis was considered the more plausible of the cardiac glycosides to have killed him.

The inquest raised more questions than it answered – most importantly, who was the dead man? Fingerprints had been taken and copies sent to police forces around the world, with no result. A cast of the dead man's head and shoulders was made and photographs circulated to every English-speaking country. No one appeared to know who he was and no one came forward to say that he was missed. The mystery of the dead man on Somerton beach has intrigued people ever since, and though many theories have been put forward, some more credible than others, none have been confirmed.*

Agatha and arrow poison

In Agatha Christie's strophanthin murders she avoided the difficulties of finding evidence of the drug in the body by having the drug itself found in the murder suspect's possession. Anya's empty bottle of strophanthin medicine would also have given a strong hint as to what the toxicologist should be looking for at autopsy.

Having strophanthin prescribed to Anya also avoids the problem of having to explain how her murderer knew about this obscure poison and where they could get hold of it. The murderer in 'The Case of the Caretaker' had to sneak it out of a pharmacy, but how they knew to take that particular drug is not explained. Nothing is explained about the murderer's knowledge of, or how they got hold of, strophanthin in 'Triangle at Rhodes'. It appears to be something they carry around with them just in case an opportunity arose to kill Valentine Chantry with it.

Skimming over the details is understandable in a short story that is racing to its conclusion and doesn't need to be bogged down with unnecessary details. But Christie usually delights in this kind of detailed information. Perhaps she was also aware, like the experts at the inquest into the Somerton man, that some of these details could be very dangerous if they were revealed publicly.

Notes

- * The hunters distinguished the potency of the poison from different sources as 'one tree', 'two tree' etc., depending on how far the animal was likely to get having been poisoned with it.
- † Published in the US under the title *What Mrs McGillicuddy Saw!*
- * Another arrow poison from Europe that Christie used is yew (see Chapter T).
- † According to Arthur Conan Doyle's novel *The Lost World* (1912), the poison could also bring down dinosaurs.
- ‡ It isn't just humans that have used this toxic plant for their protection. The African crested rat, *Lophiomys imhausi*, chews the roots and bark of the *Acokanthera* tree and smears its saliva over its fur. The long hairs that make the rodent look like a fluffy porcupine absorb the poison and it spreads over the pelt by capillary action. Any animal that tries to bite the crested rat will get a dangerous, possibly lethal, dose of poison.
- * Don't get any ideas. Toxicology was a little different at the time Christie was writing. Strophanthin would be detectable in a body using today's more sophisticated analytical techniques. Strophanthin is also difficult to get hold of and your searches will leave behind clues for investigators.
- * In 1673, a Danish missionary, Wilhelm Müller, wrote that arrows used by the Fetu were poisoned with crocodile bile. The Fetu may well have added crocodile bile to their arrow poison, but it was probably another ingredient that did the deadly work.
- * Fraser thought he was working on *S. hispidus*, but he was wrong.
- * These are usually early signs of toxicity and a warning that a patient's medication may need to be adjusted.
- * These heart conditions are mostly heart failure, when the organ no longer pumps blood efficiently, and certain types of arrhythmias or irregular heartbeats. The alternative drugs mentioned are compounds such as beta blockers.
- † Strophanthin is still in use in infinitesimal doses in homeopathic remedies. It also has a role in research for electrophysiological experiments in cardiac myocytes.
- * For comparison, think how small headache tablets are and they regularly contain 500mg (0.018oz) of the active ingredient bulked out with inert material to hold the pill together while it is stored.
- * However strong the urge to intervene and do something to help someone in obvious distress, forcing someone to vomit or pouring water down their throat can do more harm than good – please leave this to the medical professionals.

- † Today, activated charcoal is made to certain medical standards, but in the past, poison centres would begin their working day by burning lots of toast and scraping off the blackened bits ready for use.
- * Agatha Christie was familiar with this collection of poems.
- * In 2022, after an extensive DNA investigation, researchers believe the dead man was Charles Webb, an electrical engineer born in Melbourne.

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Cards on the Table

Agatha Christie's use of poisons was masterful. Her toxic choices are perfect for her fictional circumstances, being carefully targeted to their victim and the symptoms accurately described to reveal brilliant clues. But there are some poisoners that surpass even her skills – bacteria. These single-cell organisms can craft and deploy batteries of highly specific toxic weapons to their advantage and their hosts' distinct disadvantage. Some can even create red herrings to evade capture by the immune system, but their trail of destruction through a body also leaves characteristic clues to their identity.

Christie's use of bacteria in three stories shows her knowledge of the lethal potential of these microscopic murderers. A pus-covered bandage is applied to a cut to create a deadly infection in *Murder is Easy*. Anthrax is added to a shaving brush with lethal results in *Cards on the Table*. And multiple members of the bacteria kingdom are deployed to murder members of a cult in Christie's short story 'The Flock of Geryon'.*

Before we get to the details, I should warn you I am going to give away the identity of a murderer in one story and big hints of whodunit in a second story. Stop reading and skip to another chapter if you don't want to see significant spoilers to *Cards on the Table* or 'The Flock of Geryon'. In my defence, Christie identifies both these characters as murderers almost from the start of both stories. In *Cards on the Table*, the murders and murderer I am going to discuss are not the main mystery to be solved. In 'The Flock of Geryon' the whodunit is fairly obvious; the puzzle is how Poirot will prove that a series of deaths are the result of murder and not natural causes. It is this *howdunit* that will be discussed in this chapter. So, fair warning, this is your last chance to look away!



Starting with Christie's 1939 novel *Murder is Easy*, Dr Humbleby's death from septicaemia[†] is initially attributed to an unfortunate accident. His work treating the sick inevitably exposes him to a great number of potentially deadly bacteria and some of those bacteria got into a cut on his hand. The bacteria multiplied in this unclean cut, spread through his blood and finally killed him. It is only later that the murderer confesses to deliberately infecting the doctor. The bandage so carefully applied to his wound had been deliberately contaminated with pus from a cat's ear infection. The cat's ear infection could be caused by a number of different bacteria, but it is the overwhelming number of these bacteria that are allowed to multiply through Dr Humbleby's body that results in his death rather than a specific type. It is a vague and opportunistic murder unlike Christie's other biocrimes where the bacteria are very carefully chosen.

A doctor is the murderer rather than a victim in the 1936 novel *Cards on the Table*. In fact, he is one of four murderers invited by eccentric Mr

Shaitana, along with four notable detectives, to play bridge at his apartment. While everyone is apparently distracted by their cards, Mr Shaitana is stabbed. To find out which of the murderers did it, their personalities, bridge technique and past crimes are investigated. One of the bridge players, Dr Roberts, is revealed to have had two of his patients, husband and wife, die under unusual circumstances. First the husband contracted anthrax from a shaving brush, then the wife, on a holiday in Egypt to recover from her grief, fell ill and died. Though no foul play was suspected at the time of the deaths, it is later shown that Dr Roberts had the means, motive and opportunity to deliberately introduce the infections that killed them both.

In the third example, Christie crams many different bacterial species and multiple murders into one short story. The flock in 'The Flock of Geryon' (published in 1940) are the rich, vulnerable ladies who have signed up to a pseudo-religious cult. This cult encourages its followers to bequeath all their worldly goods to their charismatic leader after their death. Some of the richer members of this cult die conveniently quickly. Mrs Lloyd dies of pneumonia, from the bacteria *Pneumococcus* or *Streptococcus pneumoniae*. Ulcerative colitis, a consequence of the *Escherichia coli*, or *coli communis* as Christie calls it, claims the life of Miss Everit. Miss Lee succumbs to typhoid, and there are several other deaths.

How all these diseases are contracted is most unnatural, as Christie's detectives prove, but the diseases themselves are natural enough. Bacteria have been responsible for ending vast swathes of life on this planet. Their deadly talents have been honed by millions of years fighting to survive.

The bacteria story

For the past roughly three billion years bacteria have been the dominant organisms on this planet. They are incredibly successful and have turned up everywhere scientists have looked, even seemingly inhospitable places like highly acidic hot springs, the frozen wastes of Antarctica and radioactive waste. They can even survive the vacuum of space.

In order to adapt to the many and varied habitats and lifestyles bacteria occupy, they have evolved into all manner of shapes and styles. For

example, anthrax, *E. coli* and typhoid bacteria are rod-shaped, while tuberculosis and *Pneumococcus* are spherical. They also vary in size. Some bacteria are visible to the naked eye but most tend towards the tiny and are generally much smaller than animal cells. Despite their diminutive size, each individual bacterial cell is capable of carrying out all of life's necessary functions – sensing, moving, feeding, excreting, respiring, growing and reproducing. It is the day-to-day functions of some bacteria that can severely compromise the day-to-day existence of other organisms, including humans.

Bacteria were around long before humans existed and we have had to learn to get along with them, which, for the most part, we do extremely well. Around 90 per cent of the cells in our bodies are bacterial, though being so tiny they only take up roughly 0.1 per cent of our mass. We wouldn't survive without them. The bacteria in our gut help feed us by digesting food we can't break down ourselves, while bacteria on our skin prevent water loss as well as protecting us from harm by supporting our immune system.

Disease-causing bacteria make up a tiny lethal minority of bacterial species. Some of the molecules excreted by these pathogens interact with the human body and cause problems. For the bacteria, these chemicals are a means of survival; they punch holes in cell membranes to get at food or disrupt signals from cells to flush out fluids that carry them on to a new host. For us, the inadvertent damage done by the action of bacterial toxins can be anything from inconvenient to catastrophic. The everyday activities of three of the bacteria Christie used in her murder mysteries – anthrax, tuberculosis and typhoid – have had a considerable impact on human health and welfare through the ages.

Bacillus anthracis

Anthrax, named from the ancient Greek word for 'coal' after the black lesions that it causes on the skin, has cropped up repeatedly in human history. It has been suggested that anthrax was at the root of two of the 10 plagues of Egypt. The book of Exodus in the Old Testament describes a plague that destroyed Egyptian cattle, but not the animals belonging to the

Israelites. This was followed by another plague of festering boils that erupted on the skin of the Egyptian people and their cattle. In another example, the ‘Black Bane’ of the seventeenth century killed around 60,000 people in Europe and many more cattle.

Major outbreaks were devastating, but anthrax was also responsible for smaller-scale long-term disruption. In the past, all over Europe, fields of lush pasture appeared to be cursed. Generation after generation of animals that grazed on them died while animals on neighbouring fields were fine. Some people were also infected with the disease, but not everyone. Why some animals and some professions ran such a high risk of contracting anthrax, or ‘woolsorter’s disease’, was a mystery until the German physician and father of microbiology Robert Koch (1843–1910) took an interest.

In 1876, Koch examined the carcasses of animals dead from anthrax. In all of them he found tiny organisms and his painstaking experiments proved these organisms, *Bacillus anthracis*, were the cause of the animals’ fatal disease. His detailed investigations also showed *Bacillus anthracis* lives in soil from where it can transfer to an animal’s skin to cause an infection. This infection can then be passed on to humans who work with these animals or their skins.

Koch found that these bacteria can dramatically change depending on the circumstances. Anthrax bacteria can form spores – a heat-resistant dormant state that can be maintained for at least 70 years – but revive and return to their normal active lives when conditions became favourable.

Thanks to Koch’s insights, anthrax outbreaks could be contained and the impact minimised by burying animals who had died of the disease quickly and deeply, before the spores could form and spread. Over time, human-to-human transmission also declined as people came into contact with fewer diseased cattle.

The murderer in *Cards on the Table* could learn everything they needed to know from Koch’s published research: how to obtain anthrax (from soil), culture, purify and store its spores until the opportunity arose to sprinkle them on his victim’s shaving brush. If this sounds far-fetched, consider that in real life several governments around the world have investigated the use of anthrax as a weapon of war.

During the Second World War, tests on Gruinard island, off the west coast of Scotland, showed that anthrax spores could be exploded in a bomb and remain potent enough to kill sheep grazing in the area. It also showed that those anthrax spores continued to be a problem for a very, very long time. The island remained uninhabitable for decades. Tons of soil had to be removed and gallons of formaldehyde sprayed over the infected land before Gruinard was finally declared safe to visit, more than 50 years after the tests were carried out.

Robert Koch's name would have gone down in history for his work on anthrax alone, but the discovery of *Mycobacterium tuberculosis*, the microscopic cause of tuberculosis, was his crowning achievement.* It was also his biggest mistake. Just as Koch's observations could be used by others to do terrible things, his work on tuberculosis gave Agatha Christie a very devious method of murder, thankfully fictional on this occasion.

Mycobacterium tuberculosis

In the nineteenth century tuberculosis, or TB, was responsible for one in every seven deaths in the UK and a similarly terrifying mortality rate was experienced in many other parts of the world. The disease appeared to kill suddenly and indiscriminately. Not even those in the prime of life were safe. In 1882, a 36-year-old labourer, who seemed to be in perfect health, began to cough and complain of little pains shooting through his chest. Three weeks later he was admitted to hospital. He had lost so much weight it was like he was being consumed from within.* Four days after he was admitted he was dead, his body full of tiny greyish-yellow tubercles. Robert Koch took a sample from one of these tubercles and in it he found the cause of the labourer's disease, the bacteria *Mycobacterium tuberculosis*.

Mycobacterium tuberculosis can infect any tissue in the body, but they most commonly cause problems in the lungs. There they cause night sweats, loss of appetite, weakness, fever and pain. The bacteria live in tiny tubercles that develop into abscesses and cavities that produce an abundance of sputum, giving rise to the victim's hacking cough as they try to clear the fluid from their lungs. These cavities bleed and then

haemorrhage. The mucous starts off blood-flecked, but as the disease progresses more and more blood leaks into the lungs.

Some people can survive the early stages of a tuberculosis infection. Their body's immune system fights back against the invading bacteria, resulting in what looks like a cure. The symptoms disappear and the person can live decades of normal healthy life. But the TB bacteria have entered a latent state where they lay dormant and wait for an opportunity when the human immune system weakens so they can re-emerge and multiply. The disease then makes a dramatic reappearance and can kill its victim in a matter of months or weeks.

Having identified and isolated the cause of TB, Koch tried to deliberately weaken the bacteria so it could be introduced into a healthy person as a vaccine. The vaccinated individual would then be better able to fight off any subsequent infection by the fully virulent bacteria. But Koch went further. He claimed his injection of what he named 'tuberculin' could cure people who already had the disease. In Koch's defence, the press and the general public pressurised him into announcing his discovery before he had completed testing, but he knew it wasn't perfect. Tuberculin did not cure tuberculosis. In fact, in some cases it made the disease worse and people died as a result. We now know that tuberculin was overstimulating the immune system in some patients.

Tuberculin is useless as a treatment for tuberculosis, but it can be used to test if someone has been exposed to *Mycobacterium tuberculosis* bacteria. Someone who has never had tuberculosis would show no reaction to a small injection of tuberculin. But if someone has recovered from tuberculosis, their immune system has antibodies ready and prepared to fight off the reappearance of these bacteria. Injecting a tiny amount of tuberculin under their skin prompts an immune response and the area around the injection site becomes red and swollen. If someone has a latent tuberculosis infection the immune response to the tuberculin can stimulate the re-emergence of the disease.

Tuberculin is at the centre of Poirot's plan to expose the murderer in 'The Flock of Geryon'. He enlists the help of Miss Carnaby, a woman with no previous exposure to tuberculosis who is willing to work undercover in the cult to find out what is going on. Poirot instructs her to tell those in the

cult that she has experienced a few recurrences of the disease, but that a final treatment in a sanatorium had cured her.*

Poirot's hope is that, enticed by exaggerated rumours of Miss Carnaby's wealth, she will be the cult's next victim and, suspecting she might have a latent tuberculosis infection, they will use tuberculin and try to trigger a recurrence of her tuberculosis. It would be the safest way of getting hold of her money without raising suspicions. However, because Miss Carnaby has never actually suffered from TB, she should be perfectly safe, but there are considerable risks for everyone involved.

The risk for Miss Carnaby is that the cult leaders choose something other than tuberculin, something far more likely to kill her. Miss Carnaby would be dead and Poirot's career and reputation would be ruined. And, depending on what is chosen, there still might not be any proof to use to convict anyone.

The risk for the cult is that tuberculin might not trigger a recurrence of the disease and Miss Carnaby might not die even if it did. But even if the tuberculin doesn't work, they could always try again with something else. Tuberculin has the added advantage of only affecting the person receiving it, though if it does revitalise an existing tuberculosis infection, this person could potentially go on to infect others.* However, the dangers of starting an outbreak of a highly contagious and deadly disease doesn't seem to worry the cult. They have used other deadly bacteria to kill off their rich followers that spread far more easily. The most worrying of their bacterial choices is the use of *Salmonella enterica serovar Typhi*, also known as the typhoid bacteria.

Salmonella Typhi

Typhoid has troubled humans for thousands of years, when standards of sanitation and hygiene have been allowed to slip, typically during periods of war and famine. Some unfortunate individual would eat some infected food and the typhoid bacteria would be introduced to their gut. From there the bacteria could spread through the blood to affect organs such as the spleen and liver.

In the first week they would experience a high fever, headaches and malaise. During the second week, their temperature would stabilise, but at an elevated 40°C (104°F). An upset stomach would cause either constipation or diarrhoea, enabling the bacteria to leave the body, contaminate new places and infect new victims. Delirium might set in around this time. The third week would bring changes in body temperature, continued delirium, malnutrition and dehydration because of damage to the gut which, in severe cases, can be perforated, leading to haemorrhage and peritonitis. There can be many other symptoms and complications, from rashes to spots, and swelling of the brain to pneumonia.

Though typhoid bacteria have been around for a very long time, their unpleasant effects on the human body were easily attributed to other diseases. It took until the nineteenth century for typhoid to be recognised as a separate disease with a unique cause.* It was named typhoid meaning ‘typhus like’ because of close similarities with this disease, even though typhus is caused by a completely unrelated bacteria and spread by lice not contaminated food or water.

In ‘The Flock of Geryon’, Miss Lee’s typhoid infection is initially thought to have come from a contaminated salad. Poirot realises she was deliberately infected. She was more likely to have contaminated the salad rather than the other way around. Miss Lee could have contaminated many other things as she went about her everyday life and passed on that contamination to others, as illustrated by a real outbreak in Aberdeen, Scotland in 1964.

The outbreak started with one tin of corned beef and ultimately affected 469 people, three of whom died. Contaminated water had probably got into the tin through a small puncture. The infection had then been unwittingly spread to many other food stuffs when the corned beef was sliced on the supermarket’s meat slicer. Displaying some of these meats in the shop window exposed the bacteria to sunlight and they multiplied. A strict quarantine and widespread media coverage and health advice brought the outbreak to a halt just 28 days after the first case appeared.†

The huge public health response was warranted. Some people – between 2 and 5 per cent of those infected – can harbour the typhoid bacteria but show no symptoms. It means they are potentially able to pass the infection

on to others unaware, as was the case with ‘Typhoid Mary’.^{*} In ‘The Flock of Geryon’, the cult was lucky only one person died as a result of their typhoid plan.



Koch’s work on anthrax in 1876 was the first time bacteria had been proved to be the cause of disease. Twelve years later, Friedrich Loeffler (1852–1915) discovered how these tiny organisms caused those diseases. He had been studying *Corynebacterium diphtheriae*, the bacteria responsible for diphtheria.

Diphtheria kills its (usually) child victims by forming a membrane in the throat that suffocates them. It also causes damage to other organs in the body, such as the kidneys and heart, but the bacteria is only ever found in the throat. Loeffler realised something was being released by the bacteria that travelled throughout the body to cause problems elsewhere – in other words a toxin.

Diphtheria toxin was the first bacterial toxin to be discovered, in 1888. Since then over 300 different bacterial toxins have been identified, with a wide variety of biological targets. Let’s look at the toxins that are produced by our three Christie-related killers.

How bacteria kill

Bacillus anthracis carries out a two-pronged attack on the body using three proteins. One protein helps deliver the other two toxins into cells where they do the damage. One of the toxins disrupts cell signalling that is involved with how water is moved in and out of cells and is, therefore, named ‘oedema factor’.

Oedema is the build-up of fluid in the body’s tissues. This can be clearly seen when it affects the arms or legs, as it commonly does when someone sits or stands for prolonged periods, particularly older people. The skin can feel tight and the limbs heavy with pain in the joints or there can be restricted movement. It can be uncomfortable and may be a sign of an underlying medical condition that needs to be treated. More concerning is

oedema in internal organs such as the lungs or brain. Lung tissue swollen with fluid will cause shortness of breath, which can be life threatening depending on the degree of swelling. The brain is more or less enclosed in the skull to protect it, but the skull is also an inflexible barrier if the brain swells. Vital nerves and blood vessels can be crushed by the build-up of fluid in the brain and death can be rapid.

Anthrax's second toxic protein is called 'lethal factor' (see Fig. B.1) but what it does to be so lethal is not known exactly.* What is known is that lethal factor inactivates neutrophils (a type of white blood cell that kills invading bacteria). There is also evidence it damages endothelial cells that form the linings of structures inside the lungs, the pericardium (that surrounds the heart), lymph vessels and blood vessels. This damage causes fluids to leak and can lead to death.



Fig. B.1. Structure of anthrax toxin 'lethal factor'.

How effective these toxins are, and how likely they are to result in death, is largely down to where in the body the anthrax infection takes hold. Without treatment, inhalation anthrax kills virtually everyone it infects because of direct damage to the lungs. Aggressive treatment can save just over half of those infected this way.

Ingesting anthrax spores is almost as dangerous as inhaling them. Gastrointestinal anthrax kills just over half of its victims without treatment, but just under half, around 40 per cent, with treatment. Cutaneous anthrax is the most common form of the disease, causing the black sores described in ancient texts, and is the least concerning. Nevertheless, without treatment cutaneous anthrax kills around 20 per cent of those infected.

In 2001, several people fell ill after handling envelopes containing anthrax spores sent through the US postal system. Eleven people suffered from cutaneous anthrax infections, and 11 contracted inhalation anthrax. All of those who caught the skin infections survived, but five of the inhalation cases died.

Dr Roberts' anthrax victim in Christie's *Cards on the Table* is most likely to have contracted a skin infection from his deliberately contaminated shaving brush and was unlucky not to survive. However, there is another form of anthrax infection that might be relevant if the man cut himself while shaving, but it isn't clear from Christie's text if he did.

Injection anthrax, identified in heroin-injecting drug addicts, is rare (it has only been reported in Europe). With very few cases to examine, the details of this form of anthrax are scant. Symptoms to look out for are redness and swelling around the injection site, which may not initially raise any alarm bells, followed by shock and multiple organ failure, by which time it is often too late.



Mycobacterium tuberculosis may be one of the most well-known disease-causing microorganisms but, so far, only one toxin has been identified from the bacteria. This toxin is the tuberculosis necrotizing toxin, or TNT for short.

When tuberculosis bacteria enter the body, the invasion is countered by the body's immune response. Macrophages, large white blood cells, engulf the bacteria as they would any material they don't recognise as 'self'. Normally the macrophages would then kill and digest whatever they had swallowed up, rendering it harmless. In the case of *M. tuberculosis*, the macrophages become home for the bacteria. The immune system recruits other cells to surround the infected macrophages and a granuloma, or 'nodule', forms to protect the rest of the body from something the immune system can't get rid of.

Inside the macrophages the tuberculosis bacteria release TNT to hijack the macrophage's metabolism, taking nutrients from the surrounding cell to feed themselves and multiply. Eventually the disruption to the macrophage's normal functions causes it to die, the cell ruptures and the granuloma bursts. Tuberculosis bacteria are released to spread through the body and set up more sites of infection and form more granulomas. Eventually, the resulting tissue destruction leads to death.

Many of the details surrounding tuberculosis bacteria's life inside the human body – how they protect themselves from destruction by the immune system, how they enter and re-emerge from their dormant state, their resistance to antibiotics – are still the subject of ongoing research. And tuberculosis is not the only ancient disease still giving scientists something to think about.



Exactly how *S. Typhi* have caused so much human illness was a mystery until very recently. The bacteria's cell walls contain lipopolysaccharides (LPS), molecules that trigger a response from our immune system. If there are lots of bacteria with lots of LPS, the response can run out of control and result in septic shock.

The body's normal response to a bacterial infection is to attack the invaders on several different fronts. The immune system releases chemical signals to raise the body's temperature and the fever helps to slow the bacteria's rate of reproduction. Other chemicals are released to destroy the invaders. In severe cases this activity can escalate, resulting in damage to

the body's own cells, including the blood cells, blood vessels and organs. Damaged blood and blood vessels means less oxygen is delivered to tissues. These tissues also become less capable of extracting the oxygen. Lack of oxygen impairs function and causes knock-on effects, such as the build-up of more toxic chemicals or damage to the kidneys that prevents effective filtration of the blood. Vital organs fail one after the other. The body's attempts to control and eliminate the invaders result in so much damage that life becomes insupportable.

Dr Humbleby's death in *Murder is Easy* may well have been as a result of septic shock. The bacteria introduced into his blood from a cat's ear infection multiplied, causing a massive infection, but this infection was not typhoid. Cats can't get typhoid: only humans can be infected with these bacteria. Septic shock can be triggered by a massive infection of any bacteria, as well as viruses, fungi and microscopic parasites. So typhoid's lethality cannot be explained just by septic shock. The typhoid bacteria must be producing some other toxin or toxins to explain its symptoms.

Scientists struggled to find these typhoid toxins because *S. Typhi* don't release their toxins to float around in the extracellular fluid ready to be absorbed by the next cell they bump into. *S. Typhi* inject their toxins directly into the hosts' cells using an injectisome, a needle-like structure that can pierce the cell membrane. Once scientists knew to look inside the host cells they found plenty of compounds introduced by the typhoid bacteria to investigate.

In recent decades evidence of toxicity has accumulated around one particular protein produced by *S. Typhi*. The most persuasive evidence comes from mice injected with the protein. The mice develop typhoid-like symptoms even though mice can't catch typhoid. This protein is a type of so-called 'cytolethal distending toxin'. The variant *S. Typhi* produce has been imaginatively named 'typhoid toxin'. Typhoid toxin slices through strands of DNA, chopping up chromosomes. When a cell's DNA instructions are damaged in this way, the cell dies.

An enormous amount has been learned about how some bacteria cause disease. Identifying the specific toxic culprits is a step towards better, more directed treatment. Koch's insights led to a decline in the number of anthrax cases, but anthrax infections can still kill. Tuberculosis continues to kill over a million people every year. There are an estimated 20 million typhoid

cases annually and more than 200,000 people die from the infection. Almost a century ago, the prognosis for Agatha Christie's victims of bacterial poisoning would have been even worse.

Is there an antidote?

The obvious treatment for a bacterial infection is to take antibiotics. Substances that are highly toxic to bacteria can be swallowed safely because of the differences between bacteria cells and our animal cells. For example, animal cells are surrounded by a cell membrane. The contents of each bacterium's cell are also enclosed by a membrane, but on top of this their cell membrane is surrounded and protected by a cell wall. Bacteria's cell walls comes in two varieties, one that takes up the Gram stain^{*} and one that doesn't – and bacteria are therefore broadly divided into Gram-positive and Gram-negative bacteria.

Compounds such as the cephalosporins that target the bacterial cell wall can kill a wide range of different bacteria without damaging our cells. Other compounds interfere with essential proteins that are not found, or are very different, in animal cells, hindering the bacteria's growth or reproduction.[†] Antibiotics that interact with these more specific targets are effective against a narrower range of bacteria.

However, Agatha Christie wrote her bacteria murders in the 1930s and 1940, before antibiotics were widely available. Salvarsan, an arsenic-based antibiotic, had been released in 1910 to treat syphilis, but it was difficult to administer, gave terrible side effects and was not very effective in any case. Neosalvarsan, available from 1912, was easier to prepare and resulted in relatively mild side effects, such as nausea and vomiting, but it was even less effective. Prontosil, a sulfa drug developed out of the artificial dye industry, was available from the mid-1930s. This compound was effective against a wide range of Gram-positive bacteria, which includes tuberculosis and anthrax bacteria, but was completely ineffective against Gram-negative bacteria such as the *typhoid bacillus*. For Christie's fictional murder victims the options were few, effectiveness limited and the side effects unpleasant, sometimes very unpleasant. These compounds were, nevertheless, revolutionary in terms of medical care.

The big breakthrough against bacteria came with the development of penicillin during the Second World War. Penicillin targets the bacterial cell wall and is therefore effective against a great many bacteria. Its widespread and effective use triggered the search for more compounds with antibiotic properties.

Modern medicine has drastically increased our chances of surviving bacterial disease. But some bacteria can evade or withstand these chemical assaults and still claim hundreds of thousands of lives every year, as can be seen by the statistics for tuberculosis and typhoid above. And the threat from bacterial infections continues to grow thanks to decades of over-enthusiastic use of antibiotics. Some bacteria have evolved tricks and mechanisms to counteract the effects of these antibiotics or render them ineffective. Antibiotic-resistant tuberculosis is of particular concern.

Prevention is always better than cure, and modern attitudes to cleanliness and good hygiene prevent many infections occurring in the first place. But hand washing and regularly wiping down surfaces isn't always enough.

Vaccination can be an incredibly effective way of preventing someone from contracting a disease, or suffering only a milder infection that they are more likely to survive. This is done by introducing a dead, weakened or partial form of the infectious agent into the body. It allows the immune system to study it and prepare antibodies specific to that agent that can be rapidly mass-produced to fight off any subsequent invading microorganisms of the same type. The concept has been around for centuries but was popularised in the UK by Edward Jenner (1749–1823). He scraped pus from cowpox pustules into people's arms, thereby introducing the body to a less virulent relative of the deadly smallpox virus. When the bacterial causes of other lethal diseases were identified, these too could be deliberately weakened to create vaccines, as Koch tried to do with tuberculosis and tuberculin.

Much of the pioneering work on vaccines was carried out by Louis Pasteur (1822–1895) and his team. His method of heat-treating beer, wine and milk to kill off unwanted bacteria has become known as Pasteurisation. He and his co-investigators also developed effective vaccines against rabies and anthrax. They weakened the bacteria using techniques including exposing them to the air and adding chemicals to corrupt the protein structures within the microbes. These techniques have since been refined

and added to, increasing the number of diseases that can be prevented by vaccination.

By the time Christie was writing about murderers using anthrax, tuberculosis and typhoid bacteria, there were vaccinations available for all of them. However, problems with early versions of the tuberculosis vaccine meant a safe version didn't come into widespread use until after the Second World War. It is unlikely it would be available to the murderer working with tuberculosis and tuberculin in Christie's 'The Flock of Geryon' in 1940.

An improved anthrax vaccine, based on Pasteur's version, was available from the 1930s and was effective for animals and humans, but there was still a very slight risk that some of the bacteria remained virulent enough to cause a lethal infection. For this reason, the vaccinations were mostly restricted to cattle and the few people most at risk of contracting the disease through their occupation. In Christie's *Cards on the Table*, Dr Roberts could vaccinate himself before preparing his anthrax sample to add to the shaving brush and be fairly certain that he would be protected, but his intended victim wouldn't be.

Typhoid vaccines today still require multiple doses and the protection is short-lived, just as it was in Christie's time. It was, and is, therefore, only recommended for travellers going to regions where typhoid could be contracted. Dr Roberts' second victim in *Cards on the Table* was following good medical advice when she went to the doctor for her vaccines before a trip to Egypt. The doctor then used this opportunity to infect her with something else.

The typhoid victim in 'The Flock of Geryon', Miss Lee, could have been chosen because she had not been abroad in a while and was therefore unlikely to have received the two typhoid vaccination shots. The murderer could use a vaccine to keep themselves safe from serious infection while preparing the typhoid culture for Miss Lee.

Agatha Christie was certainly creative and knowledgeable enough to have come up with these biological plots by herself. In the case of the contaminated shaving brush in *Cards on the Table*, however, she borrowed from a real-life incident. Her other bacteria plots may also have been inspired by some very similar real-life bacteria murders. In some cases there were no contemporary real-life examples to draw on, but subsequent events have shown that her plots were certainly plausible.

Some real-life cases

Inspiration for the doctor who killed his patients with bacteria in Christie's *Cards on the Table* may have come from the real-life case of 'Gentleman Girard'. Henri Girard was a dashing man living in Paris in the early twentieth century. He used his considerable charm to inveigle himself into the lives of various people he subsequently insured and then killed. His weapons of choice were home-grown germ cultures and deadly mushrooms.

In 1909 Girard met insurance broker Louis Pernotte and persuaded him to take out an insurance policy for 300,000 francs and sign over power of attorney. In 1912, Pernotte and his family went on holiday where they came down with what looked like typhoid, but they all survived. Girard, claiming he had some medical knowledge, then started treating Pernotte with injections. Pernotte died later that year and the insurance policy was swiftly claimed. Girard also informed Pernotte's wife her husband had owed him 200,000 francs.

Just like Dr Roberts in Christie's story, Girard appeared to have got away with it until he became a suspect in another murder case. In 1918, Girard targeted a widow, Mme Monin. With a large insurance policy taken out in her name, he invited her for dinner, where she collapsed and died. After his arrest, an examination of his apartment discovered culture dishes of typhoid and other bacterial nasties along with lots of *Amanita* mushrooms.* The investigation into Girard's criminal activities took three years but he took his own life in his prison cell before the trial could begin.

Had he stood trial, Girard could have tried to blame Pernotte's death on natural causes, but Mme Monin's death would have been difficult to explain away. Christie's fictional Dr Roberts also used the 'natural causes' explanation for the death of his second victim after her fake vaccination. He also had a credible excuse for anthrax being on a shaving brush that killed his first victim.

In 1915 cheap shaving brushes imported from China were found to be infected with anthrax after one man died of the disease with no obvious exposure to anthrax spores.[†] His symptoms had started with swelling of his neck (oedema) and his health rapidly deteriorated. A post-mortem revealed a slight cut on the man's neck within the shaving region. Lengthy enquiries

with the man's family revealed he had recently started using a new shaving brush. Examination of the brush, and several others from the shop it was purchased from, revealed the presence of anthrax. Tests showed that the bacteria were easily able to survive the soap and warm water used to lather the brush.*



Dr Anderson's cult in 'The Flock of Geryon' also has real-life cult equivalents. Bhagwan Shree Rajneesh was the charismatic, Rolls-Royce-loving founder of a global cult.† The Rajneesh movement started as a commune in Pune, India, in the 1960s. By the 1980s increasing tensions with locals caused the cult to seek out new premises and settle in 'The Big Muddy Ranch' in Oregon, USA. The settlement became a town named Rajneeshpuram, but local opposition continued, particularly with the Wasco County Court, whose members clashed over the cult's desire to acquire more land.

When two Wasco County commissioners visited the ranch for a tour, they were given glasses of water to quench their thirst. The next day the two men became violently ill, one needing hospital treatment to save him. It later emerged that the glasses of water had been contaminated with cultures of *Salmonella*.

After this incident, the cult's poisoning operations were expanded. Cult members visited local restaurants and poured brown liquid over the contents of the salad bars. In total 751 people, 45 of whom were hospitalised, were affected with severe cases of food poisoning. At the time it was attributed to lapses in hygiene standards that somehow simultaneously affected several establishments. It was only after the commune collapsed that the truth emerged.

One of the cult members, Ma Anand Puja, a nurse, had been working on various virus and bacterial cultures. Apparently, she had spoken of using *Salmonella* Typhi to contaminate the restaurant's salad bars, but elected instead for *Salmonella typhimium*, a well-known cause of food poisoning.* These bacteria may be from the same genus but they can result in

drastically different outcomes. Food poisoning is serious, but an outbreak of typhoid could have been deadly.

Agatha and bacteria

Christie's use of bacteria in her murder mysteries may, on the face of it, appear far-fetched and unrealistic. But truth is often stranger than fiction, as the real-life examples show. That she used it so rarely seems a shame as the fictional opportunities are fantastic.

The fact that she didn't use bacteria in her murder mysteries after 1940 may be due to the wider availability of antibiotics from 1945. Such was the optimism that science had found a way to defeat disease-causing bacteria, bacteria-based plots might not have appeared so credible. It might also be that Christie simply went back to conventional poisons because she was more confident in the science behind them. But her biological crimes are no less inspired or scientifically accurate than her other murder plots.

Notes

- * One of the short stories published in the collection *The Labours of Hercules*.
- † Septicaemia and sepsis are sometimes confused but they are slightly different. Septicaemia is a blood infection that can lead to septic shock (see later).
- * He also discovered the cholera bacteria.
- * Such dramatic changes in appearance meant the disease was widely known at the time as consumption.
- * This was the most common treatment for tuberculosis cases at the time. The bed rest and good food given to the patients in the sanatorium improved their general health and gave their immune system the best chance of fighting the infection.
- * Tuberculosis is spread through coughing and spitting and so separating the healthy from the sick in sanatoriums also reduced the spread of the disease.
- * The bacteria responsible for typhoid was identified in 1880 by Karl Joseph Eberth (1835–1926) and confirmed by Georg Gaffky (1850–1918).
- † The huge publicity may have helped control the spread of the outbreak, but it did nothing for Aberdeen's reputation. The Queen made a high-profile visit to the Scottish city later in the year to demonstrate it was safe.
- * 'Typhoid Mary' was Mary Mallon, a cook and carrier of typhoid who infected between 51 and 122 people with typhoid fever. She was forcibly incarcerated to prevent her infecting anyone else and spent 30 years of her life isolated.

- * Lethal factor, isolated from the rest of the anthrax bacillus, has been investigated for anti-tumour activity as it attacks a key signalling protein that is often mutated in human cancerous cells.
- * A stain developed by a man named Gram.
- † There are plenty of other differences and similarities between bacteria cells and animal cells but these are the ones we are going to focus on in this chapter.
- * A large family of mushroom species that includes the death cap and fly agaric, commonly the cause of accidental poisonings.
- † Two others were infected from their shaving brushes but survived.
- * For some time after these incidents shaving brushes were required to be sterilised before sale, something that had not been considered necessary before.
- † Some of the cars, rumoured to total 93, were given to him by his acolytes.
- * A handful of Rajneesh's followers were prosecuted for these and other attacks.



The Mirror Crack'd from Side to Side

Many of Agatha Christie's legions of loyal fans will have come to love her through her written works, but there must be quite a few who were introduced to her stories through the TV or cinema. Christie's novels and short stories have been adapted for the small and big screen many times over. They have ranged from the sinister (such as the 2015 TV adaptation of *And Then There Were None*) to the slapstick (any of the Miss Marple films starring Margaret Rutherford). There have been faithful adaptations (the 1974 *Murder on the Orient Express* and the *Poirot* TV series), and films that could be best described as 'inspired by Agatha Christie', for example

Tarantino's *The Hateful Eight* owes a lot to Christie's *And Then There Were None*.

Christie's relationship with the big screen has not always been easy. As an author, she was frequently disappointed with how her mysteries and characters were presented on screen, though she often had a high regard for the actors involved.* She was also happy to make use of the film industry for her plots. Murders are committed in cinemas; locals are star-struck/suspicious/scandalised by the arrival of filmmakers in their quiet English village; actors, actresses, producers and other members of the movie industry appear as victims, suspects and murderers.

Christie's personal distrust of cinema started back in 1928 with the first adaptation of one of her stories. *The Passing of Mr Quinn* was a low-budget but modestly successful silent film. It was an adaptation of Christie's short story 'The Passing of Mr Quin'†, but the film took considerable liberties with the plot, characters and method of murder.

Although the film has not survived, the screenplay was rewritten as a novel by G. Roy McRea (probably a pseudonym).‡ In the original short story, the murder victim is a brute of a husband who is poisoned with strychnine in the port decanter. In the film, the victim is a mad professor, and also a brute of a husband, who has invented 'A poison that leaves no trace', which he conveniently keeps in an easily accessible cabinet and cheerfully shows off to anyone who shows an interest, including people who might hold a grudge against him. He may be a scientific genius but his security measures are inadequate and his instincts for self-preservation are low. Unsurprisingly, someone slips the untraceable poison into the port decanter and the professor dies after drinking it.

The police investigating the professor's death find strychnine in the port decanter, but there is none in his corpse. The professor's experiments with this poison have somehow managed to make a poison that disappears when delivered into a human body. Having strychnine both detectable and undetectable is convenient for the plot but makes no scientific sense. Christie would never have tolerated such silly non-scientific goings-on in her books.

A small amount of any compound can be difficult to track down in something as large as a human body.* It is also true that some chemicals can

be rapidly altered by enzymes in the body into something else. But even if all the original poison is broken down in the body, it will be broken down into something else that can be detected. Only fictional poisons can completely evade detection and only when convenient for the plot.

Christie was not shy of being chemically inventive to suit her plots. She made up medicines and remedies in lots of her stories, but they were usually scientifically plausible. For example, an invented drug might be emptied out of its bottle or capsule and a real-life lethal compound put in its place, as it is in *Dumb Witness*. In *A Caribbean Mystery*, Serenite pills, a fictional prescription for the treatment of high blood pressure, are left in a murder victim's room to suggest their death was a natural result of the condition.

Several of Christie's fictional compounds could potentially be used for dastardly deeds in themselves. Benvo, in *Passenger to Frankfurt*, is a compound invented by a professor (whether he is mad or not is up to the reader to decide). Benvo doesn't kill, but it afflicts its victims with a permanent feeling of benevolence. Apparently this is a bad thing. Perhaps people will be so caught up holding doors open and doing other good deeds that society will grind to a halt? Deliberately changing someone's personality without their consent is certainly not good, but the professor considers the results his compound could produce to be so terrifying that he destroys his formula and all his research notes before they can fall into the wrong hands.

Another dangerous, but rather more conventional, fictional drug is Cherubin, which makes an appearance in *Murder in the Vicarage*. According to Christie, it is a headache powder that is very effective but shouldn't be taken in large doses. A few cachets of this powder are swallowed during the course of the novel, but no one dies as a result.

Slumberyl, a sedative as the name suggests, appears in two novels, *After the Funeral* and *Curtain*. We are told it is the trade name of a barbiturate drug. Much is made of how strong Slumberyl pills are, and the dangers of taking too many (more of this later), but no one is killed with them.

Christie invented one, and only one, poison to kill off a character and it appears in her 1962 novel, *The Mirror Crack'd from Side to Side*. After more than 40 years of writing murder mysteries packed with poisons, she had not simply run out of ideas or the energy to research yet another killer

compound. Nor is it an untraceable poison used to stump the pathologist and give Miss Marple an opportunity to exercise her extraordinary abilities to solve crimes through careful character studies. It is a perfectly plausible poison called Calmo.

*

The Mirror Crack'd is set in the picturesque village of St Mary Mead, home to both Miss Marple, amateur sleuth, and Marina Gregg, Hollywood legend. Ms Gregg and her producer husband, Jason Rudd, have recently bought the big house in the village. They are aware that the locals might not be too welcoming to them as outsiders who have extensively renovated a traditional English house to suit their celebrity lifestyle. In an effort to make friends and integrate into the community, they hold an open day. For a small entrance fee that will be donated to the local branch of the St John Ambulance,* the locals can explore the redesigned gardens and peer through the big new windows at the expensive decorations inside.

A select few, local dignitaries and showbiz friends, are invited into the house to mix and mingle with Marina and Jason in person. Among the special guests is Mrs Badcock, secretary to the St John Ambulance and devoted fan of Ms Gregg. She meets her idol at the top of the stairs, they chat, she's handed a cocktail and she moves on. Very soon after drinking her cocktail, Mrs Badcock slumps on a chair, her speech is slurred and her head lolls. She also appears to be struggling to breathe and quickly collapses. Fifteen minutes later, Mrs Badcock is dead.

The police become involved when it becomes evident that Mrs Badcock's death was not due to natural causes or an overdose of her own medication. She was in excellent health and was not taking any prescribed drug or over-the-counter remedies. Ms Gregg, however, is a frequent taker of prescription medicines. Her ongoing recovery from a serious mental health breakdown a few years earlier* means there are plenty of pills and powders in her home for a would-be poisoner to slip into someone's drink. Some of the pills and pick-me-ups are also popular with Marina's friends and employees, particularly Calmo, the 'marvellous' medicine that 'bucks you up *and* soothes you down'. While you will have to read Christie's novel

to find out *who* added Calmo to Mrs Badcock's cocktail, this chapter will look at *what* Calmo might be.

Calmo

Mrs Badcock is killed by ingesting four grains (just over a quarter of a gram, or 0.009oz) of Calmo. Like the vast majority of real-life medicines, Christie's invented remedy is sold under a trade name. Compounds can be named using a standardised system that describes their precise chemical structure. This means any chemist can work out how a molecule is arranged simply from its name. It is a useful system but cumbersome for anything but the simplest compound. Abbreviated or shorter alternatives are easier to remember and print on the side of a medicine bottle.

Christie has the chemical name of Calmo, 'hy-ethyl-dexyl-barbo-quindelorytate', or 'di-ethyl-mexine etc.', read out by non-chemists who are understandably imprecise and cut things short when they start to struggle with the unfamiliar words. Neither of these names describes a real compound, but they do contain fragments of real chemical terminology mixed with made-up words. Christie has cleverly given her invented drug the appearance of scientific credibility without being specific. However, there are lots of other clues in *The Mirror Crack'd* that can help us work out if Calmo could be a real drug and what that drug might be.

The British characters in the novel are relatively ignorant of Calmo, unlike the Americans, because it is 'more prescribed in America than here'. Having one set of characters learn about the drug from another set of characters is a clever way of describing many details and offering up possible clues for the reader in normal conversation.

As is often the case with Christie, the descriptions of Calmo tend to focus on symptoms produced by the drug, both at the therapeutic and lethal dose, rather than the forensic detail of identifying it in Mrs Badcock's remains. Interviews with witnesses are used to describe what happened when Mrs Badcock took too much of the drug: difficulty standing, problems breathing and a rapid loss of consciousness. This is very different to what is supposed to happen when the drug is taken at the prescribed dose.

Characters with personal experiences of taking therapeutic doses of the drug explain that ‘it makes you feel you could do things which otherwise you might fancy you couldn’t’, it induces ‘a tranquil state of mind’. Then there are doctors and pathology reports that explain that Calmo is ‘prescribed for those under strain, suffering anxiety, depression, melancholy, sleeplessness and a good many other things’.

Calmo is clearly used for treating mental rather than physical illnesses. The use of the word ‘tranquil’ immediately suggests Calmo is a tranquilliser, but the symptoms show that in larger doses it can also act as a sedative and hypnotic (induces sleep). The late 1950s and early 1960s were awash with new drugs for a range of newly defined mental health problems, and the United States was at the centre of this frantic pharmaceutical activity. You might not think of Christie as a cultural commentator, but when it came to medicines she had her finger on the social and pharmaceutical pulse.

The story of medication for mental health

In the distant past, severe mental illness might be ascribed to the will of gods or demonic possession. Those still tethered to reality but struggling with mood, obsessions or crippling fears might blame physical rather than spiritual causes and could apply physical rather than spiritual remedies.

The ancient Greeks explained human health and well-being with their theory of the four humours – blood, black bile, yellow bile and phlegm. An imbalance of these humours caused poor physical and mental health. For example, too much black bile was associated with melancholy and depression. People were bled, blistered and fed special diets in attempts to restore their balance.*

Of course, many sought alternative and more appealing forms of relief from their daily anxieties. For thousands of years alcohol and opiates were imbibed to take the stress out of everyday living.† From the sixteenth century onwards alcohol and opium were combined into a mixture named laudanum. Taking the two drugs together enhanced their effects and provided everything from a stimulus to creativity, to pain relief and a way to make life a little easier to cope with. The downsides of alcohol and

opium were obvious, but they were readily accessible and few alternatives existed, at least until the nineteenth century.

Bromide powders, introduced in 1857 to control epileptic seizures, found wider use as a general sedative. One notable side effect of taking bromides was that it made patients feel calm.^{*} Similarly chloral hydrate (see Chapter M), available from the 1870s, sedated and calmed patients, as well as helping them to sleep. However, the amount of these compounds that needed to be swallowed to produce these beneficial effects was more or less the same amount that would produce toxic results. They were far from perfect, but in terms of their effect on mood, they were a better option than alcohol or opiates.

The next drug for the treatment of mental health problems was introduced into the medical market in 1903 under the trade name Veronal. Initially recommended as a sedative and hypnotic, it appeared to be better than bromides and chloral hydrate because, at the therapeutic dose, Veronal was more potent and produced fewer toxic side effects. It also tasted better.

Veronal is a barbiturate, specifically barbital or barbitone,[†] a derivative of barbituric acid. Barbituric acid was first prepared by Adolf von Baeyer (1794–1885) in 1864.[‡] From this basic starting point, many different barbiturate derivatives could be produced. By 1947, 1,500 variations had been made and several dozen were available on the medical market for controlling seizures, sedating the agitated and helping insomniacs to sleep. Veronal became a very popular drug for treating epilepsy and insomnia as well as depression and anxiety. As sales of Veronal and other barbiturates shot up, reports on the hazards of taking these drugs also started to appear.

Barbiturates are addictive and became drugs of abuse. They also have a very narrow gap between the therapeutic and lethal dose. Veronal was prescribed in therapeutic doses of between 5 and 15 grains (0.3–1.0g or 0.01–0.04oz), but a lethal dose was only 3.5–4.4g (0.12–0.16oz). Another barbiturate, Seconal, was first released on to the market in 1934. By the 1960s and 1970s Seconal was widely abused and was commonly known as ‘seccies’, ‘red devils’ or ‘red hearts’ because of the colour of the capsules. Because barbiturates slow a person’s reactions and increase their drowsiness, many sleepy individuals took an extra (lethal) dose without realising what they were doing. Seconal is still prescribed today in 100mg

(0.004oz) tablets for the treatment of epilepsy, as a temporary treatment for insomnia and pre-operatively in short surgical procedures. The lethal dose for an adult is around 15g (0.53oz), which is a much larger safety margin than Veronal, but it has been too narrow for many people.*

The first of what would sadly be very many celebrity deaths brought about by barbiturate overdose was Billie Carleton. The young actress died after attending a party celebrating the end of the First World War in 1918.† Christie took inspiration from Carleton's tragic case for her 1923 short story 'The Affair at the Victory Ball' as well as her 1933 novel *Lord Edgware Dies*. In this novel Carlotta Adams dies of an overdose of Veronal. And there are other references to named barbiturates in many of Christie's stories.

By the 1950s the dangers of barbiturates were well known and Christie's readers would have understood what to expect if a character in one of her novels was said to be taking them. Part of Calmo's invented chemical name in *The Mirror Crack'd* includes the fragment 'barbo', perhaps a deliberate hint to her readers as to the type of drug Calmo could be. Placing the drug in the possession of a Hollywood star and her entourage would only reinforce this association. Finally, the symptoms Mrs Badcock displays immediately before she dies – the sudden collapse, the struggle to breathe – would also link Calmo with barbiturates.

But if Christie wanted Calmo to be a barbiturate, why didn't she say so? She had used barbiturates in plenty of plots already. And she was happy to state specifically that another of her invented drugs, Slumberyl, was a barbiturate. Maybe she was thinking Calmo could be something else.

*

There were plenty of other candidates for Calmo beyond barbiturates when we look at its proposed effects and symptoms. The 1950s were a turning point in the treatment of mental health. The first breakthrough came in 1952 at the Hôpital Val-de-Grâce in Paris. This military hospital had a psychiatric wing that, like every other psychiatric institution around the world, was struggling to help its many patients. The talking therapies, particularly popular at the time, needed time the doctors simply didn't have to spare

among the hundreds of patients they were expected to manage. Talking was of no use at all for those so lost in their illness that they couldn't communicate their problems. Patients were given barbiturates, put into insulin-induced comas and shocked back to wellness with electric currents. Few got better, and the numbers of patients only ever increased in the already large and crowded asylums. Sedation was often the best a hospital could offer, to give both patients and staff some respite, or physical restraints to stop individuals hurting themselves and others.

Henri Laborit (1914–1995), a surgeon at Val-de-Grâce, suggested his colleagues in the psychiatric wing try a new drug, chlorpromazine. This drug had initially been created as a potential antihistamine, but was rejected. The drowsiness and calm detachment it produced in patients made it unsuitable for treating allergies but perfect for relaxing patients before an operation. Laborit had found the drug reduced the chances of his patients developing life-threatening symptoms of surgical shock during his operations. Perhaps, he suggested, it might also calm some of the patients in the psychiatric wing.

Twenty-four-year-old Jacques Lh was the first psychiatric patient to be given chlorpromazine in January 1952. When he was admitted to Val-de-Grâce he was severely agitated. After an injection of chlorpromazine he calmed down immediately. With repeated injections he was well enough to play bridge. After a few short weeks of treatment he was discharged. Tests on other patients produced similarly startling results. Chlorpromazine appeared to be a miracle drug. It could clear mental fog, quiet the voices in patients' heads and re-engage the catatonic with the world around them. Launched on to the medical market under the name of Thorazine, sales boomed. For the first time in history, the numbers of psychiatric patients in institutions went down.

Chlorpromazine is an excellent antipsychotic agent, but it also has its downsides. Its effects are short-lived, meaning repeated doses have to be given and a tolerance can develop. When the drug was first introduced it was over-prescribed. It was given to patients who didn't need it and very large doses were administered to those who did. In excess, the drug causes drowsiness, slurred speech, dizziness, increased heart rate, convulsions and coma. The drug also suppresses the cough reflex and unconscious patients can choke.

Christie's character Marina Gregg may have encountered chlorpromazine during her period of hospitalisation after her breakdown. It is particularly useful for treating seriously unwell patients who suffer delusions and hallucinations, for example schizophrenics. However, chlorpromazine has been described by critics as the 'liquid cosh' because of its very powerful effects. Even in its over-prescribed heyday it was unlikely to have been given to Ms Gregg, or anyone else in the Gregg household, as freely as Calmo appears to have been doled out in Christie's novel. It is therefore unlikely to be found in a bathroom cabinet in Marina Gregg's home and is not a good real-life candidate for Calmo.

Doctors realised chlorpromazine had its limitations, but it has been of huge benefit to thousands of people over the years. It also spurred the search for more drugs that could be used to treat mental illness, and promising compounds started to pop up every few years.

*

Meprobamate was synthesised by pharmacologist Frank Berger (1913–2008) in 1950 while working for Wallace Laboratories. Tests showed it was of little benefit to those suffering from serious mental health conditions such as schizophrenia, but it helped the majority of patients suffering from anxiety. Patients who had been too tense, anxious or irritable to work went back to their jobs and became productive members of society after taking meprobamate. The doses were high but the toxicity seemed low. Wallace, however, wasn't convinced there was a market for this type of drug and delayed release until 1955.

As the launch date approached, Berger talked about the new drug to his friend Nathan Kline (1916–1983), a psychiatrist, and wondered out loud if the drug should be classified as a sedative. Kline responded: 'The world doesn't need new sedatives. What the world needs is a tranquilliser. The world needs tranquillity. Why don't you call this a tranquilliser? You will sell ten times more.' And so meprobamate was sold as a new kind of drug, a tranquilliser, under the trade name Miltown, after the small, tranquil town of Milltown in New Jersey.* It became a marketing and sales phenomenon.

The celebrity world, already well acquainted with prescription drugs, embraced Miltown wholeheartedly. It appeared to offer all the benefits of barbiturates but without the problems. Movie stars and TV personalities talked openly and approvingly of the drug in interviews. Tallulah Bankhead, the film and stage actress, joked she should be paying taxes in New Jersey because of how much Miltown she was taking. Milton Berle took so many pills and talked so enthusiastically about them on his TV variety show he started calling himself 'Uncle Miltown'. At celebrity parties the 'emotional aspirins' were as readily available as peanuts at a bar and guests were served with Miltinis, a dry martini with a Miltown pill in place of the traditional olive. Los Angeles became known as Miltown-by-the-sea.

As the 1950s progressed, more and more worrying reports about meprobamate appeared. Initially marketed as 'non-habit forming' it was clear that some people became addicted to the drug. It was still safer than barbiturates, but overdoses were certainly possible. The recommended dose was one 400mg (0.014oz) tablet taken three or four times a day, or up to 1.2g (0.04oz) over a 24-hour period. The lethal dose, however, could be as low as 12g, though there were also reports of people surviving doses of 40g (1.41oz). Individuals who had swallowed too many meprobamate pills were found unconscious and unresponsive and with dangerously low blood pressure. With medical support, virtually all of them survived but, by 1959, at least three people had not been reached in time and had died as a result.

Miltown, the 'goodwill pill' with its celebrity associations and potentially lethal consequences, would appear to be an excellent fit for Christie's Calmo. She was writing at the peak of the drug's popularity, 1961. But another tranquilliser drug had recently appeared on the market that would eventually eclipse meprobamate's success.

*

Chlordiazepoxide was discovered in 1957. It had hypnotic, sedative and tranquillising properties. It restored a sense of equilibrium to those with frayed nerves, and so, when it was released on to the market in 1960, it was named Librium. Librium was the first of the benzodiazepine class of compounds.* The beneficial effects of benzodiazepines wear off after a few

weeks of use, but they are excellent for occasional or short-term use. They are not as addictive as barbiturates, but people do still become addicted to them after prolonged use. Benzodiazepines are used as daytime anti-anxiety drugs, sleep inducers, anaesthetics, anticonvulsants and muscle relaxants, depending on both the variant and the dose. They have become among the most widely prescribed drugs in the world because of their excellent safety record.

Benzodiazepines tend to have a long half-life in the body, which can lead to 'hangover' effects. Daytime sedation and oversedation can occur after taking the next day's dose on top of the residual dose from the day before. Death from a benzodiazepine overdose, however, is extremely rare. For example, in 1987, Robert McFarlane, national security advisor to Ronald Reagan, was admitted to hospital two hours before he was due to testify before a committee investigating the Iran-Contra scandal. He had swallowed between 25 and 30 benzodiazepine tablets (specifically Valium) in a suicide attempt. Though he had ingested far in excess of the daily recommended dose of just 40mg (0.0014oz), he woke up several days later very much alive.

As far as the poisoning in *The Mirror Crack'd* is concerned, benzodiazepines would seem to be a poor fit. It would take a much larger number of tablets than the 'six times the ordinary dose' stated in the novel. It would also have taken a lot longer than 15 minutes for the victim to die.

However, we cannot completely discount Librium as a possible candidate for Calmo. Sadly, there have been many deaths associated with Librium and other variants of this relatively safe class of drugs because they have been mixed with other drugs. Only one prescription compound is in Mrs Badcock's cocktail, but there is plenty of another non-prescription depressant drug in her glass – alcohol. The two compounds together reduce the chances of Mrs Badcock's survival considerably. To understand the effects of Librium, and possibly Calmo, we first need to look at nerves and how they send their messages.

How Calmo kills

Messages are sent along the length of nerve cells using electrical signals, but where one nerve cell connects to another cell there is a gap called the synapse. The electrical signal cannot jump the gap. Instead, chemicals, or neurotransmitters, are released from one nerve ending to dock at receptors on the adjoining cell (see Fig. C.1). Different neurotransmitters docking at different receptors can increase or decrease the activity of the receiving cell. Depending on the overall balance of neurotransmitters received and docked, the receiving cell will be activated – a muscle cell will contract, or another nerve will fire and pass the message further on – or it will be prevented from activating. The complexity of this system allows many subtle variations in reactions and activity of cells around the body. Fortunately for this chapter, we can focus on just one neurotransmitter, GABA (gamma aminobutyric acid).

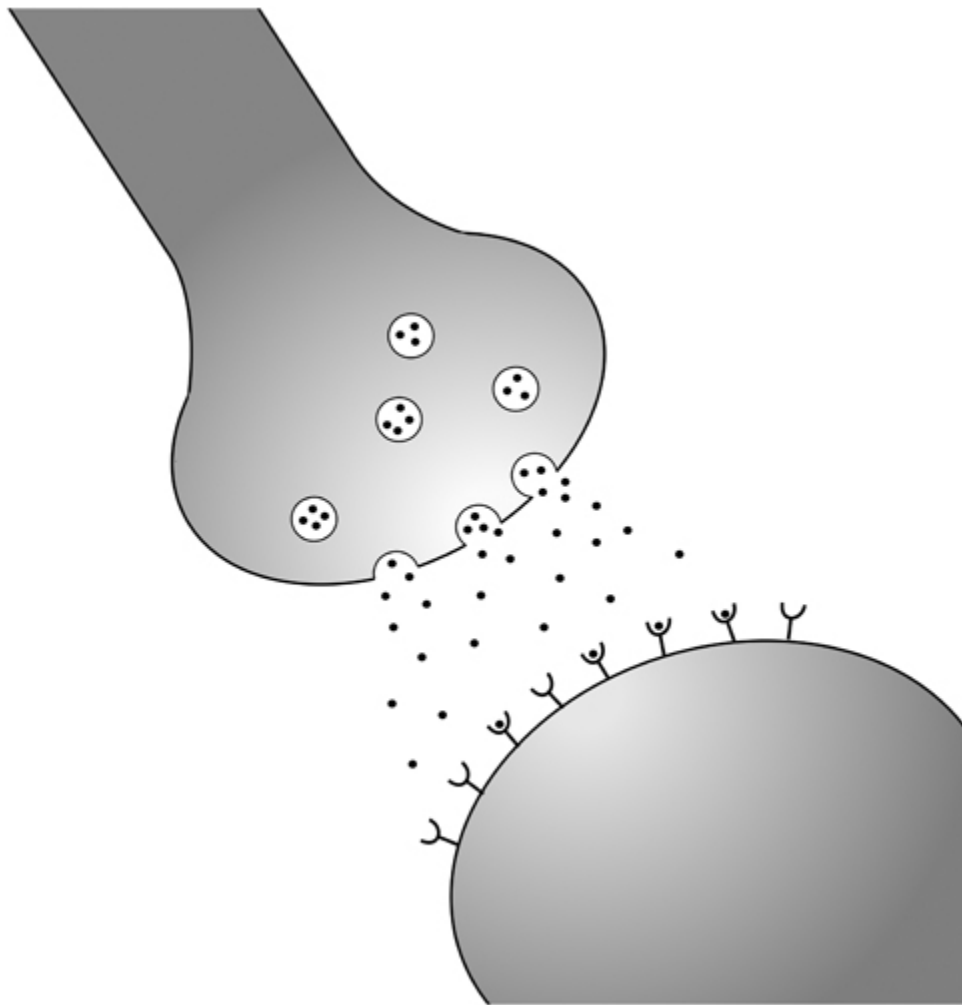


Fig. C.1. Schematic of the synapse showing neurotransmitters released from one nerve cell and docking at receptors on an adjoining cell.

GABA is the main inhibitory neurotransmitter for the central nervous system – the brain and spinal cord. When GABA binds to a receptor on a nerve cell, that nerve cell is less likely to be activated. It does this by allowing more negatively charged chloride ions (Cl^-) to flow inside the cell. This increases the negative charge inside the nerve cell and makes it harder to trigger an electrical signal. GABA therefore reduces activity in certain parts of the brain.

GABA molecules are distributed throughout the brain but there are some regions with more GABA receptors than others. These GABA receptor-rich areas are: the hippocampus (associated with memory, learning and emotion); thalamus (a relay station within the brain, passing on messages to other areas); basal ganglia (responsible for motor control, executive functions and behaviours as well as emotions); hypothalamus (manages basic functions like temperature, hunger, thirst and blood pressure as well as mood and sleep), and the brainstem (controls breathing, heart rate, balance, coordination and reflexes). By decreasing activity in the nerves in these areas, GABA relaxes us. It also helps us sleep and prevents anxiety.

Barbiturates, meprobamate, benzodiazepines and alcohol can all interact with GABA receptors. They enhance GABA's calming influence on the nerve cells in the brain (and on other nerves in the body but to a much lesser extent). All these different drugs dock at slightly different places on the receptor, meaning there is space for several different molecules to dock at the same time. And the effect they have on the receptor is slightly different in each case. Alcohol, for example, distorts the receptor slightly, making it easier for GABA to bind. Meprobamate activates the receptor directly, though weakly, producing a similar but much reduced inhibitory effect on the nerve even if GABA isn't present.

If two of these GABA-enhancing compounds are swallowed at the same time, the overall effect on the body is much greater than would be expected by simply adding up their individual effects. Together they can depress activity in the central nervous system and slow breathing. Slower breathing rates can mean less carbon dioxide is expelled from the lungs. More carbon dioxide in the blood means more carbonic acid and an increased acidity (see

Chapter F, page 99). Breathing problems also mean less oxygen is taken into the body, resulting in cyanosis – visible as a blue colouration to the skin. Combinations of some of these drugs in sufficient quantities can slow breathing to the point that it stops. However, these drugs have other effects that can contribute to or cause death.

Some of these drugs can cause oedema, where fluids build up in the muscles, lungs and brain. Suppression of the ‘cough’ reflex means that it will be difficult to clear fluid from the lungs and throat. Meprobamate can prevent the production of blood cells (a condition known as aplastic anaemia that is more common in certain types of cancers). Meprobamate can also affect heart rhythms and cause cardiac arrest.

Whether the murderer in *The Mirror Crack’d* deliberately dissolved Calmo in an alcoholic drink to maximise its chances of killing Mrs Badcock is not stated. Regular users of the drug, such as most of the household where the murder takes place, would have been aware of the enhanced effects of taking drugs like Calmo with alcohol. What they would not have been aware of is that Mrs Badcock’s medical history also made her dose of Calmo more likely to be lethal.

When barbiturates, meprobamate and benzodiazepines are taken as tranquillisers or sleeping pills, they have only a limited time of effectiveness. In the case of barbiturates, continued regular exposure to the drug means the body metabolises it faster and larger doses are needed to produce the same results. As someone who had never taken drugs like these before, Mrs Badcock would have felt their maximum effects.

The Calmo cocktail quickly overcomes Mrs Badcock; there is little time for symptoms, but the ones that are described are consistent with what might be expected for an overdose of tranquillisers. Her health declines rapidly and she is declared dead minutes after falling ill. Whatever initial medical response was given, it can’t have been continued for very long. This is surprising given that Mrs Badcock collapsed in a house full of people who are there to help raise money for the St John Ambulance. You would expect there to be at least one person present who would be able to administer first aid. With a little more perseverance, Miss Marple and the police might have been investigating an attempted murder rather than a successful one.

Is there an antidote?

The depressive effect of the tranquilliser/hypnotic on Mrs Badcock's respiration means that she was not getting enough oxygen into her lungs. If the drug was also affecting her heart, as is suggested in the text, the little oxygen that was being taken in was not being distributed around her body very effectively. The first thing to do would be to check her airways aren't blocked and administer mouth-to-mouth resuscitation. Exhaled breath still contains around 16 per cent oxygen and can be the difference between life and death for someone who isn't breathing for themselves. Chest compressions help push oxygenated blood around the body when the heart isn't doing an efficient job on its own. This is cardiopulmonary resuscitation (CPR) and would be a standard emergency response today.

People of all eras have tried their best to resuscitate the dead or dying but the benefits of combining mouth-to-mouth resuscitation and chest compressions was only researched and proved to be effective in 1957 by Peter Safar (1924–2003), an Austrian anaesthesiologist. The general public was not encouraged to learn CPR until the 1970s, but one would hope that members of the St Mary Mead branch of the St John Ambulance in Christie's 1962 novel kept up to date with the latest first aid techniques.

Tempting as it might be to try and get Mrs Badcock to vomit and remove from her stomach whatever is making her ill, if she was unconscious and her cough reflex suppressed, the vomit could easily have been aspirated into the lungs. Giving her stimulants like adrenaline or caffeine to help revive her would also have been a bad idea – it can do more harm than good. Any brief stimulation given by these drugs is followed by greater depression. And anything forced down her throat could easily end up in the lungs rather than the stomach, making things even worse. The best plan would have been to keep up the CPR until better medical help arrived.

From the late 1950s ambulances were increasingly equipped with respiration devices. Hospitals also had machines that could help a patient breathe, as well as defibrillators and electrical devices that could be used to restore a normal heartbeat. There is no specific antidote for sedative and hypnotic drugs, but once Mrs Badcock was stable, her symptoms could be treated as they presented themselves and her breathing supported until she woke up. With intensive supportive care the recovery rate from overdoses

of sedatives, tranquillisers and hypnotics is excellent. Of course, Mrs Badcock never made it to an ambulance, let alone a hospital, before being declared dead.

While the cause of Mrs Badcock's death is quickly established at autopsy, the motive behind it takes a little longer to discover. Accidental and deliberate overdoses of tranquillisers or sedatives with alcohol were all too common, even in the early 1960s, and both possibilities are considered in Christie's novel before being eliminated. Murder is a rare cause of death. Murder by poison is rarer still. The chances of being murdered using tranquillisers or sedatives and alcohol is almost unheard of, but not quite.

Some real-life cases

The first conviction for murder using barbiturates in the UK was in 1955. John Armstrong killed his infant son with Seconal tablets he had stolen from the hospital where he worked. When barbiturates were first discovered, they could be bought over the counter, but the very real dangers of this class of drugs meant access was rapidly restricted, so Armstrong had to steal them.

This availability problem also applies to Christie, who would have been aware of the difficulties of getting barbiturates into Marina Gregg's house in the quantities she describes in the story. The new tranquillisers were much safer and therefore had fewer restrictions attached to them. Even when they were available by prescription only, these prescriptions could often be refilled an unlimited number of times, allowing someone to build up a considerable stockpile by going from one pharmacy to the next. However, the poisoner in *The Mirror Crack'd* had no need of any such forward planning because the drug they used was mixed in an alcoholic cocktail. Even 'safe' prescription quantities of drugs can swiftly become lethal when alcohol is involved, as a case from 1982 shows.

Early one February morning the partially frozen body of a 42-year-old man was found behind a filling station in Nebraska. Despite the cold weather, he was not wearing a coat, scarf or boots. There were no obvious signs of violence and so the body was taken to the local medical examiner's office to thaw before an autopsy could be carried out.

The autopsy revealed no obvious cause of death and so samples were sent for toxicological examination. The deceased's blood-alcohol level showed he had been quite drunk, but not drunk enough to pass out or die from alcohol poisoning. Also in the blood was a toxic concentration of a long-acting barbiturate, though not a lethal concentration on its own. The combination, however, was enough to have killed him. The only difficulty for the medical examiner was the reason why they came to be in the man's body. Had he deliberately taken them, knowing the likely outcome? Or had he swallowed the barbiturates with alcohol, unaware that the two together were dangerous? An investigator went to the deceased's home to find out more.

The dead man's wife was very agitated during this initial interview. She described her husband as a long-time alcoholic, a poor provider and a wife-beater. A few years before, he had developed a seizure disorder for which he was taking one 30mg tablet of sodium phenobarbital each day. Asked to produce the phenobarbital bottle, she broke down and confessed to poisoning him. The day before, her husband had been in the kitchen drinking beer. She had opened the cans and slipped two or three phenobarbital tablets into each can before handing it to her husband. She estimated he had consumed between 14 and 18 cans of beer and all the phenobarbital medication available – more than 20, but fewer than 30 tablets. Even 30 tablets would give a total of 0.9g (0.03oz) of phenobarbital, below the usual accepted fatal amount of 1.5g (0.05oz). Without the beer, her husband would likely have survived.

Agatha and invented poisons

When it came to inventing a killer compound, Christie stuck to familiar territory – pharmaceuticals. Her training as a dispenser gave her an excellent understanding of what might be possible and probable with a drug in therapeutic and lethal doses, as well as technical knowledge she could use to give a sense of scientific credibility.

Calmo is a fictional but credible drug of the kind that was available, and popular, at the time Christie was writing.* It is more dangerous and fast-acting than most of its real-life equivalents, but the circumstances around

the death – the victim’s naivety to the drug and administering it in an alcoholic drink – keep the death just about within the realms of possibility and probability. Christie’s reputation for detailed descriptions of poisons and poisonings holds true even when she is making them up.

Notes

- * Christie reportedly hated the Miss Marple films with Margaret Rutherford but got on well with the actress herself. The novel *The Mirror Crack’d from Side to Side* is dedicated to Rutherford.
- † When the short story was published as part of a collection, *The Mysterious Mr Quin* in 1930, it was renamed ‘The Coming of Mr Quin’, perhaps to distance it even further from the film version.
- ‡ You can also read a scathing review of the film in the August 1928 edition of *Variety*.
- * Wannabe poisoners be warned. As detection techniques have become more sophisticated and sensitive over the years, the amount of a substance that can be detected is almost always much, much smaller than the lethal dose.
- * A UK volunteer-led charitable organisation that provides first aid training and support.
- * Marina Gregg’s character, and a big chunk of the plot of *The Mirror Crack’d*, appears to be based on the experiences of real-life actress Gene Tierney. Tierney suffered a major breakdown and struggled with mental health problems afterwards. Fans were appalled that Christie seemed to be taking advantage of Tierney’s tragic circumstances, but her agent claimed Christie knew nothing of Tierney’s situation prior to writing the novel.
- * In Shakespeare’s *Taming of the Shrew*, for example, part of Katherine’s ‘taming’ is restricting her to a meat-free diet to calm her hot temper.
- † They were also used to alleviate pain, help people sleep and render them unconscious during painful medical procedures.
- * A better reason for giving doses to British soldiers in the First World War than the widespread myth that it was to dampen their libidos.
- † It is the same compound but the name is a little different in different countries. The US favoured the ‘al’ ending for barbiturates, whereas ‘one’ was used everywhere else.
- ‡ There are two versions of the naming of the acid. One has it that Baeyer named it after a woman he was obsessed with called Barbara. The other story is that Baeyer made his discovery on St Barbara’s day.
- * Judy Garland and Tennessee Williams both died as a result of Seconal overdoses. Regulation over the prescription of these drugs has improved considerably since then.
- † At the time her death was attributed to a cocaine overdose, but it is now thought the cause of death was the barbiturates she took to cope with cocaine hangovers.
- * It was also marketed under the trade name Equanil, for the equilibrium it produced in those who took it.
- * Librium set sales records, but they were soon beaten by another benzodiazepine drug released on the market in 1963, too late for Christie to use in *The Mirror Crack’d*. It was given the trade name Valium.

- * In fact she pre-dates some of the more well-known cultural references to tranquilliser use and their dangers. The Rolling Stones were singing about 'Mother's Little Helper' (Valium) in 1965. Jacqueline Susann's novel about the mix of fame and pill taking, *Valley of the Dolls*, was published in 1966.

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‘The Lemesurier Inheritance’ (Poirot’s Early Cases)

When planning the perfect murder, the most important thing to consider is how you are going to get away with it. Agatha Christie has tried out all sorts of plans in an attempt to baffle her detectives and readers, from seemingly unbreakable alibis to murderers who have pretended to be one of their own victims. One fairly common method of misdirection is to try and shift the blame elsewhere – onto another person, a disease, an accident and even an insect. In Christie’s fictional world, wasps and bees have been unfairly implicated in the deaths of several characters.

The bee alibi is used in one of Christie's best-known novels, *And Then There Were None*, published in 1939. A bee is left in the room with the dead body of Miss Brent, who has actually been killed by an injection of cyanide into her neck. The bee in this case has more to do with the killer's obsession with making their murders fit the lines of a rhyme than genuinely trying to make the death look as if it was the result of a bee sting.* In the 1935 novel *Death in the Clouds*, a wasp is the chief suspect in the death of Mme Giselle, at least for a while. A poisoned dart found on the floor next to the victim suggests that a human rather than an insect is the real killer (see Chapter V, page 277).

Christie first tried out these six-legged red herrings in 1923. Her short story 'The Lemesurier Inheritance' found room for a bee and a wasp, along with many dead humans, within a few pages.† But it is a third insect, ants, that are used in the attempted murder of little Ronald Lemesurier.

The Lemesuriers of the story's title are an old English family blessed with a large estate and a family curse. The estate is a big house and lands in Northumberland acquired sometime around the Reformation. The curse was acquired in medieval times from a wife falsely accused of infidelity. She vowed that the firstborn would never inherit whatever the Lemesuriers had to bequeath. Since then, the Lemesurier inheritance has passed to second sons, nephews and brothers, but never to the firstborn son. In recent years the curse has been helped along by a murderer.

Poirot and Hastings encounter three of the Lemesurier family while out dining at a restaurant. Vincent Lemesurier, one of Captain Hasting's old army buddies, is there with his uncle, Hugo Lemesurier. A third Lemesurier, Roger, arrives with terrible news for Vincent. Vincent's father has been thrown from a horse and is unlikely to live. Poirot and Hastings are surprised at how much this affects the young man, only to learn from Hugo about the Lemesuriers curse.

Vincent is his father's firstborn and only surviving son. Not only is he about to lose his father but, if the curse is to be believed, he is also not long for this world. Sure enough, the very next day Vincent dies, apparently by jumping from the train that was taking him north to see his dying father.

More Lemesuriers meet their maker – via natural causes, an accidental shooting, and, importantly for this chapter, a wasp sting – until the uncle,

Hugo Lemesurier, finds himself as owner of the family estate. Now Hugo's eldest son, eight-year-old Ronald, is the next potential victim of the curse. Over the course of six months the child narrowly escapes death by drowning, falling from a window and a nasty bout of food poisoning. At Mrs Lemesurier's request, Poirot and Hastings travel to Northumberland to watch over Hugo's heir apparent.

There are no more unsettling incidents in little Ronald's life until he visits a friend of the family, a friend who happens to be a keen beekeeper. The child is rushed home after being stung by one of the insects. He is fine, there isn't even any swelling, but it makes Poirot wonder if bees and wasps haven't been convenient cover for someone artificially helping fulfil the family curse.

That night, the murderer creeps through Ronald's bedroom door. But Poirot and Hastings rush from their hiding place and overpower them before they reach the sleeping child. Gripped in the murderer's hand is a hypodermic syringe. 'What is in it? Poison?' asks Hastings on behalf of the reader. Using only the power of his little grey cells, and no analysis whatsoever, not even a sniff, Poirot replies, 'Formic acid, I fancy. ... Probably obtained by distilling ants.'

Why so many insects in one short story? The answer is all down to practicality. However convoluted and seemingly contrived Christie's methods of poisoning her characters may appear, they are always practical, at least within the world she created for those characters to live and die in. So let's look at the three implicated insects.

The killer insect story

Bees and wasps are certainly credible killers. Both are known to deliver painful, sometimes fatal, stings. Bees have even been weaponised in the past when their hives have been launched towards enemy armies, using catapults or simply pushed over as a last line of defence when soldiers have attempted to invade.

Many insects have evolved means of attack and defence, to get food or protect themselves against a perceived threat, and they quite often use chemicals to do it. Wasp stings deliver a venom rich in peptides (short

lengths of amino acids not long enough to be considered a protein) that cause pain, dilate blood vessels and increase their permeability, allowing water to leach into the surrounding tissue. The result is discomfort and oedema, or swelling, that we associate with their stings. There are also enzymes, phospholipases and hyaluronidases, which break down membranes and connective tissue, allowing the venom to spread further into the body.

Phospholipases and hyaluronidases are also found in bee venom, but around half of the mixture injected through their hollow sting is melittin, a pain-inducing peptide. This is why bee stings can hurt so much, but it is not the most toxic component in their venom. Making up around 2 or 3 per cent of dried bee venom is the most concerning compound, a peptide consisting of just 18 amino acids called apamine. Apamine is a neurotoxin affecting the central nervous system.

Every nerve cell operates a little like a battery, with positive and negative parts. Biological batteries are charged by the Na^+/K^+ -ATPase enzyme pumps we met in Chapter A. The enzymes shunt potassium (K^+) and sodium (Na^+) ions in and out of nerve cells until the interior becomes negatively charged and the exterior positively charged.* An electric current is generated when charged particles (positive or negative) move. When a nerve cell ‘fires’, channels along the length of the cell open up to allow the ions inside the cell to flood out, and the ions outside the cell to flood in. Positive and negative switch places to generate the electrical signal that travels down the length of the cell like a wave (see Fig. F.1). After the signal has been sent, the sodium-potassium pumps move everything back to their original positions so the nerve is ready to fire again.

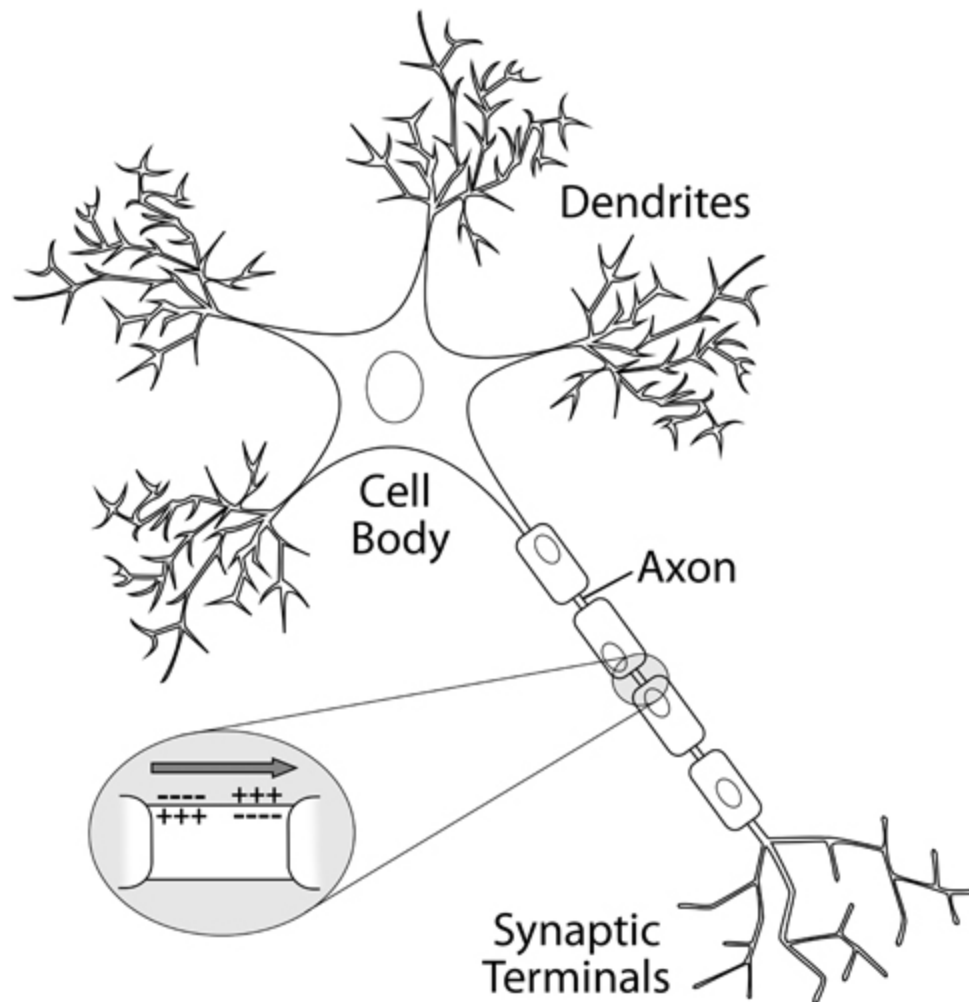


Fig. F.1. Generating an electrical signal in a nerve cell.

The apamine peptides in bee venom block the channels that allow K^+ ions in and out of the nerve cell. The result is a more excitable nerve and an overly active central nervous system. If apamine has the same toxicity in humans as it does in mice,^{*} around 300mg (0.01oz) injected subcutaneously into a person would likely kill them. A worker bee only weighs around 110mg (0.0038oz), and so can't deliver a lethal dose on its own, but this is where other chemical components within bee venom come in.

When a bee stings, isoamyl acetate, one of several alarm chemicals, is released along with everything else. Isoamyl acetate has a smell that reminds humans of bananas but it tells bees to attack. Humans may be much larger than bees, but we can still be brought down by their collective efforts. The effects of venom from multiple stings can accumulate and can

lead to hypertension, damage to the heart, changes in liver function, shock and kidney failure. It is thought that 50 bee stings can cause red blood cells to break open and spill their contents, leading to respiratory difficulties. A hundred or more stings can be fatal.

In the United States approximately 100 people per year die from the stings of bees and related insects. In England and Wales it is roughly one person per year. However, most of these deaths are not from large numbers of bees swarming to attack their victim. Some of the compounds in bee and wasp stings interact with the human immune system to produce an allergic reaction, which in sensitive individuals can lead to anaphylaxis.

Anaphylaxis is an extreme reaction from the immune system in response to foreign material in the body. Symptoms appear rapidly after exposure, within minutes if the substance has been injected, and include skin rashes or hives, swelling in the throat or tongue that can block airways, racing heart, vomiting and low blood pressure. Without rapid emergency medical treatment death can occur because oxygen can't reach the lungs, or low blood pressure is preventing it being circulated round the body. It is a very dangerous but rare medical condition – around 1–2 per cent of the world's population will experience anaphylaxis in their lifetime, but insect bites or stings are a common trigger.*

If the murderer in 'The Lemesurier Inheritance' knew their victims were allergic to bee or wasp venom, just a single sting could be enough to keep the family curse on track. Without this particular susceptibility, the murderer would need to recruit a swarm to do their dirty work. Some bee species are more suited to this task than others. Africanised bees are also known as killer bees, but not because their venom is more potent – it differs little from their European honeybee cousins.† They are also smaller and can deliver less venom with each sting. Their deadly reputation in fact comes from their aggressive nature. Attacks of hundreds of bees are not unusual. Christie could have used one bee or wasp and an allergic reaction to kill off her characters, or some angry Africanised bees to swarm them. So why didn't she?

The problem is targeting your enemy. Wasps and bees do not take direction very well. They are invaluable parts of the ecosystem, and they do incredible work supporting humans through pollination and producing

honey, but they do it on their own terms. A bee or wasp cannot be told, trained or ordered to attack, and definitely cannot be directed towards one particular human. If you have a specific target in mind, it would be easier to isolate the venom from the insects and introduce it into the victim yourself.

Wasp venom can be collected insect by insect by first immobilising the creature using sticky tape attached to the fingers. The head of the wasp is then crushed with a pin to get them to release their venom. Venom sacks can also be dissected from wasps and bees and then centrifuged to separate the venom from the tissue. The process is finicky, requires a centrifuge and, in the case of the bees, may also release the alarm pheromone, attracting many other angry bees. However, bees can be tricked into stinging using electricity. A current running through an electric grid covered with glass and placed at the entrance to a bee's hive will induce the bee to sting. The venom will be deposited on the glass and can be scraped up afterwards. Get it right and it doesn't hurt the bees and they can be milked repeatedly.

The result of all this delicate and dangerous work is a fairly potent poison, but it needs time, skill and specialised laboratory equipment. Instead, the murderer in 'The Lemesurier Inheritance' chose another insect to supply a less potent but more easily accessible venom to fill their hypodermic syringe.



Ants have made their nests on every continent on Earth except Antarctica. They seemingly fill every nook and cranny and have diversified into over 13,000 different species to do so. Some species have powerful mandibles that can give a painful bite,^{*} and several other species have formidable stings. Ant venoms, like the creatures themselves, vary considerably. The bullet ant (*Paraponera clavata*), for example, gets its name from its incredibly painful sting that apparently feels like you've been shot. Fire ants (several species of the genus *Solenopsis*) have a similarly appropriate name based on their stings. These very painful venoms are rich in toxic proteins, but they're not readily available to Christie's murderer in 'The Lemesurier Inheritance' because these ants are not native to the UK.

Ants are as abundant in the UK as they seem to be almost everywhere else, but the ones you would be most likely to encounter are from the genus *Lasius*. The little black ants that crawl over UK pavements, lawns and picnics are *Lasius niger*, members of the subfamily Formicinae.[†] Formicinae ants have much reduced stings, compared to other ant species, but much enlarged venom sacks. The venom within these sacks is very different to most other venoms. Instead of complex mixtures of peptides and other molecules, Formicinae ant venom consists of predominantly one very simple chemical compound, formic acid, named after the Latin for ant, *formica*.

Formic acid

Many natural philosophers and keen observers had noticed that ant hills gave off an acidic vapour, but in 1671 John Ray (1627–1705), noted botanist and naturalist, was the first to put into print the process of distilling large quantities of ants to obtain formic acid. He also developed a system of classifying living organisms that was an important step towards the modern binomial system we use today. Ray was clearly fascinated by living things and was more than willing to sacrifice large numbers of them to find out more about them.

Rather than fiddling around with wasps and bees, all the murderer in ‘The Lemesurier Inheritance’ had to do was replicate Ray’s centuries-old experiment. Simply coax large numbers of ants into a vessel that could then be boiled to release the formic acid as a vapour, which could then be collected via a simple condenser. A few items from the kitchen and some bits and bobs from the garden shed could be improvised into a basic still, efficient enough for the murderer’s purposes. Formic acid is a clear, colourless liquid that boils at a fraction over 100°C (213°F, almost exactly the temperature for boiling pure water) and, though slightly more dense than water, it could be easily sucked up into a hypodermic syringe.



Each molecule of formic acid consists of only five atoms: two oxygen, two hydrogen and one carbon atom (see Fig. F.2). These atoms are held together by covalent bonds formed when two electrons, one from each atom, are shared. But this sharing is not always equal. Oxygen atoms have a powerful pull on electrons, unlike tiny little hydrogen atoms. When oxygen and hydrogen bond to each other, the two shared electrons holding them together will spend most of their time around oxygen. Electrons are negatively charged and so the oxygen, with its greater share, will be slightly negatively charged. The hydrogen atom, having lost out on some of its electron share, will consequently be slightly positively charged.

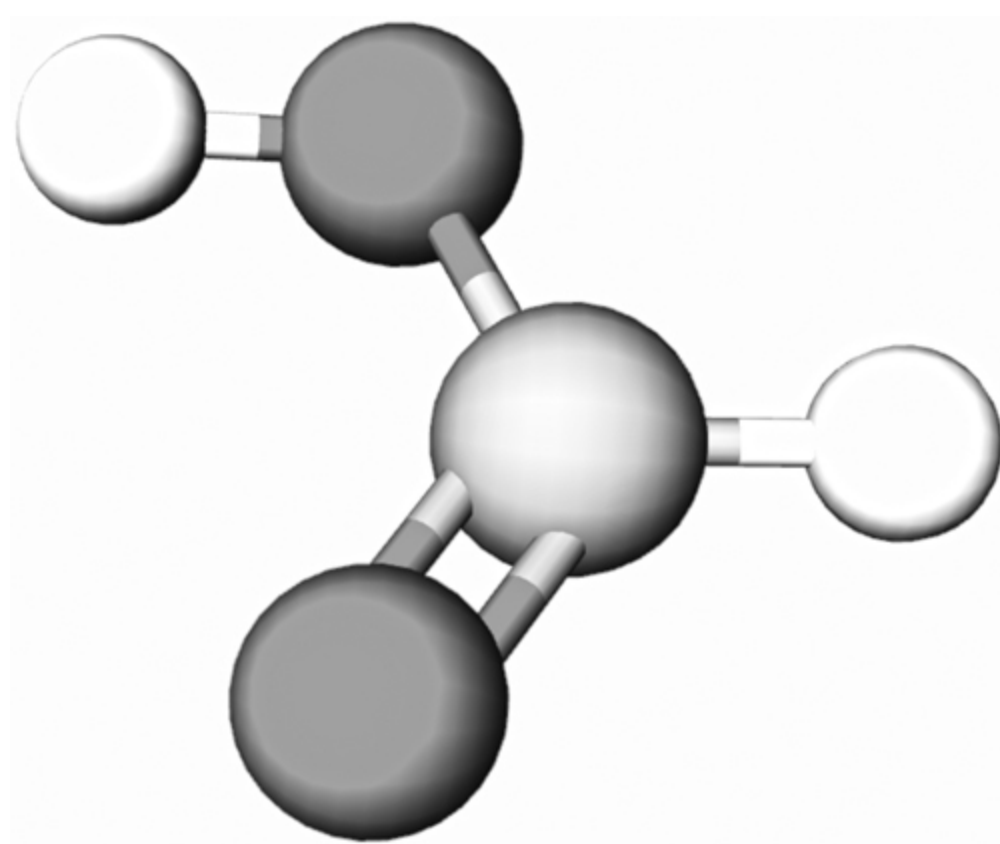


Fig. F.2. Atomic structure of formic acid. The hydrogen that can be lost to form H^+ is the white atom shown top left and bonded to the dark grey oxygen atom. The pale grey atom is carbon.

Water molecules, consisting of one oxygen atom bonded to two hydrogen atoms, also have an uneven distribution of electrons and therefore positive and negative regions within each molecule. Mix formic acid with water and the respective positive and negative regions will pair up like

magnets. Sometimes this magnetic-like pull is stronger than the bond between an oxygen and a hydrogen atom. When formic acid is dissolved in water, the water molecules can pull the slightly positive hydrogen away from the rest of the formic acid molecule as H^+ .^{*} Any molecule that can release H^+ in water is acidic and it is the H^+ component that causes the corrosive properties of these kinds of acids.[†]

The strength of an acidic solution is determined by how many H^+ ions are concentrated in the water, which depends on how much of an acidic compound is present, but also how readily the molecules of that compound will release H^+ . The pH scale used to measure the acidity or alkalinity of a solution is a measure of the concentration of these H^+ ions. The more H^+ , the lower the pH and the more acidic the solution. Though formic acid does not release its H^+ as readily as say hydrochloric acid (see Chapter H, page 131), it is still a potent acid. The 60 per cent concentration of formic acid found in *L. niger* venom has a pH of around 1.3, similar to stomach acid.



Ants have found many different uses for formic acid thanks to its corrosive nature and pungent and penetrating smell. *Lasius neglectus* spread the acid over their young to inhibit fungal growth and keep their brood healthy. *Myrmelachista schumanni* use formic acid as a herbicide to kill off plants that would encroach on *Duroia hirsuta*, a tree growing in the Amazon that is home to these ants. *Nylanderia fulva* smear the acid over their bodies to detoxify the venom used against them by their enemy the fire ant (*Solenopsis invicta*). And many species of formicine ant spray the acid around the entrance to their nests and towards creatures that threaten them as a deterrent.

Ants are not alone in using the off-putting properties of formic acid to defend themselves. Some stingless bees spray the acid as a defence and stinging nettles store formic acid and other irritating chemicals in their spicules, the hollow, hair-like structures that pierce the skin when you brush against the leaves.

Finding the same venom in such a diverse range of organisms might seem unusual. Most venoms are mixtures of all sorts of chemicals, many of

which have been refined by evolution to the creature's particular needs. Peptides, proteins and enzymes that target specific sites tend to require a lot of energy and resources to produce. By contrast, formic acid is very simple to make from everyday chemicals found in nature. Formicine ants biosynthesise their formic acid predominantly from the amino acids serine and glycine, but there are many other naturally occurring chemicals that can act as source material. The ease with which formic acid can be biosynthesised means it crops up all over the place – in pineapples, cucumbers and even people.

Formic acid is part of our metabolism, it is one of many intermediate molecules in the continuous churn of chemicals that are built up, broken down and transformed to meet the ever-changing demands of our body. It is part of a cycle that constructs vital structures such as DNA and ATP* as well as many, many others. When these structures become redundant they can be broken back down again into useful molecular building blocks, such as formic acid, or excreted as waste. Formic acid is also formed when certain foods or dangerous poisons are metabolised by our body. We're not stockpiling the stuff the way ants and stinging nettles do, but this constant making and unmaking means a small amount of the acid is always present within us.

We can tolerate quite a lot of formic acid if we ingest it because, as a molecule that is part of our daily lives, we have established routes for eliminating it. It can be broken down further into carbon dioxide and water to be excreted, or introduced into the cycle that builds up other useful molecules. Formic acid is therefore a safe preservative for the food industry to use, but that would also seem to make it a poor choice for a poisoner. However, to paraphrase the father of toxicology, the great Paracelsus, 'it is the dose that makes the poison.' If too much formic acid is introduced into our system at once, the natural routes of breaking it down and excreting it can become overwhelmed. Deliver formic acid in sufficient quantities and the results can be terminal.

How formic acid kills

Boiling up ants and injecting their venom into people is not common practice. I can find no direct real-life equivalents of what the murderer in 'The Lemesurier Inheritance' was attempting. So we have to surmise what might have happened had Hercule Poirot not stopped the murderer before he got to little Ronald Lemesurier. The effects of formic acid on the body can be determined using information from different sources: deaths from ingested formic acid, and when normal formic acid levels in the blood increase. First let's look at formic acid ingestion.

Death from formic acid is exceedingly rare, simply because it is not a substance many people have ready access to in a pure or highly concentrated form. Formic acid is also too pungent and corrosive to be swallowed by accident, but a few people do, often children. The taste and experience of swallowing even a small amount usually prevents a lethal quantity from being ingested. The few deaths that have occurred in the UK are from intentionally swallowing limescale remover or stain-removing agents that contain high percentages of formic acid. Sadly, there are a greater number of fatal cases in Kerala State in India because formic acid can be obtained there relatively easily. This is because the acid is used by small-scale rubber producers to harden the sticky sap that they collect from trees.

Concentrated formic acid can cause extensive tissue damage, blistering and scarring of the face where it splashes. Internally, it can perforate the mouth, throat and stomach by chemically corroding away the tissue and causing haemorrhage (see Chapter H, page 134, for the full grisly details of swallowing acid). Injecting acid, as the murderer tries to do in 'The Lemesurier Inheritance', would also cause tissue damage around the injection site. Injecting formic acid into the jugular vein, as the murderer intended, would be even more damaging.

The exterior jugular vein is one of the main blood vessels draining deoxygenated blood from the head to the heart. It runs down the side of the neck and is easily accessible for injection. This is why it is sometimes used to deliver drugs by paramedics in an emergency, but only if the patient is unconscious or suffering from a heart attack as the injection is very painful. Injecting corrosive acid into the vein would be excruciating.

If the murderer in 'The Lemesurier Inheritance' had succeeded in injecting a large amount of formic acid into Ronald's jugular vein it would

have easily eaten its way through the vein walls. From there it would leak into the surrounding tissue, where it would continue to break up proteins and other structures. The vein also offers a short and fairly direct route to the heart, potentially causing direct corrosive damage to this organ also. Even a smaller or more dilute dose that does less direct damage to the vein can have a pronounced effect on the body.

Acidemia occurs when arterial blood falls below a pH of 7.35. Our blood is maintained within a very narrow pH range between 7.35 and 7.45. Venous blood has a fractionally lower pH because it is carrying carbon dioxide, which dissolves to form carbonic acid, but there are limits.* In an effort to restore the blood's pH to normal, the body will exhale as much carbon dioxide as possible, reducing the amount of carbonic acid present and raising the pH. This prolonged deep and laboured breathing can lead to respiratory collapse and death.

Meanwhile, formic acid will be rapidly absorbed from the blood into the rest of the body, meaning the acid would disperse and dilute itself. This would certainly help in terms of alleviating the acidemia, but formic acid has other effects on the body, and to understand this we need to look at another poison, methanol.

Methanol, methyl or wood alcohol, is used as a feedstock or solvent in chemical synthesis and is a common solvent in things like varnish, antifreeze and paint remover. It can also be a contaminant in illegal alcohol because of poor control over the temperature used in the distillation process.

After ingestion and absorption into the body, enzymes first convert methanol into formaldehyde, which is then rapidly converted to formic acid. Eliminating formic acid, by breaking it down into carbon dioxide and water, or using it as a building block in other structures in the body, is slow compared to how fast methanol can be converted into formic acid. And so formic acid levels can build up in the body. In cases of severe methanol poisoning, there is a direct correlation between elevated levels of formic acid in the blood and increased morbidity and mortality.

An excess of formic acid can hinder our ability to use the oxygen we breathe in. Oxygen is processed by mitochondria, tiny structures inside every cell in our body, where a series of enzymes react oxygen molecules

with glucose in a controlled stepwise fashion. This is the process of respiration that releases the energy we need to keep us alive. Cytochrome oxidase is the last enzyme in the respiration sequence. Cytochrome oxidase enzymes contain iron atoms where oxygen molecules normally dock. These iron atoms can be blocked by formic acid. This means oxygen can't be processed and energy can't be released. Cells with high energy demands, such as nerves in the brain and eyes and cells in the heart and kidneys, will be most affected. Mild methanol poisoning is characterised by headache, nausea and vomiting and temporary, sometimes permanent, loss of vision. But it gets worse ...

If a cell is not getting the energy it needs to carry out its normal functions by aerobic respiration (with oxygen), it can switch to a back-up system of anaerobic respiration (without oxygen). Anaerobic respiration doesn't release as much energy as aerobic respiration, but it is useful in an emergency, such as during strenuous exercise when the lungs can't absorb enough oxygen to keep up with demand, or when the cytochrome oxidase is blocked by formic acid.

The by-product of breaking down glucose in the absence of oxygen is lactic acid, which increases the acidity of the blood. Research has shown that the H^+ ions causing the acidemia in methanol poisoning are from both formic acid and lactic acid. As the pH decreases, the formic acid binds tighter to the cytochrome oxidase, and more lactic acid is produced in a cycle termed 'circulus hypoxicus'. Blood pressure drops, red blood cells are ruptured and other cells die. Coma and death can occur from respiratory or kidney failure.

Drinking around 60–250ml (2.0–8.5 US fl. oz) of pure methanol can be fatal. Swallowing anywhere between 10 and 50ml (0.3–1.7 US fl. oz) of formic acid will have a similar result. These numbers will be a lot lower if the victim is a child, as would be the case in Christie's fictional scenario. Injecting the acid would likely mean an even smaller dose would be required to ensure the family curse is fulfilled. The extent of the damage depends on how many ants the murderer managed to boil up and how much formic acid was sucked up into the syringe. It also depends on what treatment could be given after the injection.

Is there an antidote?

When Ronald was first ‘stung’ on his visit to the beekeeper’s house, he was treated with a solution of ammonia. Bee stings are weakly acidic and a common home remedy is to neutralise the acid by washing the affected area with something alkali, like ammonia solution.* This treatment doesn’t really work because, as we have seen, it is not the acidity that is the pain-inducing part of bee venom.†

But in Ronald’s case he wasn’t stung by a bee. Ronald asserts the sting ‘didn’t really hurt, only like a pin, and I didn’t cry.’ There was also no swelling or other signs associated with a bee sting. So we can assume he was pricked with a dry syringe to set up the bee alibi for the later, lethal injection that Poirot and Hastings prevent.

Had the duo not made it in time and Ronald was injected for real, there was still a chance of Ronald’s survival. The formic acid would need to be neutralised as quickly as possible, but injecting ammonia is a bad idea, even if it was readily available. In the case of severe acidosis,‡ sodium bicarbonate, a weak alkali, can be introduced intravenously to neutralise the excess acid in the blood. Treatment for formic acid poisoning is largely supportive and reactive to new symptoms as they present themselves. If the patient survives there is, however, a very likely chance of permanent scarring that can cause ongoing health problems, particularly if it affects internal tissues.

Some real-life cases

Bees have certainly been used in warfare. Ants have also been deliberately induced to bite and sting people as part of initiation ceremonies in the Americas. And though not everyone has survived these encounters, the intention is not usually to kill. Even in a war situation, bees would be more effective as a diversion than as a way of depleting your enemies’ numbers. The Mayans are believed to have used their pottery skills to make ‘bee grenades’. Around the same time in the Middle East, ‘pots of pain’ were being crafted from clay and left outside for bees or hornets to build their

nests inside. During battle these fragile pots could be thrown towards the enemy to cause havoc. Insects have, however, been used for murder.

In his book published in 2000, *Murder, Maggots and Men*, Dr Zakaria Erzinçlioğlu tells of an autopsy he attended where two wasps were found deep in a dead man's throat, a place it would be unlikely to find them under normal circumstances. It was surmised that the insects were introduced deliberately. Generally, though, insects do not make good accomplices when it comes to murder. Using their venom would seem an easier way to achieve your deadly plans rather than trying to train insects to do the work for you. Isolating and injecting ant venom is much easier. It also shows that the murderer in Christie's story had given at least some thought to evading justice.

Boiling up ants may be a way of obtaining a poison without having to go through the risky business of buying it. It was relatively easy to obtain highly toxic compounds in 1920s Britain compared to today, but there were still poison registers. These were physical books kept in pharmacies and some other shops where sales of dangerous substances were recorded, along with who they were sold to and where that person lived. It was a system of traceability for the sale of dangerous substances. However, anyone who had overcome moral objections to murdering people probably had no qualms about lying on a poison register. Plenty of real-life murderers have used false names and credentials to obtain poison this way.

The murderer in 'The Lemesurier Inheritance' avoided any form of incriminating documentation by making their own poison, and from a source that was less likely to be spotted. Start growing poisonous plants in your garden and the police might make a link if someone close to you then dies of plant poisons, but no one is going to complain that there are fewer ants about than normal.

Despite the advantages of a formic acid injection as a form of murder, it doesn't appear to have been tried in real life. This is probably for the very simple reason that it isn't a very good plan.

Most poisoners have their victim swallow their chosen toxin because injecting it is problematic. Getting your victim to sit still while you slide a needle under their skin takes some persuading. Doctors and nurses have a better chance of getting away with such methods because injections are part of their job. Harold Shipman is believed to have killed up to 250 people

when he worked as a general practitioner, though he was only convicted of 15 murders, with injections of diamorphine. He also signed his victims' death certificates to avoid autopsies that might have discovered the needle marks.

In 1953 Kenneth Barlow, a nurse, injected his pregnant wife with a massive dose of insulin after telling her it was an abortifacient. He thought the insulin would not be traced as it would be broken down naturally and the injection marks would be impossible to find on his wife's very freckled body. However, a persistent pathologist found them with the aid of a magnifying glass and was able to isolate unnaturally high levels of insulin from the surrounding tissue.

How do you convince your victim to be injected when you have no medical training? And how do you convince the police that the very obvious mark on the skin is nothing suspicious? If you are a murderer in an Agatha Christie story, you drug your victim so they won't cry out and then you blame a bee or a wasp.

Agatha and formic acid

If you could lure someone to a place where there are a lot of bees, then sneak up on them and inject them with a syringe full of formic acid, you might initially be able to convince them the pain they felt was due to a bee sting. Seconds later, however, your plan would start to fall apart. The pain of formic acid and the damage to the tissues is not going to feel or look anything like a bee sting. If the victim has been drugged to keep them sedated, these drugs are likely to show up in toxicological tests at autopsy and will need explaining. The same problems apply to the wasp sting that was said to have killed another member of the Lemesurier family.

Poirot suspects that an earlier Lemesurier death by wasp sting might have been another case of murder. But if this was a case of murder, some poison other than formic acid must have been used that would not cause so much tissue damage or be so painful, something so convincingly like a wasp sting that no autopsy was carried out.* Alternatively, this other Lemesurier really did die from a wasp sting and the incident simply gave the murderer an idea for how to get rid of little Ronald. Both of these

Lemesurier victims, being from the same family, might share a genetic quirk making them particularly susceptible to bee and wasp stings, adding credulity to the story. The first mistake was to then use formic acid for the murder. The second mistake was to leave a large gap between the time of the fake ‘sting’ and the real injection.

If Ronald was supposed to be susceptible to a single bee sting, his reaction should have been almost immediate and his death could be expected within the hour.[†] The ‘sting’ occurred in the afternoon and Ronald was healthy and happy for the rest of the day. The real attempt to kill him wasn’t made until after midnight, a delay that would have raised suspicions. The damage and symptoms caused by the formic acid injection would then confirm that it was not bee venom that had been injected into the boy and demand further investigation.

Is this a slip in Agatha’s usually top-notch use of poisons? Perhaps the oversights are in keeping with her character’s state of mind at the time. The murderer is captured red-handed, but they are never tried for their crimes. They die soon after in a lunatic asylum. As the murderer’s mental and physical health had deteriorated they became fixated on the Lemesurier curse, and perhaps buoyed with confidence after the success of earlier murders, they overlooked some basic research into the effects of formic acid and bee stings.

Alternatively, Poirot was simply guessing at the contents of the syringe and was wrong. After all, he made no attempt to discover what was really inside. There is no doubt, however, that the formic acid could have been obtained and used to kill Ronald as described in the story, it just would never have been convincing as a bee sting. Maybe these unbelievable aspects were the reason Christie changed the poison involved in the alleged bee-sting murders that she wrote later. Whatever her intentions with the story, it is an extraordinarily creative use of an unusual poison.

Notes

* Each murder is made to fit the lines of a popular 1869 minstrel song. Look up the lyrics (warning! They are extremely racist) or read the novel to see how Christie used them for her plot.

- † The story was first published in the magazine *The Sketch* but with other short stories in the collection *Poirot's Early Cases*, in 1974.
- * Chloride ions (Cl^-) are also involved, as we saw in Chapter C (page 79), but we will ignore them here for simplicity's sake.
- * And it might not. Mice are often used as proxies to find out how compounds might interact with the human body, but they are not simply tiny humans and can have very different physiological responses to chemicals.
- * Food allergies are another common trigger, such as allergies to shellfish or nuts, but there are many other causes.
- † Africanised bees are a hybrid of Western honeybees (*Apis mellifera*) and the East African lowland honeybee (*A. m. scutellata*). They were deliberately bred for their increased honey production.
- * In the past ants have been used in place of stitches to close wounds by clamping the edges together between the mandibles. The ant is decapitated to keep the mandibles closed.
- † *Lasius niger* are common across Europe, but have also found their way to America (north and south), Asia and Australia.
- * Once released from its parent compound, H^+ gets stuck to a water molecule to form H_3O^+ , or a hydroxonium ion. The hydroxonium ion is surrounded by a cloud of more water molecules. The exact arrangement of this cloud of water molecules with H^+ at its centre is still subject to scientific discussion but, for the purposes of this and other chapters discussing acids, we will talk about the H^+ ion alone for simplicity.
- † There are several technical definitions of what an acid is, but the one that describes formic acid, hydrochloric acid (see Chapter H), oxalic acid (see Chapter O) and the other sorts of acids you may have come across in your day-to-day life are described as a source of hydrogen ions, or H^+ .
- * ATP, adenosine triphosphate, a molecule used by every cell in our body to store and release energy.
- * Venous blood is normally only pH 0.02–0.04 lower than arterial blood.
- * Ammonia solution might have been a common household substance in 1923, but it would not be so easy to get hold of it today as it is potentially quite nasty.
- † Ice is the best option for reducing the pain, but call an ambulance if the recipient of the sting is allergic to bee or wasp stings.
- ‡ Acidosis is the process of how acid builds up in the blood, acidemia is the result, but the two terms are often used interchangeably.
- * I have no idea what poison that might be.
- † This timing comes with a huge margin of error depending on what medical treatment Ronald might receive.



A Caribbean Mystery

Agatha Christie's 1964 novel *A Caribbean Mystery* sees Miss Marple on the trail of a serial wife-killer. On a holiday in the Caribbean, so kindly paid for by her devoted nephew, Miss Marple is not really enjoying herself. The relentlessly perfect weather gives her nothing to comment on and she is missing the gossip of her home village, St Mary Mead. The other guests are not particularly interesting and Major Palgrave is especially dull company. The monotonous tales of his time in India and Kenya, his oft-repeated exploits in big game hunting hold no interest for the elderly spinster. She focuses on her knitting while muttering the occasional 'yes' and 'but, of course' while the Major drones on. What she wants is a bit of variety, some

intrigue perhaps, and Miss Marple soon gets her wish. The next morning she learns that the Major has died in the night.

Major Palgrave was an older man, and he apparently suffered from high blood pressure so his death is attributed to natural causes. But Miss Marple has her doubts. She thinks about the Major, a nice man but such boring stories – apart from that one anecdote about a murderer. He had been saying something about a man who murdered at least two of his wives, but he had stopped his story abruptly and changed the subject. It would take a lot to halt or divert the Major in the middle of one of his anecdotes. Had he thought of something or spotted someone? Miss Marple believes the Major saw the murderer and tried to stop his story before he was overheard, but was too late. The Major hadn't died of natural causes – he was deliberately killed before he could expose a murderer.

Miss Marple searches for clues to the killer in the Major's murderous story. The problem is she hadn't really been paying attention. He had said the murderer would make his spouse depressed and anxious before staging her death to look like suicide. A swift burial, a quick change of name and he was ready for another wedding and another victim. There was definitely something about knocking one wife unconscious with a sleeping drug and arranging her with her head in the gas oven before turning it on. It would certainly be a clever and credible way to cover up a murder.

Gas was the most common method of suicide in the UK in the twentieth century, up until the 1960s. As something so depressingly familiar to her readers, it was a useful device to employ in Christie's fiction as it would need little explanation. In fact, suicide by gas was so well known, Christie used the idea more than once.

In her 1945 novel *Sparkling Cyanide*, Christie contrived an attempted murder by gas that was also meant to look like suicide. In this case the murderer needs to get rid of Iris Marle in order to get to her fortune. In a quick bit of improvisation, Iris is knocked out with a blow to the back of the head and her unconscious body positioned with the head lying next to the gas tap in her bedroom fire. The murderer turns on the tap and rapidly exits the room, pushing the rug on the landing up against the crack under the door to prevent fresh air drifting into the room.* Fortunately, someone else smells the gas and breaks into the room in time to rescue Iris.

Another murderer turns on the gas taps and leaves their unconscious victims to die in Christie's *Murder in Mesopotamia*, written in 1936. This time the gas poisoning was supposed to look like an accident. The story is that Dr Leidner and his wife were asleep in their flat when Mrs Leidner's ex-husband, in a fit of jealousy, broke into their apartment and turned on the gas. Fortunately, Mrs Leidner woke up, noticed the smell of gas and switched off the taps in time.

All of these plans for murder by poison gas were plausible at the time Christie was writing, but not today. This is because the gas pouring out of the fireplaces and ovens in Christie's stories was different to the gas that now supplies our boilers and stoves.

The poison gas story

The gas in Christie's stories was 'town gas', a by-product of the coal industry. Coal was the main source of fuel for industry in the nineteenth century. Some of this coal was refined into coke, a more carbon-rich form of coal used in metal processing, by heating it in the absence of air to drive off impurities. The impurities released in this process are a mixture of volatile gases that included simple hydrocarbons, hydrogen and carbon monoxide. This mix of gases can also be burnt, like coal and other fossil fuels, to release energy in the form of heat and light. Coal gas, or town gas, was piped into homes to light wall lamps, heat water in boilers and cook dinners in the oven.

The proportions of the different gases in town gas varied depending on how and where it was produced. One gas in the mix, carbon monoxide, was known to be very toxic, but this didn't prevent the widespread distribution of town gas into homes, offices and schools. Town gas was contained inside pipes, and was to be burnt when it escaped through the valves in cookers and lamps. Burning added a second oxygen atom to the toxic carbon monoxide molecules to make *non-toxic* carbon dioxide. As long as there was a good supply of air to the flame, there would be plenty of oxygen to convert all the toxic components into safe compounds.

The problem was that the pipes and fixtures meant to contain the gas sometimes leaked. And if there were blockages, or the appliance was in a

small, airless room, the supply of oxygen might not be sufficient to convert all the carbon monoxide to carbon dioxide. There was also the problem that if for some reason, malicious or desperate, a person turned on the taps and failed to ignite the gas streaming out of it, the carbon monoxide would escape. Many died as a result of poor maintenance, accidents and deliberate actions. An individual at their lowest ebb could rest their head in a gas oven and turn on the taps. The carbon monoxide would render them unconscious in a few minutes and dead within half an hour. In 1963, the year Agatha Christie must have been planning *A Caribbean Mystery*, 40 per cent of suicides in England and Wales were due to carbon monoxide poisoning. Things had started to change in the early 1950s, however, when Britain began to modernise its gas infrastructure, mainly for economic reasons.

High-quality coal was both expensive and it produced coal gas with a high percentage of carbon monoxide. Obtaining town gas by heating oil was both cheaper and released less of the toxic gas. The carbon monoxide content of town gas dropped from a high of 16 per cent to less than 7 per cent as a result. It was better, but it was still a potentially lethal amount.

Then in the 1960s natural gas was discovered under the North Sea. Natural gas contains no carbon monoxide; it is mostly methane, a non-toxic gas whose molecules are made up of one carbon and four hydrogen atoms. Methane burns extremely well but differently to town gas, so the two could not be mixed. A nationwide switch to natural gas affected 13 million consumers and involved adapting 35 million existing appliances so they could burn the new fuel. The roll-out took nine years to complete. As well as financial considerations, part of the original aim of the programme had been to reduce accidents, even though accidental deaths from town gas were outnumbered by suicides three times over. Suicides were not really considered in the plans because it was believed that anyone considering ending their life would simply find an alternative method. However, the figures suggest that this was not always the case.

There were 2,499 deaths attributed to suicide by carbon monoxide poisoning in 1960. In 1975, with the huge conversion programme nearing its end, carbon monoxide was responsible for just 23 suicides. Even more gratifying was the 35 per cent drop in *all* suicides between 1963 and 1975. The removal of carbon monoxide from domestic gas supplies was not the

sole reason for the decline, but it was certainly an important factor.* With carbon monoxide from gas ovens being such a lethally efficient way out, anyone making a ‘cry for help’ was much less likely to be reached in time. When carbon monoxide became more difficult to access, it was no longer the easy option. People had more time to reflect and reconsider.

Domestic carbon monoxide deaths are still tragically too common, even though the natural gas pumped into homes today has no toxic carbon monoxide present. Burning any carbon-based material will form both carbon dioxide and carbon monoxide, the ratio depending on how much oxygen is available to share among the carbon atoms. Poorly maintained gas appliances, blocked flues and heaters left burning in small, poorly ventilated rooms can restrict the oxygen supply. The result is incomplete combustion, or not enough oxygen to convert all the methane to non-toxic carbon dioxide.

Because it is colourless and odourless, carbon monoxide gives no warning of its presence and it doesn’t take much to kill. It has, therefore, become known as the ‘silent killer’ and is responsible for more deaths than any other poison, mostly by accident.

Carbon monoxide

Humans have probably been aware of the detrimental effects of carbon monoxide since they started making fire, but it was first documented by the ancient Greeks. Aristotle (384–322 BC) stated that ‘coal fumes lead to heavy head and death’, and there are many other references to the dangers of inhaling too much of the fumes released from burning coal since then. However, it took a long time to identify the killer component within those fumes as carbon monoxide.

Scientific investigations into different gases were very popular in the eighteenth century. It meant that several chemists, including some very notable ones, conducted experiments that produced carbon monoxide by many different methods. None, however, correctly identified it as carbon monoxide. In 1776, for example, French chemist Joseph de Lassone (1717–1788) heated a powdered mixture of zinc oxide and coke and found a gas was emitted. He noticed that this gas burnt with a blue flame but he mistook

it as a sign that he had made hydrogen, another gas that had recently been discovered. Lassone moved on to other things.

The distinctive blue flame is in fact characteristic of burning carbon monoxide (hydrogen burns with an almost colourless flame). It will be familiar to anyone who has seen a gas stove burning or the pilot light in their boiler. The blue flame shows that carbon monoxide is combining with more oxygen to make non-toxic carbon dioxide.*

Carbon monoxide wasn't correctly identified until 1800, by English chemist William Cruickshank (1745–1810). The link between carbon monoxide and coal fumes was finally made in 1842, confirming in scientific terms what Aristotle had been alluding to 2,000 years earlier. And once that link had been made, several other observations suddenly made sense. For instance, it explained why canaries had been so important in coal mining for decades. A canary's greater susceptibility to carbon monoxide meant they stopped singing before the miners became seriously affected by the gas. This alerted them to the presence of the odourless toxic gas and potentially suffocating or explosive conditions inside the mine.

The realisation that coal, coal gas and carbon monoxide were often found together also raised concerns about town gas. Hotel rooms lit by town gas often had signs requesting patrons 'turn off gas; do not blow it out' to try and prevent poisonings, as well as explosions. In 1883, the New York authorities went further and regulations were introduced limiting the amount of carbon monoxide that could be present in town gas. Carbon monoxide killed people, and efforts were made to prevent that. Efforts were also made to understand how carbon monoxide killed.

How carbon monoxide kills

Though the toxicity of coal vapours, and hence carbon monoxide, had been known since antiquity, what it did to the body could only be speculated about. The first big breakthrough came in 1857 and was provided by French scientist Claude Bernard (1813–1878). He asserted that carbon monoxide displaced oxygen in the body, causing asphyxia. Bernard was right, but it took many more decades of research to work out that it is all down to how the atoms in the carbon monoxide molecules are bonded together.

A single carbon atom and one oxygen atom share their electrons to make one molecule of carbon monoxide. There are 10 negatively charged electrons that can contribute to the chemical bonds between these two atoms, six from the oxygen and four from the carbon atom. When these 10 electrons are distributed over the whole molecule, the carbon atom ends up with slightly more than its usual four, making it slightly more negatively charged than normal. This enrichment of electrons means the carbon atoms in carbon monoxide molecules are particularly attractive to certain metals, like iron.

Iron plays many important roles in our body, the majority of them being within the oxygen cycle. Iron atoms are at the centre of every step in the entire process of absorbing oxygen from our lungs, distributing it around our body, temporarily storing it in our muscles and reacting it with glucose inside our cells to release energy.

To absorb the oxygen from the air we inhale into our lungs we have a dense network of blood vessels flowing with red blood cells. These red blood cells are packed with haemoglobin molecules, each of which has four iron atoms. Each of these iron atoms collects one oxygen molecule from the lungs to be transported and released elsewhere in the body. In our muscles, more iron atoms within myoglobin molecules hold on to oxygen molecules, providing a temporary storage site. Inside every cell in our body,^{*} oxygen binds to the iron atoms in cytochrome oxidase enzymes to break it up and combine it with glucose to release energy.

Iron performs excellently in all these oxygen-centred roles because the element has a very high affinity for oxygen, as can be seen by how easily iron rusts. Once iron has hold of oxygen it is very difficult to separate them again.[†] To prevent our bodies from rusting as we breathe in more and more oxygen, the iron in our body has to be carefully supported. Haemoglobin, myoglobin and cytochrome oxidase surround each of their iron atoms with a protective collar, an elaborate network of carbon, nitrogen and hydrogen atoms. This molecular collar is then attached to the protein that makes up the bulk of haemoglobin, myoglobin and cytochrome oxidase. The collar and protein work together to allow the iron atoms to absorb oxygen, hold it in place and then release it again as necessary.

Iron's affinity for oxygen is strong, but carbon monoxide's affinity for the iron atoms in haemoglobin, myoglobin and cytochrome oxidase is stronger. It is the combination of carbon monoxide with haemoglobin that is the biggest concern. Haemoglobin's affinity for carbon monoxide is over 200 times greater than its affinity for oxygen. It is generally assumed that if there is *any* carbon monoxide in the air breathed in it will become stuck to the iron atoms in haemoglobin. The resulting combination is called carboxyhaemoglobin and has a characteristic pink/red colour, often described at autopsy as 'cherry red'. Even if only one carbon monoxide molecule bonds to one of the four iron atoms in a haemoglobin molecule, it reduces the ability of the other three to bind oxygen. The presence of carbon monoxide also reduces the ability of haemoglobin to release its oxygen load.

Without a supply of oxygen, the myoglobin molecules cannot store it and the cytochrome oxidase cannot process it. Any oxygen reserves held in myoglobin are very quickly used up. The body can switch to anaerobic respiration and convert glucose into lactic acid, but the process simply cannot release the amount of energy the body needs to keep going long term (see Chapter F, page 101). If our cells cannot make the energy needed to carry out the second-to-second necessities of life then things start to break down.

The most energy-hungry cells will be the most affected by a depleted supply of oxygen. These are the nerves, causing the confusion commonly associated with the early stages of carbon monoxide poisoning. It hinders an individual's ability to make sensible decisions about recognising what is going on and removing themselves from danger. As the percentage of carboxyhaemoglobin increases, so does the severity of the symptoms as more and more cells become affected.

Anyone living in a city will already have a few per cent of their haemoglobin converted to carboxyhaemoglobin from the carbon monoxide emanating from car exhausts. Burning petrol or diesel, both hydrocarbon fuels, will generate some carbon monoxide. With improved engine efficiency, fuel refining and catalytic converters, the amount of carbon monoxide from these sources has dropped dramatically in recent decades. A regular smoker will also have between 5 and 10 per cent of their haemoglobin converted into carboxyhaemoglobin by the carbon monoxide

inhaled from burning tobacco.* But these relatively small percentages of carboxyhaemoglobin should not result in signs of poisoning.

Exposure to slightly higher and constant amounts of carbon monoxide over a period of time can cause depression, chronic fatigue and even hallucinations. Feelings of trepidation, sudden shivering and hearing footsteps in an old house may not be because it is haunted, but the effects produced by carbon monoxide leaking from a defective boiler or heater.† Some have speculated that the ghosts and spirits that plagued many Victorians might be down to the widespread use of town gas in heating and lighting at the time.

Low-level carbon monoxide poisoning begins when approximately 20 per cent of haemoglobin has been converted to carboxyhaemoglobin. It is around this point that breathing starts to become more difficult. At around 30 per cent carboxyhaemoglobin headaches start and flu-like symptoms are produced. Dizziness, staggering and uncoordinated movements occur as the central nervous system is increasingly starved of oxygen. Above 50 per cent carboxyhaemoglobin, an individual is likely to pass out. If not enough oxygen can reach the heart to fuel the cells that make it beat, death is not far away. At 60 per cent carboxyhaemoglobin levels, most people are likely to die.

How quickly life becomes insupportable depends on both the person and the dose. There is, of course, huge variation between individuals. A healthy adult has a better chance of survival than a baby or an older victim. Someone who is anaemic and already has a reduced number of iron atoms in their blood, for example, might succumb to a much lower dose than might be expected.

A carboxyhaemoglobin level of at least 40 per cent would be expected for death to be attributed to carbon monoxide poisoning and not some other cause. However, levels as low as 25 per cent have been known to be fatal in already debilitated victims. The highest percentage of carboxyhaemoglobin in the blood that has been survived is a staggering 73 per cent.

How quickly a person moves through the different stages of poisoning, and how much time there is to intervene and help them, also depends on many factors. Not only is the proportion of carbon monoxide in the gas escaping important, but also the kind of appliance it is escaping from. An

oven or heater will have a bigger gas supply than a wall lamp. Other factors to consider include the size of the room the gas is leaking into and how effectively fresh air has been excluded from the space.

Generally speaking, breathing air with fewer than 35 carbon monoxide molecules in every million (or 35ppm^{*}) will produce no symptoms. Concentrations between 35 and 200ppm can produce a headache in a couple of hours. Above 800ppm, a throbbing headache occurs within 10 minutes followed by dizziness and nausea and unconsciousness within the hour. At 1,500ppm, death is likely within two hours. At 6,000ppm, the time to death drops to 20 minutes. Air containing just over 1 per cent of carbon monoxide, or 12,000ppm, can kill in just 10 minutes.

What may be surprising is that carbon monoxide is in fact present in our body all the time and carrying out essential roles. In the right amount, and in the right place, carbon monoxide acts as a gasotransmitter, a small gaseous molecule produced by the body with specific roles.[†] It causes smooth muscle cells in the arteries to relax, either by opening channels through which potassium ions move or by acting on enzymes such as cytochrome P450.^{*} It also suppresses the production of endothelin, which reduces blood pressure. As with all substances, it is the dose that makes the poison.



All the carbon monoxide victims in Christie's stories were healthy adults but living in a world where smoking was common and car exhausts were particularly polluting. They would also have been exposed to large doses of the toxic gas. Two of the stories, *Sparkling Cyanide* and *Murder in Mesopotamia*, were written before 1950 when refinements to town gas started to reduce the proportion of carbon monoxide present. *A Caribbean Mystery* was published in 1964, but the poison gas murder in its pages occurred a few years earlier. Even if the victim was exposed to a 'safer' supply of town gas, it still proved fatal because even the minimum typical content of carbon monoxide at this time was 2 per cent. This was more than enough to kill quickly, especially when the victim was unconscious and their head placed next to a large supply of the gas streaming out of an oven.

The fireplaces that were the source of carbon monoxide in *Sparkling Cyanide* and *Murder in Mesopotamia* would also release a large volume of gas. For the sleeping couple in *Murder in Mesopotamia*, we are given no details about where the taps were located in relation to them, or how large the room was, or if any windows had been left open. Mrs Leidner woke up before the carbon monoxide could have much effect as she was still thinking clearly enough, and moving around well enough, to switch off the taps and save her and her husband's lives. There is no mention of any ill effects or medical attention.

We don't know the size of the bedroom where the unfortunate Iris is left to breathe in carbon monoxide in *Sparkling Cyanide*. However, we do know that Iris's head was left next to the tap and that the windows were closed and the gap under the door blocked by a rug. Quite a lot of gas must have built up in the room as some of it escaped under the door where someone passing by noticed the smell. Town gas smelled bad thanks to the natural presence of small quantities of stinky sulphur-based chemicals.* Iris was dragged out into the passageway next to an open window. Her rescuer proceeds to 'work on her' while a doctor is called. What were Iris's chances of survival?

Is there an antidote?

There is no antidote to carbon monoxide poisoning as such, but treatment is certainly available and the sooner it is given the better. The first priority is to remove the victim from the source of the carbon monoxide and get them into the fresh air, which is exactly what was done for Iris in *Sparkling Cyanide*. If she had stopped breathing, mouth-to-mouth resuscitation might be needed. Perhaps this is what was being referred to by the phrase 'work on her' in Christie's novel. What Iris needed most urgently was pure oxygen. When the doctor was called he was hopefully told Iris had been found by an open gas tap and he would know to bring a supply of oxygen with him, or call ahead to a hospital to get one ready if he didn't have his own.

Pure oxygen will maximise the uptake in whatever haemoglobin is left untainted by the presence of carbon monoxide. The oxygen will also help

displace the carbon monoxide molecules from their suffocating position on the iron atoms of the carboxyhaemoglobin proteins. It normally takes about four hours for carboxyhaemoglobin levels to halve. A hyperbaric chamber can be used, like the ones used for divers suffering from the bends, to force air into the body at three atmospheres pressure rather than the normal one. This speeds up the displacement of the carbon monoxide so that the amount of carboxyhaemoglobin will halve every 30 minutes.

Even though Iris was quickly removed from the source of gas and resuscitation started immediately, it would be unwise to predict an outcome at this stage. She may have been alive and breathing, but carbon monoxide can cause permanent damage to a number of organs, including the brain and heart, that may not become apparent for days. Brain damage can present itself as short-term memory problems, slurred speech, irritability, Parkinson's-like symptoms, even dementia. Iris, however, was lucky. A few pages later she appears to have made a rapid and complete recovery.

There was no such luck for the unfortunate carbon monoxide victim in *A Caribbean Mystery*. The murder was undetected and the murderer able to move on to more victims. Like several real-life carbon monoxide killers, he almost got away with it. At least until he met with Miss Marple's formidable powers of deduction.

Some real-life cases

Carbon monoxide's stealthy methods and high potency make it a truly terrifying poison. It can kill even the most robust people, as the murder of the 'durable Mike Malloy' illustrates.

Malloy was living and drinking in prohibition-era New York. Malloy, like many others, was struggling. He frequented an illegal speakeasy where he would complain about his lot over several glasses of alcoholic beverages of very dubious quality. There were plenty of sympathetic listeners in the bar, all with their own worries. A few of them realised that Malloy offered them an opportunity to improve their situation financially. Tony Marino, the bar owner; Red Murphy, his bartender; Francis Pasqua, the local funeral home director; and Daniel Kriesberg, a fruit vendor, conspired to insure

Malloy's life and then end that life as quickly as possible so they could receive the payout.

First they offered Malloy unlimited free drinks, thinking the poor-quality hooch they served would poison him soon enough. But Malloy appeared to thrive on the free booze. Even antifreeze and methanol (see Chapter F, page 100) on the rocks could not fell this hardened drinker. So they fed him rotten fish and other foul foods including broken glass and slivers of metal. Nothing appeared to faze him.

When he passed out one night they carried his body out into the cold street and dowsed him in ice water. He woke up and walked back inside to dry off. In desperation they paid a taxi driver \$150 to run him over. A few weeks later Malloy limped back into the bar, having been discharged from hospital after his broken bones had mended.

The conspirators then decided to rent a tiny room furnished with only a bed and a gas lamp. When Malloy passed out in the bar yet again, Murphy and Kriesberg dragged him to the room and put him on the bed. They attached a hose to the valve on the gas lamp and fed the other end into Malloy's mouth. Then they covered his face with a towel to block his nose. Carbon monoxide did what meths, rotten food and being run over could not. Five minutes after Murphy and Kriesberg turned on the gas, Malloy was dead.

A compliant doctor was brought in to assign the cause of death to lobar pneumonia, with alcoholism a contributing factor. But suspicions grew. One of the insurance agents was concerned at the speed of Malloy's internment and refused to pay out. Stories started to circulate around the Bronx bars about the man who refused to die. Months later the four conspirators were indicted for murder and Malloy's body was exhumed. A pathologist, Alexander Gettler, assured the district attorney that evidence of carbon monoxide poisoning would still be present in the body even after months of decomposition.

Due to the high number of deaths from carbon monoxide poisoning he had seen, some under suspicious circumstances, Gettler had conducted a number of experiments. He wanted to know if someone could kill a person and then try to cover up the murder by placing the body with the head in a gas oven. Using dead cats and the dead bodies of people who had not died of carbon monoxide poisoning, he enclosed them in a box and saturated the

air inside with carbon monoxide. His experiments showed that dead bodies did not absorb carbon monoxide at all.

He also needed to know how long carbon monoxide remained attached to haemoglobin in a dead body before it was broken down and the forensic evidence disappeared. In another series of experiments, Gettler collected blood from victims who had died of carbon monoxide poisoning and stored it either in a freezer or on the laboratory shelf. The samples were tested regularly to monitor the carboxyhaemoglobin levels. Importantly, the decomposition process never increased the amount of carboxyhaemoglobin in the sample, and there was only a slight decrease over time.

Analysis of Mike Malloy's body revealed ample evidence of carbon monoxide poisoning. His lungs and heart were the characteristic bright 'cherry red' of carboxyhaemoglobin and analysis of his blood revealed 70 per cent of his haemoglobin had been converted, more than enough to have killed him. The four conspirators were found guilty of murder and sent to the electric chair. Had they stuck closer to the facts and explained Malloy's death as a drunk going home, turning on the gas lamp but passing out before he could light the flame, they might have got away with it.

While establishing whether carbon monoxide is the cause of death can be relatively easy, proving whether it got into the body by accident or deliberate intent is much more difficult. Christie had her murderer in *A Caribbean Mystery* take the trouble to convince everyone his wife was depressed. Then, when she was found dead with her head in a gas oven and her body full of carbon monoxide, they didn't look any further. Christie's killer in *A Caribbean Mystery* was inspired by a real-life murderer. Though he didn't use carbon monoxide as his murder weapon, he did succeed in killing several wives by providing a plausible explanation for their deaths. He even gets a mention in the novel.



George Joseph Smith, known as the 'brides in the bath' murderer, married eight times, and all but the first of these marriages were bigamous. He walked out on most of these women after clearing out their savings. He killed three of them to inherit their money and claim on their life insurance.

Smith changed his name and address regularly, but he did not change his method of murder, and that was his undoing.

In January 1915, the Metropolitan Police received a letter encouraging them to investigate a possible murderer. Two newspaper clippings were enclosed with the letter covering the deaths of two women, Alice Smith in December 1913 and Margaret Elizabeth Lloyd in December 1914. Both women had been living in boarding houses with their husband, both women had died in their bathtub and both bodies had been found by the husband.

During the investigation, a third case of a woman drowning in her bathtub was uncovered. In 1912 George Joseph Smith, under the alias of Henry Williams, and his wife Bessie rented a house in Blackpool. There was no bath in the house but seven weeks after moving in the couple decided to rent one. Like the murderer in Christie's fictional story, Smith managed to persuade doctors that his wife was ill when she wasn't. Bessie complained to her husband of headaches but he somehow convinced her doctor to diagnose epilepsy. When Bessie was later found dead in the bath, her head under the water, legs straight and feet protruding above the surface, it was assumed she had had a seizure and drowned. Henry inherited £2,579 13s 7d from his wife's will that had been drawn up just five days before she died.*

Sir Bernard Spilsbury (1877–1947),[†] Britain's most famous forensic pathologist, was brought in to explain how these women had been killed without any sign of a struggle. He reasoned that a sharp pull on the ankles plunged the women's heads under the water very suddenly. The water that rushed into their throat when their head went under produced a sudden pressure on the vagus nerve that would cause unconsciousness and perhaps even cardiac arrest.* Smith was found guilty of murder and hanged at Maidstone prison on 13 August 1915.

Christie and carbon monoxide

Christie realised where Smith went wrong and made her fictional wife-killer far more difficult to detect. She had him change his method, as well as his name and address, meaning there was very little to link this man with his previous crimes. Each individual death was made to look like an

isolated tragedy and not a murder. The decision to drug one wife and arrange her body with her head in the gas oven was particularly ingenious. But it was not such an easy trick to pull off.

To make the poison gas death look like a convincing suicide, Christie's murderer used a narcotic to render his victim unconscious. He had to give her enough so she would not struggle, potentially leaving marks on the body, or wake up before the gas could take effect. Too much narcotic would cause an overdose and there would be no characteristic cherry red carboxyhaemoglobin, because the already dead body would not absorb carbon monoxide. Even with a non-lethal dose of narcotic, the sedative effects of the drug would combine with the effects of the carbon monoxide to make death more likely. The narcotic could suppress breathing and less carbon monoxide would be inhaled. It might mean the percentage of carboxyhaemoglobin in the blood would be lower than might be expected in the fatal poisoning of a healthy young adult.

By 1975, at the end of Christie's writing career, suicide by gas oven was extremely rare. As with some of her other poison choices, like arsenic and strychnine, it was very much of its time. Now that carbon monoxide has been eliminated from natural gas, more recent carbon monoxide killers have tried feeding a hose from a car exhaust into an enclosed space where their victim is sleeping. They have also tried blaming the death on a faulty gas appliance. But the amount of carbon monoxide in the body has not always matched the amount emitted from the appliance they tried to blame. Tampering with an appliance to deliberately increase the amount of carbon monoxide it produces will leave traces. Anyone reading *A Caribbean Mystery*, or this chapter, for homicidal inspiration will have to work a lot harder to achieve their goal. Though they may not have Miss Marple's unique but unnervingly accurate method of sleuthing to deal with, they are still unlikely to get away with it.

Notes

- * The murderer's suicide scheme was unlikely to be convincing, however, as the bruising on the back of the victim's head would have to be explained.
- * Around the same time more electric ovens were being purchased, for example.

- * If you notice an orange flame burning in these appliances, please switch off the appliance, open a window and call someone to come and fix it.
- * Red blood cells are the exception to this rule as they are the only cells in the body that do not have mitochondria where the cytochrome oxidase enzymes process oxygen.
- † Think how difficult it is to convert rust back into pure iron.
- * The presence of carbon monoxide in tobacco smoke was proved by Gustave Le Bon (1841–1931) in 1872.
- † Who you gonna call when you experience feelings of dread or strange noises in your attic? Someone who can fix your boiler might be a better choice than a Ghostbuster or a priest.
- * Parts per million, ppm.
- † Other gasotransmitters are nitrous oxide and hydrogen sulphide, both of which are also highly toxic if we breathe them in from the air.
- * The ‘P450’ in cytochrome P450 actually refers to a wavelength or the colour of the enzyme with carbon monoxide bonded to it.
- * Odourisers, very pungent sulphur-based chemicals that can be detected in tiny quantities, are deliberately added to natural gas supplies to help alert people to leaks because methane, like carbon monoxide, has no smell.
- * Around £323,000 or \$414,000 today.
- † In 1947, Spilsbury was 70 years old and had suffered a series of strokes as well as the grief of losing both his sons during the Second World War. He decided to end it all using the town gas that fed into his laboratory at University College in London.
- * A bathtub was brought into the courtroom and a woman police officer, a strong swimmer, volunteered to get in the tub and show how it was done. The officer in the tub knew full well what was going to happen to her and, unlike Smith’s victims, could prepare herself. But it took an hour to revive her after the demonstration.



Murder in Mesopotamia

There are a few stereotypes associated with Agatha Christie's literary output. First, that they are all set in large country houses, or tiny villages, with only dead bodies spoiling the picturesque view. But Christie's fictional corpses have been found on Caribbean islands, in train carriages travelling through sunny France and stuck in the snow in Yugoslavia, on boats sailing down the Nile and in aeroplanes flying over the English Channel. They have also turned up in hotels, golf courses, cinemas and restaurants.

Another assumption made about all those dead bodies littering Agatha's scenery is that they were killed by the invariable use of arsenic in the ubiquitous cups of tea. As the table of contents for this book hopefully

demonstrates, Christie's imagination for chemical methods of murder was very fertile, and she deployed plenty of other weapons as well.* It is also assumed that, whatever method she used, there will be very little blood and gore.

Murder in Mesopotamia is one of the many novels that does not fit the Christie stereotype. As the title suggests, it is set not in an English village but in the desert of Mesopotamia, a region that stretched over parts of modern-day Iraq, Kuwait, Syria and Turkey. Christie was not adding a little exotic flavour on a whim. The setting, like many of the others in her stories, was based on a place she had visited.

For this 1936 novel, Christie drew on her experiences at archaeological digs in the Middle East. She had first gone to Baghdad in 1928, where she had met the renowned archaeologist Leonard Woolley and his wife Katharine. They invited the author to return the following season, and Agatha took them up on the offer in 1930. As well as being introduced to the archaeology at Ur,* she also met Woolley's assistant, Max Mallowan. Max was charged with accompanying Christie on her homeward journey and pointing out areas of interest on the way. The two fell in love and were married later that year. From then on, Agatha would accompany her husband on his digs in Syria and Iraq.† At the start of the season she would hide herself away in a small room with her typewriter and bash out chapters of her latest detective story. But as the excavation progressed, she would leave her writing and take an active role in the archaeological work, cleaning, photographing and cataloguing finds.

As well as using her knowledge of the practicalities of archaeological excavation, there are many other details drawn from her personal experiences on digs. The fictional house in *Murder in Mesopotamia*, where the archaeologists live during the dig, is based on the design of the real house Agatha and her husband lived in during their time at Chagar Bazar.* There are bedrooms and workrooms arranged around a central courtyard area and a flat roof that is key to the puzzling circumstances surrounding one of the two murders in the book. She even based her first victim, Mrs Leidner, on the real-life Katharine Woolley, who had introduced her to archaeology and her second husband. Many of the other characters in the

novel have striking similarities to the people she met and worked with during Max's expeditions.[†]

The novel certainly doesn't adhere to the stereotypical Christie setting and its characters are not exactly the cardboard cutouts she is accused of moving around her plots like chess pieces. Nor does it adhere to Christie's reputation for 'cosy crime', featuring as it does two particularly brutal murders. Mrs Leidner is found dead in her room, killed by a vicious blow to the head, but there is no weapon in sight. The door to her room can clearly be seen from the courtyard area but no one was seen to enter or leave. The only window in the room is high above the ground. 'How', as well as 'who', dunit are mysteries so profound, not even the brilliant Hercule Poirot can work them out.[‡]

It is Miss Johnson, Dr Leidner's assistant on the dig, who realises just how Mrs Leidner's murder was so cleverly committed, and thereby learns the murderer's identity. And so Miss Johnson becomes the second victim, as the murderer desperately tries to prevent her from revealing their method or identity. Miss Johnson's nighttime glass of water is taken from her bedside and substituted for concentrated hydrochloric acid. When she wakes in the middle of the night, she swallows the acid down with grisly and fatal consequences.

The hydrochloric acid story

Hydrochloric acid, also known as muriatic acid and spirits of salt, has a long history, probably starting in the ninth century with the Islamic scholar Abu Bakr al-Razi. In his work *The Book of Secrets*, he wrote how he heated various salts together with sulphuric. With the right salts, this procedure releases hydrogen chloride gas and, if that gas mixes with moisture in the air, it dissolves to produce hydrochloric acid. But al-Razi appears to have paid no attention to this gas; he was more interested in the solid products of his reaction. It wasn't until many centuries later that a similar method was used to deliberately make strong acids.

In the thirteenth century, Latin alchemists started to distil mixtures of vitriols (sulphate-containing substances, not just sulphuric acid) and other salts together. The acids that they made would have been very strong, but

they were a mixture. These so-called mineral acids (because they came from minerals) formed an important part of alchemical practice. A mixture of roughly one part nitric acid to three parts hydrochloric acid makes *aqua regia*, at the time one of only two substances known to dissolve gold. For alchemists obsessed with making and manipulating gold, *aqua regia* (or ‘royal water’) was a boon. These alchemists were mostly very happy with their mixtures of acids and did not bother to refine them further. A procedure to make pure hydrochloric acid, separate from other strong mineral acids, was not developed and written down until the sixteenth century. It was done by mixing one specific salt, common salt or sodium chloride, with sulphuric acid, much as al-Razi had done, but these experimenters took an interest in the gas formed in their reactions, collected it and dissolved it in water to make a very concentrated solution of hydrochloric acid.

Today, hydrochloric acid is mostly obtained by dissolving hydrogen chloride gas, produced as a by-product in other industrial processes, in water. Most of this hydrochloric acid is used industrially to clean metals, by removing surface impurities, rather than dissolving the metal entirely as the alchemists usually did.* This ability to clean surfaces is probably why there is concentrated hydrochloric acid available to the murderer on the dig in *Murder in Mesopotamia*.

Hydrochloric acid would be very effective at cleaning the metal tools used on an archaeological dig, though it is also stated that it was used to clean up pottery shards. Pouring acid over archaeological finds seems a risky action. The acid would help destroy some of the dirt and debris stubbornly clinging to the ancient pottery, but it could potentially damage the pottery itself. I assume the archaeologists knew what they were doing and tested the process before dunking precious artefacts into this powerfully corrosive substance. The archaeologists probably stored a bottle of concentrated acid, most likely an industrial concentration of 20 per cent, the most stable concentration. Small amounts of this stock solution could then be measured out and diluted for pouring over pottery as needed.

Hydrochloric acid

Acids were discussed at some length in Chapter F but, to recap, most of the acids you will encounter in everyday life are acids because they release H^+ ions in water. As with formic acid, hydrogen chloride can release H^+ ions when it dissolves in water because of how its electrons are shared around the molecules.

Each molecule of hydrogen chloride is a very unsymmetrical combination of one hydrogen and one chlorine atom. The tiny hydrogen atom is dwarfed by the much larger chlorine. The electrons that buzz around the molecule, and form the bond between the two atoms, spend most of their time around the chlorine, leaving the hydrogen atom a little short. The consequence of this unequal sharing is that the chlorine end of the molecule is slightly negatively charged and the hydrogen end is slightly positively charged.

Water also has a more negative and a positive part of its molecules, as we saw in Chapter F, and these positive and negative regions are attracted to the oppositely charged ends of hydrogen chloride molecules. The attraction *between* hydrogen chloride and water molecules is stronger than the bond *within* the hydrogen chloride molecule. Unlike formic acid molecules, *all* the hydrogen chloride molecules are pulled apart to form H^+ and Cl^- ions.*

Separated from their parent compound, H^+ ions can drift around and bump into other molecules. If that molecule has a region that is slightly negatively charged, the positive H^+ ions tend to stick to it and draw the negative charge towards it, away from the rest of the molecule. The molecule's negatively charged electrons are shuffled around to compensate for those being pulled towards the H^+ ion. This redistribution of electrons can result in bonds breaking elsewhere in the molecule.

Strong acids can cause a lot of corrosive damage because there are a lot of H^+ ions around that can break a lot of chemical bonds. It is why concentrated hydrochloric acid, with so many H^+ ions that are all so readily available, is such an effective cleaning agent for metals and pottery shards. It is also part of the reason why hydrochloric acid is found in our stomach.



Inside our bodies hydrochloric acid is made and stored inside specialised parietal cells in the stomach wall. The H^+ ions, from carbonic acid in our blood formed when carbon dioxide dissolves, and the chloride ions (Cl^-) from the salt we eat, are concentrated in channels, called canaliculus, inside the parietal cells. After a meal, the canaliculi release gastric acid, a mixture of hydrochloric acid, electrolytes and enzymes such as pepsinogen.

In the stomach, the reactive H^+ ions destroy bacteria contaminating our food by disrupting the molecules that make up their cell walls and the structures within. The H^+ ions also help break up the food itself, particularly proteins, into smaller absorbable units. Proteins, long chains of amino acids, are normally wrapped up in carefully arranged shapes. The H^+ ions from stomach acid attach to protein chains, causing them to unfold. This unfolding exposes the peptide bonds that join the amino acids together. H^+ ions also convert the pepsinogen in the gastric acid into the enzyme pepsin, which breaks the peptide bonds in the proteins. The proteins that make up our stomach wall are protected from this destruction by a coating of thick mucus.

Between meals our stomach pH, a measure of the concentration of H^+ ions (see Chapter F, page 96), is usually around 1.5 to 3.0. This equates to a concentration of just 0.004 per cent hydrochloric acid at pH 3. The difference between pH 3 and pH 2 is a tenfold increase in concentration, or 0.004 to 0.04 per cent hydrochloric acid, but this is still a very weak solution. When food is introduced, the acid will be both diluted and used up in the digestion process, reducing the concentration of H^+ ions and increasing the pH up to around 6, or 0.000004 per cent H^+ ions.* This is a tiny amount of acid, but its potential to do damage can be readily understood by what happens when that acid escapes the protected environment of the stomach.

Peptic ulcers can form when the mucus fails to protect the stomach from acidic attack. Heartburn occurs when stomach acid is regurgitated into the oesophagus. These familiar conditions are a painful clue as to what can happen if higher concentrations of hydrochloric acid are present.

The strong hydrochloric acid solution in *Murder in Mesopotamia* is probably at a concentration of 20 per cent (at least 5,000 times more concentrated than stomach acid). This would equate to a pH of around

minus one.* A negative pH is terrifyingly acidic and will be horribly destructive to anyone who drinks it.†

How hydrochloric acid kills

Unlike the majority of poisons used by murderers and discussed in books like this one, corrosive acids, like hydrochloric acid, do not have a specific target in the body. The huge number of H^+ ions that are floating around in concentrated hydrochloric acid create indiscriminate havoc. They will rip electrons away from all sorts of different molecules in the body, breaking bonds, disrupting, damaging and dissolving almost anything they touch. The acid will eat its way into the body.

Swallowing hydrochloric acid introduces the H^+ ions to the proteins that make up the tissues of our mouth, throat and stomach. These proteins will be converted into acid proteinate, which dissolves in the acid. Cells, unable to withstand the chemical assault, die. Layer upon layer of tissue is eroded away, exposing blood vessels, nerves, glands and everything else in its path. Red blood cells are destroyed and the oxygen-carrying haemoglobin within them is turned into useless haematin by the acid's chemical action. The chemical corrosion in the mouth, throat and stomach will continue, often causing perforation.

Death can come quickly from shock as blood pressure plummets, or from a swelled glottis, the opening over the vocal cords, blocking the airway and causing asphyxia. Death can also come slowly. Damage from hydrochloric acid can progress for two to three weeks after ingestion, killing the victim as a consequence of the physical harm done to the body and making life insupportable. Delayed death can also come from an infection spreading internally through the perforated body. Peritonitis can kill as late as one month after the acid is ingested. And it is all agonisingly painful.

The quicker the corrosive progress through the body can be stopped, the better the chances of survival. In *Murder in Mesopotamia*, the nurse arrives on the scene very shortly after Miss Johnson swallows the acid. Rather than tell the nurse what she has swallowed, Miss Johnson uses her dying words to gasp out an oblique clue to a previous murder. The nurse has to piece

together what has happened, and therefore what treatment should be given to try and save Miss Johnson, from other evidence.

Is there an antidote?

Corrosive substances can be neutralised very effectively with the right substance. Corrosive acids can be neutralised by an alkali, and corrosive alkalis can be neutralised by an acid. Obviously, the type of corrosive substance needs to be correctly identified, as using the wrong substance could make things worse. Acids and alkalis can be quickly and easily identified using litmus or other indicators that turn a specific colour in the presence of these substances, but they are unlikely to be available at a remote archaeological dig. Fortunately there are other clues to help the nurse identify what is causing Miss Johnson so much pain and suffering. Unlike the effects of some poisons, the consequences of hydrochloric acid's chemical destruction will be clearly visible. These effects would be horrifying to see, and even worse to experience, but it does at least give very obvious and immediate clues as to what is happening and therefore what treatment is needed.

The nurse notes 'the corners of her mouth and the skin of her chin were burnt a kind of greyish white.' This damage is obviously due to something corrosive, but gives no clue as to whether it was a strong acid or alkali that caused it. Skin will be more resilient to corrosive attack than the soft tissue of the mucous membranes inside the mouth. So the nurse then looks at the damage inside Miss Johnson's mouth. What she sees isn't described, but it is likely to have been blackened, shrivelled and eroded tissues.

Strong acids transform the pigmentation of the skin and tissues from their normal colour through grey to black depending on how much blood has been altered. They also corrupt the haemoglobin in our blood, which normally contributes the pinkish hue to our tissues, into brown/black haematin that clumps into granules that look like coffee grounds. With strong alkalis, the discolouration is not as intense, giving body tissues a gelatinous off-white to grey appearance. Strong acids can also dry or dehydrate tissues, whereas alkalis give tissues a gelatinous, slippery appearance.

Miss Johnson, despite her pain, also manages to look very deliberately at the glass that has fallen to the floor. When the nurse touches her finger to some of the liquid remaining in the glass she recoils at the sharp burning pain. Although skin contact with an alkali, such as sodium hydroxide (or lye), will hurt as much as acid, alkalis tend to be slower acting and produce a soapy feeling.

Having established Miss Johnson has swallowed an acid, the nurse administers an alkali, 'a strong solution of carbonate of soda'. Sodium carbonate is sometimes better known as washing soda and is an everyday item that is likely to be readily available even on an archaeological dig in a remote spot. The sodium carbonate will neutralise the acid in a chemical reaction that produces sodium chloride, water and carbon dioxide gas. The reaction would obviously help to arrest the damage done by the hydrochloric acid, but the gas it releases will fizz and bubble. This could have been very uncomfortable on the victim's raw tissues, but probably barely registered over the overwhelming pain from the acid. Strong solutions of sodium carbonate can also be irritating to the skin, which was the last thing Miss Johnson needed in this situation, but perhaps worth the risk if it prevented greater damage from the acid. There would, however, have been a delay in treating Miss Johnson while the sodium carbonate was found and dissolved in water ready to be administered, giving more time for the acid to do its damage.

Rather than trying to neutralise the acid, an alternative course of action is to dilute it. As we have seen before, lower concentrations of H^+ ions have a higher pH and are less acidic. Dilution is the recommended first treatment for anyone who has spilled acid on their skin or swallowed it. Ideally, the acid will be diluted at least one hundred fold to minimise damage. And all you need is water, which is both easy to obtain and safe to ingest or splash all over someone's skin in large quantities.

Milk can also be used to dilute acids. Milk is mostly made up of water but with a few extras that can help neutralise the effects of the acids. There are proteins that will unfold and clump together in the presence of the acid, so the milk is curdled rather than your insides. The proteins and other components also make milk more viscous than water so it will coat tissues more effectively. But the idea that milk is slightly alkali and can neutralise

an acid is a myth. Milk is slightly acidic, though much less so than hydrochloric acid. Overall the benefits of giving milk to treat acid damage are only marginally better than just water and not worth delaying treatment if water is easier to obtain.

What must not be done is induce vomiting or carry out gastric lavage to remove the acid. This will just increase the range of damage as the acid is brought back up the oesophagus. The victim may well be vomiting anyway and fluids will have to be given repeatedly to compensate for the loss of fluid as well as to dilute the acid.

In Christie's novel, treatment with bicarbonate of soda is followed by the administration of olive oil. However, this will have no therapeutic benefit. Olive oil may soothe raw tissue but, given the excruciating pain Miss Johnson is likely to be experiencing, it will be of little comfort and is not advised. The injection of morphine she is given will be much more effective pain relief and is indeed still part of the recommended treatment after acid ingestion today. Pain relief is essential, and morphine is recommended to be given at regular intervals, not just one shot, and in amounts large enough that they could potentially depress the patient's central nervous system. The victim must be monitored closely to ensure that they have not survived the first effects of the acid only to be killed by the treatment.

Another big risk to the patient that must be treated as quickly as possible is maintaining an open airway because the throat swells as the body tries to dilute the acid and repair the damage. This swelling can potentially block the air supply. With the limited resources available to the team on the dig, it seems likely only the most crude intervention would have been possible. Perhaps an emergency tracheotomy might be tried, making an opening through the throat straight into a hopefully undamaged part of the trachea, so that air could bypass the swollen larynx and enter the lungs directly.

There is also danger from shock, where oxygen cannot be effectively pumped around the body due to damage to the circulatory system, because blood vessels and the blood itself have been destroyed by the acid. This can cause a life-threatening drop in blood pressure. Without a certain pressure in the circulatory system, the heart simply can't pump blood effectively. Shock must be treated quickly with transfusions of blood and saline. Saline drips are unlikely to be available to treat Miss Johnson, though blood

transfusions could have been tried if anyone else on the dig had a compatible blood type.

Once the patient has been stabilised, and there is no immediate risk of death, the damage can be assessed before anything else is done that could cause more harm than good. In severe cases there is a high probability that the stomach or oesophagus has been perforated, so nothing should be given by mouth until an examination has been conducted. Nutrition and hydration can be maintained intravenously until an internal inspection can be done, usually with an endoscope.*

Prednisolone can be used in the early stages of treatment of hydrochloric acid poisoning to prevent the oesophagus narrowing and making swallowing difficult. It is a steroid medication that can prevent inflammation and is used to treat autoimmune disorders as well as being given after organ transplant. Sadly, Miss Johnson does not survive long enough for these later treatments to be considered.*

We are not told exactly how long Miss Johnson lived, but it is certainly not long after drinking the acid and before a doctor can reach her. The exact cause of her death is therefore likely to be from a blocked airway that could not be kept open, or from shock, or a combination of both. It is a sad but not surprising end given the damage strong acid can do. As little as 1ml (1/4 teaspoon) of hydrochloric acid can be enough to kill. In a survey of 105 cases of ingested acid, 32 victims died as a result. Even those that survive the initial effects of acid, more than 95 per cent of them are left with permanent damage to their oesophagus. Hydrochloric acid has a high chance of killing someone, and an even higher chance of damaging their throat so they can't speak about vital clues in murder cases. But it isn't the easiest of murder weapons to deploy.

Some real-life cases

The majority of deaths caused by swallowing acid in the UK have been suicides. This is largely a thing of the past, but it was not as uncommon when Christie was writing *Murder in Mesopotamia*. In the novel, Miss Johnson's unusual death raises the question of suicide more than once, but this motive is quickly dismissed. As the nurse points out, someone would

have to be ‘in great distress of mind’ and the acid very easy to obtain for that person to contemplate such a painful act. Such actions are difficult to understand, but the alternative, that Miss Johnson drank the acid by mistake, is no easier to rationalise.

Finding a way to get someone to drink acid is difficult and why it is an exceptionally rare method of murder. Very strong concentrations of hydrochloric acid, up to 38 per cent, have an off-putting yellow tinge and cause the liquid to fume.* But large amounts of hydrogen chloride can be dissolved in water without changing its appearance. A 20 per cent solution, like the one probably used in *Murder in Mesopotamia*, would make no visible difference. The glass of acid given to the victim may look like ordinary water, but alarm bells will ring as soon as they get close enough to detect the pungent smell. If the smell doesn’t alert them to danger, or the slight stinging sensation in the eyes and nostrils as they hold the glass close to their face, then the taste and excruciating burning sensation when they take their first sip surely will.

It may be difficult to make someone drink acid, but murderers have still used it to kill, usually by throwing the liquid over their victim. Such attacks are sometimes described as ‘vitriol throwing’, not because of the hatred for the person being attacked, but because the sulphuric acid that is often used in these cases was once known as ‘oil of vitriol’.[†] The damage caused by the corrosion can result in death and even those that survive will be left permanently scarred. Christie describes such an assault in her short story ‘The Witness for the Prosecution’, which she later adapted into a highly successful play. An important witness in a murder trial shows the permanent scarring on her face from ‘Vitriol, dearie, vitriol – that’s what did that.’[‡]

Christie’s play may now be staged as a period piece, but acid attacks have sadly not dated in the same way. A spate of acid attacks in London in the late 2010s saw the law changed so acid and other corrosive substances were classified as ‘a highly dangerous weapon’ alongside knives and other more traditional weapons. From then on, threatening a person with such an attack, or carrying it out, carried a much tougher sentence.

Rather than using acid to kill, several murderers have used acid to try and destroy their victim’s body and thereby cover up their crime. The most famous of these is undoubtedly John George Haigh, more commonly

known as the ‘acid bath murderer’. Haigh is known to have killed six people between 1944 and 1949 in order to financially gain from their death or cover his tracks when relatives became suspicious.* He either hit his victims over the head with a lead pipe or shot them with a revolver. Each body was then placed in a barrel of concentrated sulphuric acid and left for a few days or weeks until it had dissolved. Haigh then tipped the human sludge into the drains down a manhole outside his rented workshop. When he moved to new premises, on Leopold Road, there was no convenient access to the drains, so he tipped the sludge over the rubble in the workshop yard.

His final victim, Mrs Durand-Deacon, he stripped of her valuables, including her coat, before dunking her in the acid. Even when a dry cleaner’s receipt for the coat was found on the premises, Haigh was confident he wouldn’t be convicted of murder. He boasted to the police, ‘I have destroyed her with acid. You will find the sludge which remains in Leopold Road. Every trace has gone. How can you prove murder, if there is no body?’†

Haigh was right that there was no body, but not every trace of Mrs Durrand-Deacon had been destroyed. What was found in the rubble in Leopold Road was 12.7kg (28lb) of human body fat, part of a human foot, gallstones and a fragment of dentures confirmed to belong to Mrs Durand-Deacon by her dentist. Haigh was found guilty and hanged.

Agatha and hydrochloric acid

The most incredible part of the murder in *Murder in Mesopotamia* is not that acid was chosen as the murder weapon, but that it was swallowed unsuspectingly. Working in the arid environment of Iraq, even during the winter digging season when the country sees almost all of its rainfall, can probably provoke quite a thirst. But to drink down a glass of concentrated hydrochloric acid, it must be a particularly powerful thirst combined with a very sleepy person so used to drinking down glasses of water by their bedside that they can do it without thinking. You would have to be very familiar with a person’s nighttime routine for this plan to have a chance of working.

Choosing concentrated hydrochloric acid is an interesting bit of improvisation on the murderer's part, given they must act quickly to prevent Miss Johnson telling what she knows, not get caught and can only use what is to hand. Though Miss Johnson's death meant she was indeed silenced, she still had the time and determination to gasp out a vital clue while writhing in agony.

When the nurse first reached Miss Johnson the only noise she made was 'an awful hoarse whisper', which is unsurprising given the damage the acid would have caused to her mouth, throat and vocal cords. But she somehow managed to pull herself together and manipulate her rapidly dissolving tissues to articulate five words, 'The window ... Nurse ... the window ...'.

Forming words with acid corroded tongue and lips cannot be easy. Forcing air past vocal cords that are corroded from the acid's effects will make a sound as raw and ragged as the cords themselves. Getting air up through the throat to the mouth, past the tissues that are swelling up and threatening to block its passage, will require considerable effort. And doing all this when in incredible pain and your dying moments shows incredible presence of mind. I don't think I could be so articulate or conscientious in the same situation. But Miss Johnson's determined altruism enables Poirot to deduce how the first murder, of Mrs Leidner, was carried out and therefore identify the only person who had the opportunity to kill her. The murderer is revealed and receives their just desserts.

The circumstances surrounding the death of Miss Johnson, the method used and her ability to reveal key clues in her dying moments were perhaps stretching credibility a little far. However, the availability of acid, its strength and effects on the body were accurately described, though in a much shortened and sanitised fashion. The treatment that was given to Miss Johnson to try and save her is also sensible based on the chemistry and what would have been available on a 1930s dig in Iraq, though modern medical recommendations are different. The injection of morphine is spot on and would be a merciful relief for the victim of one of the most horrific murders in the Christie canon. *Murder in Mesopotamia* is a delightfully gruesome deviation from the Agatha Christie standard.

Notes

- * See the appendix for a full list of Christie's murders and their methods.
- * A large Sumerian urban centre dating back nearly 5,000 years, and an important archaeological site in southern Iraq.
- † Sir Max Mallowan soon forged his own prestigious career, independent of the Woolleys, and was knighted for his services to archaeology in 1968.
- * A tell, or settlement mound, dating back 8,000 years. It was excavated by Max Mallowan over several seasons between 1935 and 1937, with the help of his team and his wife.
- † If you want to make the comparison for yourself, read *Come, Tell Me How You Live*, a very enjoyable account of Agatha's experiences on several of her husband's digs.
- ‡ Hercule Poirot, well-travelled like his creator, just happens to be in the area and is brought in to investigate.
- * There are many other uses of hydrochloric acid, from water treatment to food preparation, but far too many to go into here.
- * The difference between chlorine (Cl) and chloride (Cl-) is a negative charge.
- * All the H⁺ ions are released from hydrogen chloride when it dissolves in water so, when talking about pH, the concentration of hydrochloric acid and H⁺ ions is the same.
- * Normally, decreasing the pH by one means 10 times as many H⁺ ions, but at these concentrations the general rule starts to break down and things get more complicated.
- † Boiling proteins in a 20 per cent hydrochloric acid solution is how scientists deliberately break up proteins into their constituent amino acids for their research.
- * Though endoscopes were invented in the nineteenth century, it is unlikely one would have been packed for an archaeological expedition, but one might have been available at a hospital had Miss Johnson made it that far.
- * Though, to be fair, the novel was published in 1936 and prednisolone was not available until 1955.
- * Above 38 per cent the evaporation is too great for the solution to be stored effectively.
- † You can see why 'vitriol' has come to mean bitter abuse or caustic language. 'Vitriol' in terms of the acid gets its name from the Latin for glass, because of the glassy minerals that it was obtained from centuries ago.
- ‡ Although this is later revealed as a double bluff, the scarring effect has been created using theatrical makeup.
- * He claimed to have killed three more, but nothing was ever proved.
- † He had misunderstood the phrase '*corpus delicti*', which refers to the 'body of the crime' and not specifically a human body.



The Big Four

Agatha Christie's *The Big Four* features her most famous detective, Hercule Poirot, his sometimes sidekick in solving crimes, Captain Hastings, and the dependable Inspector Japp of Scotland Yard. There is also a fairly high body count, but this novel is not a whodunit. *The Big Four* is a thriller reminiscent of the world Ian Fleming created for his iconic spy 007, even down to the casual racism.* James Bond's debut, however, came 25 years after Christie's novel.

If Christie was mimicking anyone's style, it was more likely to be the adventure stories of Dr Fu Manchu, written by Sax Rohmer between 1912 and 1959. Like Rohmer, Christie's novel includes a Chinese villain, but

there are also American and French baddies. In a break with the tradition of casting good guys and bad guys along nationalistic lines, the fourth villain of the title's Big Four is perhaps an Englishman and the hero is, of course, a Belgian. But the villains are still intent on world domination and choose to do their plotting from inside luxuriously appointed secret lairs staffed by large numbers of nameless minions. The Countess Vera Rossakoff, a Russian aristocrat with criminal connections who captivates Poirot, could almost be a prototype Bond girl. There are also technologically advanced weapons, chases, gadgets in the form of a poisoned dart hidden in an adapted cigarette, and feats of derring-do.

The book was published in 1927, after a particularly unhappy time in Christie's life – her mother had recently died and her marriage to Archie Christie was breaking down. It is, therefore, understandably not up to her usual very high standard. Rather than try to craft an entirely new plot, Agatha's brother-in-law, Campbell Christie, suggested she rework some earlier short stories into one novel-length adventure. With Campbell's help weaving together the disjointed source material, the manuscript was completed, and thanks in part to the huge success of *The Murder of Roger Ackroyd*, and the enormous publicity around her mysterious 11-day disappearance in 1926, the book sold well. Christie's own feelings about the book are revealed in a letter to her agent Edmund Cork:

I have been, once, in a position where I wanted to write just for the sake of money coming in and when I felt I couldn't – it is a nerve wracking feeling. If I had had one MS [manuscript] 'up my sleeve' it would have made a big difference. That was the time I had to produce that rotten book The Big Four and had to force myself in The Mystery of the Blue Train.

Christie considered thrillers easier to write. The plots are less constrained by detailed clueing and characterisation as the hope is the reader will simply be swept along by events. It was a sensible choice at a difficult time. The result may not be Christie at her best, but *The Big Four* is an enjoyable adventure featuring well-loved characters and a few of the author's trademark methods of murder – poisons. One murder in particular shows Christie could still draw on her vast toxicological knowledge to include an

unusual poison rather than going for an easier, more well-known option like cyanide or arsenic. The poison is yellow jasmine and Christie uses it to bring about the perplexing death of Mr Paynter.

Captain Hastings narrates the tale of Mr Paynter, a cultured and wealthy man who, after a life of globe-trotting, had enjoyed the previous 12 years living uneventfully in Worcestershire. Seven months before his death he was joined in this quiet life by his nephew Gerald Paynter, an ‘impecunious young artist’ who becomes one of the suspects in his uncle’s murder. Another suspect is Ah Ling, Mr Paynter’s body servant. An open window in the room where Mr Paynter’s body is found also suggests the murderer may have been an outsider. A burglary might have gone wrong, but the only thing that appears to be missing is the manuscript of Mr Paynter’s memoirs. Perhaps there was something in the memoirs someone didn’t want revealed.

The death itself is a classic Christie, taking place in the study of a country house and surrounded by lots of clues. The most important clue is a newspaper Mr Paynter was holding when he was murdered and on which he had scrawled the words ‘yellow jasmine’. The initial assumption is that in his dying moments Paynter’s mind had wandered to the jasmine that grows up the walls of his home. Poirot, of course, realises the truth – that Mr Paynter has written the name of the poison he has been given.

The yellow jasmine story

There are three species of *Gelsemium* plants that Christie might have been thinking of when she wrote her story. *Gelsemium semperivens* is also known as yellow or Carolina jasmine.^{*} This plant, and its cousin *G. rankinii* with the common name Rankin’s and swamp jasmine, both grow in Central America and the southeastern states of the USA. The third *Gelsemium* species, *G. elegans*, grows in Southeast Asia. The common name for the *elegans* species in China is ‘heartbreak grass’ because of its well-known toxicity. All of these plants are climbers that produce yellow flowers, and all contain many toxic compounds.

None of these plants, however, grow in the UK, at least not without a bit of protection from the worst of the cold weather in the form of a greenhouse or a sheltered spot in a city garden. Worcestershire, the location of Mr

Paynter's country house, simply does not have the climate to allow these plants to climb all over the walls of a big house in the abundance suggested in the novel – and that is the important detail that gives Poirot the clue he needs.

Having jasmine growing all over a house in the UK is certainly possible. Winter jasmine, or *Jasminum nudiflorum*, is a common climbing plant and can be seen covered in yellow flowers from November through to March. Winter jasmine is, however, not nearly as toxic as the *Gelsemium* plant species, though still not good to eat. Poirot realises Paynter's dying message refers to exactly what it says: 'yellow jasmine', the toxic variety of jasmine that has been used to kill him and not the yellow-flowered winter jasmine growing over his house.

To confirm his suspicions Poirot dashes into a shop to purchase a book that details the toxic components of plants. He tells Hastings Mr Paynter was killed with '*Gelsemini Radix*: * Yellow Jasmine.' He then reads out the deadly details of this particular plant. The first toxic component mentioned in the book is '*gelseminine* $C_{22}H_{26}N_2O_8$, a potent poison acting like coniine'. Coniine is the toxic component of hemlock that can cause dizziness and a feeling of being whirled about before paralysing the body starting from the feet and progressing upwards to the lungs and asphyxiating the victim. The second compound listed in Poirot's book is '*gelsemine* $C_{12}H_{14}NO_2$, acting like strychnine'. * Strychnine is a potent poison popular with Christie and several real-life murderers that causes agonising convulsions (more of this later). Poirot skips over the other compounds associated with yellow jasmine to get to the poisoning symptoms that the plant can produce:

Gelsemium is a powerful depressant to the central nervous system. At a late stage in its action it paralyses the motor nerve endings, and in large doses causes giddiness and loss of muscular power. Death is due to paralysis of the respiratory centre.

It looks like Christie has simply cut and pasted this information from an existing book, maybe something in her extensive collection on toxicology, though I don't know what book. It would certainly have saved her time

when her priority was to simply finish the novel. On this occasion, however, Christie, or the original book she copied, has made some minor mistakes. The compounds are real, but the chemical formula given for them is wrong. To be fair, scientists were struggling to identify and characterise all the different toxic compounds in gelsemium plants at the time Christie was writing. And the chemical formulas she gives are closer to the real ones than some of the scientists' early estimates. The fact that these are the only minor discrepancies in her technical details shows how good she was, even when she was working under enormous personal pressures.

Gelsemium

In *The Big Four*, Christie correctly describes the toxic compounds in gelsemium plants as alkaloids. Alkaloid compounds contain at least one nitrogen atom in a ring with other atoms. This class of compounds are commonly found in plants and are often responsible for their poisonous or medicinal properties – for example morphine in poppies, quinine in cinchona bark and cocaine in coca plants are all alkaloids. More than 100 different alkaloids have been isolated from gelsemium plants where they are mostly concentrated in the roots and rhizomes.*

The first gelsemium alkaloid mentioned by Christie, gelseminine, is found in *G. sempervirens*, but at the time she was writing there was much confusion over what this alkaloid was, or even if it was a mixture of several different alkaloids. The correct formula has now been determined as $C_{21}H_{27}NO_4$, but very little research has been carried out into the effects of this compound. It is a minor component of the alkaloid mix in *G. sempervirens* and contributes little to the poisonous effects of the plant. Instead, it is the second alkaloid listed in Poirot's book, gelsemine, that is responsible for most of the symptoms produced if gelsemium plants are ingested, or if a baddie in a 1920s thriller injects an extract of the plant into a wealthy man in the study of his country house.

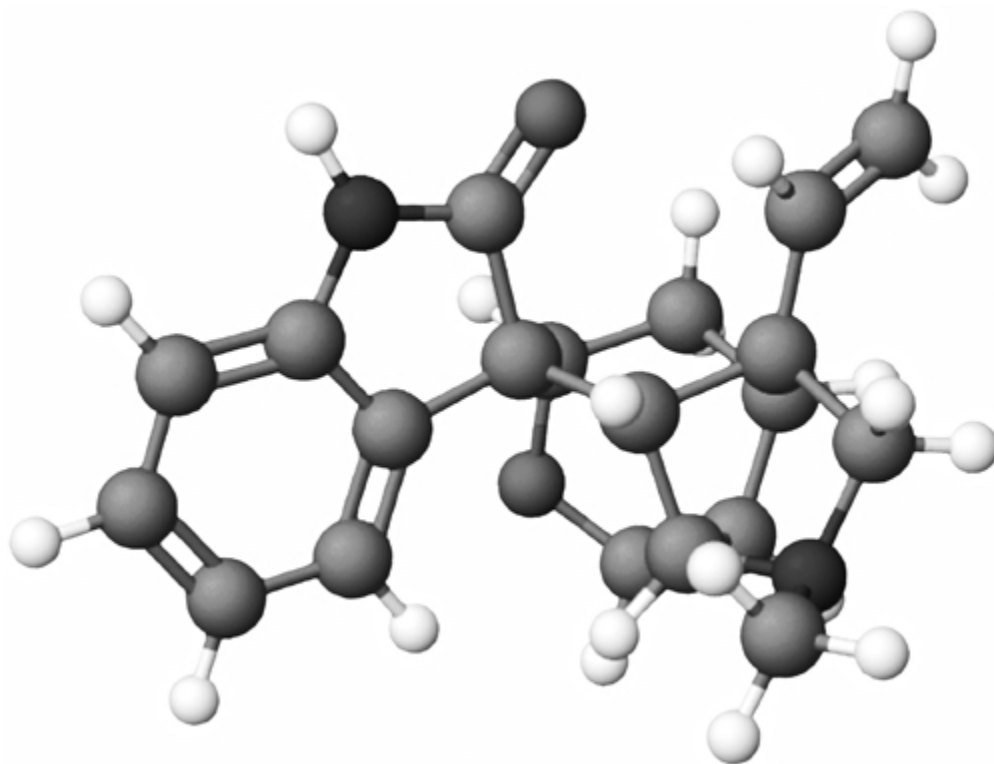


Fig. J.1. Structure of gelsemine, showing the dark grey nitrogen atom in a five-membered ring with four other pale grey carbon atoms in the centre of the molecule.

Gelsemine ($C_{22}H_{22}N_2O_2$ – the correct formula and not the one given by Christie) is found in all species of gelsemium plant and it is the most abundant of all the alkaloids in yellow jasmine (*G. sempervirens*). It is the second most abundant alkaloid in heartbreak grass, *G. elegans*, after kuomine, but it is more toxic than kuomine and responsible for most of the effects on the human body.

For a long time the effects produced by the alkaloids in extracts of gelsemium plants were considered beneficial. These extracts were made by soaking the roots and rhizomes of the plant in alcohol. The alcohol was then evaporated to leave a powdered residue containing around 6 per cent gelsemine. This gelsemine powder has been widely used medicinally. For hundreds of years it was prescribed in Chinese herbal medicine for all sorts of conditions including eczema, haemorrhoids and leprosy.* A typical powdered prescription dose would amount to between 15 and 60mg gelsemine (0.0005–0.0021oz). In the West in the nineteenth century, the powder was refined further to isolate gelsemine from the other compounds

that were present. A solution of pure gelsemine in an alcohol and water mixture, known as a tincture, was used to treat conditions ranging from tonsillitis to measles. A typical dose of this tincture was 20 minims, an old measurement equivalent to 1.2ml or 0.04 fluid ounces, which would deliver 1mg (0.000035oz) of gelsemine.

In 1879, a student at the University of Edinburgh Medical College wrote to the *British Medical Journal* about his experiments with gelsemine. He had been prescribed the drug to treat his neuralgia and had occasionally taken more than the recommended dose, with no ill effects. This made him curious as to how large a dose he could stomach and what the early symptoms of poisoning might be.

Twenty-year-old Arthur Conan Doyle gave up drinking and smoking for the week of his gelsemine experiments. He started on Monday evening with a fresh bottle of tincture and swallowed a double dose of 40 minims, delivering around 2mg (0.00007oz) of gelsemine, which produced no noticeable effects. On Tuesday night he increased the dose to 60 minims (3mg/0.0001oz), but still nothing happened. Perhaps frustrated at his seemingly slow progress, on Wednesday evening he made a big jump to 120 minims (6mg/0.0002oz) and 20 minutes later he was overcome with giddiness. His limbs felt weak, and so did his pulse, but he soon recovered. On Thursday he repeated his 120 minims dose and the giddiness was noticeably milder. However, later that night he suffered double vision and had to make a concerted effort, screwing up his face, in order to focus on distant objects.

Friday came with a dose of 150 minims (7mg/0.0002oz), which brought very little giddiness but a severe headache, diarrhoea and general lassitude. On the Saturday he swallowed 180 minims and on Sunday 200 minims (10mg/0.0004oz), resulting in more painful headaches, a weakened pulse and depression. But it was the persistent and prostrating diarrhoea that convinced him to stop his experiments. Thankfully, Doyle recovered from his ordeal and went on to write some popular crime stories.

Doyle's experiences with gelsemine are explained by its interaction with our nervous system. The individual neurons that make up our complex interconnected nervous system act like biological batteries, as discussed in previous chapters. Sodium, potassium and chloride ions are moved in and out of nerve cells to set up a negatively charged interior and positively

charged exterior. When the nerve fires, the negative and positive regions switch places, generating the electrical signals that travel along the length of the individual nerve cells like a wave (see Chapter F, page 90). At the junction between one nerve cell and the next is a gap, the synapse where chemicals, neurotransmitters, cross the gap to pass on the message. The neurotransmitter we are interested in for this chapter is glycine.

When glycine molecules dock at their receptors they cause negatively charged chloride ions (Cl^-) to flow into the cell, increasing the negative charge inside and making it harder for the nerve to fire. Glycine docking at its receptor, therefore, acts as the 'stop' message for nerve cells. These receptors are found predominantly in the brainstem, spinal cord and retina. Glycine is released along with other activating and inhibiting neurotransmitters and it is the balance between their opposite effects that keeps nerve cells operating within normal bounds.

Gelsemine attaches to glycine receptors and activates them, causing even more Cl^- ions to flow into the neuron than usual. As a consequence, nerve signals are not passed on so easily. This is potentially beneficial if the nerve is causing the pain of neuralgia, but it will also depress the activity of other nerve cells, causing the side effects noted by Doyle – weak limbs, double vision and giddiness, etc.

Gelsemine slowly fell out of medical favour as its toxicity outweighed the minimal medical benefits, especially as other more effective and less toxic alternatives became available. However, medical interest has been revived in more recent years. Gelsemine's potential as an anti-inflammatory, anti-anxiety and anticancer drug is being explored. It may be that chemical modification of the gelsemine molecule can produce safer, more effective derivatives. With a better understanding of exactly how gelsemine interacts with the body, it has the potential to be administered more appropriately and effectively.

As with many of Christie's poisons, she probably learned of gelsemine and yellow jasmine through her pharmaceutical training. Her training would also have helped her to maximise the lethal potential of the compound when she came to use it for her fictional murder in *The Big Four*.

How gelsemine kills

Increasing the dose of gelsemine will increase the effect on the nerves by activating more glycine receptors. The activity of these nerves can be hindered to the point where messages are effectively blocked. Without the appropriate signal from nerves, muscles will not contract and the victim will collapse. More seriously, signals to the muscles that move our ribs and inflate our lungs will also be blocked and the victim will struggle to breathe or stop breathing completely.

Most deaths from gelsemine are as a result of swallowing an overdose of medicine or because gelsemium plants have been misidentified as something edible. Food can also be contaminated without anyone noticing because the toxic plant has grown alongside or entwined with an edible plant and it only takes a few grams of plant material to cause poisoning symptoms.* These symptoms can appear within 20 minutes of ingestion, as described by Doyle, or up to two hours later but usually within an hour. A lethal dose can take as long as seven hours to kill, but death usually occurs around an hour later.

A murderer who needs their victim incapacitated rapidly, so they can't raise the alarm, needs the gelsemine to act more quickly. In *The Big Four*, the yellow jasmine poison is injected, bypassing the digestive system that might break some of it down into less effective compounds. A dose as low as 35mg of gelsemine given to a 70kg person (0.0012oz in a 154lb body) could be fatal. Symptoms would be expected to appear much more quickly and have a more dramatic effect, reducing the chances of Mr Paynter surviving, but medical treatment might still have saved him.

Is there an antidote?

In a study of 33 gelsemine poisoning cases in Hong Kong, all from ingesting the poison from plant material, 32 survived and made a complete recovery. The one fatality was found collapsed at home, too late for medical intervention to save them. A third of these survivors required no medical treatment at all. Another third received treatment to prevent the poison from being absorbed from their stomach into the rest of their body by giving activated charcoal (see previous chapters). The remaining third recovered after being given fluids intravenously and/or antiemetics to prevent

vomiting. Only one patient needed to be mechanically ventilated to help them breathe, and they made a full recovery.

Ventilators only became common medical equipment in the 1950s, too late for Mr Paynter, but any form of artificial respiration would have helped introduce oxygen into his lungs if his muscles were unable to do it for him. If the gelsemine was slowing his heart, atropine could have been given to stimulate it. And though there is no specific antidote for gelsemine poisoning, injections of strychnine might have helped.

Strychnine, a chemical cousin of gelsemine, interacts with the same glycine receptors in nerves. But instead of activating these receptors as gelsemine does, it deactivates them. In excess, strychnine switches off the 'stop' message to nerves and they are overwhelmed with 'go' messages. The result is overstimulation of the nerves controlling muscles, including the muscles controlling breathing. All the muscles will be painfully contracted, the whole body will be convulsed and the lungs unable to move, leading to the same cause of death as with gelsemine poisoning – asphyxia. Get the dose right, however, and in theory strychnine can counteract the effects of gelsemine.*

Because of its obvious dangers, and horrific effects on the body, the use of strychnine in medicine, and any other role, is virtually unheard of today. Things were very different in the 1920s, when strychnine was a commonly used stimulant and pick-me-up. In *The Big Four* the murderer is able to convince Mr Paynter to have an injection by telling him it is a perfectly normal stimulating dose of strychnine. In fact, the strychnine has been swapped for gelsemine.

Mr Paynter is then left alone in his study, ensuring the poison has time to take its full effect. Later, his dead body is pushed into the fireplace so it will be burnt, making the crime scene more difficult to interpret. If the murderer hoped to destroy evidence of poisoning, it is a grim and unnecessary step to take. An autopsy, particularly one carried out in the 1920s before more sophisticated analytical techniques were available, would struggle to find the tiny amounts of gelsemine needed to kill. This doesn't mean gelsemine poisoning could not be deduced or that the murderer would get away with it, as a case in India illustrates.

Some real-life cases

In 1909, Augusta Fullam and Lieutenant Henry Clark fell in love. The problem was that they were both already married. Their solution to this problem was to kill their respective partners so they could be together. Henry drew on his medical training,^{*} and a few carefully chosen passages from the books *Principles of Forensic Medicine* and *Lyon's Medical Jurisprudence*, to plan a poisoning that would look like natural causes. Augusta was his willing accomplice.

Augusta's initial attempts to dispatch her husband with arsenic produced several bouts of illness, but did not bring about his death. Taking advantage of a particularly hot summer in Meerut, north of Delhi, the pair next tried atropine,[†] possibly mixed with some cocaine, in the hope the toxic effects would look like the symptoms of heatstroke. Mr Edward Fullam collapsed, became delirious and showed an elevated temperature (though not quite in the region expected for heatstroke). He was hospitalised for weeks and, though he made some recovery, it was thought he would never work again. Plans were put in place for him to be retired back to England. This was disastrous for the lovers as it looked like they would be separated forever.

Somehow Mr Fullam was persuaded to move to Agra with his family. Conveniently for his wife, this was the same city where her lover lived. Two days after arriving in his new home, still weak from his recent illness, his wife went to the cook house and collected his food personally. This was unusual – the servants would normally do this – but it was assumed Mrs Fullam was being a particularly attentive wife to her sick husband. It was probably her opportunity to lace the food with the gelsemine that had been supplied to her by Clark.

Soon after eating, Mr Fullam felt 'dizzy and strange'. He was in pain and thought he was going to be sick. He was taken to bed, where Clark gave him three injections, to the upper arm, chest and lower arm. According to Clark, this was a combination of ether, digitalis and strychnine, prescribed and checked by another doctor, that was intended to revive Fullam if he suffered a collapse. If Clark is to be believed, the strychnine might have counteracted the effects of gelsemine given earlier but, whatever the injections were and in whatever dose, they didn't help.

Clark wrote out a death certificate attributing Mr Fullham's death to general paralysis and a relapse of his earlier illness causing heart failure. It was countersigned by a fully accredited doctor, but one who had never met Mr Fullam in life. The lovers appeared to have got away with their first murder.

Next was the turn of Clark's wife. She had proved remarkably resilient to her husband's attempts to kill her. She promptly vomited up whatever poison he gave her, so more drastic measures were needed. Clark paid four local men 100 rupees to kill his wife. Clark made sure he was out of the house meeting a friend, but this friend did not exist. The assassins then arrived and were supposed to make the attack look like a robbery gone wrong, but they bungled it. After hitting Mrs Clark over the head with a large knife three times, they fled, taking nothing with them. Mrs Clark died the following day.

Mrs Clark had obviously been murdered and so the police became involved. During their searches a box full of Augusta's love letters to Clark were discovered. Along with gushing proclamations of their love were the details of their murderous plans.* Bottles of arsenic, atropine, cocaine and gelsemine were also discovered along with confirmation of their purchase by Clark. Mr Fullam's body was exhumed and arsenic was found in the remains, but not enough to account for his death. There were no detectable traces of cocaine, atropine or gelsemine, but this is not surprising for the time. Fullam's meticulous marking of his forensic and medical jurisprudence books, together with his purchase of a surprisingly large quantity of gelsemine and the detailed plans in the love letters, were strong indicators of what had really killed Clark.

The pair were arrested and tried. The brutality of the attack on Mrs Clark, and the salacious nature of the letters that were read out in court, ensured wide publicity in India and England. It took the jury 10 minutes to find Henry Clark and Augusta Fullam guilty. Clark was hanged on 26 March 1913. Augusta, pregnant with Clark's child, was given life imprisonment. Her baby boy was born in the prison infirmary a few months later. Augusta was back in the infirmary within a year, but this time she was suffering from heatstroke. She died on 24 May 1914.

Clark's medical background would have taught him about gelsemine, as well as giving him a legitimate reason for buying the drug and an excuse to have it in his possession. Since gelsemium and its compounds fell out of medical favour, knowledge and access to these poisons has become more difficult, at least in Europe.[†] But that doesn't necessarily mean gelsemium poisonings are a thing of the past.

Russian business tycoon Alexander Perepilichnyy made his fortune in investment banking. He was doing very well for himself and his clients until the 2008/9 financial crash when he, like many others, saw huge losses on his investments. Some of his clients accused him of robbing them and demanded he repay their losses. Perepilichnyy had what has been described as a 'falling out' with some of his clients and moved to the UK in 2010. He settled into a quiet family life in a large house in the secluded St George's estate in Weybridge, Surrey.

As part of his work as an investment banker, Perepilichnyy had become aware of some very large transfers of money that had been made from Hermitage Capital Management fund to Swiss bank accounts. Concerns about some of the Hermitage fund's transactions had been raised in 2009. A tax advisor for the company, Sergei Magnitsky, blew the whistle on Hermitage being used for corrupt financial activities. Magnitsky was then arrested and died in prison under suspicious circumstances. Perepilichnyy contacted Hermitage Capital Management with information about what he knew. Hermitage in turn informed the Swiss authorities. The accounts were promptly frozen and an investigation launched. Perepilichnyy handed over a lot of financial details to this investigation.

In 2012, Perepilichnyy took out several life insurance policies that, given the amounts involved (several million pounds), required he undergo some extensive medical tests. These tests revealed that the 44-year-old was in good health with no underlying health problems. On most days he went out for a jog to keep himself fit. On 10 November he returned from a three-day trip to Paris and went out for a run around St George's. As he jogged back to his house, he collapsed just yards from his front door. He died 25 minutes later, before paramedics arrived. It was the fourth death associated with the Hermitage fund.

The police became involved and two autopsies were carried out on Perepilichnyy's body, along with extensive toxicological testing. Nothing untoward was found, nor any other clues as to the cause of his death. It seemed the millionaire's jog had simply put too much strain on his heart. However, not everyone was happy with this explanation. One of Perepilichnyy's insurers started to ask questions and sent toxicological samples to a plant expert, Professor Monique Simmonds, from Royal Botanic Gardens at Kew. Professor Simmonds had been involved in a number of high-profile poisoning cases that had involved plants, and her examinations revealed traces of gelsemine.

The inquest into Perepilichnyy's death was postponed several times but finally resumed in 2017 where evidence regarding gelsemine poisoning was discussed. Suggestions were made that the financier had ingested gelsemine from the sorrel soup he had eaten earlier in the day. If this was true, why did the poison take so long to take effect? Wouldn't early symptoms like dizziness put him off going for a run? The only evidence of gelsemine poisoning came from Simmonds's examination of material extracted from Perepilichnyy's stomach lining. The rest of his stomach contents had been 'flushed away' after the post mortem. This seems very unusual as forensic evidence would be expected to be carefully stored until an inquest had been held. If murder is even suspected, evidence is kept so it can be checked by other experts in the event of a trial. And there were other strange details about the inquest. There was no jury and over 40 documents allegedly relating to the case were unavailable. These documents were said to contain sensitive information and had to be kept secret for reasons of national security.

All sorts of suggestions, allegations and theories have been made over the death of Perepilichnyy. His wife maintains he died of natural causes. Surrey police continue to assert there was no identifiable toxic substance present in the Russian's remains. Several of the insurance companies involved with Perepilichnyy raised no concerns about the death of their client. Others allege a cover-up. There is a rumour that Perepilichnyy was on a hit list. Some have claimed the Russian was assassinated on Vladimir Putin's express orders. It is all as perplexing as an Agatha Christie whodunit.

Agatha and yellow jasmine

Christie avoided shady dealings in international finance for her thriller story. Her choice of secret, technologically advanced weaponry that can be used for world domination is more in keeping with this fictional world.* It perhaps reflects the preoccupations of the time with the development of larger and more terrifying weapons during the First World War and rapidly advancing mechanisation in the 1920s. Her use of yellow jasmine might have been seen as fantastical for the time in which she was writing. Today it is the gelsemium that provides the little touch of realism.

The confusion of yellow-flowered jasmine growing over a house with the yellow jasmine used to kill Mr Paynter is a typical Christie touch. It is a subtle clue and one that only garden enthusiasts, aware that yellow-flowered winter jasmine is not known for its toxicity, are likely to pick up on. She never explicitly points out that the yellow jasmine used to kill her character Mr Paynter is not the same species as the yellow-flowered jasmine growing over his house. Her later description of the toxic alkaloids found in gelsemium roots may leave readers eyeing the winter jasmine in their garden with more suspicion than is warranted.

The initial confusion of the two plants is deliberate on Christie's part. Failing to fully explain the differences between them is a minor quibble on my part. The details do not really matter as the important point is that Mr Paynter has been murdered and by the fourth member of the Big Four. Poirot and Hastings know they are on the right track and the adventure can continue.

What is more difficult to explain is how Paynter knew he had been poisoned by yellow jasmine so he could write it down in his dying moments for the benefit of those investigating his death. There are no distinct symptoms characteristic of gelsemium toxins that make it readily identifiable. The early onset of dizziness caused by gelsemium is unusual for plant poisons and has been suggested as a means of diagnosing gelsemium in poisoning cases in China. Although dizziness could also be attributed to the coniine in hemlock poisoning cases (see page 148), which is far more common in Europe.

To explain Mr Paynter's incredible insight, I can only assume that Christie followed yet another thriller convention. Her villain, like all good

super-villains, must have taken the time to explain to their victim how they were going to kill them in gory and graphic detail and then left before they actually died. Paynter would then have had all the information he needed to write his note after the murderer had gone.

Notes

- * Sadly fairly standard for the day but it can make for uncomfortable reading now.
- * It has lots of other common names.
- * *Gelsemini Radix* is simply the Latin for the root of the gelsemium plant.
- * Many of the toxic compounds in gelsemium plants have confusingly similar names, but gelseminine and gelsemine are the only two we need to worry about here.
- * Rhizomes are root-like growths that help some plants spread. Shoots grow up from the rhizomes to become new plants. In yellow jasmine these rhizomes can be between 15 and 20cm long (roughly 6–8in).
- * It is still prescribed as part of traditional medicine to treat pain, spasticity and skin ulcers. Because of its toxicity it is known as ‘Gou-wen’, meaning lethal kiss, and is limited to external application only.
- * Although most of the toxic compounds are concentrated in the roots and rhizomes, they are present in all parts of the plant and none of it is safe to eat.
- * This would appear to contradict Christie’s assertion that gelsemine acted like strychnine. It acts on the same receptors, but not in the same way. The mix-up may be down to the general confusion about which alkaloids were which in gelsemium and what their effects were. It might also be a mistake.
- * Henry Clark was a member of the Indian Subordinate Medical Department, formed by the East India Company to provide medical services for British military in India as well as native Indians. Though Clark was a doctor, he was not fully accredited and would not have been considered ‘top grade’.
- † The toxic component of deadly nightshade and other notoriously dangerous plants such as jimsonweed. It is chemically related to scopolamine, discussed in Chapter S.
- * I don’t know which is more nauseating, the sickly sweet sentiments of love or the cold-hearted planning of two murders.
- † In 2011 in the Guangdong province of southern China, Long Liyuan, a Chinese billionaire, died after eating cat meat stew believed to have been deliberately laced with gelsemium.
- * With generous interpretation, this beam of wireless energy that can be focused on any spot could be the precursor to Goldfinger’s famous laser beam or Blofeld’s diamond space laser.



Passenger to Frankfurt

The Mickey Finn and ‘knockout drops’ are a staple of US folklore. They have featured in films, plays and books when a character has to be rapidly and unsuspectingly rendered unconscious. These very useful substances, certainly in terms of plot, are strongly associated with the American style of ‘hard-boiled’ detective fiction, as well as thrillers and noir genres of writing. They facilitate the day-to-day work of the professional criminal underworld, making victims easier to rob, kidnap and murder. Mickey Finns, therefore, may seem out of place in Agatha Christie’s more genteel form of mass murder.

Christie wrote thrillers, as well as detective stories, as we know. She also imitated other writing styles (notably in her short story collection, *Partners in Crime*). Her stories can be serious, sinister, creepy, clever, funny, but 'hard boiled' is a phrase few, if any, would ever associate with Christie's work. The gangsters, hoodlums and mobsters that trouble many US detectives rarely turn up in Agatha's country houses to bother the local aristocracy, but Mickey Finns occasionally make an appearance. Christie's thrillers feature gangs of crooks headed by big bosses, espionage, secret codes, even the occasional chase, but the emphasis is on intriguing adventure rather than graphic violence. Christie took the key features of a thriller, including knockout drops, and adapted them to her particular brand of fiction.

Though they don't take centre stage in the same way Christie made some poisons particular features of her plots, Mickey Finns and knockout drops turn up in all manner of imaginative circumstances. Knockout drops in a glass of beer render a man unconscious in *Passenger to Frankfurt* (1970) so a spy can pickpocket their passport and slip out of the country. *The Clocks* from 1963 has a man drugged with the drops so he doesn't cry out when he is stabbed. In *Why Didn't They Ask Evans?* (1934), it is suggested that Mr Savage, a millionaire, had taken a deliberate overdose after receiving a terminal cancer diagnosis. In fact, he was deliberately drugged with knockout drops while his will was forged and he was then murdered with an overdose of the same stuff. There are yet more examples that we will come to later in the chapter.*

A substance that could be adapted so easily for so many different situations suited Christie very well, and Mickey Finns are certainly adaptable. But, before getting down to the nuts and bolts of how they work, let's explore what knockout drops and Mickey Finns actually are.

The Mickey Finn story

The origins of the Mickey Finn are a little obscure. One theory dates back to around 1870 when a discredited Scottish chemist, Michael Finn, on the run from Scottish law, was working as a bartender in the San Francisco area. The story goes that ships' captains were in search of new crew

members to replace those who had deserted in favour of trying their luck in the gold rush. Sailors drinking in Finn's bar would find themselves suddenly incapacitated, making them easy to 'shanghai' onto ships. This theory was put forward by Vincent J. Mullins, an attorney in San Francisco, but he had no documentation to support his claim.

An alternative origin story is described by Herbert Asbury, an American journalist who specialised in writing about crime and the criminal underworld.* In his book *Gem of the Prairie: An Informal History of the Chicago Underworld*, he told the tale of Mickey Finn, owner of the Lone Star Saloon in Chicago's Whiskey Row.† Finn had a history of pickpocketing, but had progressed to running a bar and employing several women (described as 'house girls' and 'streetwalkers') who aided and abetted the thieving from his patrons in return for a percentage of the takings. He had also purchased a large bottle of 'a sort of white stuff' to help the process along.

The women's job was to encourage the saloon patrons to drink one of two specially concocted knockout drinks. The first, the 'Mickey Finn Special', was a mix of raw alcohol and 'white stuff' with water. 'Number Two' was beer with added 'white stuff' and a dash of 'snuffwater'.‡ The drinkers would be rendered unconscious and easier to carry into a back room. Having been systematically stripped of any valuables, their inert body would then be left in an alleyway.

Some people have thought that having unconscious patrons lying around in bars, back rooms and alleyways would look suspicious, or at least off-putting to other potential drinkers. Instead of knocking them out, it has been suggested that a laxative was added to the drinks, but I'm not sure this would be any better for evading the scrutiny of the authorities or enticing people in for more drinks.

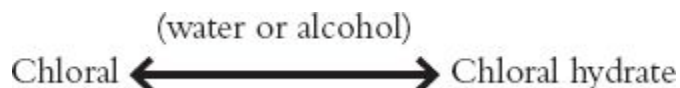
Asbury's information about the Lone Star came from 'Gold Tooth' Mary Thornton, one of the house girls at the saloon. Thornton's word may or may not be reliable, but the Lone Star was a real place that had opened its doors in 1896. The saloon's owner, Mickey Finn, was a real person whose first known appearance in Chicago was as a lushworker (someone who robs drunks) in 1893 at the World's Fair.

The ‘white stuff’ he was said to mix in his special drinks could be a lot of things. Assuming it was used to knock out his patrons and speed them towards the back room rather than the bathroom, it could well be chloral hydrate. Chloral hydrate is a white solid that has certainly been slipped into drinks with the intention of robbing the drinker, and by criminals on both sides of the pond.

Whatever the truth might be regarding the activities in the Lone Star Saloon, Mickey Finn has become a common term for administering chloral or chloral hydrate in an alcoholic drink with the intention of ‘knocking out’ the drinker. Chloral is not an abbreviation of chloral hydrate; they are different compounds. Christie used both names in her stories when talking about a substance that produced sudden unconsciousness in anyone who consumed it. This was not a mistake, however, as the two compounds are closely related and will have the same effect when added to an alcoholic drink. Chloral hydrate is a white solid, but chloral is a liquid and may be the origin of the phrase ‘knockout drops’ that were added to alcoholic drinks to create a ‘Mickey Finn’.

Chloral hydrate

Chloral is an oily liquid at room temperature with a pungent and irritating smell. When it is added to water or alcohol it undergoes a chemical reaction to form chloral hydrate. That chemical reaction can be reversed when solid chloral hydrate is dissolved in water or alcohol.



All the time water is present, there will be an ongoing chemical back-and-forth between chloral and chloral hydrate. So when Mickey Finn dissolved his solid chloral hydrate to his patron’s drinks, the unsuspecting customer would have swallowed both chloral and chloral hydrate.

Chloral and chloral hydrate were both discovered in 1832 by Justus von Liebig (1803–1873), a German scientist regarded as one of the greatest chemistry teachers and researchers of all time.* It would never have occurred to him to test the sedative or hypnotic properties of these

compounds and they were regarded as little more than chemical curiosities. However, he did note that after adding an alkali to chloral hydrate it decomposed into chloroform and formic acid, an observation that would become very important later on.

In 1847, chloroform was introduced as an anaesthetic by Scottish obstetrician James Young Simpson (1811–1870). Inhaling chloroform vapour could induce anaesthesia but it was difficult to get the dose right. Oscar Liebreich (1839–1908), a professor of pharmacology in Berlin, thought chloral hydrate might provide a better alternative. Knowing that chloral hydrate could be broken down into chloroform using an alkali, he thought the human body could provide that alkali to produce a reliable anaesthesia.

In the days before the pH scale and accurate methods of measuring acidity and alkalinity, it was commonly believed that blood was alkaline. In fact it is very close to neutral.[†] The confusion may have occurred because blood would turn litmus paper blue, usually an indication of an alkaline substance, but litmus actually turns blue under slightly acidic conditions.[‡] However, believing blood to be alkaline, it followed that if chloral hydrate was injected into the bloodstream it would be converted into chloroform, which would then produce an anaesthetic effect. According to Liebreich's thinking, there would not be enough alkali in the body's blood to convert the whole dose of chloral hydrate in one go. But he hypothesised that the body would make more alkali as the amount was depleted by the continual conversion of chloral hydrate. In this way, chloral hydrate would give a sustained anaesthetic effect over a long period of time as it was slowly converted into chloroform.

When Liebreich tested chloral hydrate on frogs, rabbits, dogs and later humans, his theories appeared to be confirmed. The drug did indeed cause individuals to fall unconscious in much the same way as they did under the influence of chloroform, and the sedation lasted several hours. Liebreich published his findings in 1869. Chloral hydrate was cheap to manufacture and produced a reliable sedation in patients. It was quickly and enthusiastically incorporated into regular medical practice.

Experiments were conducted in attempts to confirm Liebreich's theory but, try as they might, tests repeatedly failed to find any chloroform in the

blood or breath of animals given chloral hydrate. This apparent failure could be explained by the limitations of analytical techniques at the time, but more sensitive and sophisticated techniques later showed that there had been no chloroform to detect. The creation of the pH scale and accurate methods of measuring pH also revealed that blood is not sufficiently alkaline to cause the conversion of chloral hydrate to chloroform. In fact, chloral hydrate does go through a few chemical changes in the body, but chloroform does not appear at any point in the process. Liebreich had got the results he wanted, but not for the reasons he had put forward.

To administer solid chloral hydrate, whether by injection or ingestion, the drug will first be dissolved in water. This will produce the mix of chloral and chloral hydrate spoken about earlier. There are also a few other compounds produced at the same time and some of these are acidic. These compounds can cause stomach irritation, the most commonly reported side effect of swallowing the drug, and especially when chloral hydrate is taken on an empty stomach. Other compounds are deliberately added to counteract this acidity when the drug is injected.

After swallowing, chloral, chloral hydrate and the other compounds in the mix will be readily absorbed into the body proper from the gastrointestinal tract. Injecting the drug simply introduces these compounds into the tissues and bloodstream directly. It is in the blood and tissues that chloral is converted into trichloroethanol. This conversion is carried out in minutes by alcohol dehydrogenase, an enzyme that also metabolises the alcohol in alcoholic drinks. Trichloroethanol is broken down by another enzyme into inactive trichloroacetic acid and then more compounds down the line that are finally excreted. But this subsequent conversion and elimination takes a lot longer.



When a therapeutic dose of chloral or chloral hydrate is swallowed, sleep occurs usually within an hour, by which time virtually all of the chloral will have been converted into trichloroethanol. The effects of the drug are maintained for between four and eight hours, roughly the time period needed to start breaking down trichloroethanol into other inactive

substances. It is, therefore, trichloroethanol that is believed to cause most of the sedating effect.

Trichloroethanol acts similarly to the tranquillisers and barbiturate drugs we met in Chapter C by interacting with the same receptors on nerve endings. To recap, receptors for the neurotransmitter GABA are concentrated in nerves in areas of the brain that influence mood, breathing and sleep. When GABA docks at these receptors it reduces the activity of the nerve by making it more difficult for the nerve to fire. Trichloroethanol docks at the same receptors and increases the effect of the GABA molecules, making it even harder for the nerve to send its signal. Overall this produces a 'depressant' effect or sedation. Trichloroethanol, however, does not have the same effect on mood as tranquillisers and some of the other drugs that also dock at GABA receptors.

As discussed in Chapter C, taking more than one drug that docks at GABA receptors at the same time produces an enhanced effect, greater than the sum of the individual drugs. So adding chloral hydrate to an alcoholic drink, as in a Mickey Finn, produces sudden unconsciousness within half an hour of ingestion and a greater sedation. Drinks like beer, brandy and sherry are also very effective at masking the unpleasant flavour of chloral hydrate. In the early days of the drug's use, some doctors preferred this method of administration as their patients were less likely to complain about the taste of their medicine and sleep arrived more quickly.* Criminals also took advantage of the combination as their victims were far less likely to be alerted to anything untoward happening to them.

In Christie's novel *Passenger to Frankfurt*, Sir Stafford Nye knows his glass of beer has had something added to it to make him fall asleep, but comments that 'It tasted much the same as it had tasted before'. Fortunately he wakes up again with nothing worse than a hangover. In *The Seven Dials Mystery*, Gerry Wade has chloral slipped into his after-dinner whisky and soda. Like Sir Stafford, Gerry also fails to notice any unusual flavour but, unlike Sir Stafford, Gerry fails to wake up again. Swallowing chloral, and in particular chloral added to alcohol, is a risky thing to do.

How chloral hydrate kills

In the early days of chloral hydrate's medical use, its lethal effects were thought to be due to interactions with the heart and circulatory system. The death of Mrs Vandemeyer in Christie's 1922 thriller *The Secret Adversary* is attributed to an overdose of chloral, although it is also described in slightly more technical terms as 'heart failure induced by an overdose'.

Chloral and chloral hydrate are members of a family of compounds known as halocarbons because they contain halogen atoms (such as the chlorine atoms in chloral hydrate, see Fig. M.1) on a carbon backbone. These compounds have a huge range of uses from adhesives to anaesthetics, and plastics to solvents. Inhaling some of these compounds can cause something called cardiac sensitisation, which can result in sudden death because some parts of the heart are beating irregularly or out of sync with other parts.

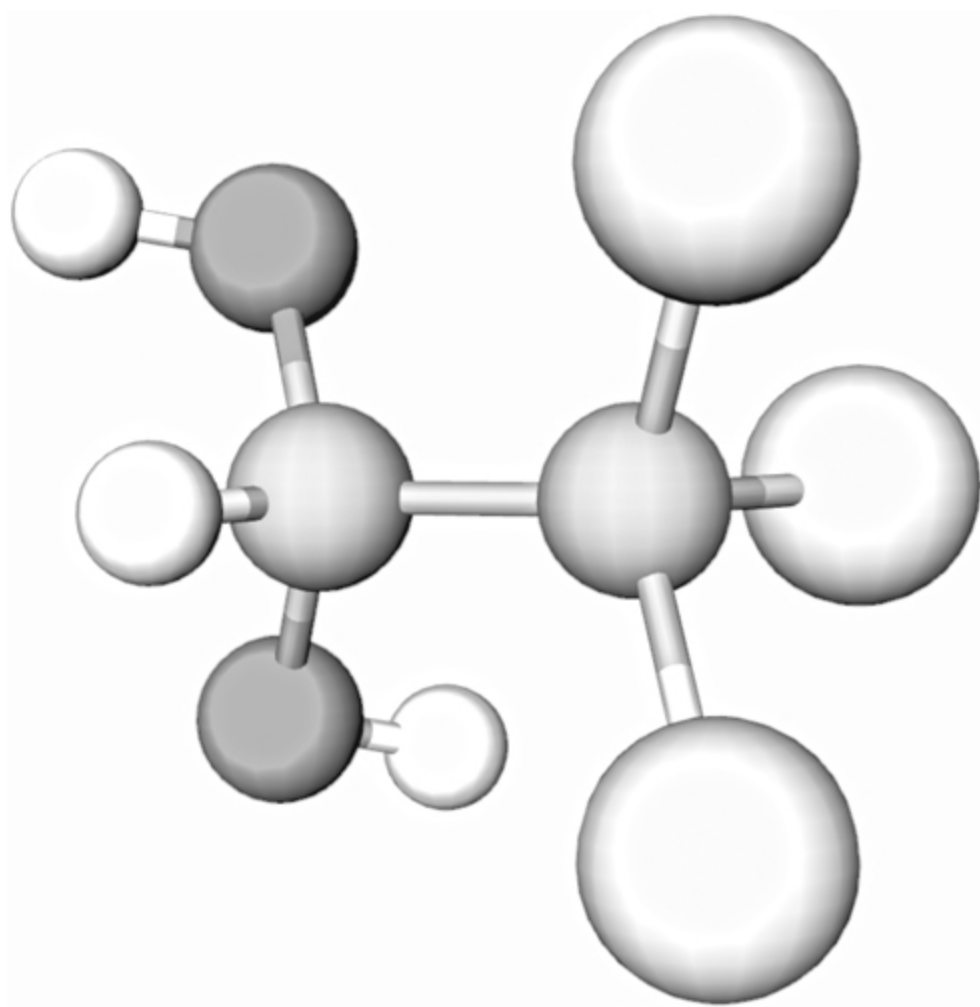


Fig. M.1. Structure of chloral hydrate with three chlorine atoms, the large white atoms on the right side of the molecule, bonded to a single pale grey carbon atom.

Though the exact mechanism for this cardiac sensitisation isn't known, many theories have been put forward, including interactions with the sodium, potassium and calcium channels. The movement of sodium, potassium and calcium through these channels enables nerves to fire and muscles to contract (see earlier chapters). Another theory suggests that these halocarbons can enhance the effects of the body's natural hormone adrenaline (aka epinephrine). Adrenaline stimulates the heart, among many other roles, and in excess can cause such severe disruption to the normal heart rhythm that it can kill (see Chapter P, page 221).

It wasn't long after the introduction of chloral hydrate as a sedative that doctors learned not to give it to anyone with 'a weak heart' or older people. Those suffering from heart failure, a condition that becomes increasingly common with age, are particularly vulnerable to chloral hydrate's effects on the heart. Undoubtedly many deaths were avoided by prescribing chloral hydrate more carefully, but deaths still occurred and not because it had been given to patients with undiagnosed heart conditions. The same interactions with GABA receptors that make it a useful drug can also make chloral hydrate, or its trichloroethanol metabolite, a dangerous poison.

A regular therapeutic dose of between 0.5 and 2g (0.02–0.07oz) of chloral hydrate would be expected to increase the depressive effect of GABA to the point where an individual falls asleep. However, taking more than just 3g (0.11oz) in a single dose can potentially push that depressive effect to dangerous levels. On average, a dose of 10g (0.35oz) is considered enough to slow breathing until it stops, but individuals respond very differently to the effects of chloral hydrate. Some have survived swallowing more than 30g (1.06oz), while others have died after taking only 4g (0.14oz). With an already very narrow margin between the therapeutic and lethal dose, combining the drug with alcohol only increases the chances of a fatal result. And there are yet more problems associated with the use of chloral hydrate, as illustrated by a 54-year-old patient in Canada.

Mr A had a few underlying health issues, but the most relevant to subsequent events were obstructive sleep apnoea, a sleeping disorder where the upper airways become partially or completely blocked for brief periods,

and a history of depression. In 2001, two weeks after losing his job, he was talking and having a few drinks with friends online. During the call he thanked them for everything they had done for him. Worried by his uncharacteristically emotional state, his friends decided to contact his local emergency services, who found Mr A unconscious in his bathroom with empty bottles of chloral hydrate nearby. He had a cardiac arrest while the emergency responders were present and he was defibrillated before being taken to hospital. Thankfully, Mr A recovered and was able to tell how he had been prescribed chloral hydrate by his family physician and given a three-month supply. But, finding it increasingly ineffective, he had resorted to taking larger and larger doses.

Adults who are prescribed chloral hydrate to alleviate sleeplessness are usually only given a two- to seven-day supply for several reasons. First, chloral hydrate is only effective short term. After a week or so of regular dosing, larger quantities are needed to achieve the same sedative results. Added to this, chloral hydrate can be addictive and the longer it is taken the more likely an addiction is to develop. Anyone with a history of addiction, particularly to alcohol, should not be prescribed chloral hydrate at all. Prescribing chloral hydrate to individuals with a history of depression is also not recommended. Just a one-week prescription can contain enough chloral hydrate for a potentially fatal 10g (0.35oz) dose.

The risks associated with chloral hydrate were recognised from the start. Since its first introduction into medicine back in the nineteenth century, chloral hydrate could only be obtained in the UK on medical prescription,^{*} something Agatha Christie was well aware of. In her 1937 novel *Dumb Witness*, the author goes into some depth about the use of chloral hydrate and how it had to be carefully prescribed. The drug is described as ‘a narcotic, and a hypnotic’ and that it is ‘Used to alleviate pain and as a sleeping draught’. This is slightly inaccurate, as chloral hydrate does not alleviate pain, but it can sedate someone while they go through a painful procedure.

Acquiring knockout drops is easy for one of Christie’s killers because they are a doctor. And though doctors are ‘statistically the most homicidal profession’ in Christie’s writing, having them be the killer every time chloral or knockout drops are mentioned would make it far too easy for fans

to spot whodunit. Christie had to be very inventive to explain how so much chloral could be acquired so easily in so many of her stories, and that is something she excelled at.

The character in *Dumb Witness* forges a doctor's prescription to get hold of enough chloral hydrate to take their own life. The murderer in her 1939 novel *And Then There Were None* is prescribed chloral hydrate to help them sleep while suffering from bouts of pain. They are able to suppress the urge to take the drug and keep it for their murderous intentions. Knowing the dangers of the drug, the doctor may not have prescribed enough for a single lethal dose, but the murderer is nevertheless able to make their stocks of chloral hydrate stretch to two victims. Their first victim had already been given one sedative, so a relatively small addition of chloral hydrate might have been enough to kill them. Perhaps to make sure, or perhaps because it happened to be convenient, the murderer dissolves the drug in brandy, masking the flavour and adding a third depressant drug, alcohol, into the mix. The victim goes to sleep and does not wake up the next morning and the murderer still has enough chloral hydrate left over for their second victim. The dose for the second victim also does not need to be a large one as it is only intended to sedate the victim enough not to notice a lethal injection of cyanide.

Christie used chloral hydrate in plenty of other plots without stating how it was obtained. Characters like Mrs Vandemeyer in *The Secret Adversary*, and Jimmy Thesiger in *The Seven Dials Mystery*, both 1920s thrillers, are professional criminals in the pay of big crime bosses and presumably have alternative illegal sources of things like chloral hydrate if they can't get hold of it on prescription. Similarly, unorthodox sources could be where the mysterious Mary Ann, a spy operating under various pseudonyms in *Passenger to Frankfurt*, obtained her chloral.

Going into the details of how criminals got hold of their lethal substances every time one is used would soon get repetitive, particularly in Agatha Christie's poison-heavy plots. But whatever cunning ruses were used, they must have been convincing in order to obtain the huge quantity sloshing about in *Sleeping Murder*. The murderer in Christie's 1976 novel adds enough knockout drops to a decanter of brandy that just a few sips would be enough to kill. But what seems like a certain death is thwarted by

the lucky presence of a police surgeon. It may have been ‘touch and go’, but Mrs Cocker’s gasps after drinking the poisoned brandy aren’t her last.

The novel doesn’t specifically say that the ‘knockout drops’ in the decanter are chloral. It is, however, a reasonable assumption based on the common association between knockout drops and chloral and Mrs Crocker’s rapid collapse after drinking the spiked brandy. Neither does it say how the doctor saves Mrs Crocker, but again we can make some educated guesses as to what he would have needed to do to save her.

Is there an antidote?

Today, the immediate response to someone whose breathing and heartbeat is disrupted would be cardiopulmonary resuscitation (CPR). As we saw earlier in the chapter, defibrillators have also been used successfully to resuscitate someone after a chloral hydrate overdose. But these techniques were only just becoming commonplace in the 1960s. Though *Sleeping Murder* was published in 1976, it was actually written in the 1940s and set in 1944.* Resuscitation techniques were known at this time but they were less efficient. Whatever the doctor did to initially stabilise Mrs Crocker, he then seems to have left her alone to sleep it off. This would also not happen today as the patient’s best chance of survival is to be transferred to a hospital where they can be monitored and given intensive medical support as needed.

For example, in 2019, a woman was found unresponsive at home. She was taken to hospital, where an initial examination showed nothing remarkable other than her obvious deep sedation. She was intubated and mechanically ventilated to ensure enough oxygen was reaching all her cells and that waste carbon dioxide was removed from her blood. Gastric lavage was carried out by inserting a tube down her nose to wash out whatever might be in her stomach. Though her pulse rate was not out of the ordinary, an electrocardiogram showed her heart was beating in an irregular way. In the following hours she had to be repeatedly revived using cardiopulmonary resuscitation. The standard protocol of drugs used to treat irregular heart rhythms did not help.

Empty packets of pills found next to the woman's bed suggested she had swallowed an overdose but blood tests showed none of these drugs were present in her system. The packets of pills by her bedside were not the culprits and her blood showed no signs of any other drug usually taken in an overdose. Only consultation with a toxicologist brought about the suggestion of chloral hydrate poisoning because of the combined sedation and cardiac effects she was experiencing. Even though there were no means of testing for chloral hydrate at the hospital to confirm the theory, treatment was started with beta blockers. The woman responded well and her heart rhythm slowly returned to normal. She was able to confirm what the toxicologist suspected: that she had swallowed chloral hydrate, somewhere between 30 and 35g (1.06 and 1.23oz) of it. Five days later she was discharged.

Beta blockers block some of the receptors for adrenaline and other related hormones that stimulate the heart. These drugs are used to treat irregular heart rhythms and protect the heart from further heart attacks as well as lowering blood pressure. They are considered one of the most important classes of drugs in the twentieth century. They would seem an obvious method of treatment for Mrs Crocker in *Sleeping Murder*, but they weren't introduced into medical use until the mid-1960s, after the book was written.

Other patients have survived chloral hydrate overdoses after treatment with flumazenil, one of the benzodiazepine family like Valium. Flumazenil also binds to the GABA receptors, just like chloral hydrate and other benzodiazepines, but unlike them it blocks rather than enhances the effects of these drugs. It can, therefore, be used as an antidote for overdoses of benzodiazepines and sometimes chloral hydrate. But, again, the doctor in *Sleeping Murder* would not have had access to this drug as it was not invented until 1981.

Mrs Crocker would have been lucky to survive if simply left alone to sleep. Perhaps Christie wanted to skip over the boring details of being taken to a hospital where medical staff could give the intense support Mrs Crocker needed to restore her back to health. Omitting certain information does not mean Christie was ignorant of it. In terms of the plot, it was more important that the doctor return to the scene of the crime and sniff out the suspicious substance in the brandy decanter and emphasize just how

dangerous it was to drink. Christie was not exaggerating those dangers. There were many accounts of the potentially fatal outcome of knockout drops dissolved into drinks, and not just in the detective fiction she was fond of reading.

A real-life case

On 27 February 1889, the Manchester morning papers printed reports of an unknown man found dead in a hansom cab in the middle of the city the previous evening. Readers were gripped by the incredible similarities between the real-life 'Manchester Cab Mystery' and the fictional *The Mystery of a Hansom Cab*, a bestselling Australian novel published in the UK in 1887.*

The novel has a hansom cab pull up at a police station in Melbourne with an unknown dead man inside. The driver tells the story of how he was flagged down by a man supporting another very intoxicated gentleman. He was instructed to drive the drunk home and the companion got into the cab too, but got out again part way through the journey. When the driver checked on his drunk passenger he found him slumped in a corner with a handkerchief tied over his face. He had died of a combination of alcohol and chloroform, working together on a heart already weakened by 'fatty degeneration'.

In the real-life UK case, paper manufacturer John Fletcher was in Manchester on business. He left his Manchester office at lunchtime, saying he was going to an auction, but instead seems to have met up with a few friends in several drinking establishments around the city. By mid-afternoon he was described as partially drunk. At 6.20 p.m., Fletcher got into a cab with a young man and the pair were driven to a pub, The Three Arrows, where they each had a glass of beer. They re-entered the cab at 7.10 p.m. and were described by the driver as quite sober and walking steadily.

At about 7.25 p.m., someone shouted to the driver that his passenger had jumped out and run. The door of the cab was open and Fletcher was alone inside with his head fallen forward on the front seat. The driver managed to rouse his remaining passenger but was told to 'Go away and leave me alone.' Worried he wasn't going to be paid, he asked a policeman to help

him extract his fare from the now-unconscious man. Thinking the man was merely intoxicated, the constable climbed into the cab and told the driver to drive to the police station. But during the journey, the constable became worried and told the driver to go to the Manchester Royal Infirmary instead. Fletcher died just before entering the hospital at 8.05 p.m.

Examination of Fletcher's body revealed all his money and valuables were missing. There were no signs of violence, but the house physician noted a very pale appearance and strong smell of alcohol. The same strong smell of alcohol was noted by the pathologist when he first opened the body for his post-mortem examination. There was no sign of disease in the body and, though there were signs of 'fatty degeneration' of the heart, this was not considered serious enough to have caused his death. Samples were therefore taken for chemical analysis.

Trace amounts of chloral were found to be present in Fletcher's stomach contents and the upper section of his small intestines, but none in any other body fluids – unsurprising, given how rapidly it is broken down in the body into trichloroethanol. Though the word 'trace' was used by the analyst, there was no doubt in the pathologist's mind that Fletcher's death was due to the combined effects of alcohol and chloral on an enfeebled heart. The pathologist pointed out that the analyst was never asked to estimate the total amount of chloral given to Fletcher, only to find its presence in the body. To do this he had carried out three separate chemical tests on each of the three samples taken from the body, suggesting there was rather more present than a mere 'trace'. It became the first documented case of murder by Mickey Finn, at least in the UK.

While the chemical analysis was being carried out, an investigation began into what appeared to be a robbery and would soon become a murder case. Armed with a vague description of the young man seen in the cab and drinking with the victim in The Three Arrows, Detective Jerome Caminada started his investigation. Caminada has been called Manchester's real-life Sherlock Holmes, and his speed and the methods he used to track down Fletcher's mysterious companion show why.

A clue led Caminada to the Locomotive Inn, a venue frequented by boxers, which started him thinking about boxers he knew. He remembered 'Pig Jack', the pseudonym of a fighting man who had once run a beerhouse where he was said to drug his customers by putting something in their beer.

Pig Jack had a son, Charles Parton, who fitted the description of the young man seen drinking with the deceased. Parton had disappeared from his home address the day after Fletcher was found dead, adding to Caminada's suspicions, but he was still able to track him down to another address. At 12.30 on the morning of 2 March, less than four days since Fletcher had died, Caminada arrived at the house where Parton was hiding out and arrested him.

By the time Parton was tried in court for murder and robbery, a considerable case had been built against him. Parton was identified by a Liverpool chemist as the man who had stolen a 0.45kg (1lb) bottle of chloral from him a few weeks before Fletcher's death. He was also seen in The Three Arrows, where he was spotted dropping a liquid out of a small bottle into a glass of beer. Furthermore, a grocer and a railway porter recalled that, on separate occasions, they had been drinking with Parton and woken up later in the evening to find a large gap in their memory and their valuables missing.

There seemed to be a pattern of slipping chloral into a drink in order to rob the victim but, in the case of Fletcher, the dose of chloral had been too much. Parton may not have intended to kill his victim, and simply not have realised how much alcohol Fletcher had already drunk over the course of the day. Nevertheless, he was found guilty of murder, but with a strong recommendation to mercy owing to his youth. A sentence of death was commuted to penal servitude for life.

Agatha and chloral hydrate

Mickey Finns may not play a major role in any single Agatha Christie story, but her many casual references to the drug, in all its guises – chloral, chloral hydrate, Mickey Finns, knockout drops – mount up to a broad, detailed and accurate knowledge of the drug. This knowledge must have come from a range of sources. Her training in dispensing would have given her the technical details about the drug and its availability, while her vast knowledge of detective stories and real-life cases would have given her insights into how these drugs could be and had been used for criminal purposes.

Snippets of information are scattered everywhere in Christie's writing. The difficulties of getting hold of chloral hydrate are addressed in *Dumb Witness* and in *And Then There Were None*. The distinctive smell of chloral is noted in *The Secret Adversary*. That the flavour of these drugs can be masked by alcoholic drinks and that their effects can be rapid and sudden are shown on several occasions. *Why Didn't They Ask Evans?* and *And Then There Were None* show that at different doses, or mixed with alcohol, the same drug can both incapacitate and kill. Its effects on breathing and the heart are noted in subtle ways that explain what is going on in *Sleeping Murder* and *The Seven Dials Mystery* but keep the plot moving. Christie's well-known skill at hiding clues in plain sight also extended to hiding scientific information about drugs and poisons in a way that readers never feel that they are in a science lesson. It is an impressive skill.

Notes

- * I have also, undoubtedly, missed a few examples.
- * His book *The Gangs of New York* was the basis for Martin Scorsese's 2002 film.
- † Whiskey Row was roughly two blocks of the west side of State Street, Chicago, where for 30 years every building and storefront was occupied by a saloon or gambling house.
- ‡ I don't actually know what 'snuffwater' is but I am guessing it is water that has had snuff soaked in it for a nicotine hit rather than the more sinister meaning of 'snuff'.
- * One of his many scientific achievements was developing a manufacturing process for beef extract, the forerunner of the Oxo cube.
- † Healthy blood has a pH in the very narrow range of 7.35 to 7.45. For comparison, water has a neutral pH of 7.
- ‡ The standard pH range for an acidic solution is between 1 and 7, and alkaline between 7 and 14. Litmus turns blue above pH 5.3.
- * It was also sometimes given in combination with bromide powders, another sedative drug, to induce sleep within half an hour.
- * The prescribing of chloral hydrate has only become more restricted over the years. Today, it is most commonly used to sedate paediatric patients for non-surgical procedures.
- * Christie kept herself busy during the Second World War by working in a hospital dispensary. In her downtime she wrote a tremendous amount, enough to put two novels aside so the royalties could go to her family if she were killed. *Sleeping Murder* and *Curtain* were stored in a bank vault and heavily insured against destruction. *Curtain* was published in 1975 and *Sleeping Murder* was published posthumously in 1976.
- * It was so popular it outsold Arthur Conan Doyle's *A Study in Scarlet*, the first Sherlock Holmes novel, also published in 1887. The author, Fergus Hume, saw none of the profits from his work, having sold his rights to the book for £50.

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‘The Chocolate Box’ (Poirot’s Early Cases)

Agatha Christie’s carefully contrived crimes would be an infuriating read if she hadn’t created some brilliant detectives to solve them. She invented not one, but two, of the most well-known fictional detectives – Hercule Poirot and Miss Jane Marple.* The eccentric but endearing Hercule Poirot was introduced to the world in Christie’s first novel, *The Mysterious Affair at Styles*, as a retired Belgian police officer and refugee from the First World War. He continued bringing criminals to justice in 32 more novels and 51 short stories, occasionally helped and hindered by his faithful sidekick

Captain Hastings. His second career as a private detective spanned over 50 years.* He possessed perhaps the most famous and magnificent moustaches in fiction and a formidable set of little grey cells that enabled him to solve seemingly insoluble crimes. 'The Chocolate Box', a short story written by Christie and published in *The Sketch* in 1923, is a tale recounted to the dependable Hastings as an example of how even the great Hercule Poirot could be 'thirty-six times an idiot!'

Back in 1893, when he was still a young detective in the Belgian police force, Poirot was asked to investigate the death of one of Belgium's prominent politicians. Paul Déroulard lived with his mother, Mme. Déroulard, and Mlle. Virginie, his late wife's cousin. On the evening he died, the politician had invited a neighbour, M. de Saint Alard, to dine with him and an English friend, Mr John Wilson, who was also staying at Déroulard's house.

After dinner the three men retired to the study for a little friendly conversation. Having shown no sign of agitation or ill health, Déroulard's face suddenly flushed bright red and he collapsed on the floor. He was dead before a doctor could arrive at the house to treat him. The official cause of death was given as heart failure but the young Mlle. Virginie was not convinced and persuaded Poirot to investigate. If Déroulard was murdered, the circumstances surrounding his death suggested only one possible method: poison.

A search of Déroulard's study revealed nothing out of place except a box of chocolates with the wrong-coloured lid. Déroulard was particularly fond of chocolates. In fact, he was the only person in the house who was. Anyone who knew this and wanted Déroulard dead would have the perfect way to give him poison without risking the health of the whole household. To find out what poison might have been used, Poirot visited the local pharmacies to see if any suspicious substances had been sold recently.

Two potential poisons were brought to Poirot's attention, both of them medicines. The first, a long-standing prescription for atropine eyedrops, had been recently refilled for Mme. Déroulard. Atropine can certainly kill, but the symptoms would be nothing like those displayed by M. Déroulard (see Chapter S, page 245). The second prescription is more intriguing. Just a day before Déroulard's death, Mr Wilson, his English guest, called at the

English pharmacy for more trinitrine tablets. But the pharmacist assures Poirot that trinitrine is not a poison.

Through a conversation between Poirot and the pharmacist, we learn that trinitrine tablets are prescribed to lower blood pressure and ‘It is given for some forms of heart trouble – angina pectoris for instance.’ The drug could also cause a very telling side effect – a flushed face, like that of M. Déroulard just before he died. Trinitrine may not be considered a poison, but the pharmacist warns against taking too many tablets, something that might easily be done as they are tiny and temptingly made out of chocolate.*

Had Mr Wilson deliberately obtained an extra bottle of trinitrine tablets, found a box of Déroulard’s chocolates (there were several in the house), scooped out the fillings and refilled them with lots of tiny trinitrine tablets? Poirot estimated 20 to 30 tablets could be hidden inside the chocolates in this way. Wilson could have then left the chocolates out for Déroulard to devour later, and simply failed to notice he had put the wrong lid back on the box. Alternatively, someone else stole Mr Wilson’s bottle of trinitrine tablets to poison Déroulard and Wilson had gone to the pharmacy to replace his missing tablets.

Plenty of people have poisoned chocolates to try and commit murder. They are often unsuccessful, however, because many poisons have an unpleasant taste that can’t be masked by the chocolate. For example, there is a half-eaten Walnut Whip in a display case in a medical museum in London, a relic from an attempted murder trial. This exhibit of a popular UK confection had been laced with a lethal dose of strychnine, but the intended victim failed to finish the chocolate because of the foul taste.* Trinitrine is one of the few poisons that might have gone unnoticed in such a situation because not only does it taste sweet, but it is also the only medicine to have been officially prepared in chocolate tablets. All the murderer had to do was make sure Déroulard ate enough of them.

Again, Christie was drawing on her extensive pharmaceutical knowledge. *The Art of Dispensing*, a book she likely studied for her apothecaries’ assistants’ exams, contains a section on trinitrine tablets (*official tabellæ*), along with a couple of recipes for making them. Each tiny tablet weighed five grains (0.324g/0.011oz) and contained just one-

hundredth of a grain (0.00065g/0.00003oz) of trinitrine, though the book uses its more common name, nitroglycerin.

The nitroglycerin story

In 1846, an Italian chemist, Ascanio Sobrero (1812–1888), took ordinary glycerine, the sweet-tasting viscous liquid commonly used in the food industry, and mixed it with a combination of strong sulphuric and nitric acid. The oily liquid that he obtained also had a sweet taste, but Sobrero warned anyone wanting to test this property for themselves. Just a trace of this new compound placed on the tongue produced ‘a most violent pulsating headache, accompanied by great weakness of the limbs.’ It was toxic, but not very toxic. A large dose given to a dog caused it to foam at the mouth and vomit, which removed most of the compound from its stomach. Nevertheless, within minutes the animal had collapsed on the floor and almost stopped breathing. The dog remained whining and trembling for two hours, but it survived. Similar results were obtained with other animals.

The effects of the new compound on the body were interesting but it was another property that dissuaded Sobrero from further experimentation. Sobrero’s new substance had an alarming tendency to explode violently, unexpectedly and at the least provocation. When he announced his discovery to the Academy of Sciences in Turin, Italy, he named it ‘piroglycerina’, but it became more widely known as nitroglycerin.

Despite his warnings, not everyone was put off experimenting with what was obviously a highly unstable and dangerous compound, including one of Sobrero’s own students, a young Swede named Alfred Nobel (1833–1896), and Alfred’s father Immanuel Nobel (1801–1872). Immanuel found that nitroglycerin was a far more powerful explosive than gunpowder, the only other explosive material available at the time. It had enormous commercial potential. All he had to do was figure out how to control the explosions.

Immanuel’s answer was to cool the reaction between the glycerol and the two acids, but the final product was still so prone to exploding that it was too dangerous to transport any distance. Immanuel therefore sold licences for his new process so that nitroglycerin plants could be set up close to the quarry or tunnelling projects where it was to be used. Behind the scenes,

Immanuel and his sons Emil and Alfred continued to experiment with ways to make nitroglycerin safer.*

Frequent unplanned explosions at their armaments factory in Stockholm, Sweden did not deter the Nobels. Buildings at one of their factories in Germany were also destroyed, twice. One explosion in Stockholm killed five people, including Emil Nobel (1846–1864), and the local authorities banned the Nobels from working within the city limits. Still Alfred Nobel would not give up. He set up laboratories on barges and boats, where he mixed nitroglycerin with different substances to try and tame it. He tested charcoal, sawdust and cement, but kieselgurh, a very absorbent clay-like material, was the best. The kieselgurh soaked up the liquid nitroglycerin to form a paste that was not only safe to transport, but could also be easily moulded and still gave a powerful explosion when detonated. Alfred called it dynamite. It made his fortune.*

In 1867, the same year Alfred Nobel patented dynamite, Scottish physician Sir Thomas Lauder Brunton (1844–1916) noticed the beneficial effects of using organic nitrites in treating angina. He originally studied amyl nitrite, a compound Agatha Christie was also aware of and used in two of her novels – to treat the heart problems of an ageing Hercule Poirot in *Curtain*, and to kill one of the victims of one of the many murderers in *And Then There Were None*. Although amyl nitrite is effective at relieving an angina attack, it is most effective when inhaled, making administering it and dosing more difficult. Nitroglycerin produced the same beneficial effects when tiny doses were given by mouth, and it soon became the treatment of choice for angina.

The nitroglycerin story came full circle when Alfred Nobel, who had suffered from poor health throughout his life, developed angina in his later years. He was prescribed the very compound that had made his fortune, but he refused to take it, knowing the terrible headaches nitroglycerin had given him during his work. He also steadfastly refused to believe it could alleviate his chest pain. He was not the only person to have doubts about nitroglycerin. Doctors and pharmaceutical companies worried that their patients would be put off placing a known explosive in their mouth, so nitroglycerin tablets were renamed trinitrine.

Nitroglycerin

Glycerine is a simple three-carbon molecule. Radiating out from each of these carbon atoms is an oxygen atom followed by a hydrogen atom. When Sobrero added his mix of sulphuric and nitric acid, these terminal hydrogen atoms were replaced by nitro groups, a nitrogen atom bonded to two more oxygen atoms (see Fig. N.1).

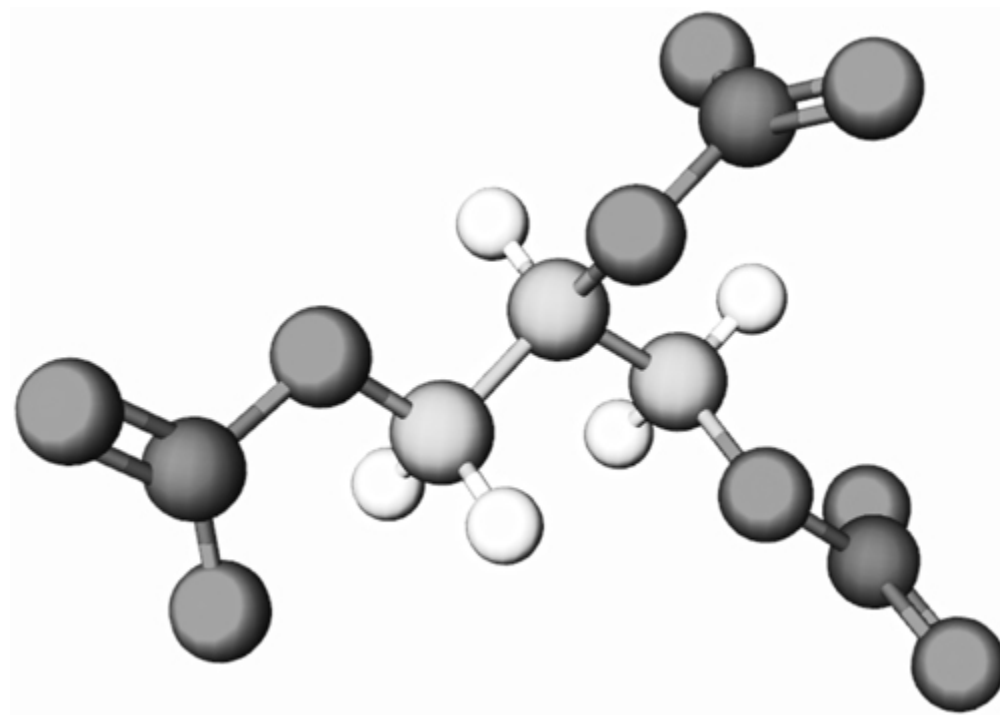
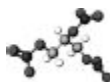


Fig. N.1. Structure of nitroglycerin. The three pale grey carbon atoms are at the centre of the molecule. Each carbon is bonded to one nitrate group (a nitrogen atom surrounded by three oxygen atoms).

The way the atoms of carbon, nitrogen, oxygen and hydrogen are positioned in nitroglycerin molecules is neat and symmetrical, but it is not the most stable arrangement. Raising the temperature, or a slight knock, causes the nitroglycerin molecules to flex and shake and bump against their neighbours. These disturbances can break weaker bonds and bring atoms into close proximity to others with which they can form stronger bonds. The molecules can fragment and the atoms within them reorder themselves to more stable arrangements – carbon dioxide, water and nitrogen molecules, for example.*

A little bit of energy has to be put in to break a weak chemical bond. Forming new, stronger chemical bonds *releases* energy, and more of it. This energy can then go on to break more weak bonds, so more strong bonds can form, releasing more energy, and so on and so on. Fragmenting one larger molecule into several smaller ones makes everything more disordered, releasing yet more energy. Many of the new, smaller, more stable molecules that are formed are gases that take up more space than the original liquid nitroglycerin molecules. If this process of breaking and remaking happens quickly enough, the combination of releasing a huge amount of energy and the outward expansion of the gases that are formed in the process can create the heat and pressure of an explosion. The disintegration of nitroglycerin molecules, and the rearrangement of the atoms into more stable configurations, can happen so quickly it can cause an explosion with a velocity of 7.8kms^{-1} (17,448mph) and reach temperatures of $5,000^{\circ}\text{C}$ ($9,000^{\circ}\text{F}$).^{*} It is also this ability to fragment that makes nitroglycerin an effective treatment for attacks of angina.



Sudden bouts of chest pain or pressure are called angina and are caused by restricted blood flow to the heart. There can be several reasons for this restriction but the most common is a narrowing of blood vessels from atherosclerosis. A build-up of arteromatous plaque in the inner layer of the artery walls causes constrictions along its length. These changes in the blood vessels can go unnoticed for years, but can eventually build up to a point where they can restrict blood flow, and therefore the oxygen supply, to the heart. Angina attacks can be triggered by strenuous exercise or a particularly stressful event, during which the heart needs more oxygen to pump faster.

To understand how nitroglycerin relieved attacks of angina, clues from several different researchers' work had to be pieced together. Research conducted by biochemist Robert F. Furchgott (1916–2009) showed that there was a chemical being produced in the endothelial cells that line blood vessels. This chemical was the signal for the smooth muscle in the vessels' walls to relax, causing the blood vessel to dilate, but he didn't know what

the chemical was. Pharmacologist Louis Ignarro (1941–), together with Furchgott, proved that this mysterious signalling chemical in the endothelial cells was nitric oxide. Research done by the physician and pharmacologist Ferid Murad (1936–2023) showed that nitroglycerin, and related drugs such as amyl nitrite, fragmented to release nitric oxide in the body.

In 1998, the Nobel Prize in Medicine or Physiology was given to Furchgott, Ignarro and Murad ‘for their discoveries concerning nitric oxide as a signalling molecule in the cardiovascular system’. The compound that had made such a contribution to the funding of the Nobel Prize, and had been prescribed to its founder, had become part of the subject of one of the prizes.

A tiny dose of nitroglycerin placed under the tongue, or amyl nitrite inhaled from an ampoule, will be rapidly absorbed into the blood vessels just under the tongue or in the lungs. In the blood, these compounds fragment to release nitric oxide, which signals for the blood vessels to dilate, giving more space for the blood to flow. The resulting decrease in blood pressure also puts less strain on the heart.

The effects of nitroglycerin and amyl nitrite can be felt within minutes but they are not confined to blood vessels serving the heart. Blood vessels all over the body can be affected, and in the head this can lead to the flushed face, pounding headaches and dizziness that those unused to the drugs, or those who take it in excess, can experience.

Today, pre-prepared blister packs, sprays and patches of nitroglycerin are bought into pharmacies, ready for dispensing. When Christie was writing, those tablets had to be made by hand, which meant handling nitroglycerin in bulk. Though it was easy to prepare nitroglycerin from chemicals a pharmacist was likely to have in stock, doing so required permission from the Home Office and, because of the risk of explosion, *The Art of Dispensing* warned, ‘Permission is not at all likely to be obtained.’ Instead, a 10 per cent solution of nitroglycerin in alcohol could be purchased from wholesalers. But a slip-up with a decimal point during calculations, or a slip of the hand when measuring out the nitroglycerin solution, and customers could be dissolving dangerous amounts of nitroglycerin under their tongues. Taking more concentrated doses of nitroglycerin can be dangerous, as Peter Eberhart found out to his cost.

One cold night in February 1905, Peter Eberhart, a farmhand from St Paul, Minnesota, was out drinking whisky with the town marshal, who showed him a bottle of nitroglycerin that had been confiscated from a bank robber. Peter bet he could drink it without suffering any ill effects and downed the lot. Seemingly unperturbed by his unusual drink, he then began his walk home but failed to arrive. Peter was found dead by the roadside the next morning. His frozen body was collected by the undertaker and placed in an outbuilding by a roaring furnace to thaw him so he could be laid out in his coffin. But the heat of the fire caused the nitroglycerin in his body to explode. The outbuilding was ‘completely wrecked’ and all that was left of Peter was a button, a boot heel and a piece of watch chain.

Whether Eberhart’s death was as a result of the whisky, explosive or extreme cold was never ascertained, but it definitely wasn’t the explosion that killed him. Nitroglycerin does disintegrate inside the body, as the Nobel Prize-winning research showed, but too slowly to cause an explosion. Nevertheless, the disintegration of the molecule can be fatal.

How nitroglycerin kills

Too much nitric oxide, released from nitroglycerin, amyl nitrite or related compounds, can cause the blood vessels to dilate to such an extent that there is a potentially fatal drop in blood pressure. Without a certain amount of pressure in the circulatory system, the heart cannot pump blood around the body and the supply of oxygenated blood from the lungs is disrupted. This causes cyanosis, characterised by a blue or grey colouration in the skin. And nitric oxide has other effects on the blood that can make this cyanosis worse.

As we saw in Chapter G (page 114), oxygen is not simply absorbed into the blood from the lungs. Inside the red blood cells, oxygen molecules bond to iron atoms in haemoglobin. Atoms of iron can lose two or three electrons to form Fe^{2+} (ferrous) or Fe^{3+} (ferric) compounds, but oxygen will only bind to ferrous Fe^{2+} iron atoms in haemoglobin. The nitric oxide released from nitroglycerin, amyl nitrite and other nitrate and nitrite compounds, can remove an extra electron from the iron atoms in haemoglobin to turn them from Fe^{2+} to Fe^{3+} , stopping them from absorbing oxygen from the lungs.

Haemoglobin molecules with Fe^{3+} iron atoms are called methaemoglobin and are a characteristic chocolate-brown colour.

The presence of methaemoglobin in the blood reduces the amount of oxygen that can be transported around the body. Without an adequate supply of oxygen to cells, they die. Cells with the highest demand for oxygen, such as nerves, die first. An inadequate supply of oxygen in nitroglycerin poisoning could be due to low blood pressure, or to a high proportion of the blood's haemoglobin made into unusable methaemoglobin, or a combination of the two. The end result is convulsions, coma, respiratory paralysis and death.

Is there an antidote?

For anyone who has taken too much trinitrine, as with many other toxic substances, the first emergency response is to keep them breathing so there is a supply of oxygen going into the body even if it is not being distributed as efficiently as usual. It is also important to remove as much ingested nitroglycerin as possible by emesis and then get the patient to swallow activated charcoal to soak up any that remains in the stomach. It is also vital to maintain blood pressure, so the heart can continue to pump effectively. This can be done by giving fluids intravenously. If blood pressure is maintained, recovery is likely.

If the patient is still struggling to breathe, it may be due to methaemoglobin in their red blood cells. The proportion of methaemoglobin can be determined by testing, and if more than 30 per cent of the body's haemoglobin has been converted, methylene blue can be administered.

Methylene blue was first prepared in 1876 and has been described as 'the first fully synthetic drug used in medicine'. In 1891, it was recommended as a treatment for malaria and in 1932 it was found to be effective in counteracting carbon monoxide and cyanide overdoses. Later still it was found to be effective in treating nitroglycerin overdoses because methylene blue converts the Fe^{3+} atoms in methaemoglobin back to Fe^{2+} .* Unfortunately, Paul Déroulard was being poisoned with nitroglycerin in 'The Chocolate Box' in 1893, long before anyone would have thought to try and give him methylene blue. In any case, the doctor arrived too late to give

him any sort of treatment. But there are still ways Déroutard could have lessened the lethal effects of the nitroglycerin in his chocolates.

It is possible to build up some tolerance to nitroglycerin, to a certain extent anyway. During the First World War it was noticed that munitions workers complained of thumping headaches on Monday, but by Friday their bodies had adapted to the nitroglycerin in the explosives they were handling. Over the weekend these workers would lose their tolerance and suffer headaches all over again the following Monday. If Déroutard had been munching on just a few poisoned chocolates each night, he might have inadvertently protected himself. But only the chocolates in a newly opened box had been poisoned, and so he apparently wolfed down his lethal dose without any inbuilt tolerance to protect him.

Another way Déroutard could have lessened the lethality of his poisoned chocolates would be to swallow them down as quickly as possible without pausing to savour their flavour. Nitroglycerin is not as easily absorbed from the stomach as it is through the membranes in the mouth. To get the maximum effect when an attack occurs, angina sufferers are advised to suck the pills, or allow them to dissolve under the tongue, rather than swallow them. But the biggest problem with trying to kill Déroutard with an overdose of trinitrine was the difficulty of getting him to swallow enough tablets to deliver a lethal dose of nitroglycerin.

Determining the fatal dose of nitroglycerin by ingestion is tricky and has been estimated at anywhere between 2 and 6g (0.07 and 0.212oz). Such a dose would fit in a teaspoon. It might sound small but most poisoners pick something far more potent. In comparison to the medicinal dose, it is an enormous amount. Trinitrine tablets each contained 0.65mg (0.000023oz) of nitroglycerin, according to *The Art of Dispensing*. Assuming the lowest lethal dose is accepted, Déroutard would have had to swallow something like 3,000 tablets, weighing a total of 900g, or 2lb, (not including the weight of the real chocolates they have been hidden inside).^{*} Trinitrine tablets are small, and chocolate is undoubtedly delicious, but that would take some considerable effort, especially as he had already eaten. But there are other things Déroutard might have consumed that night that could have made the trinitrine tablets much more potent, as a real-life nitroglycerin poisoning demonstrates.

A real-life case

In 1901, an obituary was printed in the 17 January edition of *The Centre Democrat*, notifying its readers in Bellefonte, Pennsylvania that 19-year-old Miss McW had ‘retired in her usual health on Wednesday, but could not be aroused.’ This was far from the truth. Miss McW had suffered 10 hours of convulsions, witnessed by her parents, Mr I (a local store owner and Miss McW’s ‘suitor’), and three doctors who could not administer any remedy to the patient by mouth because her jaw was locked shut. All that could be done for her were small injections of morphine. She died without regaining consciousness.

Unsatisfied with the family’s simple statement, the county prosecutor yielded to local pressure and had the young woman’s body disinterred nine weeks after her burial. The subsequent autopsy revealed that the unmarried Miss McW had been pregnant. Had an attempted abortion gone horribly wrong? There was no indication in the body of an abortion attempt or any other sign to explain the cause of death. So samples of stomach, liver, kidneys and spleen were sent for toxicological analysis.

Of the samples submitted, the most likely place to find any toxic substances would be the stomach. Dr G. G. Pond of Pennsylvania State College steam-distilled a section of Miss McW’s stomach to remove any suspicious substances from the tissues. Plant alkaloids, and several other toxic substances such as nitroglycerin, tend to be more soluble in ether than in water and so the watery distillate from the stomach was then shaken with ether. Any alkaloids would be transferred to the ether, which was then separated from the water. The ether was then allowed to evaporate and Dr Pond found a tiny amount of oily residue remained. Chemical tests showed the substance had nitrate groups present,^{*} which could be part of nitroglycerin or many other substances. To confirm the residue was nitroglycerin, Dr Pond struck it on an anvil to see if it exploded, which it did.[†]

Dr Pond estimated the total amount of nitroglycerin in the stomach was a mere 46mg (0.0016oz). This is equivalent to swallowing around a hundred trinitrine tablets, but still not nearly enough to cause Miss McW’s convulsions, let alone her death. But Miss McW’s jaw had been locked shut

for 10 hours before she died, meaning whatever nitroglycerin she had swallowed had 10 hours to be absorbed and metabolised in her body.

Nitroglycerin is absorbed slowly from the stomach, as we have seen, but 10 hours is a very long time. Nitroglycerin can also be broken down chemically under acidic conditions, but the acid of the stomach is not strong enough to have been a major factor in the elimination of the chemical. Nevertheless, the actual amount swallowed must have been much larger than the amount found by Dr Pond. But was it large enough to kill an otherwise healthy young adult? There may not have been enough nitroglycerin alone to kill Miss McW, but it was not the only thing she was given. There were the morphia injections given by the attending doctors, for example.

Drugs can interact with each other when the mechanism of one compound's interactions in the body is affected by the presence of another drug. Phenobarbitone, for example, is known to slow the effects of nitroglycerin. The evidence for the interactions of nitroglycerin and morphia, however, is scarce. In 1959, a 65-year-old woman died after swallowing only 0.6mg (0.000041oz) of nitroglycerin and an unknown amount of codeine and morphine. But there was possibly a third compound given to Miss McW that could have contributed to her death.

There are reports of deaths from the consumption of nitroglycerin and alcohol, the possible reason for poor Peter Eberhart's death on that cold night in 1905, for example. Alcohol, taken regularly in low to moderate amounts, can enhance the processes that relax the smooth muscles in blood vessels, causing them to dilate and lower blood pressure. Acute ingestion of large amounts of alcohol has the opposite effect, constricting blood vessels and counteracting the effects of nitroglycerin.

The nitroglycerin in Miss McW's body was believed to have been supplied from her suitor, Mr I, from his store in the belief that it was an abortifacient or a heart stimulant, but it is neither. It seems unlikely that Mr I would be selling pure nitroglycerin in his store, not if he valued his life, store or stock. If his supply of nitroglycerin was similar to the nitroglycerin supplied to British pharmacies, then it would be something like a 10 per cent concentration in alcohol.

The trial found Mr I responsible for 'an attempt to commit an abortion by the use of unknown instruments, death of the woman ensuing.' Those

‘unknown instruments’ may well have been nitroglycerin, the alcohol it was likely dissolved in and the morphine given to Miss McW by the doctors, combining their effects to kill her. Could a similar combination have killed Paul Déroutard in Christie’s story?

Agatha and nitroglycerin

‘The Chocolate Box’ does not mention what Déroutard consumed on the night he died other than a lot of chocolates after dinner. We are told he ‘did not drink wine of any kind’ but this does not exclude other alcoholic drinks that might have been taken to lubricate the conversation in the study after dinner.

We are also not told what was given to him when he became ill. In Agatha Christie’s world the universal antidote and restorative, seemingly given to anyone taken ill or suffering a sudden shock, is brandy. It is doubtful brandy would have helped in any of the situations in which Christie had it administered, but in Déroutard’s case it might have made a difference depending on the dose he was given. If Déroutard was not a heavy drinker, and it seems he was not, any small amounts of alcohol he did consume could have made an already serious medical situation much worse.

In Christie’s world brandy may have been doled out at every opportunity, but trinitrine tablets were not. She knew a doctor’s prescription would have to be produced before a pharmacist would hand over any nitroglycerin tablets, but anyone else in the house or visiting could have stolen those tablets.

Poirot finds an empty medicine bottle in a bathroom cabinet in the neighbour’s house, complete with the label written in English: ‘Trinitrine Tablets. One to be taken when required. Mr John Wilson.’ Alard, Déroutard’s neighbour, was clearly implicated as the murderer and Poirot had his evidence, but before he could take it any further the real murderer confessed. Poirot had misinterpreted the clue of the odd-coloured lid of the chocolate box that had started him down the trail that led to Alard. In fact, the chocolate box clearly implicated another suspect, but Poirot had overlooked it and other clues he had found, leading him to reach the wrong conclusion. All the clues are available to the reader of ‘The Chocolate Box’

and the reader of this book. Are you better than the great Belgian detective?
Can you work out who killed Paul Déroutard?

Notes

- * More about Miss Marple in Chapter T.
- * Estimates of his age at the time of his death range up to 140 years. Christie realised her error in making him so old so early in her writing career, though she can't be blamed for not foreseeing how popular he would become, and usually kept his personal chronology very vague.
- * They are no longer made out of chocolate, but the same drug is still prescribed as a spray and a skin patch as well as little white tablets for under the tongue.
- * You can still see the teeth marks!
- * Two more sons, Ludvig and Robert, were notable businessmen and engineers.
- * It is the combination of wealth accumulated from dynamite and investments in Russian oil wells that funds the annual Nobel Prizes, which were set up to assuage Nobel's guilt at producing something that had been used so destructively in conflicts.
- * As Eric Morecambe might say, the molecules of nitroglycerin contain all the right elements, but not necessarily in the right order.
- * Nitroglycerin, in combination with nitrocellulose and stabilisers, has found applications in explosive powders and as rocket propellant.
- * Methylene blue is also used in chemistry labs as an indicator and others may know it from adding it to their tropical fish tanks to treat fungal infections. It has many more uses, both medical and industrial.
- * Nitroglycerin tablets are therefore considered to have a very large margin of safety because the lethal dose is so many times larger than the therapeutic one.
- * Adding diphenylamine and ammonium chloride to the residue, along with sulphuric acid, produces a blue colour in the presence of a nitrate. This test has been used to identify nitroglycerin and nitrocellulose in gunshot residues.
- † He repeated the anvil test several times in court during the following trial.



Murder is Easy

There are plenty of ways to die in an Agatha Christie story. Some deaths are natural, but most are unnatural. Some are very bloody, but most are bloodless. And some are clichéd by murder mystery standards,^{*} but most are ingenious. Of the many strange weapons that have been deployed in her plots, hat paint has to be one of the strangest. But, as you may have come to expect by now, Christie's unusual choice is a brilliant one and backed up by some solid science. Let's take a closer look.

Hat paint was exactly what the name suggests – paint for hats: ‘one season you had a pink straw, next season a bottle of hat paint and it became dark blue – then perhaps another bottle and a black hat!’ Christie had used

hat paint herself in her younger days to keep herself looking fashionable on a modest budget. That was back in the late 1910s and early 1920s. As hat prices fell, more people could afford a new hat rather than bothering with repainting their old ones. Hat paint fell out of fashion, though it enjoyed a brief revival in the 1930s as a murder weapon in two of Agatha Christie's novels.

Christie's first hat paint poisoning features in her 1936 novel *Cards on the Table*, when it is mistakenly swallowed as Syrup of Figs. After brightening up an old gardening hat, the bottle of hat paint breaks and so the remaining paint is poured into an empty Syrup of Figs bottle. There are plenty of these empty bottles in the victim's house as they are an enthusiastic taker of this widely available laxative. The leftover hat paint, in its new bottle, is put on the top shelf of the bathroom cupboard with all the other odds and ends. Then one night, the victim pours herself a good dose of what she thinks is Syrup of Figs, but swallows down the hat paint, apparently without noticing any difference in colour or smell of her regular spoonful of medicine. The consequences of this mistake are fatal, but murder is not suspected. The bottles were easily confused and it was an unfortunate error of judgement that the bottle with the hat paint had not been relabelled. It is only later, when the death is reinvestigated, that suspicions are aroused.

Poirot and the police realise it would be easy for someone in the household, knowing full well where the hat paint had been stored, to deliberately swap the bottles and make it more likely the wrong one would be chosen. It is described in the book as a '*hopeful* kind of murder' because the victim might have noticed the switch and not drunk the hat paint, and even if she did drink it, it might not have killed her, depending on how much she drank and what treatment she received afterwards.

Hat paint poured into the wrong bottle and left for the unsuspecting victim to swallow is such a sneaky way to do away with someone that Christie used the method again just three years later in *Murder is Easy*, but with a little elaboration. As this 1939 novel demonstrates, murder is easy but getting away with it is extremely difficult. The picturesque English village of Wychwood under Ashe seems to be the perfect place to live, apart from the tendency of many of its inhabitants to die prematurely. The tiny population appears to be particularly accident prone and under a spell of

extraordinarily bad luck. People fall off window sills, in front of cars and off bridges at an alarming rate. Some get sick from septicaemia (see Chapter B, page 56) and gastroenteritis. Another has their head crushed by a stone pineapple and one swallows hat paint. All of these incidents have fatal results. Most of the villagers appear to shrug off this series of unfortunate events. Of all these deaths, only one has aroused enough suspicions to prompt any sort of investigation, that of Amy Gibbs.

Amy was a young woman working for Miss Waynfleet as her live-in maid. She had a bit of a cough and so went to the doctor for some cough mixture. Then one night Miss Waynfleet was woken by ‘an awful kind of choking scream’ and rushed to Amy’s room, but the door was locked from the inside. A passing policeman managed to climb in through the window to get to the stricken servant and unlock the door, but it was all too late. Nothing could be done for Amy and she died in hospital a few hours later.

The inquest into her death reveals that Amy had reached for her cough medicine in the dark, but got hold of a very similar-looking bottle of red hat paint instead. The fact that the incident occurred in Amy’s own room, and the door was locked from the inside, would appear to rule out outside interference. The possibility of suicide is briefly considered and dismissed. The official verdict is accidental death; Amy had simply made a terrible mistake. That is the sum total of the official investigation into her death, but there are still a few unanswered questions.

Amy’s locked door is given as a reason to exclude the possibility of someone else getting into her room. But the passing policeman who heard her screams got in easily enough through the window. Perhaps someone else – a jealous lover, a disgruntled former employer – might have done the same earlier in the day and deliberately switched the bottles. Then there is the doctor who gave Amy the cough medicine. He could have substituted something toxic and then casually switched the bottles when he was called to treat the dying victim. There are lots of suspects and plenty of opportunities to carry out the switch. The choice of hat paint also offers some clues.

Why would the unfortunate Amy, who was not the type of girl to be thrifty over her attire, keep an old bottle of hat paint around when she could have bought a cheap new hat? And why would she have chosen red hat paint when she had red hair and the colours would have clashed? If

someone had deliberately swapped the bottles in Amy's bedroom, it was someone who was, perhaps, not very fashion conscious or up to date with the latest trends, perhaps an older man. If this explanation seems a little heavy-handed, as though you are being deliberately directed to suspect certain characters in the novel over others, you are right. Christie is doing some very deliberate misdirecting.

You could also be forgiven for thinking that hat paint is a clumsy murder weapon – even some of Christie's characters comment on the fact. It appears to be an almost random choice, just something that is probably toxic and could be mistakenly swallowed with disastrous consequences. But, again, this is an Agatha Christie story and there is far more going on than at first appears. In *Murder is Easy*, Christie refers to the fact that it is the oxalic acid in the hat paint that kills Amy Gibbs. It's a compound you may not have heard of, but it has many everyday applications and is very likely to be found in the home, even today.

The oxalic acid story

Oxalic acid is also known as 'salts of sorrel' because it was first isolated from wood-sorrel plants (from the *Oxalis* family of flowering plants) by Dutch botanist, chemist and physician Herman Boerhaave (1668–1738).^{*} Sorrel can contain anywhere between 270 and 950mg of oxalic acid (0.01 and 0.34oz) in every 100g (3.53oz) of leaves, but there are plenty of other plant sources available. Oxalic acid is also found in Swiss chard (700mg per 100g), spinach (600mg per 100g) and beetroot (300mg per 100g). However, the plant most strongly associated with oxalic acid is rhubarb (500mg per 100g).

The sharp taste and laxative properties of the stems of the rhubarb plant are both due to the oxalic acid they contain.^{*} Oxalic acid, in tiny quantities, has a role in the body, though what exactly that role might be scientists have yet to determine. An excess of oxalic acid, however, has some very obvious effects on the body and none of them good.

The relatively small quantities of oxalic acid ingested from rhubarb stems, often stewed and served with sugar or custard, can irritate the gut. Eating rhubarb leaves, where the oxalic acid is more concentrated, can be

dangerous. Deaths have occurred from people boiling rhubarb leaves to eat as a vegetable, a practice that was encouraged by the British government during the First World War when there was a shortage of food, at least until the consequences became obvious. When food shortages returned in the Second World War a government campaign was launched, warning *against* the use of rhubarb leaves as a vegetable, even advising people not to feed the leaves to pigs as they might mix with other kitchen scraps. Boiling rhubarb leaves will not help reduce their toxicity because the oxalic acid is stable to heat, but it will leave your metal pans looking spotless afterwards.

The corrosive nature of oxalic acid means it has found many uses in metal polishes, paint strippers and stain removers. Its ability to dissolve the top layer of wood to expose a fresh, clean layer underneath has given it a role in wood restoration. It also has applications in the leather industry as well as the printing and dyeing industries. This wide variety of uses are all possible because of oxalic acid's chemical structure.

Oxalic acid

Pure oxalic acid is a solid white powder with a bitter taste. The molecules of oxalic acid look like bow ties with four oxygen atoms at the corners of the bows that are joined to two carbon atoms that form the knot (see Fig. O.1). Like the other acids we have met in this book – hydrochloric and formic – oxalic acid molecules can release H^+ ions when they dissolve in water. It is a more powerful acid than formic acid because it releases its H^+ ions more easily, but not as easily as hydrochloric acid. Each oxalic acid molecule can also release a maximum of two H^+ ions for every one released by hydrochloric or formic acid,* but hydrochloric acid is still the strongest acid of the three.

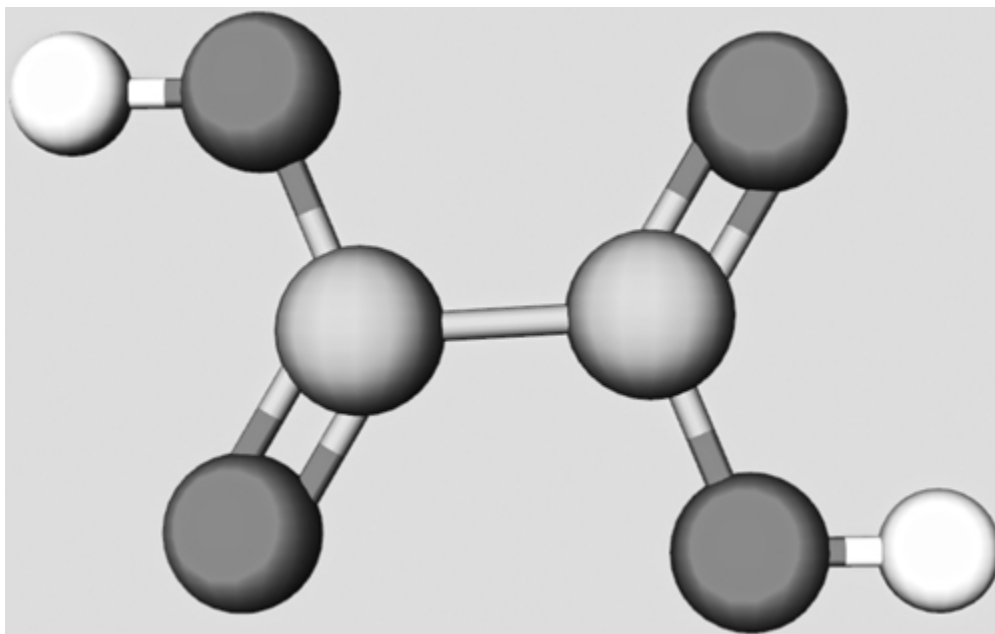


Fig. O.1. Structure of oxalic acid. The two H^+ ions are formed from the two white hydrogen atoms shown bonded to the two dark grey oxygen atoms top left and bottom right.

H^+ ions are effective at breaking up some molecules, as discussed in previous acid chapters. Oxalic acid can, therefore, break up impurities on metal surfaces to leave them gleaming, or break down layers of organic matter to restore the appearance of wood. Oxalic acid's ability to remove stains is due to how the rest of the molecule is arranged. When oxalic acid's two H^+ ions have been removed it becomes oxalate, which looks even more like a bow tie with its ends trimmed off. The symmetrical arrangement of the four oxygen atoms around the two central carbon atoms is very good at bonding to metals.

Transition metals, those in the middle of the periodic table, form highly coloured compounds that are often used as pigments in paints, dyes and inks. The colour of the natural pigment ochre, for example, comes from iron oxide, or rust, mixed with sand and clay. Iron compounds also contribute to the dark black/brown colour of traditional iron gall ink. Rust and old ink stains are almost impossible to wash out because iron oxide and the iron gallate compounds in indelible ink do not dissolve in water, even with the help of soap. These compounds are not easily broken up by acids either, because the iron atoms in them are bound very tightly to oxygen atoms. Iron, as mentioned before, has a very high affinity for oxygen.

The oxygen atoms in oxalic acid can be used to lure the iron atoms away from rust and ink. The four oxygen atoms in each oxalic acid molecule are arranged at particularly convenient positions so that an iron atom can sit comfortably between two of them (above or below the bow tie, as it were) and bind to them both at the same time. The resulting iron oxalate compound is only slightly soluble in water, but more soluble than rust or iron gallate. Oxalic acid will therefore prise the iron atoms away from rust and iron gallate compounds so the stain can be washed away. The ability of oxalic acid to bind tightly to the metal atom components of pigments means it can also be used to 'fix' dyes to a material so they *don't* wash out, or your newly painted hat doesn't stain your head on a hot, sweaty day.

A mordant, from the Latin 'to bite', is used to chemically bond a dye to a material. There are many different mordants including tannic acid, urine and oxalic acid, as well as others that contain metals such as aluminium (in alum), iron, copper and tin.* The mordant chemically reacts with the material being dyed as well as a metal atom in the dye to form a permanent physical link, a kind of chemical bridge, between the material and the dye. Two of oxalic acid's oxygen atoms on one side of the molecule will bond to the material to be dyed, and the other two oxygen atoms on the other side bond to the metal in the dye. Its inclusion in hat paint is therefore not surprising as it will help the pigment bond to the straw of the hat.

The oxalic acid in plants is also often bound to a metal. Spinach, for example, contains more oxalic acid than rhubarb, but there is no risk of poisoning if you eat lots of spinach leaves. Spinach is also often touted as a rich source of iron, and there is a lot of iron in spinach, but very little of it is absorbed by your body when you eat it. Most of the iron atoms stay stuck to oxalate, from oxalic acid, as they travel through your gut. It makes the plant safe to eat but also means the vegetable is not quite the phenomenally effective health food Popeye would have you believe.

More often, the oxalic acid in plants is bound to potassium or calcium atoms. Calcium oxalate does not dissolve in water. It forms colourless needle-like crystals that plants like rhubarb use as a defence against being eaten. These sharp crystals cause the irritation to the stomach when rhubarb leaves are eaten. When oxalic acid is ingested, from hat paint or plants, it is

the compound it forms with calcium inside the body that causes most of the problems.

How oxalic acid kills

The majority of the calcium atoms in our body, around 99 per cent, are locked up in our bones. Calcium is important for maintaining the bones' structural integrity, but the skeleton also acts as a stockpile of the element that can be drawn upon for its more important roles in the body. The other 1 per cent of calcium in the body is found bound to proteins called albumin or floating free and unattached. The movement of free-flowing calcium ions (Ca^{2+}) cause hormones to be released, blood to clot and muscles to move, among a whole host of other vital functions. And while a deficiency of calcium in bones will have serious long-term implications for health, a lack of calcium available for muscle movement has much more immediate and dangerous consequences. Muscles don't just help us walk and talk, they make our heart beat and our lungs expand to suck in air, keeping us alive from one minute to the next. Too little calcium in circulation is called hypocalcaemia and it has serious health implications.

Muscles contract when channels in the membranes of the muscle cells open to allow Ca^{2+} ions to flow inside and interact with the filaments within (see Chapter A, page 32). The presence of Ca^{2+} ions causes these filaments to slide over each other, shortening the length of the muscle fibre and contracting the muscle. Oxalic acid will readily react with these free-flowing Ca^{2+} ions to form calcium oxalate crystals, removing the calcium from circulation and generating solid lumps of calcium oxalate. If a large amount of oxalic acid has been ingested these crystals can be big enough and so numerous that they can be seen in the stomach contents at autopsy.

Symptoms of oxalic acid poisoning may start with local irritation, corrosion of the mouth, oesophagus and stomach. Oxalic acid is a strong acid and the soft tissues inside the mouth and throat will be particularly vulnerable if concentrated solutions of the acid are swallowed. Tissues can be bleached and streaked with brown, altered blood from damaged blood vessels. There will undoubtedly be a lot of pain and violent retching should someone eat a lot of it. The effects can look horrendous, but unlike the

damage done by concentrated hydrochloric acid (see Chapter H, page 134), it will be mostly superficial and is rarely the cause of death.

As the oxalic acid spreads through the body and comes into contact with Ca^{2+} ions, the symptoms will progress to muscle tremors, convulsions, weak pulse and collapse. Death from hypocalcaemia can occur within minutes. If a victim survives the initial chemical attack, they may make an apparent recovery only to die later from abnormalities in muscle function, including the heart. Even if these complications can be treated successfully and resolved, there are still further dangers to life.

Insoluble calcium oxalate salts crystallise in the most inconvenient places, such as the kidneys where they physically block tubules that filter and funnel waste and water from the body to the bladder.* The blocked tubules die and the kidney can fail, with death occurring between 2 and 10 days after swallowing the acid. A usual lethal dose of oxalic acid is between 5 and 15g (0.18 and 0.53oz), but just 1g has proved fatal.

The amount of oxalic acid that was dissolved in hat paint was likely to be quite high. The paint had to be applied to the hat with a brush as soaking the hat in a solution of dye would mean it would lose its shape. The pigment, therefore, must be fairly intense and the mordant very effective to latch onto the straw easily and not come off again. At room temperature, 90g (3.17oz) of oxalic will dissolve in a litre of water, equating to 1.35g (0.05oz) of oxalic acid in a tablespoon. This is the biggest size spoon likely to be found in the households where Christie's murders are set and, having tasted one spoonful of hat paint, the victims would be very unlikely to swallow a second. However, the concentration may be higher if ethanol was used to help dissolve the dye part of the hat paint. Ethanol would also help the paint dry faster as it evaporates more quickly than water. If pure ethanol was used to make the hat paint this could deliver just over 3.56g (0.13oz) of oxalic acid in a tablespoon. Even this increased amount is at the lower end of what is considered a lethal dose.

The two hat paint victims in Christie's stories suffer considerably before they die, consistent with a lower dose of oxalic acid. Their delayed deaths are likely due to later complications of hypocalcaemia or damage to the kidneys – though nothing is stated explicitly. Christie was evidently aware that such a poisoning might not be effective and has several characters

mention how unlucky the victims were not to survive. However, to keep these poisonings within the realms of possibility, the arrival of professional medical care is delayed in both cases. In *Cards on the Table* a doctor is called but he is out visiting another patient and cannot be reached. In *Murder is Easy* there is the delay of getting into the victim's locked room. Had help arrived sooner there might have been some very different outcomes.

Is there an antidote?

The important thing with oxalic acid poisoning is to prevent the acid binding to calcium ions in the body. If treatment is given quickly, the chances of survival and recovery are good. The easiest way to do this is to provide an alternative source of Ca^{2+} ions, and there are lots of options. There are many calcium salts that can easily be dissolved in water to release their Ca^{2+} ions – calcium chloride, calcium lactate and calcium hydroxide (lime water)* to name but a few – but these might not be readily available outside of a laboratory or hospital setting. Though the best course of action in cases of poisoning is always to let medical experts do all the treatment, in a dire oxalic acid emergency the simplest and most effective treatment might be to get the victim to drink milk, lots of milk.

The water in the milk will minimise the corrosive effects by diluting the acid, as we saw with hydrochloric acid in Chapter H (page 137). When diluted with 20 volumes of water, oxalic acid lacks corrosive properties. Milk also contains proteins that can bind some of the H^+ ions, but this still leaves the oxalate parts available to bind to free-floating Ca^{2+} ions. Even diluted, oxalic acid and oxalate will still be in the body and can still bind to Ca^{2+} ions with potentially fatal results.

Milk has always been touted as an excellent source of calcium to help keep bones and teeth strong. There is almost a gram of calcium in a litre of milk (0.035oz in 1.76 pints) and this is enough to react with almost all of the 3.5g oxalic acid calculated to be in the tablespoon of hat paint swallowed by Christie's victims. A litre of milk will certainly stop the worst of the poison's effects, but if symptoms persist then more calcium salts can

be given intravenously, by which time the victim will hopefully be in hospital under proper medical care.

What treatment is given to the victims in *Cards on the Table* and *Murder is Easy*, when it eventually arrives, is not specified. Whatever it was, it was evidently not enough to save them. Both Christie's 'hopeful' murders succeed, in that the victims died rather than the murderers got away with it. The reverse could easily have been the case, as the attempted murder of Elizabeth Rimmer shows.

Some real-life cases

In 1852, sisters Sarah and Anne Rimmer were tried for the attempted poisoning of Sarah's daughter, Elizabeth, who was just 19 years old. There had been many family rows, most of them stemming from Elizabeth being out of a job. She had left her last employment over a month before and, though she had been looking for work, she had found nothing to suit her. One row was overheard by a neighbour during which Elizabeth was told, 'If you don't go away and leave home I'll give you poison, or try some other means.' Returning one afternoon after another unsuccessful search for work, there were more threats to poison her from her aunt. Elizabeth seems to have ignored the threats and gone to bed.

Elizabeth woke up the next morning to find her mother and aunt had already had their breakfast, but they had left a pot of tea for her warming in the oven. The tea tasted unusually sweet, but she made no comment and helped herself to a second cup. She drank about half before feeling a burning sensation in her throat and stomach. Elizabeth asked her mother if anything had been put in the tea. Her mother took a sip and said some alum must have got into it by mistake.* When her mother had left the room, Elizabeth took the tea round to a neighbour, the one who had overheard the earlier poisoning threat. He took Elizabeth to a pharmacist to have the tea analysed.

Elizabeth's reaction to the tea showed that whatever was in it was clearly toxic, but there are many toxic substances and, in the nineteenth century, a lot of different chemical tests to identify them. There is a relatively simple test for oxalic acid that could be carried out with chemicals the pharmacist

was likely to have had available in his shop. The pharmacist had to first neutralise it with ammonia and then add calcium chloride dissolved in water. Tiny white crystals of calcium oxalate would appear and these crystals would dissolve in hydrochloric acid but not acetic acid. However, you had to have a rough idea of what you were looking for; otherwise it could take forever to run through all the different tests until hitting on the right one. With little more to go on than Elizabeth's symptoms and tasting the tea for himself, identifying the toxic substance could have been very difficult, but the pharmacist seems to have been up to the challenge and correctly identified oxalic acid.

The pharmacist then gave Elizabeth some magnesium, probably magnesium sulphate, also known as Epsom salts, to treat her and a note to take to a doctor who then prescribed her some medicine. Magnesium was a particularly astute treatment. Magnesium is very similar to calcium, being just above it in the periodic table. Its atoms are slightly smaller than those of calcium but they carry the same 2+ charge and have very similar chemical properties. Though administered sometime after Elizabeth had swallowed the oxalic acid, any that remained in her stomach or gut would have readily bound to the magnesium to form insoluble magnesium oxalate crystals and render the acid harmless. The pharmacist's actions may well have saved Elizabeth's life. What medicine was given to her by the doctor we do not know and it had no effect anyway because of what happened when Elizabeth returned home.

When Elizabeth walked through the door she found her mother, her aunt and two police officers in the house. Sarah, on seeing her daughter, struck her across the face and smashed the medicine bottle, saying, 'I've not done enough to you, you ought to be hung. Hanging is too good for you.' The police searched the house and found a few grains of oxalic acid in the fire grate of one of the upstairs rooms. Sarah and Anne were taken into custody.

At the trial Elizabeth was able to testify against her mother and aunt, telling of their threats. She also made accusations that the sisters had murdered another of Sarah's children in 1849 and buried the baby's body in a back garden. The spot was found and examined but only soil saturated with 'dark, oleaginous matter' and no bones were found. However, a baby's bones being so soft, nothing was expected to be found after such a period of burial in any case. Elizabeth's grandmother had also died suddenly and

been quickly buried, but again nothing could be proved. There was evidence demonstrating Elizabeth's poisoning and it was compelling enough to find Sarah and Anne guilty of attempted murder. Even though Elizabeth had survived, the sisters were given death sentences later commuted to life imprisonment.

Oxalic acid murders are rare and not always successful, as Elizabeth Rimmer's case shows. It seems an unlikely inspiration for Christie's fictional murders, if she even knew about it. If Christie did know about the case, there is not much she took from it other than underlining the difficulty of killing someone this way, and perhaps the spiteful nature behind the incident. Her oxalic acid murderers are certainly vindictive people. Her choice of hat paint instead of pure oxalic acid perhaps reflects a real-life accident she knew about. Maybe she simply heeded any warning signs she might have seen printed on a bottle of hat paint and let her imagination run away with her. She didn't have to stretch credibility too far, though, as a more recent case demonstrates.

Oxalic acid killed two people in 2015 in Manila, even though it was served with the antidote, milk. The incident took place at the Ergo Cha tea shop where Suzaine Dagohoy and her boyfriend Arnold Aydalla stopped for Hokkaido tea. This is a stewed black tea served either hot or cold, with honey, caramel or brown sugar to sweeten it, and lots of Hokkaido milk.*

Aydalla took a sip of his tea but found it tasted bad and spat it back out again. He asked Dagohoy to try it before bringing it to the attention of William Abrigo, the shop owner who had prepared the drink. Abrigo also tasted the milk tea but didn't notice any unusual flavour and prepared himself a second cup, which he then drank.

Dagohoy was the first to start vomiting and collapse. Aydalla suffered the same symptoms moments later and both were rushed to hospital. Meanwhile, Abrigo the store owner had collapsed behind the counter and was found by the shop assistant Joseph Garnacio when he returned from his lunch break. Abrigo and Dagohoy both died later in hospital, but Aydalla, after several weeks of treatment, survived.

Clearly all three had been poisoned by something in the milk tea. Tests were carried out for arsenic and cyanide but came back negative. Eventually the poison was identified as oxalic acid. Modern analytical

techniques are very sophisticated and can reveal the presence of poisons in ever smaller amounts. It can be tricky to identify a completely unknown poison even when there is a lot of it, and there must have been a lot of it in the tea prepared in the Ergo Cha tea shop. Presumably some of the oxalic acid would have reacted with the milk in the very milky tea, but there was enough left unreacted that the small sip Dagohoy took was enough to kill her.

There was a strong suspicion that the poison had been deliberately added to the tea, but by who and why? The finger was first pointed at Joseph Garnacio, the assistant. Though he hadn't prepared the fatal drink, he was seen tipping what was left in the cup down a drain. Garnacio then pointed the finger at Lloyd Abrigo, the shop owner's son, who had been to the shop the day before carrying something that smelled like bleach. While Lloyd was in the shop he had cleaned the equipment used to prepare the milk tea.

Lloyd Abrigo was charged with two counts of murder and one of frustrated murder, but the charges were later dropped. He had not prepared the lethal drink and there was no motive for killing Dogahoy or Aydalla, and targeting them would have been impossible anyway. Lloyd could not have known they would be visiting the shop at that time or that they would order the Hokkaido milk tea. If Lloyd had intended to kill his father it also seems like a risky and haphazard way of doing it as he couldn't be sure his father would drink it either. Lloyd was never charged and no one else has been arrested. The case remains a mystery.*

Agatha and oxalic acid

As poisons go, oxalic acid is a poor choice for the would-be murderer. It may be relatively easy to obtain, but the taste is unlikely to encourage your victim to ingest very much of it.[†] Treatment, if it is not delayed, can be very effective. It is the sort of poison someone might use in a rash moment of vindictiveness, which is why it was such an excellent choice for Christie in her two novels. All of oxalic acid's toxic shortcomings are acknowledged by Christie in her stories and she turns all of them to her advantage.

Neither of Christie's murderers is trying to get their hands on vast fortunes or clear their path to power or the object of their affection. They

both act because of some perceived slight. The trick they play on their victims is equally trivial – swapping a couple of bottles. Both murderers saw an opportunity to get their petty revenge and on both occasions it went further and better than either of them could have reasonably expected: two dead bodies and no hint of suspicion. The sheer improbability of it working and the trifling reasons behind the actions keep the murderers safe, at least until a few more people have been disposed of and the deaths are reinvestigated. Even the outdated use of oxalic acid in hat paint is used to Christie's and the murderers' advantage.

Notes

- * Or they have become clichéd because lots of people wanted to copy Christie's success.
- * The name oxalic acid was coined in 1787 by the legendary chemist Antoine Lavoisier (1743–1794) from the French *oxalique*, meaning 'of or pertaining to sorrel'.
- * Another acid, malic acid, also contributes to the flavour, and is present in greater quantities than oxalic acid, but malic acid is not toxic. In fact, it is involved in important processes in the body.
- * Oxalic acid is almost like two formic acid molecules joined back to back.
- * Varying the metal and the mordant can modify the colour of the dye as well as its permanency on the fabric.
- * Calcium oxalate is the main component of kidney stones.
- * Lime water is very alkali and can cause corrosive damage. Don't drink it. Not even if you have eaten a lot of rhubarb leaves.
- * Alum was a common household item at the time, being used in baking powder, as a deodorant and to stop small cuts and wounds from bleeding.
- * Hokkaido milk comes from cows grazed in the Hokkaido province of Japan and is known for its creamy, slightly vanilla flavour.
- * I am indebted to John Emsley and his book *More Molecules of Murder* for the two real-life cases I have written about here. All his books are well researched and fascinating reads. I highly recommend them.
- † Although poor William Abrigo didn't seem to notice anything amiss in his oxalic acid-laced drink.



One, Two, Buckle My Shoe

Things are not always as they seem in Agatha Christie's world. Motives can be misinterpreted, identities misidentified and murders are always mysterious. Christie's 1940 novel *One, Two, Buckle My Shoe*, has them all. What could be an international conspiracy turns out to be a simple domestic intrigue, several characters turn out to be not at all who they seemed and several deaths that could be accidents or suicides turn out to be homicides.*

The mystery starts when Hercule Poirot's dentist is found dead in his dentist's chair, shot in an apparent suicide. Poirot himself had been sitting in the chair that very morning for a check-up from Mr Morley, who seemed to be perfectly content with life. Then, just a few hours later in the middle of a

busy working day, Morley shot himself and no one can work out why. There is a chance that he was shot by someone else: a relative, someone who worked at the dental practice, or a disgruntled patient. The last person to see the dentist alive might have killed him or might at least have some useful information that could lead to the person who did. Morley's last patient was the mysterious Mr Amberiotis, who was on his first and only visit to Mr Morley. When the police ask to question him, Amberiotis tries to put them off by saying he is feeling unwell.

With their suspicions aroused, Poirot and Chief Inspector Japp decide to question Mr Amberiotis, whether he is ill or not. But, before they can see him, he also dies. His death, however, would appear to give the explanation for Mr Morley's sudden suicide. Amberiotis really had been feeling ill because he had been given a massive overdose of procaine and adrenaline while in the dentist's chair, and it killed him. When Mr Morley went to numb Mr Amberiotis's mouth for his dental procedure, he must have accidentally given him the overdose. When the dentist realised what he had done, he was overcome with guilt and fear of ruin, and promptly shot himself.

The explanation for both deaths is a little too pat for some. Poirot, for one, has his doubts. There are a few little details about the two deaths that just don't seem right. One of those little details is the injection that caused Mr Amberiotis's death. It gives Poirot and the reader a vital clue as to what is really going on.

Having trained as an apothecaries' assistant, Christie had a very detailed understanding of the mistakes that could be made with prescription medicines (see the first chapter, 'The Queen of Crime'). Dentists, of course, also dealt with very dangerous drugs on a daily basis, but her research revealed that the number and variations in formulations of these drugs were tiny in comparison to those dealt with by pharmacists. A dentist was likely to have a preferred recipe for their local anaesthetics and could vary the dose by injecting more or less as required. Mistakes were not impossible, and one patient might react differently to another, but the chances of a fatal accident were much lower.

In her novel, Christie has Gladys Neville, Mr Morley's secretary and dental assistant, explain that giving procaine and adrenaline injections is part of a dentist's everyday work. The procedure is routine and the doses

made up as standard. To make a mistake would be very unusual. This assertion is confirmed by Mr Reilly, another dentist who shares Morley's practice, who states: 'If Morley injected that Greek with the amount that they say he did, he was either drunk or else meant to kill the man.'

The anaesthetic story

Visiting the dentist can be a traumatic experience at the best of times. But the drilling, pulling and manipulation of the teeth and gums that is sometimes required has been made considerably easier to endure since the discovery of anaesthetics. For thousands of years the only way of treating pain, accidental, surgical or dental, was through the use of alcohol or the components of opium, such as morphine and codeine. These drugs were good at dulling pain, but they had other effects on the body that were not always ideal for surgical or dental procedures.

The nineteenth century saw the discovery of general anaesthetics, chemicals such as ether, nitrous oxide and chloroform that were either gases or could be easily vaporised and inhaled to render a patient unconscious and insensible to surgical procedures. It was a huge leap forward for medicine, but anaesthesia came with risks to the patient.* And in certain situations it is also not always desirable to render a patient completely unconscious. What was needed was something that would numb only the specific area of the body being worked on by the doctor or dentist without knocking the patient out cold.

Many different drugs and techniques were tried and tested on animals, patients and the experimenters themselves in search of an effective local anaesthetic. Existing pain relievers and general anaesthetics, like morphine and ether, affected the whole system and produced no localised numbing effect. Alternatives, such as electric shocks, cooling the area being operated on, cutting the nerves, or using chemicals that damaged them, either didn't work at all or left the tissues or nerves permanently damaged as a result.

The first major breakthrough in local anaesthesia was the discovery of cocaine. The leaves of the coca plant, *Erythroxylum coca*, have been chewed by South Americans for thousands of years to alleviate hunger, increase wakefulness and heighten endurance. Saliva from chewing the

leaves was also used on wounds to reduce pain. The numbing effect of chewing coca leaves was first noted by Europeans in the sixteenth century. Cocaine, the component of the coca leaves that was causing this numbness, was isolated and identified in the middle of the nineteenth century.*

Despite repeated observations of cocaine's numbing effects and recommendations for its use as an anaesthetic, it wasn't until 1884 that the theory was put into practice. That year, Karl Koller (1857–1944)† anaesthetised his own eyeball with cocaine to the extent that he could stick pins in his eye without feeling it. He reported his findings to the Ophthalmological Society in Heidelberg and recommend its use during operations on the eye. From there, cocaine anaesthesia was extended to other parts of the body and a wider range of operations.

The late nineteenth and very early twentieth century was cocaine's heyday, it being used in every branch of surgery from pulling teeth to amputating legs, but not always successfully. The drug caused tissue irritation, and more severe allergic reactions in some people. Cocaine molecules also fell apart when solutions of it were boiled to sterilise them, making it either ineffective or leaving patients at risk of infection. In the early days of cocaine anaesthesia, very high concentrations of the drug were used – between 2 and 20 per cent – though it was quickly realised that effective anaesthesia could be achieved with much more dilute solutions – just 1 or 2 per cent. Even though the use of lower concentrations had lessened the risk of poisoning considerably, mild, severe and fatal cases of cocaine poisoning still occurred in distressingly high numbers.

The difficulty was determining the proper dose as cocaine's effects on the body depended not just on the amount or concentration of cocaine, but also on how it was administered (injection, ingestion or rubbing it on the mucous membranes). The same dose could also affect different people to different degrees. Even the same dose of cocaine given to the same person could apparently react very differently on different occasions. Opinion varied widely on what might be a maximum dose.* It was also quite moreish, and many people became addicted to the drug.†

While the benefits of cocaine were obvious, it was also far from the ideal local anaesthetic. The search was on for other substances that might act like cocaine, but in a more reliable and safe way and without the risk of the

patient forming a drug habit. The starting point was cocaine itself, and debates began about whether the whole molecule was necessary to produce the anaesthetic effect or if it was only some key feature within its structure that was needed. A few small modifications to the cocaine molecule produced eucaine in 1896. This compound showed effective local anaesthetic effects and a lower toxicity than cocaine itself. It was also stable in boiling water so injections could be sterilised. However, it also produced a burning or irritating sensation in the patient.

German chemist Albert Einhorn (1856–1917) went further and found that a certain group of chemicals with some common features (see Fig. P.1) all produced anaesthesia to a greater or lesser degree. From this chemical core, atoms and chemical groups could be added or rearranged to create new potential local anaesthetics. These could then be tested to find if any gave effective local anaesthesia but with lower toxicity and less irritation. Einhorn's research culminated in the synthesis of a compound marketed by the Bayer firm as Novocain. Novocain was introduced into clinical practice in 1905 and was subsequently marketed in the United States as procaine and elsewhere as neocaine, planocaine, kerocaine and ethocaine.

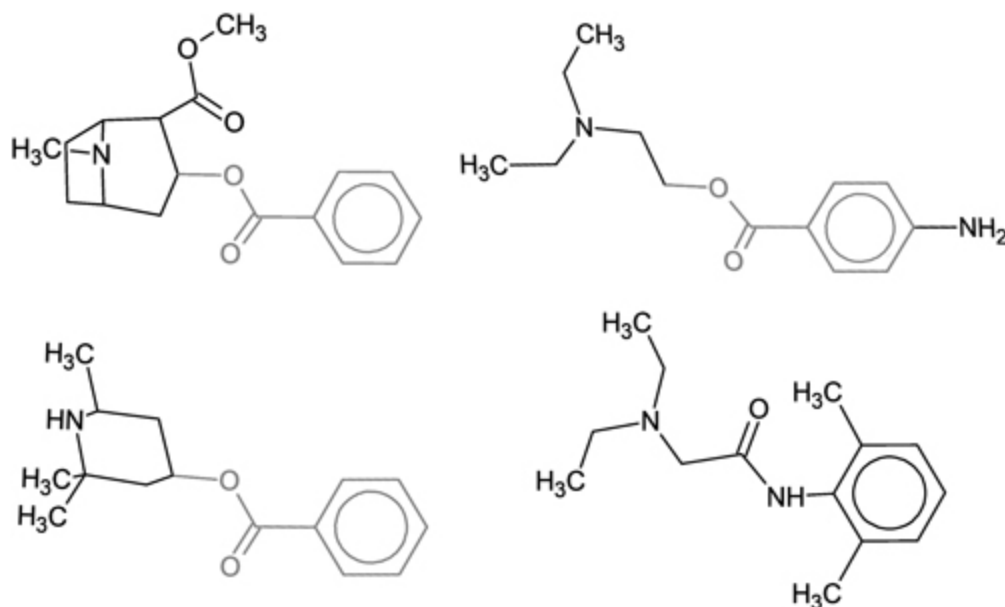


Fig. P.1. Structures of four local anaesthetics. Going clockwise from top left, they are cocaine, procaine, lidocaine and eucaine. Einhorn's common chemical feature is highlighted in grey.

Procaine

Procaine, or Novocaine, had low toxicity and was less irritating to the tissues than other local anaesthetic agents. It was soluble in water and stable enough once dissolved to allow boiling. Importantly, procaine did not have the same addictive qualities as cocaine. For these reasons, procaine soon came to dominate the field of local anaesthesia until the development of lidocaine in 1946.* Procaine would have been the anaesthetic of choice for Christie's dentist working in 1940 in the UK in *One, Two, Buckle My Shoe*.†

The sole disadvantage of procaine was that the anaesthesia it produced lasted only a short while, much shorter than cocaine. Though it wasn't known at the time, this was due to the body rapidly breaking down the procaine molecule. Part of the chemical structure of procaine is an ester, which enzymes called esterases can snap apart. Esterases are widely distributed around the body including in the kidneys, liver and blood.

Part of the reason cocaine was thought to give effective anaesthesia, for around 25 minutes, was because of a side effect. Cocaine also causes the blood vessels around the injection site to constrict, reducing blood flow and meaning the cocaine is not swept away in the bloodstream and distributed around the body. This had the effect of intensifying the degree of local anaesthesia and prolonging its duration. Procaine did not have this same effect, but other drugs could be added to it that did. Adrenaline can be added to solutions of procaine, and other local anaesthetics, to enhance their effects. The two drugs – procaine and adrenaline – that killed Mr Amberiotis in Christie's *One, Two Buckle My Shoe* were a standard combination for local anaesthetic injections.

Adrenaline

Adrenaline, also known as epinephrine, was the first chemical to be identified as a hormone. It is a compound produced naturally within our bodies that prepares us for 'fight or flight' when there is a perceived threat. Extracts of the adrenal gland had been shown to dramatically increase blood pressure in experiments conducted in 1886, but the agent that caused this effect wasn't isolated until 1895 – by several different scientists working

almost simultaneously though independently. In parallel to this research, pharmaceutical companies had been researching the effects of adrenal gland extracts in the hope of finding a commercial product. The first on the market was the Hoechst firm with 'Suprarenin', closely followed by Parke-Davis in 1901 with what they called 'Adrenalin'. Both of these products were extracted directly from the adrenal glands of animals. By 1906, Hoechst were selling man-made adrenaline and since then all the adrenaline used in medical applications is synthetically produced rather than relying on supplies from animals. Like procaine, there are several names for this compound that are still in regular use. Though adrenaline is most commonly referred to as epinephrine, we will use the word Christie used in *One, Two, Buckle My Shoe*: adrenaline.

Hormones are effective in minuscule amounts. Microgram quantities of adrenaline circulating in our bloodstream can prepare us for an emergency in seconds. The molecules of adrenaline in these few micrograms dock at a variety of receptors around the body, which trigger a range of different responses that increase our chances of successfully fighting off or running away from a threat.

A sudden release of adrenaline will cause pupils to dilate; speed up the heart and intensify its contractions; make the lungs breathe more often and more deeply; temporarily switch off a few functions that aren't immediately necessary, such as producing digestive juices and sweat; stimulate the release of glucose from the body's stores; break down fatty acids; ramp up chemicals that dull our sensitivity to pain; increase fibrinogen production for potential blood clotting; and raise blood pressure.

The effects adrenaline can produce in the body so quickly have made it a life-saving drug. Injections of adrenaline can restart stalled hearts, relax airways in bad asthma attacks and ease the symptoms of severe allergic reactions long enough for people to seek further medical help. But it is the effect on blood vessels that was of particular interest to scientists researching local anaesthetics.

Adrenaline can dilate blood vessels leading to the heart, increasing the flow of blood and oxygen to support the increased activity of the organ during an emergency response. But it *contracts* the blood vessels close to the surface of the skin. This has the double effect of increasing blood pressure and minimising bleeding in the event of damage to the skin.

Injected under the skin, or into the gums with local anaesthetics, it restricts the local blood flow, keeping the anaesthetic concentrated in a small area where it can take greater effect and prolonging its duration because it cannot be metabolised so quickly.

The amount of adrenaline needed in any of these medical settings is tiny. Adrenaline makes up less than 1 per cent of the solution injected into the heart to restart it. The EpiPens carried by people at risk of severe allergic reactions, deliver around 0.3mg (0.0003g or 0.00001oz) of adrenaline when the needle is pressed into muscle. The concentrations used in local anaesthetic injections are the weakest, typically between one in 100,000 and one in 200,000, but it can still produce a pronounced effect.

One experimenter in the early twentieth century injected himself with 1ml of a 1 per cent solution of pure procaine and found it produced 15 minutes of anaesthesia. After adding two drops of 0.001 per cent adrenaline solution to the anaesthetic, the anaesthesia lasted for *four hours*. The only apparent side effect was some slight pain at the injection site. The combination of adrenaline and substances such as procaine made local anaesthesia a simpler and more reliable procedure than ever before, greatly lessening the dangers of surgery.*

It is vital the amount of adrenaline is kept to a minimum. Too much adrenaline could cut off the blood supply to the injection site completely. If tissues are not continually supplied with at least some oxygen, and other life-sustaining supplies transported in the blood, there is a risk of gangrene. Adrenaline is also the more dangerous of the two drugs in these injections.

The 'fight or flight' response requires a lot of energy and resources and is difficult to maintain without causing damage to the body. Under normal circumstances, the body protects itself by removing adrenaline from the system as quickly as possible after the threat has gone. Enzymes rapidly break down adrenaline into other inactive substances. Too much adrenaline, as little as 100mg (0.004oz), can overwhelm the system and push its normal effects on the body to deadly extremes, the most lethal consequences being due to its dramatic effect on the heart.* The rapid response needed to defend us from danger means adrenaline acts quickly, and in the case of an overdose there can be little time for medical intervention before death occurs.

The lethal injection given to Mr Amberiotis in *One, Two, Buckle My Shoe* would have contained a large dose of adrenaline, but it was not what killed him. An adrenaline overdose would have made Amberiotis feel unwell before he even got out of the dentist's chair and dead before the taxi could deliver him back to his hotel. Instead, he left the dentist's apparently unconcerned and was alive, but unwell, hours later when Chief Inspector Japp and Poirot asked to speak to him. The adrenaline would have been a contributing factor in his death, but it was the procaine that probably killed him.

How procaine kills

Local anaesthetics work by blocking nerve signals along sensory neurons. They do this by interacting with selective sites on the sodium channels in the nerve membrane. As discussed in previous chapters, nerves are set up to 'fire' their electrical signals by pumping potassium ions (K^+) into the cell through specific potassium channels, and sodium ions (Na^+) out through sodium channels. When the nerve cell has been charged up, as it were, the sodium channels are kept closed off while the cell is at rest, but the potassium channels are normally open.

When the nerve 'fires', the electrical signal moves along the length of the cell, like a wave. Sodium channels open ahead of this wave, allowing Na^+ ions to flow into the cell and keep the signal going. To prevent sodium channels opening out of sequence or misfires, there is a minimum voltage, or threshold potential, that must be reached to open these channels. What a local anaesthetic does is raise this threshold. The nerve itself is primed and ready to fire, but because of the presence of procaine or similar local anaesthetic agents, it can't. As a result, the patient feels less pain.

The side effects and dangers of local anaesthetics come from their interactions with sodium channels in other types of nerves. Anyone who has had a local anaesthetic will know about the loss of movement that goes with the numbing effect. The difficulty of moving your mouth after dental work is because the local anaesthetic given by your dentist is also interacting with motor neurons that control your muscles. An ideal local anaesthetic would only interact with sensory nerves, but it is a small price to pay for the

benefits of numbing the pain of the procedure and the effects wear off quickly with no lasting consequences.

A greater concern is if the local anaesthetic interacts with nerves involved in vital central functions. Large doses of procaine can interact with nerves in the heart, the wider cardiovascular system causing blood vessels to lose their tone, as well as the central nervous system. The net result can include dizziness, tremors, irregular breathing, a fall in blood pressure, convulsions, cardiac standstill and circulatory collapse. The situation can be complicated by some of the products formed when the body breaks down the anaesthetic. These metabolites can interact with haemoglobin to make methaemoglobin that limits the amount of oxygen that can be carried around the body and can cause cyanosis (see Chapter N, page 192).

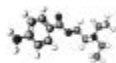
Some drugs also behave differently in some people. In the early years of procaine's use, a 23-year-old woman was injected with 3ml (just over half a teaspoon) of a 2 per cent solution of the drug with adrenaline. Following the extraction of a tooth she did not feel well and so she rested for an hour and a half after which she stood up and talked excitedly. An hour and a half after this she lay down again, complaining of dizziness. Six hours after the injection her condition became worse and two hours later she died in a coma with symptoms of cardiac weakness. Her death may have been caused by accidentally injecting a more concentrated solution of procaine, or from an allergic reaction to the normal dose, or it may have been due to something else entirely. No autopsy was carried out, so we will never know.

Allergic reactions to local anaesthetics are very rare, but they are known, and they occur almost exclusively with a certain type of anaesthetic, of the *p*-aminobenzoic ester-type, of which procaine is one. Some individuals exposed to repeated applications of local anaesthetics can form a hypersensitivity. This can be tested for by applying the local anaesthetic to the nasal mucosa for five minutes prior to injection. Signs of discomfort, beyond those you might expect for someone with a cotton bud up their nose, such as burning, swelling or irritation, indicates hypersensitivity, but it is not always a reliable test.

Dangerous as the drug may seem, and procaine is considered the most dangerous of the cocaine derivatives, it is considered very safe. In another case an experimenter injected himself in the thigh with a 10 per cent solution of procaine, six times the amount given to the 23-year-old

mentioned above. Five minutes after the injection he was feeling nauseous and vomiting. Two minutes after that he had double vision and 13 minutes further on he had a bad headache on one side. Seven minutes after this he experienced a strange sensation in his skin, but after 30 minutes he returned to normal health.

One of the biggest risks in the early days of procaine use was because of its name. Procaine sounded too similar to cocaine. Several deaths occurred after verbal orders for procaine were misheard as cocaine. No such mistake was made with Christie's Mr Amberiotis. He was deliberately injected with a large dose of procaine and adrenaline. There would be little doubt about the identity of the drugs that killed the Greek gentleman, even though neither of them leave any characteristic signs at autopsy. His recent visit to the dentist, and feeling unwell shortly afterwards, would have meant suspicion would fall quickly on procaine and adrenaline. The drugs would therefore be relatively easy to identify, but determining the exact amounts involved takes more time, as Christie points out in her novel.



Techniques for extracting toxic substances from human cadavers were developed in the mid-nineteenth century and have been improved and refined since then. The techniques available to the toxicologist in Christie's novel would have been sophisticated enough to extract both procaine and adrenaline, but working out how much of them were present would be an additional complicated step. Both compounds are rapidly broken down in the body, but death brings about a cessation of normal metabolic process. In the event of an overdose there is usually enough unchanged adrenaline left in the body for forensic toxicologists to find.

Toxic compounds are not usually extracted from the whole body. Instead, samples are taken from stomach contents or urine, plus organs such as the liver and kidneys and perhaps some sections of muscle, depending on what is available at the time of the autopsy. Comparing quantities from different samples shows how the drugs have been distributed in the body and estimates can be made about when and how the poison was administered. It would be obvious fairly early on in the toxicological

analysis of samples taken from Christie's procaine and adrenaline victim if large amounts of these drugs were present. Determining exact quantities and the original dose, however, would require further tests and calculations.

Though there can be individual variations, as we have seen, roughly 1g (0.035oz) of procaine would be considered a lethal dose for an adult. This may seem like a tiny amount, but it is hundreds of times greater than the quantities normally used to numb a tooth before it is extracted. The normal dilution for dental procedures would be between half and 2 per cent of procaine. At the higher percentage, this would equate to injecting 50ml (nearly three tablespoons) of solution into Amberiotis's gums to administer a lethal dose, which he would surely have noticed and resisted. It is more likely he was injected with a more concentrated solution. However much he was given, the result was that he felt unwell (though we are not given details of his symptoms) and died several hours later. Mr Amberiotis appears to have been rather unlucky.

Is there an antidote?

There is no specific antidote for procaine, but treating the symptoms and supporting the patient while their body breaks down the procaine naturally is usually very effective. The administration of oxygen may be enough to maintain the patient while the blood level of local anaesthetic diminishes. Falls in blood pressure can be treated simply by putting the patient in a head-down position or, if more serious, with intravenous saline or blood transfusions. Convulsion can be treated with anticonvulsant drugs, and methaemoglobinemia can be treated with methylene blue (see Chapter N, page 193). If the patient is alive an hour after their overdose, the chances are they will survive. Mr Amberiotis, however, survived several hours before succumbing.

It may be that the extra adrenaline he received, mixed in with the lethal amount of procaine, helped counteract some of the symptoms, such as the drop in blood pressure and effects on the heart. If this was the case, the relief would have been short term. Damage may have been done to the body that caused further complications and death.

Most procaine-related deaths have been as a result of individual susceptibilities to the drug or, in the years immediately after its invention, because it was used in novel ways that were not always suitable. Procaine as a poison for murder is exceptionally rare. Christie was more interested in using other aspects of dentistry for her murder mystery story (more of this later) and the use of the drugs readily available in a dentist's practice simply added complexity to her story. However, there have been real-life cases of people using both local anaesthetics and adrenaline for murder and attempted murder. Just as Christie took advantage of the drugs available in a dentist's surgery, a few murderous medical professionals have made use of drugs available in their place of work. Several medical murderers have used adrenaline to kill. A tiny few have used local anaesthetics for murder, but I can find only one case of an alleged attempted murder specifically using procaine.

Some real-life cases

Kristen Gilbert was a registered nurse working at the Veteran Affairs Medical Center in Leeds, Massachusetts. She was convicted of murdering four patients in the centre and the attempted murder of two others.^{*} Gilbert injected her victims with adrenaline because it was a natural component of the body that was quickly broken down and excreted. If used to kill an older or sick person, particularly when that person is already in hospital, it might go unnoticed. If detected, an unusually high level of adrenaline could also be explained away by the drug being injected by a doctor as part of legitimate life-saving procedures. It would seem to be the perfect murder weapon.[†] But there were plenty of other strange occurrences at the Veteran Affairs Medical Center that should have raised suspicions.

In 1991, it was noted that there was an abnormally high number of deaths at the centre when Gilbert was on duty. This was put down to coincidence, but her apparent run of bad luck continued. Records show that she activated the emergency alarm more often than all the other nurses combined. Then there was a spate of fires on her ward, 11 in total and eight of which were discovered by Gilbert. None of this was picked up by the hospital management, nor were the large number of adrenaline ampoules

delivered to the hospital, none of which had been ordered by doctors. Gilbert continued to work at the centre until February 1996.

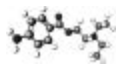
All these coincidences took on a more sinister aspect after an incident involving Angelo Vella, a 68-year-old patient suffering from chronic obstructive pulmonary disease (COPD). On 4 February 1996 Vella was admitted to the centre, suffering from shortness of breath. Soon after Gilbert arrived at work she was seen to inject him with something that made him cry out in pain so loudly that another nurse ran into the room to see what was happening. She saw Gilbert holding a syringe and Vella's heart rate at 300 beats per minute. A heart attack soon followed and the emergency alarm was raised. A chaplain was called to administer the last rites, but was not needed – Vella recovered and was able to give an account of what had happened. His heart monitor had also recorded the entire event. The tapes were carefully collected and stored.

Sensing she was under suspicion, Gilbert became hysterical and claimed a patient had attacked her. Her colleagues didn't believe her, but she was sent home anyway. She was later told not to return and that investigators had been brought in to examine the hospital records. As the investigation delved deeper, the case of Stanley Jagodowski came to light.

In July 1995, Jagodowski, a 66-year-old US army veteran with diabetes and high blood pressure, was at the centre recovering from a leg amputation. On 21 August, two nurses saw their colleague, Kristen Gilbert, enter his room with a syringe in her hand. They heard Jagodowski cry out 'Stop! Stop! You're killing me!' and Gilbert left the room almost immediately afterwards. The nurses were puzzled by the injection as none had been indicated. They found Jagodowski holding his arm, clearly in pain but otherwise fine. A few minutes later his alarm sounded when he went into sudden cardiac arrest. Despite the best efforts of the medical staff, he died three hours later. On 13 July 1998, three years after he had died, Jagodowski's remains were exhumed and tests showed the presence of an excessive level of adrenaline. There was now forensic evidence of how at least one patient had died to support the statistical and circumstantial evidence already collected.

Gilbert's trial began in November 2000. More than 70 witnesses were called and 200 pieces of evidence produced. She was found guilty, and

though she could have received the death sentence, she was given four consecutive life terms, plus 20 years, without the possibility of parole.



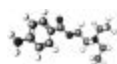
In early 1981, the tiny Community Hospital of the Valleys in Perris, California had a bad record for patient care. Medical regulators had criticised the hospital's use of vocational nurses rather than registered nurses in the intensive care unit. In March that year, the death rate was 20 per cent, twice what would be expected. The following month, the death rate jumped to almost 50 per cent.

Between 30 March and 25 April 1981, 12 patients died in the hospital's critical care ward. Curiously, these patients, both male and female and aged from 52 to 95 years old, all died during the night shift. There were no obvious medical reasons for all of these deaths and so autopsies were performed, revealing the presence of lidocaine in some of the patients.

The presence of lidocaine in a patient hospitalised due to cardiac problems is not unusual. Lidocaine, as well as being a local anaesthetic like other cocaine derivatives, is also used intravenously to suppress elevated heart rhythms. But some of the bodies from this Californian community hospital had 20 times the expected levels.

Suspicion fell on 42-year-old nurse Robert Diaz, who worked on the critical care ward at the times when the fatalities occurred, and ampoules of lidocaine were found in his home. The investigation went on for six months before Diaz was charged with murder, and he waited a further three years before he was brought to trial.

Diaz usually injected his patients around midnight. Convulsions would start within the hour and then he would leap into action to try and save the patient. Sometimes the patient survived and he became a medical hero. Often they did not. Diaz would blame the 'dumb' doctors who didn't know what they were doing, and even suggested the deaths were due to bad lidocaine. His stories were not believed and Diaz was found guilty on 12 counts of murder, though he was likely to have been responsible for many more deaths. He was sentenced to death but did not receive a lethal injection himself. He died of natural causes while on death row.



In March 1969, Joan Hill was taken to hospital, having been treated at home for four days by her husband, who was a doctor. A few days after admission she died of a supposed liver infection. Dr Hill remarried three months afterwards. Suspicions that Dr Hill had somehow been responsible for his first wife's death grew. In November, Joan's remains were exhumed and an autopsy suggested she had died of meningitis.

By 1971, Dr Hill was divorced from his second wife and on trial for the murder of his first. At the trial, his second wife alleged Hill had tried to kill her by deliberately crashing their car. Luckily she was uninjured, but she said her husband had tried to inject her with procaine while she sat in the passenger seat. She thought his plan had been to make it look like she had died from the trauma of the crash, but luckily she managed to fend him off. A mistrial resulted, but there were further revelations.

Dr Hill was alleged to keep bacterial cultures in his home, the implication being he had deliberately infected his first wife and she had died as a result of his medical neglect. Before a second trial could get underway, Hill was shot by a hired gunman, who in turn was killed by a police officer. It was believed that Joan Hill's father had hired the gunman. There were also suggestions that Hill was alive and well and living in Mexico.*

Agatha and procaine

Agatha Christie's idea for a murder mystery story centring on a dead dentist started from an idea about dental records. Dental records are sometimes the only way a body can be identified because our teeth are unique and particularly difficult to destroy.[†] As long as the victim has made a recent visit to the dentist, their records can be used to identify them. Christie's initial thoughts ran to using these records as a way of hiding a homicide, or obscuring the relationship between the victim and the murderer, through misidentification. These ideas seem to have grown and dentists became something of a preoccupation for the author.

There are several entries in Christie's notebooks centred on dentistry, including a list of 12 'latest dentist ideas'. As well as having a body misidentified, she considered the possibilities offered by the dentist's chair, where someone is particularly vulnerable. It could be an opportunity to kill someone by infecting them with a deadly disease or by giving them an overdose of local anaesthetic.

The novel *One, Two, Buckle My Shoe* manages to combine several of these ideas – misidentification from dental records and using local anaesthetic as a murder weapon. Dental records are swapped so a murder victim cannot be correctly identified and the dentist is bumped off as well, just to make sure. The murderer could then use this opportunity to get rid of another inconvenience, Mr Amberiotis, who was blackmailing the murderer. Taking the place of the dead dentist, they could make use of the lethal drugs already on hand to give Amberiotis an overdose. Any deficiencies in their dental skill while going through the pretence of working on Amberiotis would be covered by the massive dose of procaine – he wouldn't feel a thing.

The murderer must have done some reading up on local anaesthetics to get the injection right, or rather, sufficiently wrong. Through this reading they might have learned that the adrenaline component of the injection would be the more lethal but might also kill a little too quickly. In a busy dental practice, the murderer would want time to get away from the scene before suspicions were aroused. Better still, if your victim dies several hours later in another part of town, you can have an alibi ready prepared. And by using a dental anaesthetic as the murder weapon, suspicion should naturally fall on the dentist.

Christie used the knowledge she gained of local anaesthetics, not just to set up a puzzling murder and create brilliant clues for her readers, but she also avoided one of the big problems of writing murder mysteries. There is often a point in the plot where someone has to withhold information or lie, to ensure the mystery isn't solved too quickly and the story can fill out the required word count. Christie has Amberiotis die before he can talk to Poirot and Japp and give them the information that would lead directly to the truth. She does this in a way that is convincing within the scenario she created and with her usual scientific accuracy.

Notes

- * Published in the US under the title *The Patriotic Murders*.
- * Modern anaesthetics are much safer and procedures have been refined over the years to reduce the risks still further.
- * The active ingredient was first extracted by the German chemist Friedrich Gaedcke (1828–1890) in 1855, though credit is often given to Albert Niemann (1834–1861) who did the same in 1860 and gave it the name cocaine.
- † A good friend of Sigmund Freud, the founder of psychoanalysis and enthusiastic user of cocaine (for a time).
- * Cocaine, in appropriate formulations and applications, still has a role in some surgical procedures today.
- † Christie was well aware of the dangers of cocaine and had two of her characters die from its effects. However, in all her references to cocaine, it is as a drug of abuse rather than its medical applications.
- * Lidocaine produces rapid local anaesthesia and far fewer people have allergic reactions to it. It is therefore safer than procaine and now the more commonly used local anaesthetic in dentistry.
- † Procaine has also found a role in treating certain heart arrhythmias.
- * Another benefit was that adding adrenaline also reduced blood loss during the procedure. In the 1912 edition of *The Art of Dispensing*, adrenaline is described as being used by surgeons to perform ‘bloodless operations’.
- * The total amount of adrenaline injected for medical reasons will vary with need, but is typically between 200 and 500mcg (0.2 and 0.5mg, or 0.000007 and 0.000018oz) in total, a tiny fraction of the lethal dose.
- * Gilbert was probably responsible for more deaths. Some estimates have put the number of victims as high as 350.
- † Though adrenaline was her preferred method of murder, she also used injections of potassium chloride and insulin, both of which are also naturally found in the body.
- * By comparison, some of Christie’s more outlandish fictional plots look almost run-of-the-mill.
- † This was especially true in Christie’s day, before DNA analysis was available.



‘Philomel Cottage’ (The Listerdale Mystery)

Agatha Christie’s phenomenal popularity in print often overshadows her equally impressive success on stage. As Julius Green pointed out in his book *Agatha Christie: A Life in Theatre*, she was ‘the most successful female playwright of all time. She also wrote some books.’ Her closely plotted mysteries with her ability to disguise obvious clues and bring about sudden last-minute twists and revelations keeps readers and theatre-goers equally entertained. The Christie convention of a limited number of suspects, often confined in fixed settings such as country houses, is ideal for

the stage. It is a formula that has seen her play *The Mousetrap* break every theatre record going.

Christie's penchant for poisons would also appear to be ideal for the theatre. There is no need to risk accidents with daggers or embarrassment with guns misfiring. The wardrobe department doesn't have to worry about getting the fake blood out of the costumes every night, and the actor playing the victim can have immense fun drawing all the attention to themselves with convulsions and agonising screams.

Several people spotted the theatrical possibilities of Christie's detective stories, even before she did. In 1928, Charles Laughton became the first of many actors to don the famous moustaches as Poirot in *Alibi*, a play adapted by Michael Morton from Christie's novel *The Murder of Roger Ackroyd*. Perhaps slightly aggrieved at how her story had been altered, and wishing to try her own hand, Christie decided to write her own theatrical murder mystery.*

By 1928, Christie had already written several original plays, but none of them were murder mysteries and none of them had been performed professionally. Her first theatrical success was *Black Coffee*, which opened at Swiss Cottage, London in 1930.† Fully aware of what made her published stories so popular, her first play included many detective story tropes to pull in a crowd: a country house, a dead body in the library, a poisoning, a limited number of suspects and a famous Belgian detective to solve the mystery.

All the action in *Black Coffee* takes place in the library of the celebrated inventor Sir Claud Amory's country house. His family and a few visitors have gathered there for their after-dinner coffee. When one character complains of a headache, a box is brought down from the top of one of the bookcases. The box belonged to Edna, a former resident and hospital dispenser (like Christie). After the war, she had bundled all the leftover medicines into the box, put it on top of a convenient bookcase and promptly forgot about it.‡ Edna moved on, but the box remained. Now the entire collection of potentially dangerous drugs is brought out and discussed for the benefit of the audience and any potential poisoners present in the library. One of the guests, Dr Carelli, gives a running commentary on the lethality of each tube and bottle as it is shown to the audience. Strychnine:

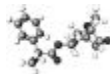
seven or eight tablets would result in a painful death. Atropine: the effects were sometimes mistaken for food poisoning and also resulted in a painful death. Hyoscine hydrobromide: half the tablets in the tube would result in 'No pain – just a swift, dreamless sleep with no awakening.' He concludes the box contains enough poison to 'kill, roughly, twelve strong men'.

We are just a few minutes into the play and we can already guess that someone is going to be poisoned. We even have a list of the poisons that might be used and a brief outline of the symptoms to look out for. Several characters handle, and sometimes surreptitiously pocket, some of the tubes and bottles from the box, ensuring the field of suspects is kept wide open. Audience members are advised to watch very closely because it is all, of course, classic misdirection – like a conjurer distracting you with a bright handkerchief so you don't notice a rabbit being manipulated into a top hat.

After the box, minus a few of its lethal contents, is put back on the shelf, Sir Claud enters and announces to the assembled guests that the house has been locked up and a detective is on his way. Sir Claud drinks his coffee, and complains about its bitter taste, before proceeding to explain why his house has been made secure. Someone has stolen the plans for one of his inventions. That person, Sir Claud believes, still has the plans on their person. As he outlines how he intends to recover his stolen property, he becomes increasingly tired, yawning and dragging out his words. He finally slumps in his chair just as Hercule Poirot makes his entrance. Sir Claud is dead and, from the symptoms displayed before he died – 'No pain – just a swift, dreamless sleep with no awakening', described earlier – we know he was poisoned with hyoscine hydrobromide. All everyone's favourite Belgian detective has to do, with the assistance of Captain Hastings and Inspector Japp, is identify the poisoner and return the missing plans before the final curtain falls.

Black Coffee completed only 67 West End performances before it closed in June 1931. It was, however, revived in repertory theatre and there was even a film version made starring a clean-shaven 34-year-old English actor Austin Trevor as the retired, moustachioed Belgian detective.* This relatively modest success, at least compared to her later plays, obviously did not put Christie off writing more murder mystery plays or setting them in English country houses. But she never put Poirot on stage again. Christie

was conscious that there would never be an actor who truly personified the Belgian detective that had lived inside her and her readers' heads for so many years. Poirot also became a focus for the audience, meaning other characters did not get the attention they deserved. In all her subsequent plays, even when she was adapting one of her works that featured Poirot or Miss Marple, the iconic characters were written out. Christie was, however, happy to reuse *Black Coffee*'s method of murder in a short story written just three years after the play.



The story of 'Philomel Cottage', written in 1934, centres on a newlywed woman who discovers her recently acquired husband is a serial killer. Sitting down to enjoy coffee after their evening meal, the wife realises she is just minutes away from becoming his next victim. She sneaks out of the room to phone a friend, asking him to come to her remote cottage and help her, before returning to her husband and trying to stall for time. To divert her husband's homicidal attention, she tells the story of how she has been married before and how she poisoned her two previous husbands with hyoscine in their evening coffee. Her husband, who has already commented on the bitterness of his coffee that evening, becomes increasingly alarmed. A knock at the door signifies the arrival of the friend along with a policeman, who enters the living room to find the husband dead in his chair.

Christie adapted this short story for the stage as a play called *The Stranger*. In 1935, Frank Vosper, an actor, playwright and amateur criminologist, acquired the theatrical rights to the story. His version is based on both Christie's play and the short story and is titled *Love From a Stranger*. He made a few changes, most notably building up the role of the tormented serial killer husband, a role Vosper took for himself when the play moved to the West End.

Love From a Stranger had a respectable run in the West End (149 performances) but was a notable flop when it transferred to Broadway (only 29 performances). Nevertheless, a film version was released in 1937, but without Vosper in the lead role.* During his return voyage from New York, Vosper was said to have fallen through a porthole while drunk at a party.

But the position of the porthole meant it was unlikely to have been an accident. His body, with one leg and a lot of cash missing from it, washed up near Eastbourne. The coroner recorded an open verdict.

In Vosper's play version of 'Philomel Cottage', the wife's story of killing her first husband does not involve poison. The comment in the original short story, about her new husband's coffee tasting bitter, is also omitted. Whatever has been added to his drink that causes him to collapse, if anything has been added at all, is left for the audience to guess. Hyoscine may not be named, but the play does contain one very vague reference to a real-life hyoscine murder (see 'A real-life case' later in this chapter) and a description of symptoms: choking, paralysing numbness and feeling cold (similar to those in Christie's original short story), gasping and unable to breathe and paralysis. Of course, you wouldn't want to hold up the action with detailed scientific discussions of toxic substances, especially at the tense climax of a play. But it seems a shame to leave it out altogether when hyoscine has such sinister associations.

The hyoscine/scopolamine story

Hyoscine has many names. In Colombia it has acquired the nickname 'Devil's Breath' for its effects on the body and its use by criminals to incapacitate their victims. In the US its official name is scopolamine (hence the title of this chapter) because it is found in plants belonging to the *Scopolia* genus.* In parts of South America it is known as burundanga, perhaps a corruption of *Brugmansia*, a highly toxic genus of plants that are native there. Hyoscine, used by Christie and the World Health Organization as the international non-proprietary name,† is a compound extracted commercially from *Hyoscyamus niger*, more commonly known as henbane. Despite the title 'S is for Scopolamine', Christie's and the WHO's preferred name hyoscine will be used throughout this chapter.

All of the different plant genera mentioned above belong to the same *Solanaceae* family that includes many other notorious members such as deadly nightshade (*Atropa belladonna*), jimsonweed (*Datura stramonium*) and mandrake (*Mandragora officinarum*). Many of these plants have become part of folklore, with stories that offer warnings of the dangers of

eating them as well as myths of their origins and how, in the right hands, they were thought to confer special powers.

For example, the name deadly nightshade is a clear warning of this plant's lethal potential. Its official name, *Atropa belladonna*, also contains hints of its dangerous properties for the classically trained. *Atropa* comes from *Atropos*, one of the three fates of ancient Greek mythology. The first fate, *Clotho*, was said to spin the thread of life. Her sister, *Lachesis*, measured the thread of life. The third sister, *Atropos*, cut it. A single berry of deadly nightshade contains enough toxins to kill an adult, and they have been used for that purpose for thousands of years.

Mandrake plants have become the stuff of legend and have been mentioned in many literary works from Shakespeare to Harry Potter. Their strange, bifurcated root, resembling a pair of legs, was once thought to indicate they were an intermediate between plants and animals. When the plants were pulled from the ground they were thought to scream so loud and piercingly it could drive a person mad or even kill them. Rather than leave well alone, people went to extraordinary lengths to harvest mandrakes. They stuffed their ears or enacted careful rituals that would protect them from the shrieks. Dogs were tied to the plant by a length of rope and scared or lured into running away and pulling the plant from the ground while their human owner was out of earshot. Such convoluted methods were obviously considered worthwhile as preparations of mandrake had medical applications, such as inducing sleep, and mystical associations because it could produce hallucinations.

Henbane, the main source of hyoscine, also has a long and dark history. It may be the bane of hens but it has caused quite a lot of problems for people as well. The plant is said to have spawned from Cerberus, the three-headed dog that guarded the entrance to the underworld in ancient Greek mythology. Where the dog's urine fell to the ground, the plant grew. Henbane applied to the skin in an ointment can give you a sense of flying, and sometimes produce hallucinations – perhaps why this and related plants were popular with witches and practitioners of black magic. Parts of the henbane plant have also been put into beer and cigarettes to treat asthma. Henbane seeds were used by the Babylonians to relieve toothache. These same seeds have been accidentally swallowed by children, causing them to

become ‘foggy and confused’. Their parents thought they were possessed by evil spirits.

Henbane also has some theatrical connections, though Christie may not have known this when she included its primary toxic component in her play *Black Coffee*. It has been suggested as the poison poured in Hamlet’s father’s ear in Shakespeare’s play.^{*} Named hebona or hebenon in the text, these words are not so different from henbane for a time when spelling was particularly free and easy. The ghost of Old Hamlet also goes into some detail about the symptoms produced by this poison when he confronts his son. One symptom in particular stands out with regard to henbane plants:

*And a most instant tetter bark’d about,
Most lazar-like, with vile and loathsome crust
All my smooth body.*

It sounds like Old Hamlet is describing nasty eruptions on his skin, and hyoscine can cause a rash in some susceptible people. Another link comes from Pliny’s *Natural History*, a book Shakespeare almost certainly read, which is full of ancient remedies for all sort of ailments. To cure earache, Pliny recommended using the juice of henbane, opium and rose oil among other ingredients. The mixture was to be warmed up and syringed into the aching ear. It would explain Shakespeare’s strange choice of the ear as the site for applying the poison.[†]

Christie made several casual references to belladonna in her writing, but only one fleeting reference to mandrake and none at all to henbane.[‡] *Datura* plants get several mentions, specifically in regard to stories from India, where parts of the plant were said to be used to drug, disorientate and send spouses mad. More usually she wrote about the compounds within these plants that cause their mystical, medical and murderous effects: atropine and hyoscine.^{*} Both of these compounds can be found in the plants mentioned, but in differing proportions depending on species, season and local growing conditions. The dried leaf of an average henbane plant, for example, contains 0.04–0.05 per cent alkaloids, of which the dominant toxin tends to be hyoscine.

Atropine

Before we get to hyoscine, we should discuss atropine first as it has already been mentioned several times in this book and is closely chemically related to hyoscine (see Fig. S.1). In fact, the similarities are so close that the two drugs produce similar symptoms in the body. It also makes sense to start with atropine because in plants, atropine is converted into hyoscine in a single chemical step.

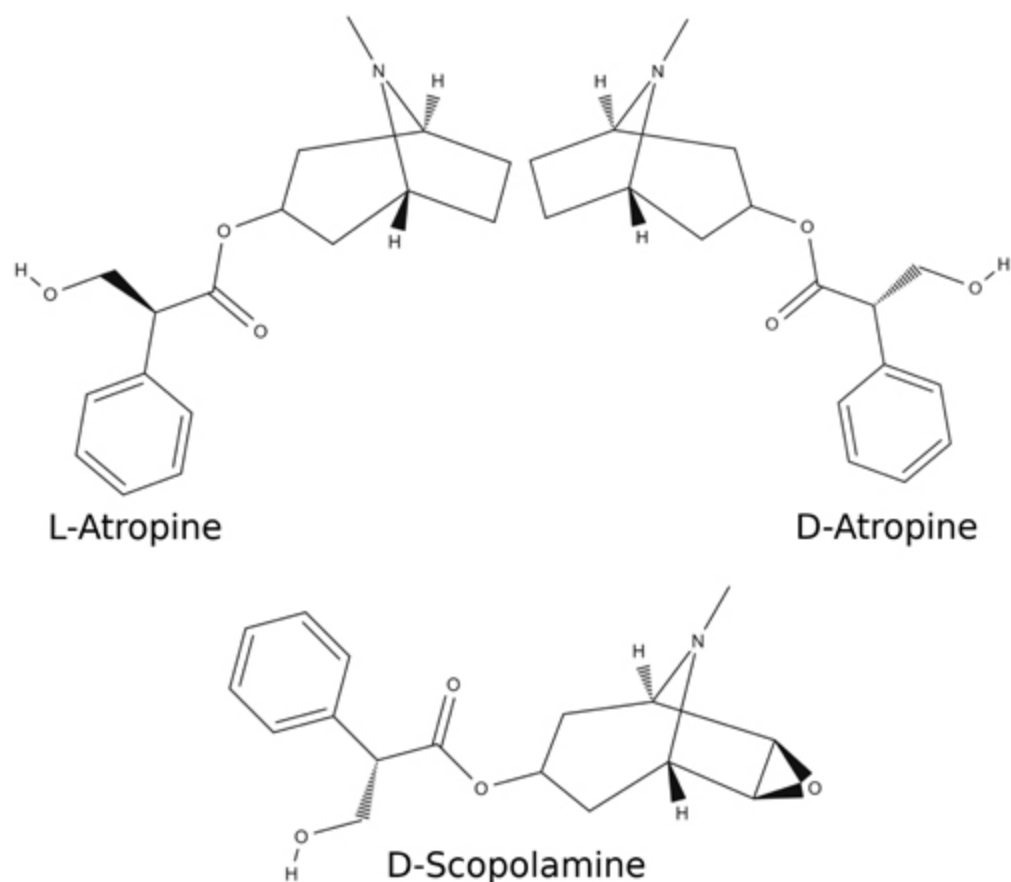


Fig. S.1. The structures of atropine (top) and hyoscine/scopolamine (bottom).

The subtle chemical differences between atropine and hyoscine account for the slightly different effects they produce and their different toxicities, as we shall see. The first difference between hyoscine and atropine is that atropine is an equal mixture of two ‘hands’ of the same molecule, while hyoscine is just one hand.

Some molecules can be ‘handed’, or the technical term chiral, because their atoms can be arranged in different orientations. The easiest analogy to explain this is a pair of hands. Hands have identical components – fingers, thumb, palm, back – but they are arranged differently so that the left hand is the mirror image of the right. Some molecules contain atoms, or groups of atoms, that can be arranged like fingers and thumbs around a central carbon to produce left or right ‘hands’. If the atoms or groups can be arranged to form mirror images that cannot be superimposed,^{*} the molecule is said to be chiral.

The two hands of a chiral molecule are chemically identical, but they interact with other handed molecules differently. Think of trying to put a left hand in a left glove and a right glove. One will fit better than the other. This is important because a lot of biologically significant molecules – DNA, glucose, some amino acids, to name but a few – are handed and can interact with other handed molecules differently. A classic example of this is thalidomide, a chiral drug that was prescribed to pregnant women as a mix of both hands to treat their morning sickness. One hand of the molecule caused serious malformations in the developing foetus.[†] Biology excels at producing one hand of a molecule or another as needed, but chemists still struggle to do the same and often have to go through lengthy separation techniques to sort out left from right compounds made in the lab.^{*}

The left ‘hand’ of atropine is thought to be responsible for most of the symptoms produced in the body. These symptoms are similar to the body’s ‘fight or flight’ response discussed in Chapter P (page 221). Heart rate is increased, digestive processes are put on hold drying up saliva, and pupils dilate. This is why atropine is sometimes given to stimulate a slow or sluggish heartbeat and why it is given as eyedrops, usually by an optician so they can examine the interior of the eye more easily, but it is also given to treat some forms of glaucoma.

In plants, left-handed molecules of atropine can interact with an enzyme that adds an extra oxygen atom to make hyoscine. This is the second small but significant difference between atropine and hyoscine. The extra oxygen atom slightly changes the shape and chemical behaviour of the molecule. In a human body full of chemical structures that are highly specialised to

recognise and interact only with certain molecules, hyoscine will fit slightly better or slightly worse into the same receptors as atropine.

Hyoscine

Hyoscine molecules interact with receptors in the autonomic nervous system, the part of the nervous system that deals with our unconscious actions, like breathing, heartbeat and digestion. Specifically, hyoscine interacts with a certain type of acetylcholine receptors, called muscarinic receptors.[†] These receptors are mostly found in the parasympathetic nervous system, the branch of the autonomic nervous system responsible for rest and digest. This branch works in tandem with the sympathetic nervous system, which is responsible for the 'fight or flight' response.

Acetylcholine is a neurotransmitter that passes messages from one nerve cell to an adjoining nerve or muscle cell or gland. When acetylcholine molecules dock at muscarinic receptors they stimulate the receiving cell into action, be that a nerve firing, a muscle cell contracting or a gland secreting. When hyoscine docks at these receptors it deactivates them, suppressing the message, and the receiving cell becomes less likely to be activated. The overall effect this produces in the body depends on the dose.

Small amounts of hyoscine dull the nerves slightly. It makes you less anxious, less likely to react violently to situations and it smooths stomach troubles that are mainly caused by overstimulation of parts of the digestive system. It is therefore an ideal drug to use in travel sickness pills. Hyoscine can be absorbed through the skin, so patches can be worn to slowly deliver the drug and someone already suffering from seasickness can still benefit from it without worrying about vomiting up a pill before it has taken effect. It is also useful as medication to control the diarrhoeal phases of irritable bowel syndrome (IBS).

In the early twentieth century small doses were being used by opticians. The interaction between nerves and muscles in the iris causes the pupil to contract or dilate according to need. Hyoscine dilates the pupils, allowing an optician to get a better look inside the eye. But it can also affect nerves that control the lens of the eye, making it difficult to focus on close objects.

Hyoscine is also a powerful sedative and in the past it has been used, sometimes in conjunction with morphia, to control violent lunatics, hysterical cases and for the treatment of nervous disorders such as delirium tremens (a rapid shaking that can affect alcoholics if alcohol is withdrawn suddenly). In 1916, Texan obstetrician Robert E. House administered hyoscine to a woman in labour to induce what is called a 'twilight sleep'. In addition to the sedative effects, the drug was thought to have the benefit of causing short-term memory loss and the trauma of a difficult delivery would be forgotten.* Dr House also discovered another interesting side effect: his patient became extremely talkative. Thinking the woman was sedated, he had gone to talk to the husband but was surprised to hear her clear voice answering his questions from the next room. This talkativeness combined with another symptom, a lack of judgement over what was said, led to experiments with hyoscine as a 'truth serum'. However, it soon became apparent that those given it revealed no new information. To complicate things, they also suffered hallucinations, headaches, blurred vision and a 'fantastically, almost painfully, dry "desert" mouth', making it difficult for them to tell anything, let alone the truth. Unable to discern fact from the fanciful, the experiments were abandoned.

Other experiments with hyoscine were considered successful. Given to parachutists in the Second World War, low doses would have perhaps alleviated their anxiety about their stressful and dangerous occupation. But the drug also slows reflexes and spoils your sense of distance, skills that would seem important for a parachutist.

Increasing the dose further, an individual may experience hot, dry, red skin; disorientation; aggression; rapid pulse and respiration; urinary retention; muscular stiffness; fever; convulsions; and coma, and all on top of the other symptoms previously described.

Sensitivity to hyoscine varies enormously from one person to another. As little as 6mg (0.0002oz) can interfere with the nerves that control the heart in more susceptible individuals. Children have died as a result of swallowing just 10mg (0.0004oz). Others have swallowed more than 400mg (0.14oz) of hyoscine and survived. With people being so wildly different in their susceptibility to the drug, it might not be a surprise that

there is no official lethal dose of hyoscine, but it is certainly more potent than its chemical relative atropine.

How scopolamine kills

Of the wide variety of symptoms produced by hyoscine, pushing some of them to extremes will obviously have serious impacts on a person's health and can potentially cause death. A rapid pulse means the heart is working hard and has high oxygen demands to keep the muscles pumping. This can be managed short term during extremely stressful episodes or when exercising, but can be pushed to dangerous levels. Toxic doses of hyoscine can also increase the rate of respiration, increasing the amount of oxygen being drawn into the body, but in larger doses it can cause respiratory depression with potentially fatal consequences.

The drug also lingers in the body for quite a long time. It can take between 5 and 10 hours for half of the hyoscine molecules to be eliminated, meaning that even low-dose medicines can have a cumulative effect if taken too frequently. This may all sound terrifying, but anyone still alive 24 hours after poisoning is likely to recover. Fewer than 1 in 100 poisoning cases results in death. Initial symptoms can appear 20 to 30 minutes after swallowing hyoscine but it normally takes a few hours to kill, giving time for intervention and treatment.

Is there an antidote?

As with the majority of poisoning cases, the important thing to do once the patient is stable is try and prevent any more of the poison from being absorbed into the body and making things worse. Hyoscine can be washed from the skin and mucous membranes. It can be soaked up from the stomach using activated charcoal and removed by gastric lavage. Saline catharsis can also be appropriate. This is where a patient is given salts to swallow that don't dissolve very well in water. This causes water from the body to move into the stomach and intestines in an effort to dissolve them. This increases the bulk of the salts and encourages peristalsis – the wave of muscle contractions and relaxations that normally moves food through the

gut. These efforts can be continued for some time because hyoscine typically slows down gastrointestinal activity. Treatment can also be given for the hyoscine that has already been absorbed into the body.

Physostigmine can act as an antidote to hyoscine poisoning because it can counteract the depressive effects hyoscine has on nerves. Normally, when the neurotransmitter acetylcholine (the ‘go’ message) docks at its receptor it is quickly broken down again so that the receptor is clear to receive more messages. This is done by an enzyme called cholinesterase. Physostigmine is a compound that blocks the cholinesterase enzyme, meaning the acetylcholine molecules hang around for longer and will have more time to find a receptor that isn’t blocked by hyoscine.

Intravenous injections of physostigmine can rapidly counteract life-threatening symptoms produced by hyoscine. But physostigmine is itself a poison because it can cause overstimulation of acetylcholine receptors. This overstimulation results in almost the exact opposite symptoms of hyoscine poisoning, including a dangerously slow heartbeat. The antidote for physostigmine also has to be on hand in case the heart slows too much, and that antidote is atropine, hyoscine’s chemical cousin.

Agatha Christie certainly knew about physostigmine – she discusses it extensively in her novel *Curtain*, and it is used as a poison, under its alternative name eserine, in another of her novels, *Crooked House*. Her hyoscine victims never have the opportunity to receive this treatment, however, as they are dead before any kind of medical help can be given. Christie’s use of all of these compounds – physostigmine, atropine and hyoscine – as murder weapons shows her variety. But hyoscine was the first, perhaps because of its very high-profile use in a real-life murder that Christie knew of.

A real-life case

In 1900, Dr Hawley Harvey Crippen, a Michigan-born homeopathic doctor, had moved to England to work in the London office of The Munyon Company, an American firm selling patent homeopathic remedies. He was soon joined by his operatically inclined wife Cora. Cora’s musical ambitions outstripped her talent and she found only moderate fame

performing in England's music halls under the stage name Belle Elmore. She also became an enthusiastic supporter of The Music Hall Ladies Guild. Belle had expensive taste and an overbearing personality. Crippen was seen as the henpecked husband who became increasingly close to his typist, Ethel le Neve. By 1905, Crippen and le Neve had started an affair.

On 10 January 1910, Crippen purchased five grains (0.324g/0.011oz) of hyoscine hydrobromide.* On 31 January, the Crippens hosted their friends Paul and Clara Martinetti for dinner at their home at 39 Hilldrop Crescent. When the visitors left in the early hours of 1 February, the couple had appeared to be on good terms. The following day, Crippen went to work as usual. Belle was never seen again.

Two days after the dinner party, The Music Hall Ladies Guild received letters, apparently from Belle, resigning her position as treasurer with immediate effect because she had returned to the United States, where a relative had been taken ill very suddenly. Later that month Crippen attended a ball given by the Ladies Guild Benevolent Fund with his mistress, who was wearing Belle's jewellery and furs. By mid-March, Crippen was telling mutual friends that Belle had been taken ill while abroad and was not expected to pull through. Ethel moved into Hilldrop Crescent and a notice of Belle's death appeared in the stage newspaper *Era* on 26 March.

Crippen and Ethel's behaviour generated a lot of gossip but nothing more until John Edward Nash and his actress wife returned from a trip to the States in June. They were shocked to hear about Belle's apparent death as they had heard nothing about it from their mutual friends in New York. Unconvinced by Crippen's answers to their questions, they took their concerns to Detective Inspector Dew at Scotland Yard. Inspector Dew then went to question Crippen.

Crippen admitted that he had lied about Belle's illness and subsequent death. He explained that she had left him and he had invented the story to cover his embarrassment. Dew conducted a thorough search of Crippen's house and found nothing untoward. But when Dew returned to the house three days later, Crippen and Ethel had disappeared.

Descriptions of the fugitives were issued while the police went over Crippen's house with a fine-toothed comb. They even dug up parts of the garden and found ... nothing. On a third search of the house some loose

bricks were noticed in the cellar floor. They lifted easily and the police started to dig. Twenty-three centimetres (9in) below the surface they found, along with pieces of a man's pyjama jacket, some long brown hair that had been bleached blonde, with some strands still wrapped around a curler. There were also pieces of human flesh from the buttock and thigh; most of the internal organs; a large piece of skin from the upper abdomen and chest area; and a smaller piece of skin from the lower abdomen. There was no head, limbs, bones or sex organs. An arrest warrant was issued.

Augustus Pepper, Master of Surgery, and Bernard Spilsbury,^{*} a young pathologist at St Mary's Hospital, set about seeing what they could find from the paucity of material the police had brought to them. The long, bleached hair, still wrapped around a curler, indicated that the remains belonged to a woman. The size of some of the pieces of flesh, and the fat around the heart, indicated the remains belonged to someone of stout build. A scar on the skin from the lower abdomen indicated a woman who had had an operation to remove her ovaries. All of these details corresponded to what was known about Belle. They also believed the remains had been buried for between four and eight months, covering the period of Belle's disappearance. But there was no sign of disease or degeneration of the organs that could have caused death. Samples were therefore passed on to another expert, William Willcox, a lecturer in forensics at St Mary's Hospital and senior analyst to the Home Office.

Teasing out what were likely to be tiny amounts of plant alkaloids from the chemically macerated human remains Willcox had been given required skill and time. After a painstaking series of washing and extracting with different solvents to separate any alkaloids present from everything else, Willcox eventually extracted a total of two-sevenths of a grain (18.6mg/0.0007oz) of an alkaloid. He then had to identify the alkaloid.

Working before the advent of sophisticated methods of analysis that assist chemists today, Willcox held down the feline resident of the hospital where he worked and placed a drop of the extracted liquid into the cat's eye. The pupil dilated, a characteristic sign of both hyoscine and atropine. He conducted tests for other common plant alkaloids, but these were negative. After more chemical tests he eventually became convinced that the 18.6mg he had extracted were hyoscine.

Most of the hyoscine had been found in the intestines, followed by the liver, then other organs, indicating that the drug had been ingested, rather than injected, and that this had occurred a few hours before death. Willcox calculated that half a grain of hyoscine must have been ingested by the deceased for this much to remain in the samples he had analysed. A fatal dose of hyoscine was believed to be between a quarter and half a grain. It was therefore reasonable to assume that the cause of death was hyoscine poisoning. It was the first case of murder by hyoscine that any of the experts involved had come across.

The gruesome discoveries, the disappearance of a minor music hall performer and Crippen fleeing with another woman ensured intense newspaper interest, both in Britain and abroad. The hunt for the fugitives began.

On 22 July Captain Kendall of the SS *Montrose*, en route from Antwerp to Quebec, made use of the newly installed Marconi telegraph equipment. He sent a message to Scotland Yard: 'Have strong suspicions that Crippen London cellar murderer and accomplice are among saloon passengers. Accomplice dressed as a boy. Voice manner and build undoubtedly a girl.' The pair were travelling under the name of Roberts, father and son. Inspector Dew boarded a faster ship, the *Laurentic*, and arrested the pair when the *Montrose* docked in Canada.

At his trial at the Old Bailey, Crippen maintained that the remains belonged to someone else and his wife had simply walked out on him. What he couldn't explain was how part of his pyjama jacket had come to be buried with these remains. The forensic analysis clearly showed the presence of hyoscine in what was left of the body, and in quantities that would be expected to be lethal. Crippen's purchase of hyoscine weeks before his wife disappeared certainly looked suspicious, but he claimed the purchase was for homeopathic remedies he prepared at home and posted to his patients to treat nervous disorders. Medical witnesses asserted hyoscine formed no part of homeopathic remedies and no purchasers of Crippen's homemade preparations could be found.

It took the jury less than half an hour to find Crippen guilty and he was hanged on 23 November 1910. He protested his innocence until the end.*

Ethel le Neve was tried separately and acquitted on charges of being an accessory. But the story doesn't end there.

Spilsbury had identified the remains from the cellar primarily by the scar on the abdomen. But what if this scar was instead a fold in the skin? Spilsbury was an extremely self-assured individual who was never willing to admit to mistakes. His authoritative presence in the courtroom meant his evidence was rarely doubted and juries would believe anything he said. It has been alleged he may have been responsible for several miscarriages of justice. But considering Spilsbury had published research on scars a few years before the Crippen case, it seems unlikely he would have made a mistake on this occasion. Several other experienced scientists also agreed with his conclusions.

As time and science has moved on, more sophisticated techniques have become available to identify human remains. A tiny portion of the tissue removed from the cellar in Hilldrop Crescent was preserved on a microscope slide and stored at the Royal London Hospital. A team led by highly respected toxicologist John Trestrail III decided to apply modern techniques to analysis of this historical evidence. In 2007, Dr David Furan extracted DNA from the specimen, and genealogist Beth Wills tracked down three of Cora Crippen's descendants so their DNA could be compared. It was not a match.

Did Crippen give his wife hyoscine to dampen her sexual ardour, or to treat her drinking without her knowledge, but give her an overdose in error? Did she not react to the hyoscine in the way he expected and he panicked and killed her some other way? Or did she simply walk out on him as he claimed? Ten years after Crippen's execution, a woman calling herself Belle Rose was registered as living with Cora's sister in New York. The woman had entered the United States from Bermuda in 1910, shortly after the disappearance in England of Belle Elmore. If Belle Elmore was alive and well in her sister's house in the States, who was that buried in the cellar in 39 Hilldrop Crescent? It seems unlikely we will ever know what happened after Crippen's dinner guests left their home in the early hours of 1 February 1910.

Agatha and scopolamine

Agatha Christie was well aware of the Crippen case – few in the UK could have been unaware of it, even years afterwards. The publicity had been extensive and a wax model of Crippen’s head was displayed in Madame Tussaud’s chamber of horrors for decades. Christie used the Crippen case, along with several other real-life murder cases, as inspiration for the plot of *Mrs McGinty’s Dead*.^{*} The case also provided her with a poison many of her readers would have been aware of, even if it was only by name.

Her accurate descriptions of hyoscine’s sedative properties tells the audience watching *Black Coffee* what to look out for. Several characters comment on a bitter taste in their coffee, in this and other stories, and many of the plant alkaloids do indeed have a bitter taste that real-life poisoners have often struggled to obscure.

Pure hyoscine is a viscous liquid with a bitter taste. Liquid hyoscine can be made into a solid, hyoscine hydrobromide, the full name of the drug mentioned in Christie’s *Black Coffee* and the form still most commonly used in medicine even today, by adding hydrobromic acid. Solids are easier to store, weigh and manipulate into tablets, but the addition of hydrobromide won’t help the very bitter taste. Whether Christie’s murderers knew of hyoscine hydrobromide’s bitter flavour and deliberately added the drug to coffee to try and mask the taste isn’t clear. It also wasn’t completely successful, as Christie’s victims comment on the bitter taste of their coffee, but it isn’t bad enough to stop them drinking the whole cup.

Anyone reading or watching an Agatha Christie murder mystery will be alerted by mention of an unusual flavour in anything someone has eaten. These characters can’t be expected to live long and they can be studied more closely for symptoms that may provide clues as to ‘whodunit’. But, Christie is so good at hiding the truth in plain sight that even those with extensive knowledge of poisons, their symptoms and use in real-life murders will struggle to work it out before Poirot.

Notes

^{*} There was also a particularly upsetting film adaptation of Christie’s short story ‘The Coming of Mr Quin’ the same year (see Chapter C).

[†] In 1931, the play moved to the St Martin’s Theatre, where Christie’s *The Mousetrap*, the most successful run of any play by anyone anywhere, now has its home.

- ‡ Why a hospital dispenser was allowed to take dangerous drugs home with her from work is never discussed.
- * Trevor claimed he only got the role because he could speak with a French accent. Sadly, we can no longer see his performance as the film is considered lost.
- * It was taken by Basil Rathbone, who would later achieve fame as Sherlock Holmes. There is also a TV version with Bernard Lee, who would go on to play the first M in the James Bond films, in the role.
- * Christie used the name scopolamine once in her short story 'The Girdle of Hyppolita' as an example of a compound that could have been used to drug a kidnap victim and leave them with little memory of their ordeal.
- † A system of assigning a single internationally recognised name to a compound in an attempt to avoid confusion when there are several different names in use in different parts of the world.
- * Christie knew *Hamlet* well as she named her play *The Mousetrap* after the play within a play that features in Shakespeare's masterpiece.
- † Not the ideal route for a poison to enter the body, being obstructed by wax and with few veins nearby to absorb the toxin into the bloodstream, but not impossible either. Rather than its lethal effect on the body, Shakespeare was probably more interested in the dramatic effect it would have on his audience.
- ‡ That I have spotted. Correct me if I am wrong.
- * There are other compounds in these plants that can affect the body, but these are the ones we are concerned with.
- * If not superimposed, the two versions won't match up no matter how much you turn them around. You can't turn a right hand around to make it look like a left hand – they are different.
- † Separating the two hands would have made no difference as the molecules can convert from one to the other in the stomach. The drug is still used to treat leprosy but given with stern warnings against getting pregnant while taking it.
- * As a result of the thalidomide disaster, all chiral drugs now have to be separated and tested independently for any unforeseen side effects.
- † They are called muscarinic receptors because they interact more strongly with muscarine, a toxic compound from some mushrooms. The other type of acetylcholine receptor is nicotinic, so named because it interacts more strongly with nicotine.
- * It also means the drug has been used for criminal activity. People have woken up to find their possessions missing or that they have been the victim of rape but with no recollection of events. They only discover later that their drink or food had been laced with scopolamine.
- * Such a large amount that the pharmacy had to get it from the wholesaler and Crippen had to call back three days later to collect it.
- * The same Spilsbury we met in Chapter G (see page 123). The Crippen case established his reputation as Britain's foremost criminal pathologist.
- * An appeal made to the court did not succeed.
- * To underline the theatrical connections, *Mrs McGinty's Dead* features a subplot where Ariadne Oliver, a largely biographical portrayal of Christie herself, is collaborating with a playwright over an adaptation of one of her crime novels.



A Pocket Full of Rye

Agatha Christie killed *a lot* of characters and yet we rarely consider her to be a morbid or macabre writer. Helping to keep the mood light and the focus on the problem-solving aspects of her mysteries, Christie's victims are often not very likeable. It is a strategy that saves on words of mourning and simultaneously lengthens the list of possible suspects to include anyone who may have been offended, thwarted or crossed by the victim during their lifetime. Such is certainly the case with the first victim in Christie's 1953 novel, *A Pocket Full of Rye*.

Rex Fortescue's unscrupulous, possibly illegal, business practices have earned him a lot of money and a few enemies. His sons have had enough of

his recent erratic behaviour, which has led to some particularly rash decisions and is threatening to bankrupt the family business. His daughter has reason to dislike her father because he has forbidden her to marry the man she is in love with. Rex's young wife is fed up with her married life and has embarked on an affair. Life for a lot of people would be much happier with Rex out of the way. When the unlikeable businessman dies after being taken suddenly and seriously ill in his office one day, few of his nearest and dearest seem very upset.

As a slightly corpulent man in his sixties with a very stressful job, a heart attack might be suspected of causing Rex's sudden and rapid demise, but neither of the two doctors who attended him in his final hours thought this was the case. Nor did they believe the sudden death was due to any other natural cause. Mr Fortescue's premature end must, therefore, have been brought about by *unnatural* causes, probably poison.

Any sudden death under unusual circumstances means the police will become involved, and in this case it is Inspector Neele leading the investigations. He soon learns that Mr Fortescue, before he died and between convulsions, said there was something in the tea made for him by his secretary, Miss Grosvenor. The traditional mid-morning cup of tea taken by a large proportion of the UK population would be the obvious vehicle for administering poison to an Englishman. The usually capable and efficient secretary was also not very prompt in calling for medical help. It doesn't look good for Miss Grosvenor, but she is one of the few people who has no motive for killing Rex Fortescue.

Christie used poisoned tea in more than one plot, but in *A Pocket Full of Rye* it would be a far too easy and obvious solution. Before the reader can start to worry that the Queen of Crime was having an off-day when she plotted this particular novel, we learn that Miss Grosvenor and the tea are completely innocent. As is so often the case with Christie, the symptoms of the poisoning provide the clues to what really happened.

We learn from Dr Bernsdorff, one of the medical staff who treated Fortescue before he died, that if Rex's symptoms had started immediately after drinking the tea, the tea couldn't have been poisoned. Despite what you may have seen in TV murder mysteries, very few poisons take effect so quickly. Cyanide and nicotine are both mentioned as fast-acting poisons, which they are, but their symptoms are very different to those shown by Mr

Fortescue. Dr Bernsdorff is privately convinced that Rex was poisoned with taxine, the toxic component of yew trees.

Dr Bernsdorff admits that taxine is an unusual poison and that to his knowledge it had never been used deliberately. The tests to confirm his suspicions haven't been carried out yet, but he is confident in his assertion because he saw a case of taxine poisoning just a few weeks earlier. The doctor had treated two children after they had drunk the 'tea' they had made from yew berries for their dolls' tea party.

If Dr Bernsdorff is right, it means Mr Fortescue was mistaken in thinking that the poison was in his mid-morning tea. Tea is also unlikely to mask the slightly bitter flavour of taxine and Rex would be unlikely to drink it. Bernsdorff's medical opinion is that it would have taken at least one hour, perhaps longer if Mr Fortescue had a large breakfast, before the symptoms from taxine poisoning began to present themselves. Inspector Neele learns that Mr Fortescue did indeed eat a large breakfast on the morning of his demise at his home, Yewtree Lodge.

A visit to Yewtree Lodge establishes that it is aptly named, having yew hedges and an ancient yew tree in its grounds, providing plenty of raw material for the would-be poisoner. Detailed enquiries into the breakfast habits of the occupants of the house reveal that the taxine must have been added to the marmalade. Rex Fortescue was the only one to eat marmalade while everything else at breakfast was also eaten by someone else in the house. The means of poisoning Mr Fortescue are quickly identified and confirmed when a jar of marmalade, with some of its marmalade missing, is found thrown into a hedge. The opportunity to poison Rex was open to anyone in the house; family members, servants, even passing poisoners who noticed a door left open and a convenient jar of marmalade sitting in the pantry.

While the police go about establishing alibis and testing the contents of jars of marmalade, Miss Marple arrives to solve the mystery simply by talking to the suspects while she knits. This is not her first foray into detecting crime. Her first appearance had been in Christie's 1927 short story 'The Tuesday Night Club', which became the opening chapter of *The Thirteen Problems* where an already old Miss Marple solved 13 crimes while barely moving from her seat. By 1953, despite her advanced years, she was going out and about in search of murderers.

Christie's obvious affection for her Miss Marple character meant she reappeared regularly. Her character softened as time went on, but she hardly aged over her 41 years of amateur sleuthing in 12 novels and 20 short stories. Christie would have undoubtedly included her in more stories but, not being a professional detective like Poirot, it was difficult to contrive ways for a private citizen of limited means and mobility to stumble across enough crime to justify her appearance.

Miss Marple's arrival at Yewtree Lodge in *A Pocket Full of Rye* is explained by another death in the household. One of the maids is found strangled, a maid that Miss Marple had trained in the art of domestic service. Out of loyalty to the girl, she turns up to find out who killed her. Miss Marple's ability to simply walk in and take up residence in a strange house in the middle of a murder investigation may seem astonishing, but as a comment on how older women become invisible to the society around them, it is very much on the nose. Miss Marple's knowledge, mental acuity, powers of observation and understanding of the modern world are also frequently undermined and underestimated by those around her, in a way that many women will be able to relate to. Her methods of deduction, however, by careful observation of character and comparisons with people and incidents from her tiny home village of St Mary Mead, defy logic or analysis, but she is invariably correct. What can be analysed in *A Pocket Full of Rye* is Christie's methods of poisoning and her intriguing choice of taxine.

Christie was certainly correct when she wrote that taxine was a rarity in terms of criminal poisoning. There may not have been a well-known real-life case to inspire her, but taxine from yew trees is a particularly appropriate choice for a writer renowned for her use of poisons. The scientific name for yew trees is *Taxus*, taken from the Roman name for the trees. The Romans are thought to have got the name from the Greeks. *Toxon* was the ancient Greek word for bow, and the best wood for bows is yew. The Greek word *toxicon* was for the poison used to tip their arrows. It is also the source of the English word 'toxic'. Both yew trees and their toxic components have a long association with death.

The yew story

Species of the genus *Taxus* are slow-growing, long-lived trees commonly known as yew. They are ancient in every sense of the word. Fossils of *Taxus jurassica* date back 140 million years to the mid-Jurassic period. The modern English yew (*Taxus baccata*) is 15 million years old.* Today, there are 12 species of yew growing over much of Europe, North Africa, North America, China, Japan and the Philippines.

The trees can live to an incredible age, but it can be difficult to determine the exact age of an individual yew tree. After 400 years of growth their trunks begin to hollow out, erasing the rings that are commonly used to date trees. There are, however, many examples of living trees that are believed to be several thousand years old. Some of the oldest yews in Europe can be found in English churchyards, and are therefore associated with the dead. However, their position on holy ground may have originally had more to do with their longevity and resilience, symbolising immortality.

Such ancient trees have become part of human history and folklore. Yews feature in several religions and mythologies. They have been identified as 'World Trees' that link heaven, earth and the underworld from their leaves down to their roots. According to ancient Greek mythology, Hecate, the goddess with expert knowledge of plant lore, medicines and poisons, had yew growing in her garden along with many other poisonous and healing plants. Yew also has a connection to the goddess and huntress Artemis, who meted out justice with her bow and arrows that she sometimes tipped with poisonous yew when she directed them at humans.* Yew trees have also found practical as well as spiritual significance.

Yew artefacts dating back to the Stone Age show that the wood has found many different uses over the centuries, but the most common of these by far was for making bows. In the medieval period the most fearsome weapon was the longbow and the Welsh and English were particularly well known for their prowess with these powerful weapons. Local yew provided the wood for the excellent-quality bows used by English and Welsh archers, but the very best longbows were made from foreign yew. Between the thirteenth and sixteenth centuries so many yew trees were cut down to supply wood to English armies that 500 years later the trees have still not recovered.† The slow-growing nature of *Taxus* and the continuous human

manipulation of the land in much of Europe means that in many places they are a threatened species.

The longbow was a lethal weapon, but there were other ways yew could kill. In *Richard II* Shakespeare wrote of ‘double-fatal yew’ in reference to yew’s poisonous as well as military uses. ‡ The Greek philosopher Theophrastus (c.371–c.287 BC) was the first to write about the toxicity of yew. Pliny the Elder (23–79) warned that the wine carried in traveller’s vessels made from yew wood was poisonous. Other writers around the same time tell of how Gallic tribes defended their lands from Roman invasion with arrows tipped with yew poison. Catuvolcus, ruler of the Eburones, * a tribe located in northeastern Gaul, poisoned himself with yew ‘juice’ in 53 BC rather than suffer the humiliation of Roman captivity. The Greek physician and pharmacologist Dioscorides (c.40–c.90 AD) claimed the yews in some locations were so potent they could kill without even coming into contact with their victim: ‘the yew growing in Narvonía [Spain] has such a power that those who sit or sleep under its shade suffer harm or in many cases may even die!’ Dioscorides may well have been exaggerating, but there are many others with tales of feeling more than a little unusual in the presence of yew trees.

In 1970, 71-year-old Dr Kukowka, a retired medical professor, had been gardening under the shade of some yew trees for a couple of hours when he was suddenly overcome by dizziness, nausea, headache and restlessness. He became disorientated, lost his sense of time and broke out in a cold sweat. He then began to hallucinate. Images of vampires, vipers and ‘diabolic scenes’ appeared to him that eventually gave way to those of paradise, heavenly music and an ‘indescribably happy’ mood. His doctor examined him after his experience but could find no signs of poisoning, salt deficiency, overheating or anything else that might explain his delusions. Kukowka found he could reproduce these effects by revisiting the yew trees, but his experiences became too frightening to continue his experiments.

Of the many different chemicals that can be found in yew, there is one that has been considered as a possible cause of these disturbing experiences. Ephedrine is present in tiny quantities in yew trees and it is known to

constrict blood vessels. However, pharmacology studies have not been able to confirm any psychoactive substances or effects from *Taxus* trees.

Compounds that could induce terrifying hallucinations are one thing, but compounds that can kill are quite another. Fortunately for researchers, medical staff, detectives and crime writers, the lethal components of yew have been isolated and identified.

Taxine

In 1856, a German pharmacist, H. Lucas, extracted a white powder from yew, which he named taxine. This is the substance Agatha Christie names in *A Pocket Full of Rye*. However, three years after the publication of the novel, taxine was revealed to be not a single compound, as Lucas and Christie had believed. Two German scientists, Graf and Boeddecker, used chromatography to tease out the components of taxine.

Chromatography, literally meaning ‘colour writing’, was invented in 1855 to separate out different coloured plant pigments. As a child you may have done something similar to separate ink spots into rainbows of different coloured components. Almost any mixture of chemicals, be they a pigment, or chemical extracted from a plant, or sampled from a person during an autopsy, can be separated into its components this way. The sample under analysis is dissolved in water or another liquid and allowed to soak along a piece of paper or through a column of some other solid material. Some chemicals in the sample will dissolve more easily than others and travel further in the liquid, seeping through the paper or column.

Once the different chemical components have been separated, they can be analysed and identified individually. This identification might be made by comparison with known chemicals. For example, if a green and purple ink both show a blue spot at the same position after being treated under the same chromatographic conditions, then the same blue colouring must have been used to make both the green and purple ink. Different blue pigments made by different manufacturers can then be treated under the same conditions and the distance they travel compared with the blue dot from the sample.

In a suspected poisoning case, material extracted from the body can be separated by chromatography. The distance travelled through a column by each of the components in the extract can then be compared with the distance travelled by known toxic chemicals under the same conditions. Each individual component can also be tested and analysed by a variety of techniques to confirm that it is the same chemical and not something else that coincidentally travels the same distance.

Chromatography has become much more sophisticated over time and its use much more widespread, particularly since the 1950s. It is now a standard technique for separating substances in medical, criminal and chemical research settings. Working in 1956, Graf and Boeddecker found a certain combination of liquid, solid material and speed separated taxine into at least 11 similar but different alkaloids.

As techniques became more sensitive, many more compounds have been separated from yew extracts, but identifying these alkaloids has not been easy. The chemical structure of the most common yew alkaloid was not determined until 1991, for example, though that didn't stop scientists testing the effects of these different compounds on the body.

Today, over 400 different alkaloids have been identified in yew. The most well known of these is paclitaxel, even though it is only found in minute quantities in the bark of one specific species of yew, the Pacific yew (*Taxus brevifolia*). This is because of its use in medicine.

Christie wrote in her novel that taxine had no medical use, and this was certainly true for conventional medicine in 1953, but in the 1960s paclitaxel was found to be a very effective treatment against certain cancers.* Paclitaxel interferes with the mechanism cells use to divide and stay healthy. If cancer cells cannot divide, the tumour cannot grow, and if cells cannot stay healthy, they die. Marketed under the name Taxol, it is used to treat breast, lung and prostate cancers.†

Intriguingly, the Bower manuscript, dating to the first half of the sixth century, mentions a treatment for abdominal cancer containing parts of the yew tree mixed with clarified yak butter. Though there may have been some anticancer compounds within this preparation, without proper refinement they would only have been present in very tiny quantities that were greatly outweighed by the more toxic components. The same would be true of any

remedy that relied on crude extracts of yew plant, but that doesn't seem to have deterred people from prescribing yew for all manner of ailments over the centuries.

In India, a few pieces of yew bark were cooked with salt and ghee to make a drink called *jya*, which was said to bestow spiritual strength and vitality. It has also been suggested as an ingredient in witch's salves and visionary 'sleeping salves'. In the Greco-Roman world, yew poison was used as an antidote to adder bites. In Coptic medicine yew extract was used to treat skin problems. In addition to these remedies, since the early Middle Ages right up to the twentieth century, yew preparations have been used to treat parasites, diphtheria, epilepsy, rheumatism, arthritis and tonsillitis. One particularly common use was to stimulate menstruation and induce abortion. These remedies often called for a dose of unrefined yew that was close to, or in excess of, the lethal dose. Many patients would have died of their treatment.

Many of the problems caused by yew extracts would have been due to the most abundant and toxic of the major yew alkaloids, taxine B. It constitutes approximately 30 per cent of the total alkaloids extracted from English yew, *T. baccata* though some of the many other alkaloids present have similar effects on the body.

How taxine kills

The effects of taxine B on the body are mostly due to interactions with sodium (Na^+) and calcium ion (Ca^{2+}) channels in heart cells. As discussed in previous chapters, it is the movement of Na^+ and Ca^{2+} in and out of the cardiac myocytes in the heart that causes bundles of thick and thin fibres to slide over each other, shortening them and making the cell contract. The coordinated contractions of the cardiac myocytes result in the overall movement of the heart, squeezing the chambers and pushing blood around the body.

A higher concentration of Ca^{2+} ions inside a myocyte makes it contract more strongly. The increased flow of Ca^{2+} ions into the heart cells caused by taxine can make them beat twice in rapid succession, or beat at a higher rate in some parts of the heart independently of others. The heart can slow

down, speed up, the different chambers of the beating heart can become uncoordinated and an irregular heart rhythm can develop. All of this means blood is not being pumped efficiently. In some cases an irregular rhythm can lead to cardiac arrest.*

An estimated 50–100g (1.76–3.53oz) of yew leaves, and slightly smaller quantities of the seeds, contain enough taxine to kill an adult. Old yew clippings can be more toxic than fresh because the reduced water content increases the concentration of alkaloids. Even eating the bark can kill. The difficulty for Christie's taxine poisoner, and any other taxine poisoner, is persuading their victim to eat all these leaves, bark or seeds without them becoming suspicious. The murderer in *A Pocket Full of Rye* has the cunning plan to add the poison to a jar of marmalade, knowing that their chosen victim, Rex, and only Rex, is likely to eat it and be killed. But only one part of a yew tree is likely to go unnoticed in a jar of marmalade – the red-orange berries.

In *A Pocket Full of Rye*, Christie makes repeated references to yew berries in relation to poisoning, but the red fruits of yew trees are neither berries nor toxic. The fleshy red parts of yew that surround each of the yew's seeds are known as arils and they are the only part of the plant that is safe to eat – just as long as they really are red and ripe, not green, and you take care to spit out the seed without damaging it with your teeth. This is not necessarily a mistake on Christie's part. She does not say whether the yew berries that killed her fictional victims still contained their seeds or not. She is also quite specific that it is *taxine* added to the marmalade and therefore an extract of yew, rather than fragments of the plant itself.

Just 240–520mg (0.008–0.018oz) of yew taxines would have enough of an impact on the normal function of the heart to kill an adult. More than the minimum lethal dose must have been added to the jar of marmalade to ensure that there would be enough spread on Rex's breakfast toast to give him a single lethal dose. Anything less would cause illness, but not necessarily death, and thereby alert the victim that there were designs on his life. Rex's murderer may not have bothered to spend time refining their extract, making the amount needed to be added to the jar even larger. Even so, the amount would still be so small and nondescript in appearance that it could be mixed with the top layer of marmalade without making it look any

different to the rest of the jar. The slightly bitter flavour of the marmalade would also help mask the bitter taste of the taxine – as Christie correctly states in her novel. Rex would eat his hearty breakfast and head off to his office none the wiser, only to be suddenly struck down later in the morning.

Symptoms of taxine poisoning typically start to appear 30 to 90 minutes after ingestion. In the novel, Christie stretches this to almost two hours but explains the late onset of symptoms as due to the victim eating a large breakfast that would have slowed absorption of the alkaloid into the body. Christie is also correct in saying that once the taxine does take effect it does so ‘very suddenly and drastically’. With little or no warning, taxine can cause nausea, vomiting and abdominal pain from the inflammation of the gastrointestinal tract. There can also be diarrhoea, dizziness and numbness, dilated pupils and pale skin. Initially, breathing speeds up but then becomes slower and shallower. As the circulation becomes more disordered, consciousness fades, there is collapse, muscle weakness, convulsions and coma.

Christie skips over any vomiting and diarrhoea; her poisonings are usually quite clean and tidy even if they are painful and unpleasant in other ways. She generally ignores any nausea or the sudden appearance of any bodily fluids, and goes straight to the convulsions. Rex is described as ‘contorted with agony’ and his ‘convulsive movements were alarming to watch’. He is only able to gasp out a few words at a time, suggesting he is also having problems breathing. There is, however, no loss of consciousness and Rex still has enough breath in him to continue complaining about poisoned tea until shortly before he dies.*

The description of Rex’s symptoms is consistent with taxine poisoning, but none of them, not even the ones Christie left out, are *characteristic* of taxine poisoning, because there are no unique or tell-tale signs for yew toxins. This leads to the very real problem of diagnosing victims and treating them appropriately. Rex is, eventually, taken to hospital, but what could have been done for him when he got there?

Is there an antidote?

If Rex was wheeled into hospital suffering convulsions and complaining about poisoned tea, the first thing to do, if his life wasn't in immediate danger and after controlling the convulsions, would be to remove any poison that hadn't already been absorbed into his body and prevent his condition from getting any worse. As mentioned several times already, gastric lavage washes poison out of the stomach and swallowing charcoal helps trap any remaining poison molecules within its pores.

A patient's vital signs are monitored as standard procedure when they arrive in hospital, alerting medical staff to changes in things like blood pressure, blood oxygen levels and heart rate. But none of these changes is likely to suggest yew poisoning to the average doctor. Drugs and treatment are given depending on the symptoms presented and this will vary between individuals. There is a case of a woman surviving taxine poisoning after being given large doses of lidocaine, but this is rare. Other patients have pulled through after their elevated blood potassium levels were corrected,^{*} but there is no standard treatment and no specific antidote for taxine poisoning.

The difficulty in treatment is made clear in the novel because, despite the best efforts of medical staff, Rex does not survive. It takes Rex around three and a half hours from swallowing taxine to dying – well within the expected range of 90 minutes to 24 hours.

Christie also acknowledges the difficulty in diagnosing taxine poisoning. Her fictional doctor is only so confident in his assertion because he has seen two recent cases to compare with Rex's condition. Confirmation has to wait for the results of a post-mortem examination, but none of the interactions between taxine and the body, nor the symptoms that result, leave any identifying trace. The mechanism of death through heart irregularities leaves no visible sign at autopsy. A recent case in Canada illustrates the problems.

A real-life case

In 2021, two friends were chatting over a video call when one of them suddenly started struggling to breathe. Her friend urged her to call an ambulance, but she was unable to respond and collapsed. The friend then

called the collapsed woman's mother who then called an ambulance. The paramedics arrived and attempted to resuscitate the woman but without success.

The woman's sudden death prompted an investigation. An autopsy revealed rib fractures at the front of the chest where the paramedics had tried to resuscitate her, but there were no other signs of injury. Her lungs showed clear signs of pulmonary oedema, which is consistent with acute cardiorespiratory arrest, but it is also a sign of a lot of other things. All in all, there were no physical signs that could account for the woman's death and so samples were collected for toxicological analysis.

Toxicological screenings are usually looking for common poisons, such as prescribed and recreational drugs. Yew toxins and their metabolites are not part of routine blood analysis and so none would be expected to be detected at this point. In this case, the routine blood analysis revealed only the presence of caffeine. There was no alcohol, prescription medication or illicit drugs present. The absence of common chemicals that cause overdoses meant the toxicologists and pathologists had to look elsewhere. Fortunately, there was a clue in the stomach contents.

In the stomach were 180g (6.35oz) of plant material. These chopped up green leaves and small twigs were sent to a forensic botanist for identification. After the plant fragments were positively identified as those of *T. baccata*, the English yew, toxicological tests for the taxines could be carried out. When the deceased's blood was reanalysed specifically for yew toxins, taxine alkaloids were found. These results were compared with extracts of the plant material from her stomach to confirm she had been poisoned by the leaves and twigs that she had eaten.

A search of the victim's home found plant material identical to those found in the stomach contents and they looked as though it had been used for cooking. Interviews with the woman's family members revealed she often foraged fresh herbs and other plant material from her local environment. It appears she had either accidentally foraged the wrong leaves or collected the yew leaves and twigs in the mistaken belief that they were edible. Based on all the available information, the coroner determined that the woman's death was accidental.

The majority of yew poisoning cases are determined not by symptoms or even blood analysis, but by finding plant fragments in the stomach or

intestines of the victim at autopsy. Confirmation of taxine poisoning usually comes afterwards if the blood and other material is reanalysed for the presence of taxines or their metabolites.

Taxine B is metabolised in the body to 3,5-dimethoxyphenol (3,5-DMP) and finding this in the body can therefore indicate taxine poisoning, but is not conclusive. The metabolite 3,5-DMP is also found in other plants and it is not always present in yew poisoning samples, as was the case for the woman in Canada. Today, even though they are not routinely used, there are standard samples of taxine B and 3,5-DMP available for comparison with blood and other bodily samples. But none of these would have been available for comparative testing when Christie was writing.

This is a potential problem for the taxine poisoning in *A Pocket Full of Rye* where no plant material should be present because it would have deterred the victim from eating his poisoned marmalade. The sharp-eyed doctor that treated Mr Fortescue, and was convinced of taxine poisoning, must have alerted the pathologist to look out for yew plant fragments or yew toxins in the body. Extracts could have been made from yew plant material for comparison with any alkaloids isolated from the body using the chromatographic techniques described earlier.

Christie almost never goes into the details of how forensic toxicology tests are carried out. But the identification of taxine poisoning in the toxicology report handed to Inspector Neele in the novel would be well within the scientific capabilities of the day.

Agatha and taxine

Yew's deadly associations are strong, diagnosing yew poisoning is difficult and no standard treatment is available, but the actual number of deaths caused by yew are surprisingly low. A survey conducted in 1998 found 11,197 yew poisoning cases worldwide from all species of yew, and no deaths. A paper published in 1992 could only find 11 fatal cases reported in the literature for the previous 31 years. And as there were no high-profile, or even low-profile, cases of taxine poisoning for Christie to draw on when she was writing, where might she have got her inspiration from? A line in the novel may provide a clue.

Miss Ramsbottom, an elderly relative living in Yewtree Lodge, mentions that ‘Children eat yewberries sometimes’, and that it ‘Makes them very ill too. I remember a case when I was a child. It made a great impression on me.’ It is also the case of two children who died after drinking ‘yew tea’ that alerts Dr Bernsdorff to the possibility that Rex Fortescue was also poisoned by yew. Perhaps Christie was recalling an incident from her own past?

Christie’s use of taxine is scientifically credible in this story but very unusual. Her readers might be vaguely aware of the plant’s toxicity or its associations with death from its familiar presence in graveyards, but it is a rarity in terms of criminal poisoning. She, like her fictional doctor, may also have been ‘tired of the inevitable weed-killer’ in crime fiction and fact. The classic, but less commonly used, taxine may have offered the author ‘a real treat’ to write about.

Notes

- * For context, modern humans haven’t been around for one million years yet.
- * According to Homer.
- † For example, between 1512 and 1592 the Fürer company held the yew privileges for southern Germany and parts of Austria. In that time it is estimated the company exported 1,600,000 yew staves. And there were other companies operating all over Europe during the same period.
- ‡ Along with henbane (see Chapter S, page 241), yew extract has been suggested as the poison dropped in Old Hamlet’s ear.
- * ‘Eburones’ is also a name derived from an old word for yew.
- * It took decades to carry out the necessary clinical trials and develop a method of mass production to make it available to patients.
- † Rather than removing the bark and killing thousands of Pacific yew trees, a different compound is extracted from the clippings of the more common European yew and then chemically converted into Taxol on a large scale. 3,000kg (6,613lb) of needles are needed to obtain just 1kg (2.2lb) of the paclitaxel precursor, but it doesn’t kill the trees.
- * The therapeutic doses of paclitaxel in cancer treatments are too small to affect the heart.
- * Some authors around the turn of the twentieth century mention a ‘particularly happy facial expression’ on *Taxus* corpses, but Christie makes no mention of it.
- * The movement of potassium ions in and out of cells is inextricably linked with the movement of sodium and calcium ions (see Chapter A).



Death in the Clouds

The inspiration for Agatha Christie's stories came from many sources. People she had met reappeared as victims and suspects in her pages. Places she had visited set the scenes where murder and mayhem could occur. Poisons were plucked from her pharmaceutical training and collection of toxicology books. True crime stories provided motives and methods. Plots were also picked from her extensive knowledge of crime fiction.*

Agatha, and her sister Madge, were connoisseurs of detective fiction. More than once in her novels Christie, or at least her famous detective Poirot, gives her opinion on what is good, bad and overrated in the world of whodunits. Poirot was not overly impressed with the much-loved works of

Sir Arthur Conan Doyle, but Christie was happy to borrow several ideas from the Sherlock Holmes case files and give them her own twist. *Death in the Clouds* * would appear to be Christie's take on 'The Speckled Band'.

Doyle's short story is a classic locked-room mystery. Two people are murdered in a room where no human could have got in or out. The dying words of the second victim, 'speckled band', are the solution to the mystery.
Warning *Spoilers*

The speckled band is in fact a trained venomous snake, an Indian swamp adder. This creature can slither through openings too small for a person, to bite and kill the unsuspecting occupant of the locked room on the murderer's behalf.

In Christie's *Death in the Clouds*, the victim also dies from the effects of snake venom, but the locked room is an aeroplane in mid-flight. How these two crime writers tackled these somewhat similar ideas shows the different ways they approached the puzzles they set.[†]



When Doyle wrote 'The Speckled Band' in 1892, the locked-room mystery was already a well-established subgenre of crime fiction. Over the 40 years that Doyle wrote the Sherlock mysteries, he cemented many other detective fiction tropes, such as the eccentric private detective and the less-intelligent sidekick, even though he didn't invent them. By the time Christie wrote *Death in the Clouds* in 1935, her readers knew these tropes inside and out and Agatha could have a little fun with them.

Doyle's locked-room mystery excludes humans and keeps clues to a minimum, allowing Holmes to show off his brilliant deductions from minimal information. By contrast, Christie has lots of clues and lots of humans locked into her room, or plane, giving us a long but clearly defined list of suspects. Among the limited number of possible murderers is a famous detective who, following yet another well-worn cliché, just happens to be in the right place at the right time to be helpful in a murder enquiry. However, where Holmes sees everything, murder is committed right under Poirot's magnificent moustaches and he sees nothing. Here, the eccentric

Belgian detective is swaddled in scarves and coats to save him from draughts, and has his eyes closed for most of the flight while he endures the discomfort of travel sickness. He only rouses himself when he hears mutterings about doctors and an unresponsive, possibly dead, passenger.

A doctor is identified among the passengers and confirms that Mme Giselle is not merely resting, but she has ceased to be. A mark on her neck appears to indicate foul play, but another passenger points to a dead wasp in their coffee cup as the possible culprit. A sudden and terrible reaction to a wasp sting could certainly have caused death (see Chapter F, page 92), but the doctor won't commit to a definite diagnosis until after an autopsy.

Then Poirot spots something on the floor: 'a long, peculiar-looking thorn with a discoloured tip.' At one end of the thorn is 'a little knot of teased fluffy silk, orange and black' to give it a passing resemblance to a wasp. The dead wasp in the coffee saucer is dismissed as one of the many red herrings that have been packed onto the plane along with passengers and crew. Mme Giselle has been murdered using yet another detective story trope – a poisoned dart from a blowpipe of the South American hunters.

The blowpipe in question is found hidden behind one of the passenger's seats, the seat that had been occupied by an immaculately dressed little Belgian. It would seem the evidence points to Poirot using his little grey cells to commit a crime rather than solve it – an accusation that was levelled at Sherlock Holmes more than once. Of course, we know our heroes cannot go to the bad so easily and we can sit back to enjoy watching the detectives disentangle themselves from the wrong end of the law by proving someone else is guilty of the crime.



Blowpipes and poison-tipped darts get mentioned in many murder mysteries, including the Sherlock Holmes tale 'The Sign of Four'. This short story would appear to be where Christie got her information about these weapons. Doyle describes the blowpipe as a 'short, round piece of wood, like a ruler'. Christie's blowpipe is the thickness of a pen and short enough to hide down the back of an aeroplane seat. Both authors were

wrong, as blowpipes are typically much thicker than a pen and, though they vary in length, tend to be at least 1m (39in), and often longer.* The two authors perhaps thought blowpipes were more like peashooters.

Where Doyle used a blowpipe as an important clue, Christie used it as yet another red herring. Her diminutive blowpipe wasn't used to fire the poisoned dart – someone would have noticed in the cramped confines of the aeroplane had anyone tried. Instead, the dart was simply pushed into the neck of the victim as the murderer casually walked down the centre aisle of the plane.

A single dart pushed or fired into a body is unlikely to kill, so the hunters that use them cover the tip with poison (see Chapter A, page 24). Doyle's dart in 'The Sign of Four' is dipped in a strychnine-like substance, determined by the symptoms displayed by the victim. Christie's victim shows no helpful characteristic signs; it is up to a toxicologist brought in to investigate the sudden death who determines Mme Giselle was killed by one of the two poisons on the tip of the dart.

The first poison identified is yet another trope, the classic curare that Christie's readers would expect, but there is only a trace amount on the thorn and not enough to kill. It is the second poison that killed Mme Giselle, 'the venom of *Dispholidus typus*, better known as the boomslang or tree snake.'

Christie always went out of her way to avoid the obvious, but snake venom is a particularly unusual choice. I can only guess that she was inspired by Doyle or something she had picked up in her extensive background reading into toxic compounds. Her choice of snake is even more surprising. It isn't a very well-known venomous snake, certainly not outside of its native Africa.* But its venom does offer certain advantages to the murder mystery writer.

The snake venom story

All snakes are predators that eat relatively large animals, and some of those animals are more than capable of fighting back against the almost defenceless snake. Snakes have no limbs, claws or canine teeth to hold or subdue their prey, so they must either pick easy targets, like fish or frogs

that pose little threat, or immobilise their victim before it can damage them or escape.

The majority of snakes physically immobilise their prey by constricting them, but around 20 per cent of the more than 3,700 snake species in the world are venomous. Venom is biologically expensive to produce in terms of energy and resources but it does offer some advantages over the physical exertion of constriction.

Venoms are full of chemicals that target specific biological sites to cause rapid and major disruption to a body even in small doses. This is necessary because a single bite has to deliver sufficient toxins to kill the prey rapidly, either while still in the snake's jaws or not very far away. Other compounds, such as enzymes that break up cell walls or break down proteins, are toxic but also help the venom spread through the body. In addition, these destructive proteins can speed up the process of digestion, which is helpful, as snakes cannot chew or tear their food into more digestible lumps.

There can be anywhere between 20 and over 100 different compounds in an individual snake's venom. Most of these compounds, by number and volume, are proteins (both enzymatic and non-enzymatic). Then there are metal ions that can activate these proteins or cause disruption in their own right. Some snake venoms also contain lipids, carbohydrates and biogenic amines, such as histamine and neurotransmitters that affect nerve signals. All of these individual components can disrupt the normal working of the body, but many work together to make things even worse for the victim.

The exact composition of venom delivered from a snakebite varies because there are many different snakes. Different snakes from the same species can also have different mixtures of compounds in their venom because of their age, sex, geographical location and availability of prey. Rather than reel off lists of individual components, these arrays of compounds are sometimes loosely grouped together based on their biological effects: neurotoxins (toxic to nerves), haemotoxins (toxic to blood) and cytotoxins (toxic to cells).

Evolution has refined snake venom towards the snake's preferred food, but they can still affect humans. However, because humans are many times bigger than a snake's usual food, it is less likely to kill us and will take longer to do so. There are also many other factors, such as natural habitat and behaviour, to take into consideration when evaluating the danger

venomous snakes present to people. David Warrell, a professor of tropical medicine, classified venomous snakes into four groups based on their medical significance: really dangerous snakes that bite a lot of people; really dangerous snakes that hardly ever bite anyone; not very dangerous species that bite a lot of people; and not very dangerous species that hardly bite anyone.

There are an estimated five million snakebites to humans annually, with around half of them resulting in poisoning and more than 100,000 dying as a result of that bite. There are also hundreds of thousands of others left with permanent disabilities. Most of these bites occur in tropical and subtropical countries where snakes are a significant health problem. For comparison, Europe has 11 venomous snake species, out of the 27 that are native there, and they are mostly in the ‘not very dangerous species that hardly bite anyone’ club,^{*} whereas Africa and Asia have a significant number of snakes that fall into the ‘really dangerous snakes that bite a lot of people’ category.

India, the supposed origin of Doyle’s swamp adder in ‘The Speckled Band’, has 216 species of snake of which four are venomous – krait, cobra, Russell’s viper and saw-scaled viper. The Indian swamp adder is an invention. The description of the snake’s physical appearance and the rapid action of the venom suggests the cobra, *Naja naja*, is the closest real-life example of Doyle’s fictitious reptile.

Agatha Christie’s boomslang snake, however, is real and one of the roughly 150 potentially lethal snake species in Africa.[†] *Dispholidus typus* can be found in almost every part of sub-Saharan Africa in woodland, savannah and semi-desert. The common name ‘boomslang’ is Afrikaans or Dutch for ‘tree snake’ because that is where they spend most of their time.

Boomslangs are large snakes averaging between 1.2 and 1.5m long (roughly 4–5ft) when fully grown, but one specimen was found over 2m. They have small egg-shaped heads with large dark eyes.[‡] Most boomslangs are green in colour, making them easy to confuse with the other 20 species of green snake in Africa, but they also present in grey, brown and other colours. Very few African green snakes are dangerous[§] and the boomslang is a timid creature that will generally flee from anything too big to eat. If cornered it will inflate its neck and will only bite if provoked. Its fangs are

also towards the back of its mouth, which would seem to make it more difficult for a boomslang to bite into something as big as a person.

But there are records of boomslangs biting humans. Their head may be small and the fangs situated well back in the mouth, but these snakes can open their mouths almost 180°. They can sink their six surprisingly large fangs (three on each side of the mouth and approximately 3.5mm/0.14in long) into an arm or leg on first contact. If a boomslang got lucky and lodged its fangs into an unwary human living in the first half of the twentieth century, the chances are that person would still be unconcerned. That is because boomslang venom glands, when compared to well-known front-fanged species like cobras, seem insignificant, and there are no ducts to channel the venom to the fangs. However, unlike cobras that tend to strike fast and retreat, when boomslangs bite they hang on to their victim. This allows their limited supplies of venom to run down grooves in their fangs and into the wound. So the important question becomes how toxic is that venom?

Boomslang venom

Boomslangs feed mostly on chameleons, but they will also eat other reptiles, mammals, birds and eggs that can be found in trees. Their venom might be fatal for these small creatures, but as recently as the 1950s it was generally assumed that the venom from rear-fanged snakes did not pose a significant threat to humans, making Christie's choice even more curious. Even experts had been lulled into a false sense of security around these snakes, and sometimes to their cost.

In 1954, K. P. Schmidt, a well-renowned herpetologist working at the Field Museum in Chicago, was sent a snake for identification. As he took the snake from its container he was bitten on the thumb. The snake was a juvenile boomslang and only one fang penetrated a few millimetres into his skin. He and his fellow herpetologists did not think there was any real danger, so Schmidt carried on with his normal day and decided to write down any effects he experienced as a result of the bite. He stopped making notes the following morning because he felt he was recovering. Things changed quickly. By noon he was struggling to breathe. Offered medical

help, he refused because it would interfere with the symptoms he was documenting. He died just before 3 p.m., 24 hours after he was bitten.

Christie was right about the potential dangers of these snakes and it was not a lucky guess on her part. The first probable recording of a fatal boomslang bite was back in 1912 in the book *The Snakes of South Africa: Their Venom and the Treatment of Snake Bite* by F. W. Fitzsimons. He wrote about a man who had been bitten by a boomslang and subsequently died, though at the time his death was attributed to an infection introduced into the wound during attempts to treat him. Looking back on the case and the symptoms he experienced, Fitzsimons believed it was more likely the man had died from the effects of the snake venom.

Obviously not everyone had read Fitzsimons's book, and Christie appears to have only skimmed it, as she was clearly familiar with some parts but not others. She states in her 1935 novel that the effects of boomslang venom on humans was unknown, which was not true, and her fictional toxicologist makes assumptions based on accounts of bites to other large animals. He claims that a hyena injected with boomslang venom 'died before the needle could be withdrawn' and a 'jackal died as though shot by a gun'. Christie's phrases appear to have both been taken from the same paragraph in Fitzsimons's book where he is describing an experiment on a jackal.

The jackal died very quickly, but according to Schmidt's and Fitzsimons's accounts of human bites, the venom took much longer to kill. Some animals are more susceptible to snake venom than others, but how quickly a venom takes effect also depends on where the venom is introduced into the body. The jackal had the venom injected directly into a vein. Deliver snake venom directly into a vein, or worse still into an artery, and the flow of blood will spread the venom to all parts of the body very quickly. For example, in an experiment carried out in the late 1930s, boomslang venom injected into a vein killed a horse in two minutes. From the wound in Mme Giselle's neck, Christie's toxicologist would know the venom entered the jugular vein and would be able to assert that she must have died almost instantly.

Normally, a gland-full of venom can only be delivered as deep as the snake's fangs can reach. With front-fanged snakes this can be quite deep but you have to be very unlucky for the snake to hit a big blood vessel. Rear-

fanged snakes may not even make it all the way through the skin, depositing their venom into the dermis. Venom deposited in adipose tissue or muscle fibre, as in Schmidt's case and most others, will take a while to diffuse into the bloodstream, especially if the bite is to an extremity like the leg or thumb. In *Death in the Clouds*, Mme Giselle's rapid demise is, therefore, scientifically reasonable given the boomslang venom is introduced directly into the jugular vein in her neck.

Schmidt's experience is, however, the more typical outcome of being envenomated by a boomslang. The impact around the bite site is usually minimal, though some swelling has occasionally been observed. The most common symptoms include nausea, vomiting, colicky abdominal pain and headache. Then the victim will notice they are bleeding, from old and recent wounds including the puncture wounds made by the snake's fangs, as well as the gums, eyes and under the skin.

There is yet more bleeding internally. The colicky pain mentioned earlier is from bleeding in the gastrointestinal tract. Schmidt had noticed blood in his urine because his kidneys were bleeding. The difficulties he experienced in breathing were because his lungs were filling with blood. His autopsy showed there was also bleeding from his heart and brain. This uncontrolled bleeding can lead to shock, which can be seen in a patient as a fall in blood pressure, collapse, cold, cyanosed and sweaty skin, and impaired consciousness. Usually, it takes boomslang venom between 24 and 48 hours, sometimes longer, to kill an adult human, which is slow by snake standards.*

Of all the symptoms described here, Christie's toxicologist mentions only the bleeding under the skin. If Mme Giselle showed this symptom, it must have been hidden under her clothes because it is very noticeable. Fitzsimons's book describes huge purple patches with swelling that appeared on the body and face of his assistant, James Williams, after he was bitten by a boomslang. Williams's appearance was described as a 'terrible and never-to-be-forgotten sight' that left him almost unrecognisable. Williams was close to death for six days but he slowly recovered.*

Fitzsimons also mentioned bleeding from the wound where Williams was bitten. This should also have occurred in Christie's victim where the thorn was inserted in her neck. Her head lolled forward, perhaps obscuring

the blood from view, but it should have been noticed when the steward tried to wake her or when the doctor examined her. This might be an oversight by Christie, or perhaps she assumed death would have been so rapid, these symptoms would not have had time to develop, which is a possibility. Even if she had included this bloody detail, it would not have made a difference to the storyline. Obvious bleeding from the neck might have raised the alarm sooner, though it would still be too late to save Mme Giselle (see *Is there an antidote?*, page 289). An autopsy would still be needed, but the cause of death might have been determined sooner and the poison identified more quickly. Even so, there would still be a murder to solve and the murderer would be no easier to spot.

The huge difference in the times between envenomation and death in the fictional and real-life case can be explained by where the venom entered the body, but what about the amount of venom? The venom needed to kill the fictional Mme Giselle had to fit on the tip of a sharp thorn. The toxicologist speaking at her inquest claims that the boomslang snake is ‘one of the most deadly and poisonous in existence’, and there is some justification to this claim. Animal experiments have shown that boomslangs can kill with smaller quantities of venom than cobras or the much-feared black mamba. Schmidt was bitten by a young snake, meaning the venom glands might not have been full-sized yet. The dose he received also came from a single venom gland as only one fang penetrated his skin. And yet it was enough to kill him.

Using Warrell’s criteria, boomslangs are clearly in the ‘really dangerous snakes that hardly ever bite anyone’ group. The exact lethal dose of boomslang venom for a human is not known and difficult to calculate. Based on the venom’s toxicity in mice, a lethal dose for a person would be 9 or 10mg (0.0003 or 0.0004oz), but it could be as little as 2 or 3mg (0.00007 or 0.0001oz). To explain how such tiny amounts of snake venom could do so much damage to something as big as an adult human, we have to look in detail at the venom itself.

How boomslang venom kills

A typical adult boomslang has around 7.6mg (0.00027oz) of a colourless, transparent, syrupy venom in each of their two venom glands. Around 60 per cent of this weight is water. Removing the water leaves behind a whitish, flaky, crystalline solid. Dried snake venom can be rehydrated with water or saline and it will still be a potent poison, which is lucky for Agatha Christie. She decided not to have her murderer go through the difficulty of training a snake to commit murder as Doyle did.* Instead, they could use dried snake venom on the tip of a thorn, knowing it would dissolve in the blood and do its deadly work when the thorn pierced Mme Giselle's jugular.

Around 90 per cent of dried boomslang snake venom is made up of proteins that fit into the haemotoxic and cytotoxic categories mentioned earlier. These proteins can be subdivided further into seven protein families: three-finger proteins; phospholipase A₂; cysteine-rich secretory proteins; disintegrin-like proteolytic fragments; C-type lectin-like proteins; snake venom serine protease; and snake venom metalloproteases (SVMPs), just in case you wanted to look them up. The proteins responsible for most of the symptoms shown by victims of boomslang envenomation, i.e. the excessive bleeding, are thought to be due to the SVMPs, though the exact details have not been worked out.

In a healthy body, blood clots when there is damage to a blood vessel. Clots plug the gaps, stopping blood from leaking out and preventing bacteria and other pathogens from getting in. These clots are typically made up of fibrin, long, fibrous strands that tangle together a bit like a fishing net to trap platelets and other cells, forming a solid clot.

Fibrin does not float free in the blood because it would snag together and clots would form all the time. Clots in free-flowing blood can get stuck, restricting blood flow, and therefore oxygen supply, to parts of the body.* So fibrin goes about as fibrinogen, which doesn't tangle and is converted into fibrin as and when it is needed to form a clot. There are several interconnected chemical steps that are needed to trigger this fibrinogen to fibrin conversion, known as the coagulation cascade. Interfering with any of these steps can potentially disturb the normal clotting behaviour of the blood.

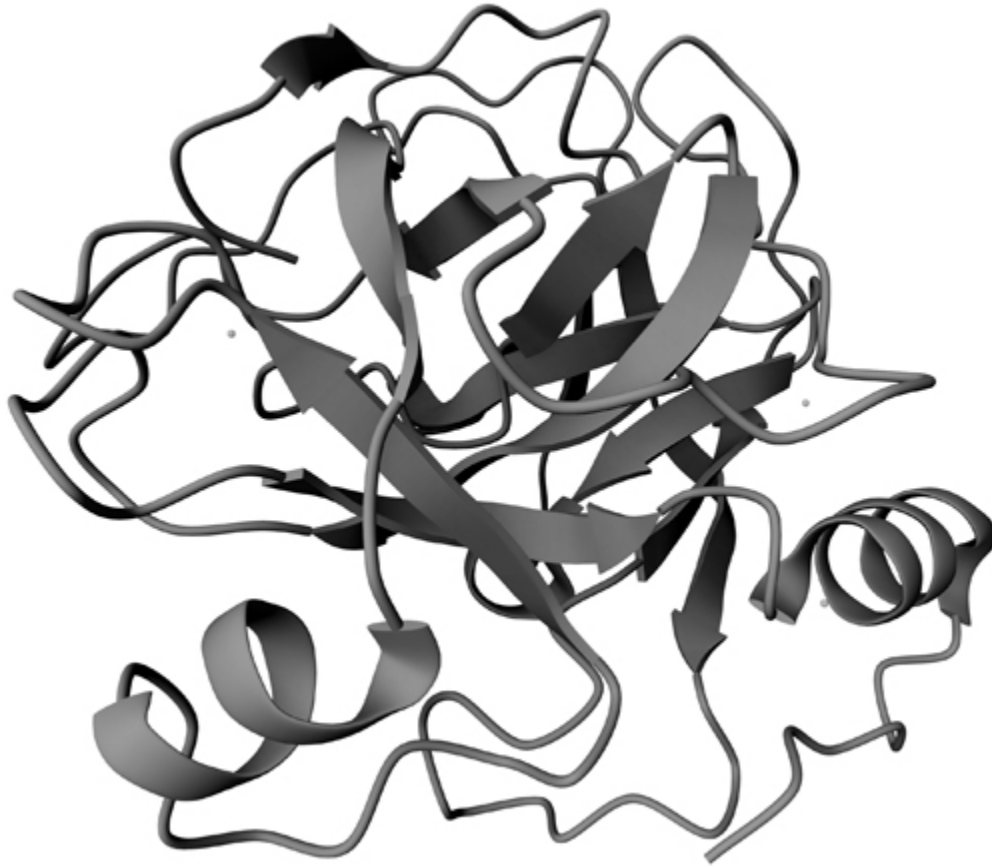


Fig. V.1. One of the snake venom metalloproteases (SVMP) from Bothrops jararacussu, the South American pit viper. To date, no SVMP from boomslang venom has been fully characterised.

The action of the SVMP components in snake venom result in the conversion of fibrinogen to fibrin. Introducing even a small amount of venom into a vein, as is the case with Mme Giselle, would cause the sudden and rapid conversion of lots of fibrinogen to fibrin and the formation of many clots. These clots could then go on to cause blockages in the blood vessels of the heart and lungs, preventing these organs from working and leading to a quick death. This may be what Christie was referring to in *Death in the Clouds* when her toxicologist states the poison, ‘acts on the heart, paralysing its action’. She isn’t wrong, but it is because of an indirect effect of the venom on the blood.

A slower introduction of the venom, such as in Schmidt’s bite to his thumb, would not cause the fibrin to tangle up all at once, but the stores of fibrinogen are used up faster than the body can replace it. Without fibrinogen, the body can’t form new clots when it needs to and there is no

way to stop new bleeds. This process is known as venom-induced consumptive coagulopathy. In the case of boomslangs, the bleeding that results from using up fibrinogen stores may not become apparent in a victim for several days.

Members of the other protein families in boomslang venom make the situation worse. Snake venom serine protease and C-type lectin-like proteins mentioned earlier have also been shown to affect coagulation of the blood. Three-finger proteins and cystein-rich secretory proteins can disrupt the walls of the blood vessels and make bleeds more likely. Phospholipase A₂ proteins break up phospholipids, molecules that make up cell membranes, causing inflammation and oedema.

In addition to the specific effects of boomslang venom, a small number of people are hypersensitive to snakebites, as others are to bee or wasp venom. Anaphylactic shock is a severe, life-threatening reaction (see Chapter F, page 92). Though Mme Giselle's death was very quick, it seems unlikely it was due to this hypersensitive reaction. One of her fellow passengers or the stewards were likely to notice signs of physical distress, such as difficulty breathing. But could anything have been done even if they had noticed?

Is there an antidote?

The usual first aid given to anyone suffering from shock, illness or deliberate poisoning in the Golden Age of detective fiction was to give them brandy. Alcohol was once also considered an effective treatment for snakebite, but it had no benefits whether it was poured over the wound or down the throat of the victim. More common is the widespread belief that the venom should be cut or sucked from the wound. Cutting into the wound will only cause more damage and may introduce infection. And snake venom can disperse through the body better than you can suck it out. Others may suggest using a tourniquet to stop venom spreading. An injection site on the neck would rule this out immediately, but even if someone has been bitten on a limb it is still inadvisable;* tourniquets, by design, cut off the blood supply, which the limb needs to function. Some people have had to

have limbs amputated because of the first aid they received rather than because of the snakebite.

The best advice is to leave well alone. Make the patient comfortable, reassure them and minimise their movements, and therefore the spread of the venom, while getting them to a medical facility as quickly as possible. Every case is different, but even with severe bites from very venomous snakes there is usually some time to get the victim to hospital. This is not so for poor Mme Giselle in *Death in the Clouds*. There is a doctor on board the plane with her, but even if he was a snake expert and had a well-stocked medical bag in his hand luggage, there is little that could be done. The only thing that will save someone who has received a lethal dose of snake venom is the appropriate antivenom.

First of all, it is important to establish that the patient really has received a dangerous dose of venom. This isn't always obvious in the case of snakebites. If a snake is trying to kill its prey it will inject a lot or all of its venom during the bite. A snake merely defending itself may bite as a warning but inject little or none of its precious venom.

Furthermore, the same amount of venom delivered into different people can have different outcomes. How susceptible an individual is to the effects of snake venom will depend on their age, size and state of health. Small children, older people and those with compromised health will be most susceptible. The victim in Christie's novel is middle-aged and apparently in good health.

If the patient is showing only mild symptoms or is simply shocked or terrified from being bitten by a snake, it is unwise to administer antivenom. In some people antivenom can trigger anaphylactic shock.* Patients should be closely monitored by medical staff for clear signs of envenomation before deciding to administer antivenom. In the case of slow-acting boomslang venom, this should focus on looking for clotting abnormalities and continue for 48 hours.

The second crucial factor is giving the *appropriate* antivenom. Antivenom contains a collection of antibodies specific to the antigens (the proteins and other chemicals that do the damage) in the venom. These antibodies stick to the venom antigens and inactivate them. Antibodies are produced naturally by the immune system, but it takes time. By injecting

small amounts of snake venom into large animals, such as horses, antibodies can be harvested from their blood, refined and stored in refrigerators, ready to be introduced directly into a patient when needed.[†] The antivenom dose will depend on the status of the patient and the product used, but several ampoules are usually required. These should be administered slowly via intravenous drip.

The antibodies produced by the horse will be specific to the venom they have been exposed to and, because snake venom is highly specific to the species and even the individual snake, different antivenoms have to be produced. In order to administer the right antivenom, the snake has to be correctly identified. Capturing or trying to kill the snake that bit you is dangerous and a waste of valuable time and energy that could be better spent getting treatment. Medical staff are also not very happy about venomous snakes being brought into hospitals, especially live ones. A photograph, description of the snake or even simply the symptoms being experienced by the victim can help narrow down the possibilities.

Some polyvalent antivenoms are produced that counteract the effects of the venom from several commonly dangerous species, but these are ineffective against the venom of the boomslang. Monovalent boomslang antivenom, a freeze-dried preparation, is what is needed, but it was developed in the 1940s, too late for Mme Giselle. Even if it had been available, it is produced by only one company based in South Africa. They will send it anywhere in the world as quickly as possible, but there is no way it could have reached Christie's murder victim in time.*

Such a potent poison, killing in just minutes, when delivered directly into a vein, would appear to have distinct advantages for anyone wanting to dispatch a rich relative or inconvenient acquaintance quickly and with minimum fuss. It does, however, pose considerable difficulties for the murderer, which is perhaps why snake venom homicides are exceptionally rare.

Some real-life cases

The first problem when selecting snake venom for your murder weapon is acquiring it. Snake venom is not available from retailers. You have to get it

from the snake, and venomous snakes are not readily come by in every part of the world. Having located a source of snakes, there is then the problem of interacting with them without being bitten yourself. One man decided to hire an expert to do the dangerous work for him.

In 2012, a man in India hired a contract killer and engaged the services of a snake charmer to kill his parents over a property dispute. The hired killer forced the elderly couple to sit in the back of one car and their driver was told to sit in another. He then drove the couple to an isolated spot. On the return journey, the snake charmer, who was sitting in the front passenger seat, removed a cobra from a box and forced it to bite the couple. They collapsed immediately and their driver was told to take them to a hospital. They arrived within an hour of being bitten, but they were both dead.

How the contract killer, snake charmer and son expected to get away with murder is difficult to understand. Admittedly, the hospital where the couple were taken did not have the facilities for toxicological analysis that would have identified snake venom in the bodies. But the killers instructed the driver to say the couple had been bitten by a snake. The fang marks were clearly visible, but their position on the victims' arms was suspicious. Cobras can rear up to quite a height before they bite, but bites as high as the arm are rare. Inevitably, the police became involved, they discovered the property dispute and the whole plot quickly unravelled from there.



Another murderer in China used snake venom to kill his wife, and avoided many of the pitfalls the Indian murderer fell into, but still didn't get away with it. This killer decided to do everything himself rather than involve any third parties. He bought 13 kraits, members of the *Bungarus* genus, some of which are among the most venomous snake species in the world. From these snakes he extracted roughly 2ml (0.07 fl. oz) of bungarotoxin and injected half of it into his wife's hip using a syringe. Ten minutes later she was suffering from blurred vision and five days later she was dead. An autopsy revealed the damage the venom had done to the woman's body, the injection mark was found on her hip and the purchase of the snakes

discovered. As with the case in India, no toxicological analysis was carried out, either because the facilities weren't available or the tests were not deemed necessary for prosecution, but it is possible.

Snake venom can be identified from wound extracts, tissue surrounding the bite, blood, urine and even vitreous humour (the fluid in the eye). This would have been useful to the toxicologist in Christie's novel, but it wasn't possible in 1935. Venom was first detected in a human body in 1977, using a technique called enzyme-linked immunosorbent assay (ELISA). This technique relies on the fact that antigens bind strongly to specific antibodies, the same idea as with the antivenoms mentioned above. An unknown snake venom can be identified by finding out which of a range of known snake-venom antibodies it binds to. By 1988, the technique had been refined so it was possible to detect viper venom directly from a tiny 200 microlitre sample of blood in less than 20 minutes.

Fortunately for the toxicologist in *Death in the Clouds*, he didn't have to try and detect the venom in Mme Giselle's body as he had the thorn. Even so, there wouldn't have been any tests available to identify the specific snake venom from this sample either. Residues on the thorn could have been given to animals to see its effects, but the haemorrhaging effect is not specific to boomslang venom. More detailed information about the toxicity of the venom and exactly how it clots the blood could perhaps help identify the poison, but this information was only discovered after Christie's novel had been published. However it was done in *Death in the Clouds*, identifying the poison that killed Mme Giselle only created more questions.

Agatha and venom

Christie's poisoner might have hoped that by using an obscure poison it might go undetected. Once boomslang venom has been detected, however, it dramatically reduces the number of suspects. As Christie herself points out, 'Only about two people in a hundred would be likely to have any knowledge of it and be able to lay their hands on it.' At least, that is what you are meant to think. A few of the fellow passengers on board are well travelled and may have had the opportunity to acquire snake venom in far-flung parts of the world. Christie also explains that snake venom is

researched for medical applications, expanding the list of suspects even further to include anyone with medical connections.

The huge variety of compounds within snake venoms, many of which have a very dramatic effect on human physiology in tiny quantities due to very specific interactions, makes them biologically and medically interesting. Christie specifically mentions that cobra venom has been investigated as a possible treatment for epilepsy. No snake-based epilepsy treatments have ever made it to market, but there are drugs based on key components of snake venom that are used to treat hypertension and reduce clotting in people with acute coronary syndrome. The cytotoxic effects of some snake venom proteins also make them interesting drug leads for the treatment of cancer, as well as antibiotic and antiviral agents.

In ‘The Speckled Band’ Doyle only has one real suspect – the trick is being able to prove how they did it. His snake murder was a creative invention meant to show off Sherlock’s deductive skills. In Christie’s story she explains early on *howdunit* – so the mystery is *whodunit*. Her use of snake venom is much more scientifically accurate, and is all the more impressive because of the scarcity of information that was available when she was writing the book. She only stretches credibility when it comes to the ease of identifying the venom in question.

Christie, like Doyle, was writing genre fiction with no obligation to facts or real-world experiences. All that matters is that her stories work on their own terms. Slavish attention to scientific detail could easily detract from Christie’s light-hearted whodunits. However, her attention to scientific accuracy in all her stories only adds more clues, red herrings and interest to the puzzles she set her readers.

Notes

* I’m not accusing Christie of plagiarism. She may have taken ideas from other writers, but her finished stories are her own creation.

* Published in the US under the title *Death in the Air*.

† The murder mystery author Arthur B. Reeve wrote another variation on this theme in ‘The Death Cry’, published in 1935. His locked room was in a hotel with the only access via a narrow window ledge. *Spoilers* Reeve’s murder weapon was not a snake but, I kid you not, a blood-sucking cat with canines coated in a deadly poison.

- * Readers were quick to point out Christie's mistake and she confessed to the error indirectly in one of her later novels, *Mrs McGinty's Dead*, through the character of Ariadne Oliver.
- * Christie did visit South Africa as part of a world tour. Perhaps she learned about the snake there.
- * This is not to say you should be complacent if a snake bites you in Europe. Deaths are not unheard of. If you are bitten seek medical attention as soon as you can.
- † There are approximately 600 different snake species to be found across the African continent.
- ‡ Is this what attracted Christie to the boomslang? A strange coincidence in head shape between the snake and her famous private detective?
- § One very dangerous green snake that is easily mistaken for a boomslang is the green mamba, *Dendroaspis angusticeps*.
- * A young South African man bitten by a boomslang died from cardiac arrest just eight hours later, but he had been bitten twice, presumably while handling the snake, and may have had an intercranial bleed.
- * Fitzsimons described his assistant's recovery as 'little short of miraculous' and attributed it to Williams's 'vigorous constitution, the result of very abstemious habits'.
- * Snakes can be trained but not much more than accustoming them to being fed with certain foods or being handled without biting their handler. Snake brains are much simpler than mammal brains. Commands like 'fetch', 'roll over' or 'bite the lady in seat number two' are probably beyond them.
- * This is thrombosis and can be very serious.
- * Fitzsimons supported the use of a tourniquet for boomslang bites, but then he also recommended injections of strychnine. His advice on snakebite treatment is very much out of date.
- * People with a history of previous exposure to horse serum and those with a history of infantile eczema or asthma are prone to this reaction. It can be treated with adrenaline.
- † Antivenoms are expensive. They require the maintenance of horses, or other suitable animals, and high-tech laboratories that can process the blood and isolate the antibodies. Then there is the cost, inconvenience and dangers of identifying, capturing, keeping and milking different snakes for their venom. Once produced, most antivenoms have to be kept cold until they are delivered to patients.
- * Even if their lives are saved, snakebite victims are often left with permanent aftereffects that can require prolonged, complicated and expensive treatment.

Appendix: Christie's Causes of Death

This table lists all the Agatha Christie novels and short stories in order of publication, and the cause of death in each of them. Agatha Christie's plays are not included, nor are the titles published under the name Mary Westmacott. Not all Christie books that were published in the UK were published in the US (and vice versa), especially the short-story collections, and these sometimes vary in composition. An interesting note: *Three Blind Mice* has never been published in the UK as it provides the basis for Christie's play, *The Mousetrap*, which is still running; the book's publication would give away the identity of the murderer.

In the table below, please note:

* Suicide ** Attempted murder
*** Medication withheld **** Invented drug

UK title	Methods of murder	US title
<i>The Mysterious Affair at Styles</i>	Strychnine	<i>The Mysterious Affair at Styles</i>
<i>The Secret Adversary</i>	Chloral hydrate Cyanide* Morphine**	<i>The Secret Adversary</i>
<i>The Murder on the Links</i>	Stabbed Morphine**	<i>The Murder on the Links</i>
<i>The Man in the Brown Suit</i>	Electrocuted Strangled	<i>The Man in the Brown Suit</i>

UK title	Methods of murder	US title
<i>Poirot Investigates</i>	Shot	<i>Poirot Investigates</i>
The Adventure of the 'Western Star'	Shot	The Adventure of the 'Western Star'
The Tragedy at Marsdon Manor	Blood poisoning	The Tragedy at Marsdon Manor
The Adventure of the Cheap Flat	Strychnine	The Adventure of the Cheap Flat
The Mystery of Hunter's Lodge	Shot*	The Mystery of Hunter's Lodge
The Million Dollar Bond Robbery	Hit on head	The Million Dollar Bond Robbery
The Adventure of the Egyptian Tomb	Trinitrine	The Adventure of the Egyptian Tomb
The Jewel Robbery at the Grand Metropolitan		The Jewel Robbery at the Grand Metropolitan
The Kidnapped Prime Minister		The Kidnapped Prime Minister
The Disappearance of Mr Davenheim		The Disappearance of Mr Davenheim
The Adventure of the Italian Nobleman		The Adventure of the Italian Nobleman
The Case of the Missing Will		The Case of the Missing Will

		The Chocolate Box The Veiled Lady The Lost Mine
<i>The Secret of Chimneys</i>	Shot	<i>The Secret of Chimneys</i>
<i>The Murder of Roger Ackroyd</i>	Arsenic Veronal* Stabbed	<i>The Murder of Roger Ackroyd</i>

UK title	Methods of murder	US title
<i>The Big Four</i>	Cyanide Throat cut Yellow Jasmine Electrocuted Run over Stabbed	<i>The Big Four</i>
<i>The Mystery of the Blue Train</i>	Strangled	<i>The Mystery of the Blue Train</i>
<i>The Seven Dials Mystery</i>	Chloral hydrate Shot	<i>The Seven Dials Mystery</i>
<i>Partners in Crime</i>		<i>Partners in Crime</i>
A Fairy in the Flat/ A Pot of Tea		A Fairy in the Flat/ A Pot of Tea
The Affair of the Pink Pearl		The Affair of the Pink Pearl
The Adventure of the Sinister Stranger		The Adventure of the Sinister Stranger
Finessing the King/ The Gentleman Dressed in Newspaper	Stabbed Electrocuted Hit on head	Finessing the King/ The Gentleman Dressed in Newspaper
The Case of the Missing Lady Blindman's Buff	Stabbed Arsenic** Ricin	The Case of the Missing Lady Blindman's Buff
The Man in the Mist The Crackler		The Man in the Mist The Crackler
The Sunningdale Mystery		The Sunningdale Mystery
The House of Lurking Death		The House of Lurking Death
The Unbreakable Alibi		The Unbreakable Alibi
The Clergyman's Daughter/The Red House		The Clergyman's Daughter/The Red House

UK title	Methods of murder	US title
The Ambassador's Boots		The Ambassador's Boots
The Man Who Was No. 16		The Man Who Was No. 16
<i>The Mysterious Mr Quin</i>	Strychnine	<i>The Mysterious Mr Quin</i>
The Coming of Mr Quin	Shot	The Coming of Mr Quin
The Shadow on the Glass	Shot	The Shadow on the Glass
At the 'Bells and Motley'	Drowned Drowned Poison gas** Shot Strangled	At the 'Bells and Motley'
The Sign in the Sky		The Sign in the Sky
The Soul of the Croupier		The Soul of the Croupier
The Man from the Sea		The Man from the Sea
The Voice in the Dark		The Voice in the Dark
The Face of Helen		The Face of Helen
The Dead Harlequin		The Dead Harlequin
The Bird with the Broken Wing		The Bird with the Broken Wing
The World's End Harlequin's Lane		The World's End Harlequin's Lane
<i>The Murder at the Vicarage</i>	Shot Sedative**	<i>The Murder at the Vicarage</i>
<i>The Sittaford Mystery</i>	Hit on head	<i>The Murder at Hazelmoor</i>
<i>Peril at End House</i>	Shot Cocaine*	<i>Peril at End House</i>
<i>The Thirteen Problems</i>	Arsenic	<i>The Tuesday Club Murders</i>
The Tuesday Night Club	Hit on head Stabbed	The Tuesday Night Club
Ingots of Gold		Ingots of Gold
The Blood-Stained Pavement		The Blood-Stained Pavement
The Idol House of Astarte		The Idol House of Astarte

UK title	Methods of murder	US title
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Motive v. Opportunity The Thumb Mark of St. Peter	Atropine Cyanide Drowned	Motive v. Opportunity The Thumb Mark of St. Peter
The Blue Geranium The Companion The Four Suspects A Christmas Tragedy The Herb of Death The Affair at the Bungalow	Pushed down stairs Bludgeoned Digitalis Drowned	The Blue Geranium The Companion The Four Suspects A Christmas Tragedy The Herb of Death The Affair at the Bungalow
Death by Drowning		Death by Drowning
<i>Lord Edgware Dies</i>	Stabbed Veronal	<i>Thirteen at Dinner</i>
<i>The Hound of Death</i>	House collapse Lightning	
The Hound of Death	Shot	
The Red Signal The Fourth Man The Gypsy The Lamp	Strangled Poisoned	
Wireless	Starvation Natural causes	
The Witness for the Prosecution	Heart attack Crowbar	
The Mystery of the Blue Jar	Cyanide**	
The Strange Case of Sir Arthur Carmichael	Hit by bus Hit by tube train	
The Call of Wings	Supernatural	
The Last Seance		
SOS		
<i>Murder on the Orient Express</i>	Stabbed	<i>Murder in the Calais Coach</i>

UK title	Methods of murder	US title
<i>The Listerdale Mystery</i> The Listerdale Mystery Philomel Cottage	Hyoscine	
The Girl in the Train	Hit on head	
Sing a Song of Sixpence	Cyanide	
The Manhood of Edward Robinson	Arsenic	
Accident	Fell off cliff	
Jane in Search of a Job	Stabbed	
A Fruitful Sunday		
Mr Eastwood's Adventure		
The Golden Ball		
The Rajah's Emerald		
Swan Song		
<i>Why Didn't They Ask Evans?</i>	Pushed off cliff Morphine	<i>The Boomerang Clue</i>
	Shot	
	Chloral hydrate	
<i>Parker Pyne Investigates</i>		<i>Mr. Parker Pyne, Detective</i>
The Case of the Middle-aged Wife		The Case of the Middle-aged Wife
The Case of the Discontented Soldier		The Case of the Discontented Soldier
The Case of the Distressed Lady		The Case of the Distressed Lady
The Case of the Discontented Husband		The Case of the Discontented Husband
The Case of the City Clerk		The Case of the City Clerk
The Case of the Rich Woman		The Case of the Rich Woman
Have You Got Everything You Want?		Have You Got Everything You Want?

UK title	Methods of murder	US title
The Gate of Baghdad	Stabbed Cyanide*	The Gate of Baghdad
The House of Shiraz	Fell from balcony	The House of Shiraz The Pearl of Price
The Pearl of Price	Strychnine	Death on the Nile The Oracle at Delphi
Death on the Nile		
The Oracle at Delphi		

Three Act Tragedy	Nicotine	Murder in Three Acts
Death in the Clouds	Snake venom Cyanide	Death in the Air
The A.B.C. Murders	Hit on head Strangled Stabbed	The A.B.C. Murders
Murder in Mesopotamia	Hit on head Carbon monoxide** Hydrochloric acid	Murder in Mesopotamia
Cards on the Table	Stabbed Anthrax Septicaemia Drowned Shot Veronal Hat paint	Cards on the Table
Dumb Witness	Phosphorus Chloral hydrate*	Poirot Loses a Client
Death on the Nile	Shot Stabbed	Death on the Nile
Murder in the Mews Murder in the Mews The Incredible Theft Dead Man's Mirror Triangle at Rhodes	Shot* Shot Strophanthin	Dead Man's Mirror Murder in the Mews Dead Man's Mirror Triangle at Rhodes
Appointment with Death	Digitoxin Shot*	Appointment with Death
Hercule Poirot's Christmas	Throat cut	Murder for Christmas

UK title	Methods of murder	US title
<i>Murder is Easy</i>	Hat paint Pushed out of window Pushed in canal Septicaemia Run over Arsenic Hit on head	<i>Easy to Kill</i>
<i>And Then There Were None/Ten Little Niggers</i>	Drowned Cyanide Hit on head Hanged* Shot Run over Axe Chloral hydrate Amyl nitrite*** Poisoned Starvation Exposure Medical malpractice	<i>And Then There Were None/Ten Little Indians</i>
	Stabbed Strychnine Cyanide Stabbed Shot Heart attack Stabbed	<i>The Regatta Mystery</i> The Regatta Mystery The Mystery of the Baghdad Chest How Does Your Garden Grow? Problem at Pollensa Bay Yellow Iris Miss Marple Tells a Story The Dream In a Glass Darkly Problem at Sea
<i>Sad Cypress</i>	Morphine	<i>Sad Cypress</i>

UK title	Methods of murder	US title
<i>One,Two, Buckle My Shoe</i>	Shot Medinal Procaine and adrenaline	<i>The Patriotic Murder/An Overdose of Death</i>
<i>Evil Under the Sun</i>	Strangled Arsenic	<i>Evil Under the Sun</i>
<i>N or M?</i>	Shot	<i>N or M?</i>
<i>The Body in the Library</i>	Strangled Digitalin**	<i>The Body in the Library</i>
<i>Five Little Pigs</i>	Coniine	<i>Murder in Retrospect</i>
<i>The Moving Finger</i>	Cyanide Stabbed	<i>The Moving Finger</i>
<i>Towards Zero</i>	Hit on head Heart attack Shot with	<i>Towards Zero</i>

	arrow	
<i>Death Comes as the End</i>	Pushed off cliff Poisoned Shot with arrow	<i>Death Comes as the End</i>
<i>Sparkling Cyanide</i>	Cyanide Carbon monoxide**	<i>Remembered Death</i>
<i>The Hollow</i>	Shot Cyanide*	<i>The Hollow/Murder After Hours</i>
<i>The Labours of Hercules</i> The Nemean Lion The Lernaean Hydra The Arcadian Deer The Erymanthian Boar The Augean Stables The Stymphalean Birds The Cretan Bull The Horses of Diomedes The Girdle of Hyppolita	Strychnine** Arsenic Stabbed Hit on head Atropine Shot*	<i>The Labours of Hercules</i> The Nemean Lion The Lernaean Hydra The Arcadian Deer The Erymanthian Boar The Augean Stables The Stymphalean Birds The Cretan Bull The Horses of Diomedes The Girdle of Hyppolita

UK title	Methods of murder	US title
The Flock of Geryon The Apples of Hesperides The Capture of Cerberus	Influenza/ typhoid/ gastric ulcer/ tuberculosis** Fell	The Flock of Geryon The Apples of Hesperides The Capture of Cerberus
<i>Taken at the Flood</i>	Hit on head Morphine Shot* Arsenic Fell off cliff Cyanide Strangled Hyoscine Shot Hit on head Heart attack Crowbar	<i>There is a Tide ...</i> <i>The Witness for the Prosecution and Other Stories</i> Accident The Fourth Man The Mystery of the Blue Jar The Mystery of the Spanish Shawl (aka Mr. Eastwood's Adventure) Philomel Cottage The Red Signal The Second Gong Sing a Song of Sixpence SOS Where There's a Will (aka Wireless) The Witness for the Prosecution
<i>Crooked House</i>	Eserine Digitalis Car crash	<i>Crooked House</i>

UK title	Methods of murder	US title
	Neglect Strangled Strangled Strophanthin Shot Pushed down stairs Hit on head	<i>Three Blind Mice and Other Stories</i> Three Blind Mice Strange Jest The Tape-Measure Murder The Case of the Perfect Maid The Case of the Caretaker The Third Floor Flat The Adventure of Johnny Waverly Four-and-Twenty Blackbirds The Love Detectives
<i>A Murder is Announced</i>	Shot Narcotics Strangled	<i>A Murder is Announced</i>
<i>They Came to Baghdad</i>	Stabbed Hit on head Stabbed Stabbed Cocaine	<i>They Came to Baghdad</i> <i>The Under Dog and Other Stories</i> The Under Dog

Formic acid**	The Market Basing Mystery
Arsenic	The Lemesurier Inheritance
Hit on head	The Cornish Mystery The King of Clubs The Submarine Plans The Adventure of the Clapham Cook
	The Plymouth Express The Affair at the Victory Ball

UK title	Methods of murder	US title
<i>Mrs McGinty's Dead/ Blood Will Tell</i>	Hit on head Poisoned Strangled	<i>Mrs McGinty's Dead</i>
<i>They Do It With Mirrors</i>	Shot Aconitine** Crushed Drowned	<i>Murder with Mirrors</i>
<i>After the Funeral/Murder at the Gallop</i>	Hatchet Arsenic**	<i>Funerals are Fatal</i>
<i>A Pocket Full of Rye</i>	Taxine Cyanide Strangled	<i>A Pocket Full of Rye</i>
<i>Destination Unknown</i>	Poisoned	<i>So Many Steps to Death</i>
<i>Hickory Dickory Dock</i>	Morphine Poisoned Hit on head Medinal	<i>Hickory Dickory Death</i>
<i>Dead Man's Folly</i>	Strangled Drowned	<i>Dead Man's Folly</i>
<i>4.50 from Paddington</i>	Strangled Arsenic Aconitine	<i>What Mrs McGillicuddy Saw</i>
<i>Ordeal by Innocence</i>	Hit on head Stabbed	<i>Ordeal by Innocence</i>
<i>Cat Among the Pigeons</i>	Shot Hit on head	<i>Cat Among the Pigeons</i>
<i>The Adventure of the Christmas Pudding</i>	Stabbed	
<i>The Adventure of the Christmas Pudding</i>	Hit on head Pushed down stairs	
<i>The Mystery of the Spanish Chest</i>	Shot	
<i>The Under Dog Four and Twenty Blackbirds</i>	Shot with arrow	
<i>The Dream</i>		
<i>Greenshaw's Folly</i>		

UK title	Methods of murder	US title
	Cyanide**	<i>Double Sin and Other Stories</i>
	Shot with arrow	Double Sin Wasps' Nest The Theft of the
	Supernatural Shot	Royal Ruby (aka The Adventure of the Christmas Pudding)
		The Dressmaker's Doll Greenshaw's Folly The Double Clue
		The Last Seance Sanctuary
<i>The Pale Horse</i>	Thallium Hit on head	<i>The Pale Horse</i>
<i>The Mirror Crack'd from Side to Side</i>	Calmo**** Cyanide Shot	<i>The Mirror Crack'd</i>
	Sleeping tablets	
<i>The Clocks</i>	Strangled Stabbed	<i>The Clocks</i>
<i>A Caribbean Mystery</i>	Depressant drug Stabbed	<i>A Caribbean Mystery</i>
	Atropine** Drowned	
	Carbon monoxide	
<i>At Bertram's Hotel</i>	Shot Car crash*	<i>At Bertram's Hotel</i>
<i>Third Girl</i>	Pushed out of window Stabbed	<i>Third Girl</i>
<i>Endless Night</i>	Cyanide Drowned Stabbed	<i>Endless Night</i>
	Strangled	
<i>By the Pricking of My Thumbs</i>	Morphine	<i>By the Pricking of My Thumbs</i>

UK title	Methods of murder	US title
<i>Hallowe'en Party</i>	Drowned Stabbed Cyanide*	<i>Hallowe'en Party</i>

<i>Passenger to Frankfurt</i>	Shot Strychnine**	<i>Passenger to Frankfurt</i>
	House collapsed Poisoned Starvation Natural causes Cyanide** Hit by bus Hit by tube train	<i>The Golden Ball and Other Stories</i> The Listerdale Mystery The Girl in The Train The Manhood of Edward Robinson Jane in Search of a Job A Fruitful Sunday The Golden Ball The Rajah's Emerald Swan Song The Hound of Death The Gypsy The Lamp The Strange Case of Sir Arthur Carmichael The Call of Wings Magnolia Blossom Next to a Dog
<i>Nemesis</i>	Poisoned Crushed Strangled	<i>Nemesis</i>
<i>Elephants Can Remember</i>	Shot Hit on Head	<i>Elephants Can Remember</i>
<i>Postern of Fate</i>	Digitalis Hit on head	<i>Postern of Fate</i>

UK title	Methods of murder	US title
<i>Poirot's Early Cases</i>	Stabbed Cocaine	<i>Hercule Poirot's Early Cases</i>
The Affair at the Victory Ball	Arsenic	The Affair at the Victory Ball
The Adventure of the Clapham Cook	Hit on head Formic acid**	The Adventure of the Clapham Cook
The Cornish Mystery The Adventure of	Stabbed Trinitrine	The Cornish Mystery The Adventure of
Johnnie Waverly The Double Clue The King of	Shot	Johnnie Waverly The Double Clue The King of
Clubs The Lemesurier	Cyanide**	Clubs The Lemesurier
Inheritance The Lost Mine	Heart attack Stabbed	Inheritance The Lost Mine
The Plymouth Express The Chocolate Box The	Strychnine	The Plymouth Express The Chocolate Box The
Submarine Plans The Third Floor Flat Double		Submarine Plans The Third Floor Flat Double
Sin		Sin
The Market Basing Mystery		The Market Basing Mystery
Wasps' Nest The Veiled Lady Problem at Sea		Wasps' Nest The Veiled Lady Problem at Sea
How Does Your Garden Grow?		How Does Your Garden Grow?
<i>Curtain: Poirot's Last Case</i>	Arsenic Morphine Shot Cyanide Hit on head Physostigmine Amyl nitrite***	<i>Curtain: Poirot's Last Case</i>
<i>Sleeping Murder</i>	Strangled Sleeping tablets Chloral hydrate**	<i>Sleeping Murder</i>

UK title	Methods of murder	US title
<i>Miss Marple's Final Cases</i>	Shot	
Sanctuary Strange Jest	Strangled	
Tape-Measure Murder The Case of the	Strophanthin	
Caretaker	Stabbed	
The Case of the Perfect Maid		
Miss Marple Tells a Story		
The Dressmaker's Doll In a Glass Darkly		
<i>Problem at Pollensa Bay</i>	Shot Cyanide	
Problem at Pollensa Bay	Poisoned**	
The Second Gong Yellow Iris	Hit on head	
The Harlequin Tea Set The Regatta Mystery The Love		
Detectors Next to a Dog Magnolia Blossom		
	Fell off cliff Shot* Stabbed Poisoned**	<i>The Harlequin Tea Set</i> The Edge The Actress

	While the Light Lasts The House of Dreams The Lonely God Manx Gold Within a Wall The Mystery of the Spanish Chest The Harlequin Tea Set
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UK title	Methods of murder	US title
<i>While the Light Lasts</i>	Fell off cliff	The House of Dreams
The Edge	Stabbed Shot*	The Actress
Christmas Adventure		The Lonely God
Within a Wall		Manx Gold
The Mystery of the Baghdad Chest		
While the Light Lasts		

Selected Bibliography

Below is a selection of interesting books related to the topics I've discussed in this book. A full list of citations (including many more detailed references from the academic literature) can be found on my website, www.harkup.co.uk.

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